

IdleWorlds Fantasy Skin

Technical handover, integration guide, maintenance manual and security review

Extension	idleworlds-fantasy-skin v1.6.0 (Chrome MV3 content script)
Source commit	0590816 (branch main)
Prepared	17-18 September 2026 · PDF generated 2026-09-18
Documents	9 – README.md, 01-project-overview.md, 02-feature-inventory.md, ...
Source of truth	handoff/reskin-technical-handover/*.md in the repository

Nothing in this package was verified against the live game. Statements that need a check on the running application, its server or with the IdleWorlds team are marked **Verify** or **Assumption**.

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IdleWorlds Fantasy Skin — Technical Handover

A complete technical handover, integration guide, maintenance manual and security review for the **IdleWorlds Fantasy Skin**: a Chrome extension that reskins `idleworlds.com` into a dark-fantasy interface without access to the game's source.

Snapshot	Commit <code>059081671bf3af4490a025121f25f13ada142b7e</code> on <code>main</code> , extension version 1.6.0
Prepared	17-18 September 2026, from a line-by-line reading of every shipped file, plus generated inventories and a full test run
Audience	IdleWorlds developers who will audit, maintain, modify or officially integrate the skin
Status of this package	Untracked files in <code>handoff/reskin-technical-handover/</code> ; not committed . No other file in the repository was changed

How to read this package

Part	File	Read it for
1	Project overview	Purpose, scope, UX, how it integrates with the game, architecture diagrams, module map, repository layout, asset inventory and provenance, dependencies, browser floor, network surface, build and CI, documentation drift
2	Feature inventory	All 55 features (F-01...F-55), each with what, why, where, dependencies, runtime, states and fallbacks, responsive, motion, interaction, accessibility, edge cases and tests
3	CSS architecture	Delivery pipeline, cascade model, <code>!important</code> strategy and removal procedure, design tokens, every stylesheet section by section, selectors, media and container queries, animations, fragile rules, conflicts, browser behaviour, performance, recommendations
4	Constraints and workarounds	The 24 constraints that shaped the code, design versus workaround, and how full access would replace each
5	JavaScript documentation and safety review	Threat model, complete network/storage/sink/listener census, 29 per-feature reviews (J-01...J-29) with explicit safety statements, the findings register , the hardening plan
6	Integration and maintenance	Install, kill switch, integration options, start-up order, every constant, customisation how-tos, all assumptions about the game , break/diagnose matrix, troubleshooting, migration plan, owner decisions
7	Testing and verification	The 47 suites, current results, manual test plan (MT-01...MT-30), responsive, accessibility, cross-browser, performance, regression and security testing, recommended tests, the final checklist
A	File inventory	Every tracked file (556): size, whether it ships, runtime use, description (generated)
B	CSS rule index	Every CSS rule (1,373): selectors, specificity, context, properties, <code>!important</code> , host/skin hooks, fragility, comment (generated)
C	CSS dependency surface	Game hooks, skin hooks, fragile patterns, <code>:has()</code> , <code>!important</code> by property, custom properties, queries, animations, pseudo-classes and pseudo-elements (generated)
D	DOM contract	Every attribute (all 141 names), class, element, inline property, event, storage key and highlight the skin owns, with owner and teardown
Tools	<code>tools/</code>	Reproducible generators: <code>css-inventory.mjs</code> , <code>css-superseded.mjs</code> , <code>hook-census.mjs</code> , <code>js-api-census.mjs</code> , <code>file-inventory.mjs</code>
Evidence	<code>evidence/npm-test-2026-09-17.log</code>	The full <code>npm test</code> output (local path redacted)

Evidence labels used throughout:

Label	Meaning
[Source]	Confirmed in the cited file and line
[Generated]	Produced by a script in <code>tools/</code>
[Test run]	Observed in the test run

Label	Meaning
[Project record]	Stated in the project's own notes; not re-measured
[Assumption] / [Verify]	Needs confirmation on the live game or with the owners

Nothing in this package was verified on the live game.

EXECUTIVE SUMMARY

What it is. An MV3 extension with:

- **permissions:** `storage` only, on two host patterns;
- **two content scripts:** the bundle, in the isolated world at `document_idle` ; and a 64-line page-world probe that detects React hydration;
- **no background worker, UI, telemetry or runtime dependencies.**

It turns the game's rendered DOM into the fantasy look through six reversible mechanisms:

1. 7 injected stylesheets;
2. semantic `data-iw-*` role attributes;
3. four classes;
4. owned inline styles;
5. appended decorative elements;
6. CSS highlight registrations.

How it works.

- **Observation.** A single budgeted `MutationObserver` (60 elements per frame) emits events to classifiers and renderers.
- **Classification.** They recognise surfaces from the game's structure and **English copy**, because the game ships almost no stable hooks.
- **Data.** Item data comes from the public `items.json` . The dashboard Village frame reads the authenticated `/api/player` endpoint.
- **Reversibility.** Everything is removed by a storage-key kill switch.

Engineering quality. The code is unusually disciplined for an overlay:

- compare-before-write everywhere;
- measured performance fixes;
- per-module attribute namespaces;
- exact restoration of native inline styles;
- 47 regression suites with negative controls.

`npm test` passed (exit 0, 3 min 26 s).

The main costs are structural, and belong to the extension approach:

- **Fragility against game copy and markup changes:** 37 documented assumptions (§6.5).
- **A heavy cascade:** 3,879 `!important` declarations, at least 350 of them provably dead.
- **Very large art:** a 142.8 MB zip, 60.8 MB of it unreachable; sprite atlases that decode to about 126 MiB.

Safety. No High-severity issue was found. The census demonstrates:

- no dynamic or remote code;
- no third-party requests;
- no cookie or credential access;
- no messaging or globals;
- no synthetic interaction with the game;

- game text treated strictly as text.

The disclosed issues are below.

KEY FINDINGS

Full register: §5.6.

ID	Severity	Finding	Recommended action
SEC-04	MEDIUM	The dashboard Village frame polls the authenticated <code>/api/player</code> endpoint (session cookies + league header) every 5 min	IdleWorlds decision; add backoff (H-14) or remove (B-08)
SEC-07	MEDIUM	A real browser profile (saved logins, autofill, wallet) and business documents sit inside the working tree (git-ignored)	Remove before sharing any copy of the folder (H-20)
FUN-01	MEDIUM	The game's "cycle XP display" button is disabled and hidden by skin CSS, so a game control is unreachable	Design decision B-01; add operability test T-01
CONT-01	MEDIUM	Skin-authored gameplay statements (World Boss notice, orb and drop copy) appear as if they were game text	Replace with game-provided copy (B-06)
A11Y-01	MEDIUM	Text shown through CSS generated content over hidden native text, and mirrors not hidden from assistive technology: duplicate or mismatched announcements	B-10; screen-reader test
PERF-01 / PERF-07	MEDIUM	Infinite decorative animations; sprite atlases up to 1280×19,584 px	H-17; H-24
SIZE-01	MEDIUM	60.8 MB unreachable <code>exact-v3</code> art in the release	H-18 (§6.8)
SEC-01	LOW	Two catalogue fields (<code>craft_level</code> , <code>tier</code>) are concatenated unescaped into the tooltip HTML	H-01 (one-line fix) + H-02 + H-03
FUN-04	LOW	After a kill-switch round trip, collapsed panels can lose their title	H-08 + test T-04
TEST-01	LOW	The quest signature suite tests a stale copy of the module logic	T-09

Decisions needed from IdleWorlds: API use, authored content, the Toolkit link, the XP-cycle control, art rights, AI-generated art, reduced-motion intent, integration path, distribution, and font attribution (§6.9).

QUICK FACTS

Metric	Value
Runtime JavaScript	35 files, 11,869 lines (33 modules + entry + page probe); 0 import cycles
Stylesheets	14 active (+1 inactive), injected as 7 <code><style></code> elements; 1,373 rules; 5,888 declarations; 3,879 <code>!important</code> ; 37 <code>:has()</code> rules; highest specificity (0,30,3)
Bundle	<code>dist/content.bundle.js</code> : 668,309 bytes, unminified; 48.6% of it is CSS text
Skin DOM vocabulary	141 <code>data-iw-*</code> / <code>data-fs-*</code> attribute names; 4 classes on game nodes; 6 storage keys; 6 event types
Network	2 game endpoints (<code>items.json</code> , <code>/api/player</code>) + packaged assets; 0 third-party runtime origins
HTML sinks	3 <code>innerHTML</code> writes (1 constant; 1 fully escaped; 1 with SEC-01)
Tests	47 suites + sprite-window audit; all passed on 2026-09-17
Tracked files	556 (148.4 MB); 325 ship (144.6 MB); zip 142.8 MB
Effective browser floor	Chrome 111 (the manifest declares none)

REGENERATING THE GENERATED PARTS

Run from the repository root:

```
node handoff/reskin-technical-handover/tools/css-inventory.mjs
```

```
node handoff/reskin-technical-handover/tools/css-superseded.mjs
```

```
node handoff/reskin-technical-handover/tools/hook-census.mjs
```

```
node handoff/reskin-technical-handover/tools/js-api-census.mjs
```

```
node handoff/reskin-technical-handover/tools/file-inventory.mjs
```

The scripts are dependency-free (Node.js only). They read `src/`, `assets/`, `manifest.json` and `git ls-files`, and refuse to report "clean" when they find nothing.

NOTES

- `handoff/technical/` contains **separate, earlier draft documents produced by another tool**. This package did not modify or rely on them.
- The repository's own engineering notes (`CLAUDE.md`, `docs/traps/*.md`) remain the best record of *why* individual decisions were made; this package cites them where relevant.
- Do not commit or distribute `build-tools/header-live-profile/`, `output/`, `tmp/` or `release/` (§1.7.2).

1. PROJECT OVERVIEW

Snapshot. Everything in this chapter describes commit `059081671bf3af4490a025121f25f13ada142b7e` (branch `main`, extension version **1.6.0**), inspected on **17 September 2026**. The source tree (`src/`, `build-tools/`, `tests/`, `manifest.json`, `package.json`) was clean at that commit. Line numbers in links refer to that commit.

Evidence labels used throughout this package

Label	Meaning
[Source]	Confirmed by reading the file and line cited.
[Generated]	Produced by a script in <code>tools/</code> from the source files, and reproducible.
[Test run]	Confirmed by running <code>npm test</code> in the project on 2026-09-17 (see Chapter 7).
[Project record]	Stated in the project's own notes (<code>CLAUDE.md</code> , <code>docs/traps/*.md</code> , asset READMEs). These are historical measurements that were not re-measured for this handover.
[Assumption] / [Verify]	Cannot be confirmed from the repository. It needs a check against the live IdleWorlds application, its server, or its owners.

1.1 PURPOSE

The **IdleWorlds Fantasy Skin** is a Chrome **Manifest V3 browser extension** that restyles the live web application at `https://idleworlds.com` (and `https://www.idleworlds.com`) into a dark-fantasy RPG interface. It replaces the game's flat Tailwind "dashboard" look with forged-metal frames, parchment-and-brass typography, illustrated item and skill artwork, per-zone colour palettes, and denser, more game-like layouts for the main surfaces (header, navigation, skill cards, quests, inventory, world bosses, village, market and the activity feeds).

It is built by a third party **without access to the IdleWorlds source code**. Everything it does is layered over the DOM that the IdleWorlds React application renders in the player's browser.

The project states its governing rule as **"Presentation only"** (`CLAUDE.md`):

- 1. never change what a game control does;
- 2. never reparent (move) a node React owns — decoration may only be *appended*;
- 3. every attribute, element and inline style the skin adds must be reversible by its teardown;
- 4. one `MutationObserver` only;
- 5. never destroy information the game paints (active tab, equipped item, red/green state).

This handover shows (Chapters 2, 3 and 5) that the implementation follows these rules closely. It also records the specific places where the effect goes further than "presentation". One example: a game button is hidden by CSS and becomes unreachable (finding `FUN-01`). Another: the skin performs an authenticated read of a game API endpoint (`SEC-04`).

1.2 SCOPE

In scope (what the skin changes):

Surface	What the player sees	Primary implementation
Whole page	Dark "Ashen Iron" palette, self-hosted fonts (Cinzel, Barlow, Crimson Text), flattened Tailwind radii, forged control plates, themed scrollbars, focus rings	<code>base.css</code> , <code>ui-system.css</code>
Per-zone theming	Nine environment palettes (glacial, infernal, verdant, ...) chosen from the player's current zone. Zone-specific header painting (desktop strip and mobile portrait crop), frame filigree, separators and button art	<code>HeaderRenderer.js</code> , <code>zoneThemes.js</code> , <code>SkillsArtService.js</code> , <code>base.css</code>

Surface	What the player sees	Primary implementation
Header	Live player header reorganised: crest, smoked-glass identity block, utility rail, vitals/buff tile grid	<code>HeaderRendererer.js</code> , <code>header.css</code>
Navigation + zone bar + announcement	One merged framed "chrome" box. Route tabs with the active tab preserved, zone title and actions, and an added outbound Toolkit link	<code>UIFoundation.js</code> , <code>HeaderChrome.js</code> , <code>ui-system.css</code> , <code>header.css</code>
Every <code>.panel</code>	The shared "forged frame" (textured ground, gold hairline, four-corner filigree). A per-panel collapse control, remembered per panel	<code>UIFoundation.js</code> , <code>CollapsibleFrames.js</code> , <code>ui-system.css</code> , <code>collapsible.css</code>
Current Action / Action Log / World Chat	Framed activity panels. The Current Action bar is smoothed into continuous motion; the feeds are restyled	<code>UIFoundation.js</code> , <code>ProgressCadence.js</code> , <code>ui-system.css</code>
Skill Actions cards	The "V2" skill card: medallion with XP ring, level badge, action title with base-XP chip, material/source/detail frame, unmet-requirement note, themed square action button with discipline icon, fitted label, smoothed activity fill, long-action countdown, recipe pager	<code>DOMWatcher.js</code> , <code>SkillPanelRenderer.js</code> , <code>SkillCardDesignController.js</code> , <code>SkillsArtService.js</code> , <code>skillpanel.css</code> , <code>skillcard-v2.css</code> , <code>card-buttons.css</code>
Quests	Three-zone quest card with a themed sigil (objective item icon, completion %), reward chip, sprite-framed Turn In / Skip, and a hovercard on the objective item	<code>QuestPanelRendererer.js</code> , <code>skillpanel.css</code>
Inventory	Each item row replaced visually by a rich overlay: sprite icon (with +N badge), category colour, stat chips, owned-item details (loadout, sockets, rolled stats), requirements, quantity. The native action buttons are kept and restyled. Themed panel chrome with a single-row filter bar	<code>InventoryRendererer.js</code> , <code>InventoryModel.js</code> , <code>itemDisplay.js</code> , <code>AtlasService.js</code> , <code>inventory.css</code>
Item information	Rich item hovercard (stats, acquisition, wiki link) on inventory icons, quest objectives, village buildings, boss rewards, and item names detected in the game's prose	<code>TooltipEngine.js</code> , <code>NameScanner.js</code> , <code>ItemDatabase.js</code> , <code>tooltip-engine.css</code>
World Bosses & Zone Control	Illustrated encounter cards, compact Prejoin/Queued labels, a "Possible Rewards" disclosure derived from item data, and an animated "Dominion Ward" visual for zone control	<code>WorldBossPanels.js</code> , <code>ui-system.css</code>
Market	Smoked-glass listing rows	<code>UIFoundation.js</code> , <code>ui-system.css</code>
Village route	Housing hero with house art and tier pips, add-on slot cards with building art, art in the install picker, and building hovercards	<code>VillagePanels.js</code> , <code>villageBuildings.js</code> , <code>ui-system.css</code>
Dashboard (new)	A Village scene frame added below Skill Actions: five building plots plus a ledger of per-building and total stat benefits	<code>VillageScene.js</code> , <code>VillageLedger.js</code> , <code>village-scene.css</code>
Pop-ups / modals	Generic forged frame on any viewport-covering dialog	<code>OverlayFramer.js</code> , <code>overlay.css</code>
Small controls	Theme-atlas "ghost metal" art on nav tabs, filters, pagers, row actions, the chat Send button and the loadout I/II toggle	<code>CompactButtons.js</code> , <code>compact-buttons.css</code>

Out of scope (the skin does not do these) [Source]:

- It implements **no gameplay**: no actions, purchases, sales, joins, equips or crafting. It never calls a game action endpoint, and never clicks or submits on the player's behalf.
- It has **no background service worker, popup, options page, telemetry, analytics, remote code, or server of its own** (`manifest.json`).
- It does not reach the Equipment Window, Mailbox, Notifications, Settings, Supporter Pack, Invite Friends or Profile panels. The one exception: if they open as a full-screen modal, the generic overlay frame applies (`docs/traps/classification.md` coverage note).
- It is not integrated into the IdleWorlds build. It cannot use the game's Tailwind configuration, component props, stores or events.

1.3 INTENDED USER EXPERIENCE

- **Instant, non-destructive activation.** The skin boots once the page has hydrated, typically within ~0–4 s of load (`HydrationGate.js`). It decorates what is on screen and follows every React update. If the game changes a value, the skin re-

reads it: the skin never computes or stores game state.

- **Every native control still works.** Buttons keep their handlers, labels and disabled states. Where the skin hides the game's own text (to redraw a cleaner label), that text stays in the DOM as the accessible name. The exception is FUN-01, where a control is hidden entirely.
- **State stays legible.** Active tabs, equipped items, met/unmet requirements, disabled buttons, team colours and quest readiness are read from the game's own classes and attributes and restyled, not flattened (rule 5).
- **Reduced motion is honoured.** Under `prefers-reduced-motion: reduce`, transitions and animations are disabled (skin-wide, and page-wide through `base.css`; see FUN-03).
- **Phones and tablets are first-class.** The skin's own breakpoints are aligned to the game's Tailwind breakpoints (640/768/1024/1280). Component-level container queries are used where panel width is not a function of viewport width.
- **Reversible at any moment.** Setting `chrome.storage.local['iw-skin-enabled'] = false` removes the whole skin from the live page without a reload. Setting it back to `true` rebuilds it (see §6.1.4).

1.4 HOW THE RESKIN INTEGRATES WITH IDLEWORLDS

1.4.1 Delivery vehicle

The integration is entirely client-side, through two content scripts declared in `manifest.json` (lines 13–37) [Source]:

#	Script	World	Runs at	Frames	Purpose
1	<code>dist/content.bundle.js</code>	ISOLATED (default content-script world)	<code>document_idle</code>	top frame only	The whole skin: styles, classifiers, renderers, data services
2	<code>src/page/hydration-signal.js</code>	MAIN (the page's own JavaScript world)	<code>document_start</code>	top frame only	Watches React's root for hydration completion and writes one attribute, <code>data-iw-page-hydrated</code> , on <code><html></code>

- **Permissions:** `storage` only. **Host permissions:** `https://idleworlds.com/*`, `https://www.idleworlds.com/*`.
- **Web-accessible resources:** `assets/*` plus four PNG globs for button atlases. These are required because the stylesheets are injected into the page, so image and font URLs are fetched with `chrome-extension://<id>/assets/...` URLs that the page must be allowed to load.

1.4.2 The integration contract (what the skin relies on)

The skin reads the rendered DOM and relies on these observable facts about IdleWorlds. Each is documented as a fragility in §6.5:

1. The app is a React (Next.js) single-page app that **hydrates** server HTML. `hydration-signal.js` reads React's private root fields (`__reactContainer$...` → `stateNode.current.memoizedState.isDehydrated`) [Source].
2. It ships very few stable hooks: `.panel`, `.compact-panel`, `.compact-row`, `#current-action-panel`, `data-skin`, `header`, and some named classes such as `.header-icon-btn`, `.stat-chip` and `.header-player-name` [Source; Project record].
3. Most surfaces are identified from **English copy**: "Inventory", "Skill Actions", "World Bosses", "Village Add-ons", "Zone N:", "Turn In", "Reward:", "Base reward:", "Lv N - X% • ... to go", "Message World Chat", "Current tier: N", filter labels, action verbs and more [Source].
4. The whole panel stack is rendered **twice** (a wide `hidden xl:grid` copy and a narrow `xl:hidden` copy). Which copy is visible swaps at Tailwind's `xl` breakpoint, 1280 px [Source: Viewport.js comment; Project record].
5. Progress bars are `width: N%` inline styles rewritten once per tick [Source].
6. `https://idleworlds.com/items.json` is a public item catalogue with fields such as `item_id`, `name`, `category`, `subcategory`, `tier`, `effects_raw` and `atk` [Source; build-tools/audit-items-contract.mjs].
7. `GET /api/player?section=<route>&scope=core` returns `player.housing.tier` and `player.villageAddons`. `{totalSlots, installed[]}` for the logged-in session. The league is selected by an `x-idleworlds-league` request header that the page normally adds by patching `fetch` [Source: VillageScene.js:43-66, 312-316; Project record: docs/traps/village.md].

1.4.3 How presentation is applied

The skin changes what the player sees through six mechanisms only [[Source](#)]:

Mechanism	Where it lives	Reversible by
Injected <code><style data-iw-style></code> elements (7 of them, in fixed order)	<code>StyleInjector.inject()</code>	<code>StyleInjector.removeAll()</code>
Semantic role attributes on game nodes (<code>data-iw-ui="section-frame"</code> , <code>data-iw-skill-role="action-button"</code> , ...) — 141 distinct <code>data-iw-*</code> / <code>data-fs-*</code> names [Generated : <code>tools/hook-census.mjs</code>]	Classifiers/renderers	each module's <code>clear*()</code>
Classes on game nodes (only a few: <code>fs-skill-panel</code> , <code>fs-skill--<type></code> , <code>fs-skills-section-frame</code> , <code>fs-quest-panel</code>)	<code>SkillPanelRenderer</code> , <code>QuestPanelRenderer</code>	<code>clearPanelChrome</code> , <code>clearCard</code>
Inline style properties on game nodes, owned property-by-property (<code>InlineStyleOwner</code>) or written as custom properties (<code>--iw-...</code> , <code>--fs-...</code>)	<code>BackgroundPainter</code> , <code>SkillPanelRenderer</code> , <code>InventoryRenderer</code> , <code>UIFoundation</code> , <code>HeaderRenderer</code> , <code>QuestPanelRenderer</code> , <code>SkillCardDesignController</code> , <code>SkillsArtService</code>	owner <code>restore*()</code> / <code>removeProperty</code>
Skin-owned elements appended/inserted into game containers (overlays, medallions, labels, toggles, the Toolkit link, the Village scene, ...)	many renderers	each module's <code>clear*()</code> removes them
CSS Custom Highlight API registrations (<code>iw-item-name</code> , <code>iw-skill-ingredient-met</code>) — paint without DOM nodes	<code>NameScanner</code> , <code>SkillPanelRenderer</code>	<code>CSS.highlights.delete</code>

No game node is moved or replaced, and no game text node is rewritten. The smoke suite asserts that game text nodes are never detached [[Test run](#)].

1.5 ARCHITECTURE

1.5.1 Component diagram

1.5.2 Layers and responsibilities

Layer	Modules	Responsibility	Owns DOM?
Entry & lifecycle	<code>content.js</code> , <code>HydrationGate.js</code> , <code>hydration-signal.js</code>	Read the enabled flag, wait for hydration, boot/teardown, keep the kill-switch listener alive	No (only <code><style></code> elements through <code>StyleInjector</code> , and the hydration latch)
Runtime utilities	<code>Runtime.js</code> , <code>StyleInjector.js</code> , <code>InlineStyleOwner.js</code> , <code>Viewport.js</code> , <code>ProgressCadence.js</code>	Error isolation, extension storage, asset URLs, animation-frame scheduling, style injection, reversible inline styles, rendered-state checks and breakpoint epoch, tick measurement	No
Observation	<code>DOMWatcher.js</code>	The single <code>MutationObserver</code> : collects dirty rows, panels and subtrees, then emits typed <code>CustomEvent</code> s on <code>document</code> within a 60-element-per-frame budget. It also runs skill-type detection	No
Data services	<code>ItemDatabase.js</code> , <code>AtlasService.js</code> , <code>SkillsArtService.js</code> , <code>InventoryModel.js</code> , <code>itemDisplay.js</code> , <code>normaliseItemName.js</code> , <code>aliases.js</code> , <code>zoneThemes.js</code> , <code>villageBuildings.js</code>	Item catalogue, sprite geometry, art indexes, pure parsing and formatting, and generated lookup tables	No (<code>AtlasService</code> and <code>SkillsArtService</code> paint inline backgrounds and custom properties on hosts their callers pass in)
Classifiers & renderers	<code>UIFoundation.js</code> , <code>HeaderRenderer.js</code> , <code>HeaderChrome.js</code> , <code>InventoryRenderer.js</code> , <code>SkillPanelRenderer.js</code> , <code>SkillCardDesignController.js</code> , <code>QuestPanelRenderer.js</code> , <code>WorldBossPanels.js</code> , <code>VillagePanels.js</code> , <code>VillageScene.js</code> , <code>VillageLedger.js</code> , <code>CollapsibleFrames.js</code> , <code>CompactButtons.js</code> , <code>OverlayFramer.js</code> , <code>BackgroundPainter.js</code>	Recognise game surfaces, write roles, append decorations, and restore everything on teardown	Yes, each in its own attribute namespace
Item information UI	<code>TooltipEngine.js</code> , <code>NameScanner.js</code>	One shared hovercard element and delegated listeners. Item-name detection in prose, using live <code>Range</code> s and CSS highlights	<code>#iw-tip</code> (skin-owned) plus ARIA attributes on triggers
Presentation	<code>src/styles/*.css</code> (14 active sheets, concatenated into 7 style elements)	All visual treatment	—

1.5.3 Boot sequence

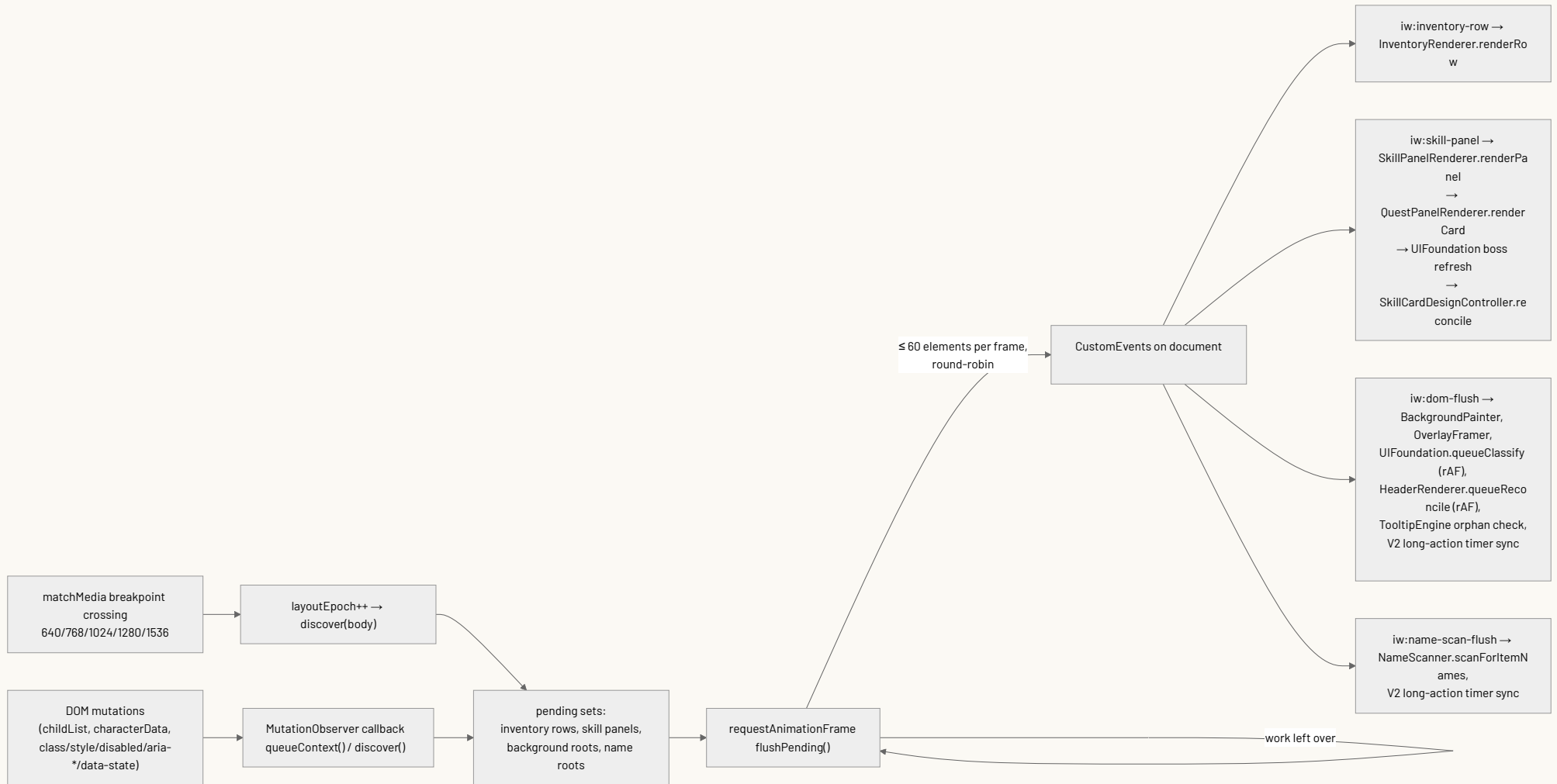


Syntax error in text
mermaid version 11.15.0

Why the order matters [Source: content.js comments; static-invariants test]:

- Consumers must be subscribed **before** `startWatcher()`. The watcher's initial `discover(document.body)` emits events synchronously on the next animation frame, and a missing listener would lose them.
- `ui-system` must be the **last** style element. Its generic control rules are tuned to win ties against every earlier sheet.
- `initSkillCardDesignController()` runs on **every** boot, after the page-lifetime listeners. Its `iw:skill-panel` listener is registered after `SkillPanelRenderer`'s, so the V2 pass sees the `fs-skill-panel` class the renderer has just added.
- `setRuntimeActive(true)` runs first; teardown calls `setRuntimeActive(false)` **last**. Every `guard()` / `guardEach()` callback is inert while the runtime is inactive, but teardown's own guarded cleanup must still run.

1.5.4 The reconciliation loop (steady state)



Every renderer is written to be **idempotent**: it compares before writing, caches its "which node plays which role" resolutions, and revalidates those caches cheaply. This keeps a quiet page quiet. `tests/flush-quiescence.test.mjs` asserts **zero** mutations and flushes on a page whose game timers are stopped [[Test run](#)].

The classify pass inside `UIFoundation.queueClassify()` has a fixed order that later steps depend on [[Source](#): `UIFoundation.js:1362-1401`]:

- 1. `classifyMainNav` → 2. `classifyZoneBar` → 3. `classifyActivityPanels` → 4. `classifySectionFrames` → 5. `classifyHeaderChrome` → 6. `classifyDailyBoost` → 7. `classifyBossCards` → 8. `classifyMarket` → 9. `classifyVillagePanels` → 10. `reconcileVillageScene` (async) → 11. `decorateCompactButtons` → 12. `decorateCollapsibleFrames` (**last**: it reads every earlier mark).

1.5.5 Event bus

All inter-module communication inside the bundle happens through a handful of `CustomEvent` s dispatched on `document` (`bubbles: false`), plus direct function calls along the import graph. No event carries game data beyond element references and counters [[Source](#)].

Event	Dispatched by	detail	Consumers (registration order)
<code>iw:inventory-row</code>	<code>DOMWatcher.flushPending</code> (<code>DOMWatcher.js:45</code>)	<code>{ row, reason: 'reconcile' }</code>	<code>InventoryRenderer.renderRow</code>
<code>iw:skill-panel</code>	<code>DOMWatcher.flushPending</code>	<code>{ panel, skill, reason }</code> where <code>skill</code> is the detected discipline or <code>'unknown' / 'locked'</code>	<code>SkillPanelRenderer.renderPanel</code> → <code>QuestPanelRenderer.renderCard</code> → <code>UIFoundation</code> (boss cards inside the resolved World Bosses root) → <code>SkillCardDesignController.reconcile</code>
<code>iw:dom-flush</code>	<code>DOMWatcher.flushPending</code>	<code>{ roots }</code> (subtrees that changed, for background painting)	<code>content.js</code> (<code>BackgroundPainter + OverlayFramer</code>), <code>UIFoundation.queueClassify</code> , <code>HeaderRenderer.queueReconcile</code> , <code>TooltipEngine</code> (orphaned-anchor check), <code>SkillCardDesignController.syncTimersOnTick</code>
<code>iw:name-scan-flush</code>	<code>DOMWatcher.flushPending</code>	<code>{ roots }</code>	<code>content.js</code> → <code>NameScanner.scanForItemNames</code> ; <code>SkillCardDesignController.syncTimersOnTick</code>
<code>iw:atlas-updated</code>	<code>AtlasService._loadMissing</code> (<code>AtlasService.js:225</code>)	<code>{ revision, gearReady, itemReady }</code>	<code>InventoryRenderer.reconcileAll</code> , <code>TooltipEngine.refreshOpen</code>
<code>iw:item-db-updated</code>	<code>ItemDatabase._index</code> (<code>ItemDatabase.js:224</code>)	<code>{ revision, generatedAt, source, count }</code>	<code>content.js</code> (rescan names), <code>InventoryRenderer.reconcileAll</code> , <code>TooltipEngine.refreshOpen</code> , <code>VillageScene.reconcileVillageScene</code>

Warning — the event names are a public surface of the page. They are dispatched on the page's `document` . The page's own scripts can therefore listen for them, and could also dispatch them. The worst effect of a forged event is extra reconciliation work: every consumer re-reads the DOM itself and trusts nothing in `detail` beyond element references. See §5.3.

1.6 MODULE MAP

The runtime is 35 JavaScript files (**11,869 lines**) and 15 stylesheets, of which 14 are active [[Source](#); line counts from `wc -l` at the snapshot commit]. The table lists every JavaScript file. "Writes" names the DOM namespace each module owns. Appendix D gives the complete attribute-by-attribute ownership.

File	Lines	Role	Imports (local)	Events it consumes	DOM it writes
src/ conte nt.j s	217	Entry point: <code>start()</code> → <code>gate</code> → <code>boot()</code> / <code>teardown()</code> ; kill-switch listener; consumer binding	15 modules, 10 stylesheets	<code>iw:dom-flush</code> , <code>iw:name-scan-flush</code> , <code>iw:item-db-updated</code>	nothing directly (delegates)
src/ page/ hydra tion- signa l.js	64	MAIN-world hydration probe (not bundled)	—	—	<code>html[data-iw-page-hydrated]</code>
Hydr ation Gate. js	56	Waits for the latch (4 s cap) and removes it	—	—	removes <code>data-iw-page-hydrated</code>
Runt ime.j s	177	<code>guard</code> / <code>guardEach</code> , <code>warnOnce</code> , runtime-active flag, <code>chrome.storage.local</code> wrappers, <code>onStorageChanged</code> , <code>assetUrl</code> , <code>raf</code>	—	<code>chrome.storage.onChanged</code>	—
Styl eInje ctor. js	42	Injects/removes <code><style data-iw-style></code> ; rewrites <code>../assets/</code> URLs	Runtime	—	<code><style data-iw-style></code> in <code><head></code>
Inli neSty leOwn er.j s	144	Factory for reversible, property-level inline styles	—	—	inline properties (on behalf of callers)
View port. js	107	<code>isRendered</code> / <code>pickRendered</code> / <code>preferRendered</code> ; breakpoint epoch via <code>matchMedia</code>	—	<code>MediaQueryList</code> change	—
Prog ressC adenc e.js	115	Pure tick-interval estimator for smoothed progress bars	—	—	—
DOMW atche r.js	484	The single <code>MutationObserver</code> ; dirty queues; 60-per-frame flush; skill-type detection	Runtime, Viewport	DOM mutations	— (dispatches the four watcher events)
Item Datab ase.j s	287	Fetches, caches and indexes <code>items.json</code> ; 6 h refresh chain	<code>normaliseItemName</code> , Runtime	—	page <code>localStorage</code> (one legacy <code>removeItem</code>)
Atla sServ ice.j s	466	Loads gear/item sprite indexes; resolves an item to a sprite; paints icon hosts; <code>+N</code> badges	aliases, <code>normaliseItemName</code> , Runtime	—	inline <code>background-*</code> on skin-owned hosts; <code>data-iw-atlas</code> ; <code>span.iw-icon-badge</code>
Skil lsArt Servi ce.j s	289	Skills UI atlas geometry → CSS custom properties; theme button art variables	Runtime, <code>revised-v5/index.json</code> (bundled)	—	<code>--fs-*</code> / <code>--iw-*</code> custom properties, <code>data-iw-skills-*</code> , <code>data-iw-button-atlas</code> , <code>data-iw-compact-atlas</code>
Inve ntory Model .js	260	Pure parser for owned-item detail text	—	—	—

File	Lines	Role	Imports (local)	Events it consumes	DOM it writes
itemDisplay.js	185	Pure item → stat chips/rows/tier/category helpers	—	—	—
normaliseItemName.js	20	Pure name normaliser (NFKD, punctuation, case)	—	—	—
aliases.js	52	Temporary cloak-material name aliases	—	—	—
zoneThemes.js	62	Generated: zone 1–34 → one of 9 themes	—	—	—
villageBuildings.js	63	Generated: 34 buildings + 5 houses → art files	—	—	—
TooltipEngine.js	688	Shared item hovercard #iw-tip; delegated hover/focus/keyboard; positioning	ItemDatabase, AtlasService, itemDisplay, StyleInjector	iw:dom-flush, iw:atlas-updated, iw:item-db-updated; DOM input events	div#iw-tip; aria-controls / aria-haspopup / aria-expanded / aria-describedby on triggers
NameScanner.js	456	Finds item names in game prose; CSS highlight iw-item-name; hover hit-testing	ItemDatabase, TooltipEngine, Runtime	(called by content.js); mousemove, click, scroll	CSS.highlights entry only (no DOM)
InventoryRenderer.js	760	Inventory row overlay, native-child suppression, panel chrome classification	DOMWatcher, AtlasService, ItemDatabase, TooltipEngine, StyleInjector, itemDisplay, InventoryModel, Runtime, InlineStyleOwner	iw:inventory-row, iw:item-db-updated, iw:atlas-updated	data-fs-*, data-iw-inventory-*, div.fs-inv-row, div.fs-inv-rule, inline display:none
SkillPanelRenderer.js	1577	Skill card structure annotation (three-zone), button styling, readout/ingredient neutralisation, medallion art	DOMWatcher, StyleInjector, Runtime, InlineStyleOwner, SkillsArtService	iw:skill-panel	data-fs-skill, data-iw-skill-*, data-iw-btn-state, data-iw-req-state, ...; classes fs-skill-panel, fs-skill--<type>, fs-skills-section-frame; skin spans; highlight iw-skill-ingredient-met
SkillCardDesignController.js	548	"V2" skill card layer: level readout, glyph/label, fill smoothing, long-action countdown, material body, requirement note	DOMWatcher, Runtime, ProgressCadence, Viewport	iw:skill-panel, iw:dom-flush, iw:name-scan-flush	html[data-iw-skill-card-design], data-iw-skill-v2-*, .iw-skill-v2-* nodes, --iw-skill-v2-* inline properties
QuestPanelRenderer.js	466	Quest card detection, roles, sigil medallion, objective tooltip trigger	DOMWatcher, StyleInjector, Runtime, SkillsArtService, AtlasService, ItemDatabase	iw:skill-panel	data-fs-quest, data-iw-quest-*, class fs-quest-panel, span.fs-quest-sigil, --fs-quest-*; trigger attributes on the objective <p>

File	Lines	Role	Imports (local)	Events it consumes	DOM it writes
UIFoundation.js	1468	Classifier hub: nav, zone bar, activity panels, section frames, daily boost, bosses, market, village; Toolkit link; runs the ordered classify pass	DOMWatcher, StyleInjector, Runtime, ProgressCadence, Viewport, WorldBossPanels, VillagePanels, VillageScene, CollapsibleFrames, HeaderChrome, CompactButtons	iw:dom-flush, iw:skill-panel	data-iw-ui, data-iw-tab, data-iw-state, data-iw-panel(-part/-header), data-iw-zone-*, data-iw-boss, data-iw-market, data-iw-skill-boost, data-iw-progress-reset, --iw-progress-duration, a[data-iw-nav-link="toolkit"]
HeaderRenderer.js	465	Header roles, crest, header art variables, per-zone surface and theme	DOMWatcher, StyleInjector, Runtime, zoneThemes, SkillsArtService	iw:dom-flush	data-iw-header*, data-iw-zone, span.fs-header-crest, --iw-header-* / --iw-nav-* / --iw-zone-*; on <html>: data-iw-zone-theme, --iw-zone-atlas, --iw-corner-filigree, --iw-zone-separator, button-art variables
HeaderChrome.js	144	Resolves nav + notice + zone bar into one merged frame	Viewport	(called by UIFoundation)	data-iw-chrome
OverlayFramer.js	188	Detects viewport-covering modal scrims and their cards	Runtime	(called by content.js on iw:dom-flush)	data-iw-overlay
CollapsibleFrames.js	295	Collapse toggles on frames; persisted preferences	Runtime	(called by UIFoundation); click on its own buttons	button[data-iw-collapse], data-iw-collapse-head/-title/-spine, data-iw-collapsed; storage iw-collapsed-frames
CompactButtons.js	77	Marks small controls and appends three art layers into each	—	(called by UIFoundation)	data-iw-compact-button, data-iw-compact-selected, span[data-iw-compact-layer] ×3
WorldBossPanel.js	364	Boss/zone-control cards: roles, art, rewards disclosure, Dominion Ward	AtlasService, ItemDatabase, SkillsArtService, TooltipEngine, Runtime, world-bosses/rewards.json (bundled)	(called by UIFoundation); toggle on its <details>; MediaQueryList change	data-iw-encounter, data-iw-boss-*, data-iw-control*, skin nodes data-iw-boss-owned, p.iw-boss-notice; storage iw-boss-rewards-collapsed; <boss>
VillagePanels.js	380	Village route housing hero and add-on slot cards	ItemDatabase, Runtime, villageBuildings	(called by UIFoundation)	data-iw-village*, skin nodes data-iw-village-owned, --iw-village-sprite; trigger attributes on building-name <p>
VillageScene.js	346	Dashboard Village frame; reads /api/player	Runtime, Viewport, villageBuildings, ItemDatabase, VillageLedger	iw:item-db-updated; (called by UIFoundation)	section.iw-village-scene (all nodes data-iw-village-scene-owned)
VillageLedger.js	415	Building benefit ledger inside the scene	ItemDatabase, itemDisplay, Runtime	click on its own buttons	nodes created through VillageScene's own(); data-iw-vs-*; storage iw-village-ledger

File	Lines	Role	Imports (local)	Events it consumes	DOM it writes
BackgroundPainter.js	142	Replaces the game's navy surface colours with the skin's warm dark ink	Runtime, InlineStyleOwner	(called by content.js on iw:dom-flush)	inline background-color / border-color !important, data-iw-painted

1.7 REPOSITORY LAYOUT

1.7.1 Tracked files (556 at the snapshot commit)

Path	Tracked files	What it is	Ships in the release zip?
manifest.json	1	MV3 manifest	Yes
package.json, package-lock.json	2	Dev tooling manifest; no runtime dependencies	No
dist/content.bundle.js	1	Committed build output (668,309 bytes, unminified). CI fails if it is stale	Yes
src/	50	35 JS files (§1.6)+ 15 CSS files (14 active; header.claude.css inactive)	Only src/page/hydration-signal.js (the bundle contains the rest)
assets/	337	Art, fonts, sprite indexes (§1.8)	Yes, minus .md, .html, prompt(s).txt/json
build-tools/	26	Build, packaging, vendoring, art import/generation, audits, fixture renderer	No
tests/	47	Standalone Node test suites (Chapter 7)	No
claude/	51	Development probes and DevTools capture snippets, perf harness, mobile audit, two archived CSS drafts (header.chatgpt.css, inventory.pre-forge.css) and two captured layout JSONs	No
docs/	19	docs/traps/*.md (14 engineering post-mortems), docs/superpowers/ (2 plans + 2 specs), docs/zone-control-motion-notes.md	No
handoff/	7	Earlier handoff specs for the collapsible and (retired) rearrangeable panels: HTML, PDF and a spec script each, plus a README	No
.github/workflows/ci.yml	1	CI	No
Root Markdown	13	README.md, CLAUDE.md, STABILITY_NOTES.md, VISUAL_NOTES.md and nine version notes(V1.5.0_UI_FOUNDATION.md ... V1.6.0_MOBILE_LAYOUT_AUDIT.md)	No
.gitignore	1	See below	No

1.7.2 Untracked and ignored local content (present on the inspecting machine)

Warning — do not commit or share these.

Path	Size	Why it matters
build-tools/header-live-profile/	—	A real Chromium browser profile used by audit-header-live.mjs. .gitignore says it holds saved logins, autofill and wallet state. It must never be committed, zipped or uploaded.
output/	~80 MB	Previews, motion recordings, image-generation work files, PDFs. .gitignore describes it as holding business/legal documents.
tmp/	~333 MB	Audit and perf-harness output(tmp/mobile-audit/ , tmp/perf/)
release/	~137 MB	idleworlds-fantasy-skin-v1.6.0.zip (142,788,866 bytes)
build-tools/fixtures/	—	Rendered visual fixtures(npm run fixtures)
node_modules/	~47 MB	Dev dependencies
assets/skills-ui/buttons/themes-v2/ , .../source-v4/	~70 MB	Superseded button-art generator inputs(ignored)
handoff/technical/ , handoff/reskin-technical-handover/	—	Untracked documentation drafts. The first was produced by another tool and left untouched. The second is this package

1.8 ASSET INVENTORY

Sizes are raw bytes of tracked files [**Generated:** git ls-files + stat]. The release column is derived by re-running the allowlist logic in build-tools/package-release.mjs (325 files, 144.6 MB raw; the zip is 142.8 MB because PNG/WebP do not recompress).

Asset family	Files	Size	How it reaches the page	Runtime consumer	Provenance (as recorded in the repo)
assets/gear_icons_atlas.png + gear_icons_manifest.json	2	25.9 MB	fetch() of the manifest(JSON) from the extension; atlas via inline background-image: url(chrome-extension://...)	AtlasService	Vendored by vendor-assets.mjs from raw.githubusercontent.com/BustedCipher/idleWorlDs-game-sprites-BC pinned at commit eadbc0fe1eae549184dd740470b69fc76974a411
assets/item_icons_atlas.png + item_icons_index.csv	2	12.5 MB	fetch() of the CSV; atlas as inline background	AtlasService	Same pinned repository
assets/fonts/*.woff2	6	142 KB	@font-face in base.css	all text	Downloaded from Google Fonts by vendor-assets.mjs (Cinzel 600-700 variable, Barlow 400/500/600/700, Crimson Text italic). [Assumption — licence] these families are published under the SIL Open Font License; the repository does not include OFL.txt (§4)
assets/skills_ui_atlas.webp + skills_ui_index.json	2	92 KB	JSON fetch() ; atlas via CSS custom properties	SkillsArtService ; default --iw-zone-atlas	Project-generated art (see docs/superpowers/*retina-skill-theme-atlases*)

Asset family	Files	Size	How it reaches the page	Runtime consumer	Provenance (as recorded in the repo)
assets/skills_icons_atlas.webp + skills_icons_index.json	2	73 KB	JSON <code>fetch()</code> (the index names the atlas)	<code>SkillsArtService.paintIcon</code> (skill medallion)	Project art
assets/skills_panel_texture.webp, skills_nav_prev/next.svg	3	5 KB	CSS <code>url()</code> and custom properties	frames, tooltip, overlay, skill nav	Project art
assets/skills_xp_plaque_wide.webp	1	10 KB	not referenced at runtime (a CSS comment records the plaque being dropped)	—	Project art; removal candidate
assets/skills-ui/theme_*.webp, panel_corners_*.webp, separator_fLOURISH_*.webp	27	7.9 MB	Inline custom properties on <code><html></code> (<code>HeaderRenderer.applyZoneTheme</code>)	every frame, inventory title separator, skill UI atlas	Imported by <code>import-skill-themes.mjs</code> from the sibling repository <code>../idleWorlds-game-sprites-BC/assets/skills-ui-atlas/families</code>
assets/skills-ui/zone-theme-map.json	1	small	not at runtime (compiled into <code>zoneThemes.js</code>)	tests, importer	same sibling repository
assets/inventory/panel_corners.webp, separator_fLOURISH.webp	2	51 KB	<code>base.css</code> defaults for <code>--iw-corner-filigree</code> / <code>--iw-zone-separator</code>	un-themed fallback	Generated by <code>make-panel-corners.mjs</code> (<code>npm run art</code>) [Source : <code>package.json</code>]
assets/header/*.webp	15	1.5 MB	Inline custom properties on header, nav, announcement and zone bar	<code>HeaderRenderer</code> + <code>header.css</code>	Project art; audited by <code>audit-header-assets.mjs</code>
assets/header/zones/zone_N.webp, zone_N_mobile.webp (N = 1...34)	68	8.4 MB	<code>--iw-header-surface(-mobile)</code> inline on the header	<code>HeaderRenderer.applyZoneSurface</code>	Imported by <code>import-zone-headers.mjs</code> from <code>../idleWorlds-game-sprites-BC/assets/zone-headers/</code> ; <code>make-zone-surfaces.mjs</code> can generate procedural placeholders for missing zones
assets/skills-ui/action-icons/*.svg + index.json	11	44 KB	CSS <code>url()</code> per discipline in <code>skillcard-v2.css</code>	V2 action-button glyph	Converted from the project owner's Illustrator EPS by <code>import-action-icons.mjs</code>
assets/skills-ui/buttons/revised-v5/*.png + index.json	13	9.3 MB	Custom properties on <code><html></code> (<code>SkillsArtService.applyThemeVariables</code>); <code>index.json</code> is bundled into the JS	skill/quest/boss action and chevron buttons	First four themes from reference images supplied by the owner; remaining five generated with a built-in image-generation tool (<code>README.md</code> in that folder; prompts in <code>sources/</code> , which are not tracked)

Asset family	Files	Size	How it reaches the page	Runtime consumer	Provenance (as recorded in the repo)
assets/skills-ui/buttons/compact-ghost-v3/*.png	13	2.9 MB	--iw-compact-atlas on <html>	compact-buttons.css	Built by build-compact-buttons.py from generated source sheets
assets/skills-ui/buttons/card-v6/{action,nav}-atlas.png	6	3.8 MB	card-button-atlas.css (generated) custom properties	V2 action and nav plates	Generated with the built-in image-generation tool; windows built by build-card-buttons.mjs
assets/skills-ui/buttons/exact-v3/<theme>/*.png	111	60.8 MB	Referenced only in the else branch of SkillsArtService.applyThemeVariables (SkillsArtService.js:178-181)	none — unreachable. All nine themes that zoneThemes.js can produce are listed in revised-v5/index.json → themes, so the revised branch is always taken	Produced by make-exact-button-states.ps1 ; superseded by revised-v5
assets/quest-frames/approved-frames.png	1(+3 docs)	1.6 MB	CSS url() (skillpanel.css:2777) with hard-coded percentage windows	quest sigil ring	Built-in image-generation tool (prompt.txt)
assets/world-bosses/encounters.png, zone-control-contested-v3.png, rewards.json	3 (+README)	5.7 MB	CSS url() ; rewards.json bundled	boss art, Dominion Ward crest, reward fallback data	Built-in image-generation tool, 2026-09-07(README)
assets/village/building_1...34.webp, house_1...5.webp, cobblestone.png, buildings.json	41	3.8 MB	24 (<code>ci.yml</code> !); the local version is not pinned (<code>engines</code> absent)	everything	
<code>esbuild</code>	<code>^0.28.2</code>	<code>build-tools/build.mjs</code>			
<code>jsdom</code>	<code>^25.0.0</code>	~half the test suites			
<code>playwright</code> (+ <code>npx playwright install chromium</code>)	<code>^1.47.0</code>	real-browser suites, fixtures, audits, perf harness			
Python 3 (+ imaging libraries) [Verify exact packages]	—	<code>build-compact-buttons.py</code> , <code>build-revised-buttons.py</code> (art rebuilds only)			
Windows PowerShell	—	<code>make-exact-button-states.ps1</code> , <code>make-source-preview-button-states.ps1</code> (art rebuilds only)			
Sibling repositories <code>../idleWorlds-game-sprites-BC</code> , <code>../idleWorlds-art-source</code>	—	<code>import:*</code> scripts only			

1.10 BROWSER PLATFORM REQUIREMENTS

The manifest declares no `minimum_chrome_version`. The build targets `chrome109` (`build.mjs`:89, 101), but esbuild does not lower the modern CSS features the stylesheets use. Their counts below are occurrences in the 14 active sheets [**Generated**]. Minimum versions are **general platform knowledge** [**Assumption – verify against current compatibility tables**]:

Feature	Where	Min. Chrome	Behaviour on older Chrome
<code>content_scripts[]</code> . <code>world:</code> "MAIN"	manifest	111	Inference: the signal cannot see React's root, so no latch is written and the gate waits its full 4 s
<code>color-mix()</code> (145 uses)	theme edge scale, V2 card, buttons	111	Declarations are invalid; custom properties that depend on them become <i>invalid at computed-value time</i> , so many borders and fills fall back to their initial values

Feature	Where	Min. Chrome	Behaviour on older Chrome
:has() (58)	frames, header, V2 card, collapse	105	Rules are dropped
Size container queries, cq1 (6 / 7)	inventory filter row, nav row, V2 card, village	105	Responsive font fitting and phone card layouts are lost
CSS Custom Highlight API (CSS.highlights, ::highlight)	NameScanner, SkillPanelRenderer	105	The JS feature-detects it and paints nothing
Element.checkVisibility()	Viewport.js, V2 controller	105	Viewport.js falls back to rect size
dvh	tooltip max-height	108	Declaration dropped
caretPositionFromPoint / caretRangeFromPoint	NameScanner	128 / long-standing	Falls back to caretRangeFromPoint
WeakRef, FinalizationRegistry	InlineStyleOwner	84	—
backdrop-filter (14), prefixed + unprefixed mask	glass plates, rings	76 / prefixed long-standing	—

Effective minimum: Chrome 111 (desktop and Android Chrome). Firefox and Safari are **not supported**. The manifest uses `world: "MAIN"` and the `chrome.*` namespace, and there is no cross-browser test. Chromium derivatives (Edge, Brave, Opera) are **untested** [Verify].

Recommendation (non-behavioural):

- Add `"minimum_chrome_version": "111"` to `manifest.json`.
- Change the esbuild target to `chrome111`, so the declared and the real minimum agree.

See §4.

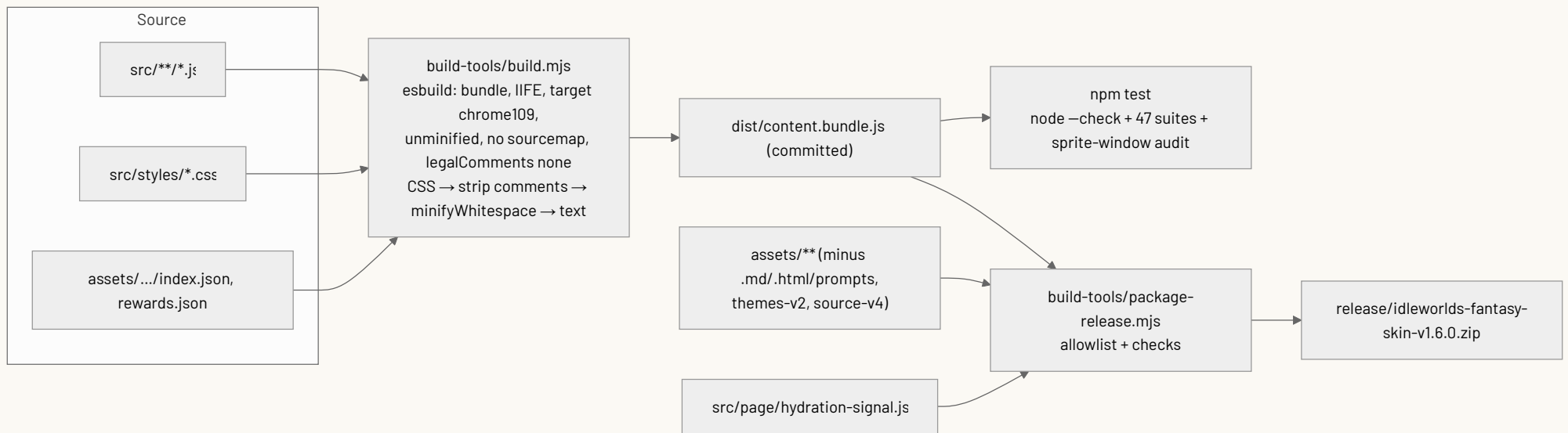
1.11 EXTERNAL RESOURCES AND NETWORK SURFACE

Resource	Type	When	Initiated by	Credentials	Data sent	Data kept
https://idleworlds.com/items.json	fetch (GET, cache: 'no-cache', conditional)	At boot, if no fresh cache (6 h max age), then every 6 h; retry after 15 min on failure	ItemDatabase._fetchFresh (ItemDatabase.js:165)	Default same-origin. Same-origin on idleworlds.com, so the page's cookies are sent. On www.idleworlds.com the request is cross-origin, so no cookies are sent, and it depends on the server's CORS/redirect behaviour [Verify]	none (URL only)	The item catalogue in chrome.storage.local
/api/player?section=<route>&scope=core	fetch (GET, credentials: 'same-origin', cache: 'no-store', 10 s abort)	On the dashboard when the Village frame's anchor exists, the tab is visible, and the snapshot is stale: every 5 min; 30 s retry; ≤10 s after a route change	VillageScene.reconcileVillageScene (VillageScene.js:312)	Session cookies (same origin) + x-idleworlds-league header	route section, league	In memory only: housing tier, slot capacity, installed buildings
chrome-extension://<id>/assets/...	CSS url(), , JSON/CSV fetch	Boot and on demand	StyleInjector-injected CSS, AtlasService, SkillsArtService, VillageScene	n/a	none	—
https://idleworlds.com/wiki/items/<slug>	Link (target="_blank" rel="noopener noreferrer")	Only when the player clicks	TooltipEngine	n/a	—	—

Resource	Type	When	Initiated by	Credentials	Data sent	Data kept
<code>https://idleworldstoolkit.com</code>	Link (<code>target="_blank"</code> <code>rel="noopener noreferrer"</code>)	Only when the player clicks	<code>UIFoundation.ensureTookitLink</code>	n/a	—	—
<code>raw.githubusercontent.com</code> , <code>fonts.googleapis.com</code> / <code>fonts.gstatic.com</code>	Download	Build time only (<code>npm run vendor</code>)	<code>vendor-assets.mjs</code>	—	—	committed files
<code>idleworlds.com</code> (live page, CSS)	Headless browser	Developer tooling only (<code>claude/audit-mobile.mjs</code> , <code>build-tools/audit-*live.mjs</code> , <code>npm run audit:items</code>)	<code>scripts</code>	The <code>audit-header-live.mjs</code> profile may be logged in [Verify]	—	<code>tmp/</code>

No runtime request goes to any origin other than the game's own origins and the extension itself **[Source:** every `fetch()` call site, listed in the Chapter 5 census].

1.12 BUILD, PACKAGING AND CI



- `npm run build` (`build.mjs`) first checks 10 `REQUIRED_ASSETS` : 4 sprite files and 6 fonts. It fails if any are missing, unless `--allow-missing-assets` is passed, which is what `npm test` uses. The CSS plugin normalises line endings and strips comments with a quote-aware scanner. It then minifies whitespace (target `chrome109`) and returns the CSS as a text module. The bundle has a **soft budget of 300,000 bytes**: exceeding it prints a warning, and the current bundle is 668,309 bytes [[Test run](#)].
- `npm run package` (`package-release.mjs`) builds from an **allowlist**. It fails when:
 - the versions in `manifest.json`, `package.json` and `content.js` disagree;
 - a content-script file is missing;
 - a `web_accessible_resources` pattern matches no packaged file;
 - the bundle references a source map.

It warns that the manifest has **no icons** (the Chrome Web Store requires a 128×128 icon).

- **CI** (`ci.yml`) runs on `ubuntu-latest` with Node 24 (`permissions: contents: read` , 10-minute timeout):
 1. `npm ci`;
 2. installs Playwright Chromium;
 3. runs `npm test`;
 4. runs `git diff --exit-code dist/content.bundle.js`.

CI never writes to the repository.

- **Versioning**: `1.6.0` appears in `manifest.json`, `package.json` and `content.js` (`VERSION`). The package step enforces agreement between them.

1.13 DOCUMENTATION DRIFT FOUND DURING THIS INSPECTION

These statements in the project's own documentation or comments no longer match the source. None changes runtime behaviour, but each can mislead a maintainer [[Source unless noted](#)]:

#	Where	Says	Actually
D-01	<code>CLAUDE.md</code> ("Cascade")	The generic <code>button:not(...)</code> rule in <code>ui-system.css</code> is (0,4,1)	Computed specificity of <code>ui-system.css:1019</code> is (0,30,3) because every <code>:not()</code> argument adds its specificity [Generated: Appendix B]
D-02	<code>README.md</code> lines 15, 106, 117	<code>npm test</code> "runs five suites"; "all five runtime stylesheets"	47 suites plus the sprite audit; 7 <code><style></code> elements
D-03	<code>content.js:130-132</code> comment	<code>initSkillCardDesignController</code> restores "the persisted New/Current choice"	The toggle and stored preference were removed; the controller always sets <code>data-iw-skill-card-design="new"</code>
D-04	<code>UIFontFoundation.js:929-933</code> comment	DOMWatcher's <code>attributeFilter</code> is <code>['class', 'style', 'disabled', 'aria-disabled']</code>	Eight entries (adds <code>aria-pressed</code> , <code>aria-selected</code> , <code>aria-current</code> , <code>data-state</code>)
D-05	<code>OverlayFramer.pickCard</code> comment	Picks the "outermost" visible <code>.panel</code>	Picks a panel that contains no other candidate, i.e. the innermost
D-06	<code>Viewport.js</code> comment; <code>README.md</code> line 26	Targets "Chrome 109+"	Effective minimum is Chrome 111 (§1.10)
D-07	<code>ItemDatabase.js:17</code> vs 163	"~5.5 MB (4,458 items)" vs "fresh ~230 KB download"	Probably decoded size vs compressed transfer; [Verify] against the live endpoint
D-08	<code>assets/world-bosses/README.md</code>	Active Zone Control art uses <code>zone-control-red-v3.png</code> , <code>blue-v3.png</code> , <code>-contested-v3.png</code>	Only <code>zone-control-contested-v3.png</code> is tracked and referenced (<code>ui-system.css:1564</code>)

#	Where	Says	Actually
D-09	CLAUDE.md (Rearrange removed)	:not([data-iw-order-handle]) exclusions are kept deliberately	Still true; recorded here because it inflates specificity (see D-01)
D-10	HeaderRenderer.js:155, 232, 462	—	Two statements share one line in three places, e.g. <code>if (!profile) return {...}; setRole(profile, 'profile');</code> (L232) and <code>export function initHeaderRenderer() { inject('header', css);</code> (L462). The code is correct, but it reads like a bug (probably an automated-edit artefact)

2. COMPLETE FEATURE INVENTORY

This chapter lists **every** visual, interactive and behavioural feature of the skin as its own entry. Each entry uses the same rows so features can be compared:

Row	Meaning
What	What the player sees or experiences
Why	The problem it solves or the design intent (from code comments and docs/traps)
Where	Modules, functions, stylesheets and line ranges
Depends on	Game DOM hooks, other skin modules, data, assets
Runs	When it executes and how often
States & fallbacks	Loading, empty, error, and degraded behaviour
Responsive	Breakpoints and container queries
Motion	Transitions and animations, and reduced-motion behaviour
Hover / focus / keyboard	Interaction states
Accessibility	Semantics, names, hidden content, target sizes
Edge cases & findings	Known limits and linked findings (register in Chapter 5)
Tests	Suites that pin it (Chapter 7)

Evidence labels ([Source], [Generated], [Test run], [Project record], [Assumption]/[Verify]) are defined in Chapter 1. Unless marked otherwise, every statement here is [Source].

2.0 FEATURE INDEX

ID	Feature	Primary modules	Primary stylesheets
F-01	Activation and hydration gate	content.js , HydrationGate.js , hydration-signal.js	—
F-02	Kill switch and teardown	content.js , every clear*()	—
F-03	Style injection and asset URL rewriting	StyleInjector.js	all
F-04	Mutation reconciliation and flush budget	DOMWatcher.js	—
F-05	Rendered-copy selection and breakpoint epoch	Viewport.js	—
F-06	Error isolation and console reporting	Runtime.js	—
F-07	Global dark-fantasy theme and typography	—	base.css
F-08	Surface colour repainting	BackgroundPainter.js	—
F-09	Per-zone theming	HeaderRenderer.js , zoneThemes.js , SkillsArtService.js	base.css , card-button-atlas.css
F-10	Generic control language	—	ui-system.css , base.css
F-11	Ghost-metal compact button art	CompactButtons.js	compact-buttons.css
F-12	Forged section frames	UIFoundation.classifySectionFrames	ui-system.css

ID	Feature	Primary modules	Primary stylesheets
F-13	Collapsible frames	<code>CollapsibleFrames.js</code>	<code>collapsible.css</code>
F-14	Player header reorganisation	<code>HeaderRenderer.js</code>	<code>header.css</code>
F-15	Per-zone header painting	<code>HeaderRenderer.applyZoneSurface</code>	<code>header.css</code>
F-16	Main navigation rail	<code>UIFoundation.classifyMainNav</code>	<code>ui-system.css</code> , <code>header.css</code>
F-17	Toolkit link	<code>UIFoundation.ensureToolkitLink</code>	<code>ui-system.css</code>
F-18	Announcement bar	<code>HeaderRenderer.classifyAdjacent</code>	<code>header.css</code>
F-19	Zone bar	<code>UIFoundation.classifyZoneBar</code>	<code>header.css</code> , <code>ui-system.css</code>
F-20	Merged header chrome	<code>HeaderChrome.js</code>	<code>ui-system.css</code>
F-21	Current Action panel and progress smoothing	<code>UIFoundation</code> activity panels, <code>ProgressCadence.js</code>	<code>ui-system.css</code>
F-22	Action Log panel	<code>UIFoundation</code>	<code>ui-system.css</code>
F-23	World Chat panel	<code>UIFoundation</code>	<code>ui-system.css</code>
F-24	Daily XP Boost chip	<code>UIFoundation.classifyDailyBoost</code>	<code>ui-system.css</code>
F-25	Skill card detection and discipline identity	<code>DOMWatcher.detectSkillType</code>	—
F-26	Skill card structure and three-zone layout	<code>SkillPanelRenderer.annotateStructure</code>	<code>skillpanel.css</code>
F-27	Skill command button styling	<code>SkillPanelRenderer.styleButton</code>	<code>skillpanel.css</code>
F-28	Skill medallion artwork and identity text	<code>SkillPanelRenderer</code> , <code>SkillsArtService</code>	<code>skillpanel.css</code>
F-29	Level/XP readout presentation	<code>SkillPanelRenderer.neutraliseReadouts</code>	<code>skillpanel.css</code> , <code>skillcard-v2.css</code>
F-30	Material (ingredient) lists and met highlighting	<code>SkillPanelRenderer</code>	<code>skillpanel.css</code>
F-31	V2 card: medallion XP ring and level badge	<code>SkillCardDesignController</code>	<code>skillcard-v2.css</code>
F-32	V2 card: themed square action button, glyph and fitted label	<code>SkillCardDesignController</code> , <code>SkillsArtService</code>	<code>skillcard-v2.css</code> , <code>card-buttons.css</code> , <code>card-button-atlas.css</code> , <code>skillpanel.css</code>
F-33	V2 card: activity fill smoothing	<code>SkillCardDesignController.smoothActionFill</code>	<code>skillcard-v2.css</code> , <code>card-buttons.css</code>
F-34	V2 card: long-action countdown	<code>SkillCardDesignController</code>	<code>skillcard-v2.css</code>
F-35	V2 card: detail body and requirement note	<code>SkillCardDesignController</code>	<code>skillcard-v2.css</code> , <code>skillcard-v2-runtime-safe.css</code>
F-36	Recipe pager (previous/next)	<code>SkillPanelRenderer</code> , <code>SkillCardDesignController</code>	<code>skillpanel.css</code> , <code>skillcard-v2.css</code> , <code>card-buttons.css</code>
F-37	Skill Actions section frame	<code>SkillPanelRenderer.ensureSkillActionsFrame</code>	<code>skillpanel.css</code> , <code>ui-system.css</code>

ID	Feature	Primary modules	Primary stylesheets
F-38	Quest cards	<code>QuestPanelRenderer.js</code>	<code>skillpanel.css</code> (quest section)
F-39	Quest objective item hovercard	<code>QuestPanelRenderer.ensureObjectiveTrigger</code>	<code>skillpanel.css</code>
F-40	Inventory row overlay	<code>InventoryRenderer.renderRow</code>	<code>inventory.css</code>
F-41	Inventory panel chrome and filters	<code>InventoryRenderer.classifyInventoryChrome</code>	<code>inventory.css</code>
F-42	Item catalogue service	<code>ItemDatabase.js</code>	—
F-43	Sprite icons and upgrade badges	<code>AtlasService.js</code>	<code>base.css</code> , <code>inventory.css</code> , <code>tooltip-engine.css</code>
F-44	Item hovercard (tooltip)	<code>TooltipEngine.js</code>	<code>tooltip-engine.css</code>
F-45	Item-name detection in game text	<code>NameScanner.js</code>	(no stylesheet; see FUN-02)
F-46	World Boss encounter cards	<code>WorldBossPanels.js</code> , <code>UIFoundation.classifyBossCards</code>	<code>ui-system.css</code>
F-47	World Boss "Possible Rewards"	<code>WorldBossPanels.rewards</code>	<code>ui-system.css</code>
F-48	Zone Control "Dominion Ward"	<code>WorldBossPanels.ensureDominion/updateDominion</code>	<code>ui-system.css</code>
F-49	Market listing rows	<code>UIFoundation.classifyMarket</code>	<code>ui-system.css</code>
F-50	Village route panels	<code>VillagePanels.js</code>	<code>ui-system.css</code>
F-51	Dashboard Village scene	<code>VillageScene.js</code>	<code>village-scene.css</code>
F-52	Village ledger	<code>VillageLedger.js</code>	<code>village-scene.css</code>
F-53	Pop-up and modal framing	<code>OverlayFramer.js</code>	<code>overlay.css</code>
F-54	Reduced-motion support	many	all
F-55	Responsive and touch behaviour	<code>Viewport.js</code> , many	all

2.1 LIFECYCLE AND PLATFORM FEATURES

F-01 Activation and hydration gate

What	Nothing visible. The skin appears within roughly 0–4 s of page load, once React has finished hydrating the server-rendered page.
Why	Appending skin nodes while React was hydrating made React throw minified error #418 (hydration mismatch) and re-render the whole tree [Project record : <code>CLAUDE.md</code> , <code>HydrationGate.js</code> header].
Where	<code>src/page/hydration-signal.js</code> (MAIN world, <code>document_start</code>); <code>HydrationGate.waitForPageHydration</code> ; <code>content.js</code> <code>start()</code> lines 196–217.

Depends on	React's private root fields: the <code>__reactContainer\$<random></code> expando on the container, then <code>.stateNode.current.memoizedState.isDehydrated</code> . The candidate containers are <code>document</code> , <code><html></code> , <code><body></code> , <code>#__next</code> and each child of <code><body></code> .
Runs	Once per page load. The signal polls every 50 ms with a <code>setTimeout</code> chain for up to 15 s. The gate polls the latch every 50 ms for up to 4 s.
States & fallbacks	The signal writes <code>data-iw-page-hydrated</code> as follows: "1" when <code>isDehydrated</code> <code>=== false</code>; "unknown" when the root is found but the field is not a boolean; - nothing after 15 s. The gate resolves <code>{state: '1' 'unknown' 'timeout' 'disabled', waitedMs}</code>, then removes the latch. On timeout the skin boots anyway. <code>console.log('IW Fantasy Skin boot after page hydration: <state> (<ms>ms)')</code> records the measurement.
Responsive / Motion	—
Hover / focus / keyboard	—
Accessibility	—
Edge cases & findings	- If hydration finishes between 4 s and 15 s, the latch is written after the gate has gone and stays on <code><html></code> until teardown (FUN-13). - The page could write the latch itself and make the skin boot early (SEC-06). - A React upgrade that renames these private fields degrades every load to the 4 s timeout. <code>HYDRATION_GATE_ENABLED = true</code> is a compile-time switch.
Tests	<code>hydration-gate.test.mjs</code> (protocol, timeout, disabled path); <code>extension-e2e.test.mjs</code> (both worlds load from the real unpacked extension).

F-02 Kill switch and teardown

What	Setting <code>chrome.storage.local['iw-skin-enabled']</code> to <code>false</code> removes the whole skin from the open page without a reload, and setting it to <code>true</code> rebuilds it. There is no user-facing toggle (no popup or options page). The key is changed from DevTools in the extension's content-script context (§6.1.4).
Why	A/B debugging against the same live game state, and a guarantee that the skin is fully reversible (rule 3) [Source: README "Kill switch"].
Where	<code>content.js</code> <code>applyEnabled</code> , <code>boot</code> (L119–153) and <code>teardown</code> (L163–189); <code>Runtime.onStorageChanged</code> .
Depends on	<code>chrome.storage.local</code> (permission <code>storage</code>).
Runs	On every change to the key, for the page lifetime. Boot is idempotent (<code>booted</code> flag). Teardown before boot only marks the runtime inactive.
States & fallbacks	A missing key or a failed read means enabled . Teardown order: watcher → item-DB refresh timer → tooltip hide → name-scan highlight → inventory → V2 card → skill panels → quests → UI foundation (header chrome, collapse, Village scene, compact buttons, boost, bosses, Village panels, nav, frames) → header → background paint → overlays → all style elements → hydration latch → <code>setRuntimeActive(false)</code> → <code>console.log('... disabled – active presentation removed')</code> .
Responsive / Motion	—
Hover / focus / keyboard	—
Accessibility	Restores every native <code>aria-*</code> value that the skin overwrote. Exception: the attributes listed in FUN-09 are removed unconditionally.
Edge cases & findings	- Page-lifetime listeners (tooltip, name scanner, event-bus consumers) stay bound but inert. <code>#iw-tip</code> stays in the DOM, hidden (FUN-10). - Collapsed-panel titles can disappear after a round trip (FUN-04). - The enabled value is read before the up-to-4 s hydration wait, and the change listener is only registered after it. A change made during that window is not seen until the next change.
Tests	<code>smoke.test.mjs</code> (disable/enable/disable round trips, native inline styles restored exactly, no duplicate tooltip/storage surfaces); <code>compact-button-boot.test.mjs</code> (real bundled reactivation path); <code>extension-e2e.test.mjs</code> .

F-03 Style injection and asset URL rewriting

What	Seven <code><style data-iw-style=""></code> elements appended to <code><head></code> , in fixed order.
Why	- Manifest-declared CSS cannot be removed at runtime. - Injecting text lets the kill switch remove every rule, including the global theme [Source: README rule 5]. - Rewriting <code>url(.. /assets/...)</code> to <code>chrome-extension://<id>/assets/...</code> makes art load from the extension, not a third-party origin that the page's CSP could block [Source: <code>vendor-assets.mjs</code> header].
Where	<code>StyleInjector.js</code> (<code>inject</code> , <code>rewriteAssetUrls</code> , <code>removeAll</code>); <code>content.js</code> <code>injectPresentationStyles</code> (L57-72); <code>UIFoundation.injectUIFoundationStyles</code> .
Depends on	<code>chrome.runtime.getURL</code> ; <code>web_accessible_resources</code> for <code>assets/*</code> .
Runs	Once per boot. <code>inject(id)</code> is a no-op if that id is already injected.
States & fallbacks	- Without <code>chrome.runtime</code> (tests), <code>assetUrl</code> returns the relative path. - The rewrite regex only matches <code>url(.. /assets/...)</code> with optional quotes. Any other form, such as <code>url(/assets/...)</code> or <code>url(assets/...)</code>, is left untouched and would 404 on the game's origin.
Edge cases & findings	<code>removeAll()</code> removes every <code>style[data-iw-style]</code>, including one another script might add with the same attribute. <code>textContent</code> is used, so no HTML is parsed. - Injection order is load-bearing: <code>ui-system</code> must be last (<code>static-invariants.test.mjs</code>).
Tests	<code>static-invariants.test.mjs</code> (order, count of 7); <code>smoke.test.mjs</code> (all style elements removed on disable).

F-04 Mutation reconciliation and flush budget

What	Nothing visible directly. Every skin decoration follows React's updates within a frame or two.
Why	React replaces and updates nodes constantly (timers, XP ticks, chat). One budgeted observer keeps the skin current without feedback loops or long frames [Project record: <code>docs/traps/performance.md</code>].
Where	<code>DOMWatcher.js</code> : <code>startWatcher</code> (L358 observer), <code>queueContext</code> , <code>discover</code> , <code>drainGlobalBudget</code> , <code>flushPending</code> , <code>emit</code> .
Depends on	<code>.compact-row</code> / <code>[class*="item-row"]</code> (inventory rows) and <code>.compact-panel</code> (skill, quest, boss, village and shop cards).
Runs	Observer. Observes <code>document.body</code> with <code>childList</code>, <code>subtree</code>, <code>characterData</code>, and <code>attributes</code> filtered to <code>class</code>, <code>style</code>, <code>disabled</code>, <code>aria-disabled</code>, <code>aria-pressed</code>, <code>aria-selected</code>, <code>aria-current</code> and <code>data-state</code>. Flush scheduling. Mutations are sorted into four pending sets (inventory rows, skill panels, background roots, name roots). One <code>requestAnimationFrame</code> flush processes at most 60 elements in total, round-robin across the sets. Leftovers reschedule to the next frame. Breakpoint crossings (640/768/1024/1280/1536) trigger <code>discover(document.body)</code>.
States & fallbacks	Disconnected elements are dropped from the queues. Name roots already covered by an ancestor root are skipped.
Edge cases & findings	<code>data-iw-*</code> writes and <code><html></code> attribute changes are invisible to the observer (by design). Code that changes state only through them must run its own pass. - A same-value <code>class</code> / <code>setAttribute</code> / <code>textContent</code> write still emits a record, so every renderer compares before writing. - Page scripts can dispatch forged <code>iw:*</code> events (SEC-05).
Tests	<code>flush-quiescence.test.mjs</code> (zero flushes on a quiet page); <code>smoke.test.mjs</code> ; <code>route-swap-reclassify.test.mjs</code> ; <code>viewport-swap.test.mjs</code> .

F-05 Rendered-copy selection and breakpoint epoch

What	Nothing visible. The skin decorates the copy of each panel that is actually on screen.
Why	The game renders its panel stack twice: a <code>hidden xl:grid</code> copy and an <code>xl:hidden</code> copy. Which one is visible swaps at 1280 px, and a breakpoint crossing mutates nothing [Source: <code>Viewport.js</code> header; Project record].
Where	<code>Viewport.js</code> : <code>isRendered</code> (<code>checkVisibility()</code> , else a non-zero rect), <code>pickRendered</code> , <code>preferRendered</code> (falls back to every candidate when none is rendered, as in jsdom), <code>startLayoutWatch</code> / <code>stopLayoutWatch</code> (<code>matchMedia</code> ('(min-width: Npx)') change listeners), <code>getLayoutEpoch</code> .
Depends on	Tailwind breakpoints 640/768/1024/1280/1536.
Runs	Consulted by every classifier cache. The epoch increments on each breakpoint crossing and is never reset.

Edge cases & findings	Caches keyed on elements must also check the epoch, or they keep decorating the now-hidden copy.
Tests	<code>viewport-swap.test.mjs</code> , <code>header-hidden-duplicate.test.mjs</code> , <code>inventory-root.test.mjs</code> .

F-06 Error isolation and console reporting

What	If one element is malformed, only that element stays un-skinned. The console receives at most one warning per failure label.
Why	The game mutates continuously, so an exception thrown per frame would flood the console and break every later renderer in the same flush.
Where	<code>Runtime.js</code> , <code>warnOnce</code> (L132), <code>guard</code> (L145), <code>guardEach</code> (L160).
Runs	Around every consumer, renderer and teardown step. <code>guard</code> returns <code>false</code> without running <code>fn</code> while the runtime is inactive.
States & fallbacks	Console output from the skin: - boot and hydration lines (<code>console.log</code>); <code>warnOnce</code> warnings prefixed <code>[IW Fantasy Skin] <label>;</code> <code>AtlasService</code> warns once per item name without an icon (<code>[AtlasService] No icon found for "<name>"</code>); <code>ItemDatabase</code> warns on load and refresh failures. No error is sent anywhere.
Edge cases & findings	<code>_warned</code> and <code>AtlasService._missingWarned</code> are unbounded <code>Set</code> s keyed by label and by item name. Both grow slowly and are cleared only by a reload.
Tests	<code>smoke.test.mjs</code> (malformed rows do not stop others).

2.2 GLOBAL LOOK

F-07 Global dark-fantasy theme and typography

What	Palette. Near-black warm ink replaces the game's navy/slate palette, with brass/ember accents. Fonts. Cinzel headings, Barlow UI text, Crimson Text italic for flavour text. Shapes. Rounded corners flattened to 2–4 px, and dark forged grounds on every <code>.compact-panel</code> and <code>.compact-row</code>. Controls. Themed scrollbars, a gold focus ring on controls, and forged text inputs.
Why	Establishes the "Ashen Iron" identity with the fewest DOM writes: everything is a stylesheet, removed as one unit.
Where	<code>base.css</code> (713 lines): <code>@font-face</code> x6; <code>:root</code> tokens <code>--iw-ink*</code>, <code>--iw-th*</code>, <code>--iw-line*</code>, <code>--iw-gold*</code>, <code>--iw-ember*</code>, text, semantic, tier and font tokens, radii, <code>--iw-action-tick</code>; game token overrides (<code>--background</code>, <code>--foreground</code>, <code>--card</code>, <code>--popover</code>, <code>--primary</code>, <code>--accent</code>, <code>--ring</code>, <code>--border</code>, <code>--input</code>, <code>--muted</code>, <code>--sidebar*</code>, <code>--panel-bg</code>, <code>--panel-border</code>, <code>--surface*</code>), all <code>!important</code>; - body ground; heading fonts; forged <code>.compact-panel</code> / <code>.compact-row</code> ground; Tailwind radius flattening (<code>[class*="rounded-..."]</code>); button/input base; <code>::-webkit-scrollbar</code>; <code>.tier*</code> colours; a global reduced-motion rule.
Depends on	Tailwind utility class substrings (<code>rounded*</code> , <code>hover:underline</code> , <code>px-</code>); the shadcn-style CSS variables the game defines [Assumption: token names inferred from the overrides]; <code>.compact-panel</code> , <code>.compact-row</code> .
Runs	Static CSS, active while injected.
States & fallbacks	Font files are self-hosted with <code>font-display: swap</code> , so the fallback serif/sans shows until they load.
Responsive	None in <code>base.css</code> beyond inherited tokens.
Motion	<code>@media (prefers-reduced-motion: reduce) { * { transition: none !important; animation: none !important; } }</code> . This applies to the whole page , including the game (FUN-03).
Hover / focus / keyboard	Buttons: <code>filter: brightness(1.08)</code> on hover. <code>:focus-visible</code> gets <code>outline: 1px solid gold</code> with 2 px offset (<code>!important</code>) on buttons, <code>[role=button]</code> , inputs, selects and textareas.
Accessibility	The focus ring is 1 px. WCAG 2.2 "Focus Appearance" (AAA) is not met; minimum contrast is [Verify]. Scrollbars are 7 px.

Edge cases & findings	- Overriding the game's own design tokens with <code>!important</code> restyles components the skin never classifies. That is useful, but it depends on the game's token names. - The <code>.iw-ico</code>, <code>.iw-btn*</code> and <code>.iw-xp*</code> rules are unused (MAINT-01).
Tests	<code>static-invariants.test.mjs</code> (breakpoints aligned to Tailwind, injection rules); fixture renders (<code>npm run fixtures</code>).

F-08 Surface colour repainting

What	Game surfaces painted in one of 20 known navy hex colours are repainted in the nearest-luminance warm ink (<code>#0B0C0A</code> , <code>#14130F</code> , <code>#191711</code> or <code>#1E1B15</code>). Borders in those colours become <code>#342D20</code> .
Why	Some grounds come from inline styles or Tailwind arbitrary values that no stylesheet can target generically [Source : module header].
Where	<code>BackgroundPainter.js</code> : <code>GAME_NAVIES</code> , <code>SURFACE_SELECTOR</code> , <code>isSurfaceCandidate</code> , <code>paintElement</code> , <code>paintBackground</code> , <code>clearBackgroundPaint</code> .
Depends on	Exact computed colours: an exact-hex whitelist. Candidates: <code>body</code> ; any element matching <code>.compact-panel</code>, <code>.compact-row</code>, <code>[class*="item-row"]</code>, <code>[role=dialog/menu/listbox]</code>, <code>main</code>, <code>section</code>, <code>article</code>, <code>aside</code>, <code>header</code> or <code>nav</code> ; children of <code><body></code> ; grandchildren of <code><body></code> with more than one child element.
Runs	- At boot on <code>document.body</code> . - On every <code>iw:dom-flush</code>, for each changed background root (or the whole document when there are no roots). - Each run is a TreeWalker over the root, with <code>getComputedStyle</code> on candidates only.
States & fallbacks	Writes inline <code>!important</code> through <code>InlineStyleOwner</code> , and records <code>data-iw-painted="1"</code> . React rewriting a stock colour gets repainted on the next flush.
Edge cases & findings	- Alpha is ignored, so translucent navy becomes opaque (FUN-11). - The walker runs a <code>closest()</code> per element (PERF-04). - Skin-owned trees (<code>.fs-inv-row</code>, <code>.fs-skill-header</code>, <code>.iw-tip</code>) and <code>.fs-skill-wrapper</code> are skipped.
Tests	<code>smoke.test.mjs</code> (native inline background restored exactly on teardown).

F-09 Per-zone theming

What	The page's accent palette follows the zone the player is in. There are nine environment themes: glacial, infernal, verdant, forged-metal, celestial, voidborn, runic-arcane, lunar-spectral and tempest-oceanic. For each theme the following swap: - the theme palette (<code>--iw-th-*</code>); - the skills UI atlas (<code>theme_<t>.webp</code>); - the four-corner frame filigree (<code>panel_corners_<t>.webp</code>); - the separator flourish (<code>separator_flourish_<t>.webp</code>); - the button art (revised-v5 action/chevron atlases, compact-ghost-v3 small-control atlas, card-v6 V2 plates).
Why	Ties the chrome to the game world, so each zone "feels" different [Project record : <code>docs/traps/zone-theming.md</code>].
Where	<code>HeaderRenderer.applyZoneTheme</code> writes <code>html[data-iw-zone-theme]</code> plus inline <code>--iw-zone-atlas</code>, <code>--iw-corner-filigree</code> and <code>--iw-zone-separator</code> on <code><html></code>, and calls <code>SkillsArtService.applyThemeVariables(html, theme)</code>. That call writes <code>data-iw-button-atlas="revised-v5"</code>, <code>data-iw-compact-atlas="compact-ghost-v3"</code>, <code>--iw-compact-atlas</code>, and <code>--iw-<kind> / --iw-<kind>-paint</code> for 12 button-art kinds. <code>zoneThemes.js</code> (generated map: zone 1-34 → theme). <code>base.css</code> (9 <code>:root[data-iw-zone-theme="..."]</code> palettes, the derived <code>color-mix</code> edge scale, the un-themed defaults). <code>card-button-atlas.css</code> (generated per-theme V2 plate windows). <code>skillpanel.css</code> L3642-3885 (themed button states, the revised-v5 crossfade).
Depends on	The zone label, read from <code>[data-iw-ui="zone-title"]</code> (F-19). Fallback: a text scan for <code>^zone\s*\d+\s*</code> : over every <code>div</code> , <code>span</code> , <code>p</code> , <code>strong</code> .
Runs	On every header reconcile (a rAF after each <code>iw:dom-flush</code>). It writes only when the theme key or button readiness changes.
States & fallbacks	Unknown zone (nothing read yet, or a zone outside 1-34): <code>data-iw-zone-theme="default"</code>, inline variables removed, <code>base.css</code> defaults. Leaving the Game route (zone bar unmounts): the last zone read is kept (<code>lastZoneNumber</code>), so other routes stay themed.

Motion	Theme changes apply instantly; no transition.
Edge cases & findings	• <code><html></code> is outside the observed <code><body></code>, so these writes can never trigger a flush. • The <code>exact-v3</code> art branch is unreachable (SIZE-01). • The zone text scan on non-Game routes is a cost (PERF-02). • A zone added after 34 falls back to <code>default</code> until <code>zone-theme-map.json</code> and <code>zoneThemes.js</code> are regenerated.
Tests	<code>zone-themes.test.mjs</code> (map in step with the JSON; every asset exists); <code>button-art-states.test.mjs</code> , <code>revised-button-atlas.test.mjs</code> , <code>card-button-art.test.mjs</code> , <code>compact-button-atlas.test.mjs</code> ; <code>corner-filigree-density.test.mjs</code> ; <code>separator-symmetry.test.mjs</code> .

F-10 Generic control language

What	Every game <code><button></code> and <code>[role=button]</code> that is not specifically themed gets a "carved slate" surface: a faint edge, a bevel gradient and an inset shadow. Hover lights an ember inner glow, and disabled controls are desaturated and darkened. Search inputs get a recessed look.
Why	Gives unclassified controls (modal buttons, steppers, chat tools) a consistent material without changing their geometry or their text colour , which is often live state (rule 5) [Source : comments at <code>ui-system.css</code> L1008-1050].
Where	<code>ui-system.css</code> : L1019 (rest), the hover rule, L1057-1062 (min-height floors for classified controls only), the disabled rule, and <code>input[type=search]</code> . <code>base.css</code> has the matching <code>:not()</code> chains.
Depends on	The <code>:not()</code> opt-out chain lists every control that paints its own art: <code>.iw-item-ref</code> , <code>[data-iw-collapse]</code> , <code>[data-iw-compact-button]</code> , <code>[data-iw-inventory-control]</code> , <code>[data-fs-preserved-action="control"]</code> , <code>[data-iw-ui="nav-tab"]</code> , <code>[class*="chat-name-"]</code> , <code>[class*="hover:underline"]</code> , the skill level-progress/nav-button/action-button roles, <code>[data-iw-quest-role]</code> , plus more (Appendix B, <code>ui-system.css</code> :1019).
Runs	Static CSS.
Hover / focus / keyboard	Hover: <code>border --iw-line-hot</code> , inner glow. Focus: from <code>base.css</code> . Disabled (<code>:disabled</code> , <code>[aria-disabled="true"]</code>): <code>filter: saturate(.58) brightness(.84)</code> .
Edge cases & findings	• Specificity is (0,30,3) (D-01). A new control that paints its own art must be added to the <code>:not()</code> chain, never beaten with higher specificity (MAINT-02). • Geometry floors apply only to positively classified controls; the reason is recorded as Audit S3.5.
Tests	<code>skill-card-system.test.mjs</code> (cascade checks with all sheets); <code>revised-button-atlas.test.mjs</code> .

F-11 Ghost-metal compact button art

What	Small text and icon controls wear themed "ghost metal" art with three states (idle, hover, pressed) that crossfade: • navigation tabs; • inventory filters and pager; • zone actions; • the World Chat Send button; • preserved inventory row actions; • the loadout I/II pair; • Change Zone and its timer icon. The selected loadout and the active filter keep a visible accent.
Why	These controls are too small for the large action frames. A three-slice atlas keeps the end caps undistorted at any label width.
Where	<code>CompactButtons.js</code> (<code>decorateCompactButtons</code> , called by <code>UIFoundation.queueClassify</code> step 11), <code>compact-buttons.css</code> , <code>--iw-compact-atlas</code> from F-09.
Depends on	• Matching. <code>CONTROL = button</code>, <code>a[data-iw-ui="nav-tab"]</code>, <code>a/[role=button][data-fs-preserved-action="control"]</code>. <code>kind()</code> then classifies by earlier skin marks, by label (<code>^(I II)\$</code> pairs, <code>^change zone\$</code>) and by timer glyphs. • Selection. Read from <code>aria-pressed</code> / <code>aria-selected</code>, else <code>data-state="active"</code>, else a class substring (<code>bg-/text-/border- + orange amber primary accent ember</code>).
Runs	Every classify pass (RAF after each <code>iw:dom-flush</code>), over every matching control.

States & fallbacks	- The layer spans(<code>span[data-iw-compact-layer="idle hover clicked"]</code> , <code>aria-hidden</code>)are appended inside the native control. - They stay <code>display:none</code> unless <code>html[data-iw-compact-atlas="compact-ghost-v3"]</code> is present, so there is no art without a theme. - A control that stops matching has its layers removed.
Motion	Hover layer fades in over 180 ms; pressed layer over 70 ms. Both are disabled under reduced motion.
Hover / focus / keyboard	<code>:hover</code> and <code>:focus-visible</code> show the hover layer; <code>:active</code> shows the pressed layer; <code>:focus-visible</code> has a 2 px accent outline. Disabled controls hide the hover and pressed layers and drop to 0.45 opacity.
Accessibility	Layers are <code>aria-hidden</code> ; the native label and handlers are untouched. The active nav tab text is forced light for contrast.
Edge cases & findings	Appending children into a native <code><button></code> changes <code>button.children</code> . Game code that reads <code>children[0]</code> of these buttons would see the same first child, because layers are appended last [Assumption : the game does not index children].
Tests	<code>compact-button-atlas.test.mjs</code> , <code>compact-button-reconcile.test.mjs</code> , <code>compact-button-boot.test.mjs</code> .

F-12 Forged section frames

What	Every top-level game panel(Inventory, Skill Actions, Quests, Market, Leaderboards, World Bosses, Village, Salvaging, Zone Selector, the activity panels, and the nav rail)is drawn as one shared "forged frame". The frame has a textured ground with a per-zone wash, a 1 px themed edge, a gold hairline along the top, and four corner filigrees at a fixed size. Headings get the Cinzel frame title at <code>--iw-frame-title-size</code> (18 px).
Why	One treatment so every panel reads as the same object; previously three different grounds competed [Source : comment L27-41].
Where	<code>UIFoundation.classifySectionFrames (data-iw-ui="section-frame" , section-title);</code> <code>ui-system.css</code> L44-134 (frame, <code>::before</code> hairline, <code>::after</code> corner border-image , children raised to z-index 1), the ≤ 1024 px padding and corner scale, the nav rail opt-out <code>:not(:has([data-iw-ui="nav-tab"]))</code> , and layout columns stripped bare.
Depends on	Panels. <code>.panel</code> , via <code>preferRendered</code> , excluding: overlays; elements owned elsewhere(<code>header</code> , the zone bar, <code>[data-iw-panel]</code> activity panels); and panels nested in a frame(leaf frames only, via <code>:not(:has(...))</code>). Headings. Orphan headings are matched by name(<code>inventory market leaderboards quests worldbosses village salvaging skill actions zone selector</code>)and walked up to 5 levels.
Runs	Each classify pass, using a heading index built once per pass.
States & fallbacks	A panel without a recognised heading still gets the frame, because frames key on <code>.panel</code> rather than on copy [Project record : <code>docs/traps/panel-frames.md</code>].
Responsive	≤ 1024 px: padding comes from <code>--iw-frame-pad-mobile-*</code> and the corner shrinks to 27×25 px.
Accessibility	Decorative pseudo-elements only, with <code>pointer-events: none</code> .
Edge cases & findings	- A frame inside another frame is left bare (the layout-column case). - A collapsed frame hands its corner art to <code>collapsible.css</code> through a <code>:not([data-iw-collapsed="1"])</code> opt-out, because the base rule is <code>!important</code> on every declaration.
Tests	<code>panel-frame-nesting.test.mjs</code> , <code>route-swap-reclassify.test.mjs</code> , <code>collapsible-frames.test.mjs</code> .

F-13 Collapsible frames

What	A 22 px chevron button at the right of each frame's heading row folds the panel to a single title bar; clicking again expands it. The choice is remembered per panel across reloads. Panels covered: Skills, Inventory, Quests, Village, World Bosses, Current Action, Action Log, World Chat, Zone Control and the Village scene.
Why	Players with many panels want to hide the ones they are not using. Only the skin's own appended control changes state; React's nodes are never moved or wrapped [Source : <code>CLAUDE.md</code> ; <code>docs/traps/collapsible-frames.md</code>].
Where	<code>CollapsibleFrames.js</code> : <code>decorateCollapsibleFrames</code> (the last classify step); <code>markSpine</code> , <code>ensureToggle</code> , <code>applyState</code> , <code>loadPreferences</code> , <code>persist</code> , <code>clearCollapsibleFrames</code> . <code>collapsible.css</code> .

Depends on	Frames. <code>FRAME = [data-iw-ui="section-frame"]</code>, <code>[data-iw-inventory-root="1"]</code> or <code>.fs-skills-section-frame[data-iw-skills-ui-ready="1"]</code> (leaf frames only). Title. <code>[data-iw-ui="section-title"]</code> → <code>[role=heading]</code> → <code>h1-h4</code>. Head. The frame's direct child that contains the title. Fold target. The closest leaf <code>.panel</code>, else the frame. Storage key. <code>panel:<slug></code> or <code>title:<lowercase title></code>.
Runs	Each classify pass. Preferences load once per activation from <code>chrome.storage.local['iw-collapsed-frames']</code> .
States & fallbacks	Collapsed: <code>data-iw-collapsed="1"</code> on the target; <code>[data-iw-collapsed="1"] > *:not(toggle):not(head) → display: none;</code> - inside the head, every element not on the spine is hidden, leaving the title plus the toggle; - a single top-left corner filigree is drawn. Expanded: attributes removed. Toggles whose target disappears are removed along with their marks.
Responsive	Under <code>(pointer: coarse)</code> , a <code>::before</code> extends the hit area by 5 px on each side (32 px target).
Motion	The chevron rotates between -135° and 45° ; disabled under reduced motion.
Hover / focus / keyboard	Native <code><button type="button"></code> , so Enter and Space work. <code>:focus-visible</code> outline. The click handler calls <code>preventDefault()</code> and <code>stopPropagation()</code> , so a heading row that is itself clickable in the game does not also react.
Accessibility	<code>aria-expanded</code> reflects state; <code>aria-label</code> and <code>title</code> read <code>Collapse <name></code> or <code>Expand <name></code> . No <code>aria-controls</code> .
Edge cases & findings	- The title can vanish from collapsed bars after a kill-switch round trip (FUN-04). - The Village ledger and boss rewards use their own disclosure namespaces, so the document-wide sweep never adopts them. - When V2 cards are folded under a collapsed Skill Actions frame, <code>skillcard-v2.css:109</code> hides them explicitly.
Tests	<code>collapsible-frames.test.mjs</code> (38 checks: every parent frame folds alone, persistence, teardown).

2.3 HEADER, NAVIGATION AND ZONE CHROME

F-14 Player header reorganisation

What	The game's player header becomes one framed banner laid out as a grid of three parts: Identity region: an ornamental crest, a smoked-glass block with the "IdleWorlds" brand between rules, the player name, the profile title, "Combat Lv N · Zone N", and a "Players online: N" line with a teal dot. Utility rail: the game's icon buttons as equal-width plates. Status grid: vitals and buff tiles in two columns (combat, boost, timers, gold). On wide screens the combat and boost tiles form a plaque.
Why	Gives the header the RPG identity, and makes the stat tiles readable at a glance without changing any value [Project record: <code>docs/traps/header-chrome.md</code>].
Where	<code>HeaderRenderer.js</code>: - resolution: <code>findLiveHeader</code>, <code>classifyProfile</code>, <code>ensureCrest</code> (prepends <code>span.fs-header-crest</code>), <code>classifyUtilities</code>, <code>classifyStatus</code>, <code>tagStatusCards</code>, <code>statKind</code>; - painting: <code>applyHeaderVars</code>, <code>reconcile</code>, <code>queueReconcile</code>, <code>clearHeaderRenderer</code>. <code>header.css</code> §0–§5, §9, §10.
Depends on	The live header: a <code><header></code> whose text contains <code>combat lv N</code>, <code>players online: N</code> and <code>atk N ... def N ... hp N</code>, chosen with a visibility tie-break. Name and profile: <code>h1.header-player-name</code> (or <code>h1</code>); <code>.min-w-0.overflow-hidden</code> for the profile block. Utilities and status: <code>button.header-icon-btn</code>; <code>.grid.grid-cols-2</code> or <code>.stat-chip</code>. Assets: the header art files under <code>assets/header/</code>.
Runs	<code>queueReconcile</code> on every <code>iw:dom-flush</code> , coalesced to one rAF. The resolution is cached and revalidated by connectivity and roles. Stat cards are re-tagged on every reconcile.

States & fallbacks	- If no qualifying <code><header></code> exists (logged-out page, changed copy), nothing is tagged and the header stays native. - A missing profile, utilities or status block leaves that part unstyled. - Roles are written as <code>data-iw-header="root layout profile profile-name profile-meta profile-online brand profile-title identity-region utilities utility-button status-grid status-card announcement zone-shell"</code>, plus <code>data-iw-header-card</code> (1-based index) and <code>data-iw-header-stat</code> (the kind).
Responsive	≤1280 px: one column; the identity block stretches; 42 px utilities. ≤860 px: crest 108×116; utilities in a flex row; status grid in two auto columns. ≤768 px: mobile header painting (F-15). ≥861 px: combat/boost plaque via <code>:has([data-iw-header-stat="boost"])</code>.
Motion	Hover transitions on utility plates; disabled under reduced motion (§10).
Hover / focus / keyboard	Utility plates brighten on hover. Native buttons keep their focus behaviour, with the <code>base.css</code> focus ring.
Accessibility	The crest is <code>aria-hidden</code> . "Players online" is a real <code><button></code> in the game and stays one; only its chrome is removed. The player-name colour is left to the game (rule 5): it may encode a cosmetic or rank.
Edge cases & findings	- Detection depends on English copy (MAINT-03). - Teardown walks <code>querySelectorAll('*')</code> to remove the header custom properties, which costs O(all nodes) once per disable. <code>--iw-header-frame-bar</code> maps to no asset (MAINT-04). - Two statements are joined on one source line (D-10).
Tests	<code>header-compact.test.mjs</code> , <code>header-hidden-duplicate.test.mjs</code> , <code>smoke.test.mjs</code> . Header assets are audited by <code>build-tools/audit-header-assets.mjs</code> (not in <code>npm test</code>).

F-15 Per-zone header painting

What	The header background shows a painting of the player's current zone: a wide 4:1 strip on desktop, and a portrait crop at ≤768 px.
Why	Makes location visible at a glance and ties the header to F-09 theming.
Where	<code>HeaderRenderer.applyZoneSurface</code> (L90-102) writes <code>--iw-header-surface</code> and <code>--iw-header-surface-mobile</code> inline on the header root, plus <code>data-iw-zone="<N> fallback"</code> . <code>header.css</code> §2 paints the variable with <code>cover</code> , and §9 swaps to the mobile variable at ≤768 px.
Depends on	The zone number (F-09 fallback chain); <code>assets/header/zones/zone_1...34(_mobile).webp</code> ; <code>header_surface.webp</code> .
Runs	Each header reconcile. It writes only when the zone key changes or the variable is missing.
States & fallbacks	Zone unknown or above <code>ZONE_SURFACE_MAX</code> (34) → <code>header_surface.webp</code> . The last zone read is retained on non-Game routes.
Responsive	Desktop/mobile art switch at 768 px. The art choice is CSS-only, so a resize needs no JavaScript.
Edge cases & findings	<code>make-zone-surfaces.mjs</code> can generate procedural placeholders for zones without art. The imported paintings come from a sibling repository (§1.8, [Verify] rights).
Tests	<code>zone-headers.test.mjs</code> (all 68 files present and referenced).

F-16 Main navigation rail

What	The route tabs (Game, Market, Leaderboards, Village, Dungeon) become one row of 32 px forged plates in uppercase Barlow. The active tab gets an ember gradient and a glowing underline.
Why	Clear route affordance that preserves the game's own active state (rule 5).
Where	<code>UIFoundation</code>: <code>resolveMainNav</code>, <code>classifyMainNav</code> (cached by <code>mainNavResolutionValid</code>: epoch, connectivity, roles, tab count); <code>applyMainNavState</code>, <code>deriveTabActive</code>, <code>warmBackground</code>. CSS: <code>ui-system.css</code> L699-780 (rail), L1081-1090 (≤768 px), L1113-1135 (≤1279 px container font fitting). <code>header.css</code> §6 (L661-725) styles the same roles earlier in the cascade; it is overridden property-by-property (§3.9 Conflicts, X-01).
Depends on	Tabs whose label is one of <code>NAV_LABELS = ['game', 'market', 'leaderboards', 'village', 'dungeon']</code> ; the rail is a <code>.panel</code> .

Runs	Each classify pass.
States & fallbacks	Active-state priority: 1. the tab's own semantic state (<code>aria-selected="true"</code> , <code>aria-current="page"</code> , or an active-looking class); 2. the route in <code>location.pathname / hash</code> ; 3. on the first classification only, a "warm" computed background colour; 4. the previous state. Writes <code>data-iw-ui="main-nav main-nav-shell nav-tab"</code> , <code>data-iw-tab="<label>"</code> and <code>data-iw-state="active"</code> .
Responsive	• ≤ 1279 px: the rail becomes an inline-size container <code>iw-nav-row</code> , tabs do not wrap, and the font size is <code>clamp(7.5px, (100cqi - fixed)/k, 11.5px)</code> with a measured <code>k</code> (28.1; 23.25 at ≤ 767 px). • ≤ 768 px: full-width rail, tabs flex. • ≤ 560 px (<code>header.css</code>): horizontal scroll with a hidden scrollbar.
Motion	Colour/background transitions on hover; none under reduced motion.
Hover / focus / keyboard	Hover lightens the plate. Tabs are the game's own links or buttons, so keyboard behaviour is native.
Accessibility	Active state is conveyed by the game's semantics when present; the skin adds no ARIA.
Edge cases & findings	• A tab label outside <code>NAV_LABELS</code> (a new route) is not styled as a tab. • The route fallback reads <code>location</code> only; it never writes to it.
Tests	<code>header-compact.test.mjs</code> , <code>menu-rows.test.mjs</code> , <code>route-swap-reclassify.test.mjs</code> .

F-17 Toolkit link

What	A "Toolkit ↗" link at the end of the nav rail opens <code>https://idleworldstoolkit.com</code> in a new tab. It is hidden below 768 px.
Why	The skin author's companion calculator site. Opening in a new tab keeps the idle session running [Source: comment at <code>UIFoundation.js</code> L292-299].
Where	<code>UIFoundation.ensureToolkitLink</code> (L289-304); <code>ui-system.css</code> L785-815, L1090, L1134.
Depends on	The resolved nav track (F-16).
Runs	Each classify pass. The link is created once per track, with every attribute set before insertion, so it costs one mutation record.
States & fallbacks	Not created if the rail is not found. Its label is not in <code>NAV_LABELS</code> , so it can never be marked active.
Responsive	≤ 767 px: <code>display: none</code> (the phone bar read as too crowded) [Project record, 2026-09-16].
Hover / focus / keyboard	A native <code><a href></code> : focusable, activates with Enter. Hover changes colour, border and background.
Accessibility	Its <code>title</code> says it opens in a new tab. The visible "↗" is CSS <code>::after</code> content.
Edge cases & findings	A third-party destination inserted into the game's navigation needs a product decision (SEC-09). <code>rel="noopener noreferrer"</code> means the target cannot reach the game tab and receives no referrer.
Tests	<code>header-compact.test.mjs</code> , <code>menu-rows.test.mjs</code> , <code>flush-quiescence.test.mjs</code> (no repeated writes).

F-18 Announcement bar

What	The game's short announcement line under the navigation becomes a teal gradient bar with a four-point star ornament.
Why	Distinguishes system notices from navigation and zone chrome.
Where	<code>HeaderRenderer.classifyAdjacent</code> → <code>findAnnouncement</code> (<code>data-iw-header="announcement"</code> , <code>--iw-announcement-frame</code>); <code>header.css</code> §7. When merged, it is <code>HeaderChrome</code> 's <code>data-iw-chrome="notice"</code> (F-20).
Depends on	Position: the next sibling of the nav rail's <code>.panel</code> (or of the nav or its parent) whose text is ≤ 260 characters and is not the zone bar.
Runs	Each header reconcile.
States & fallbacks	No announcement → nothing is tagged.
Accessibility	The star is a <code>::before</code> pseudo-element; the text is unchanged.

Edge cases & findings	Identification by position and length: a short, unrelated element in that slot would be styled as an announcement [Verify live].
Tests	<code>header-compact.test.mjs</code> .

F-19 Zone bar

What	The "🌙 Zone N: Name" bar shows the zone title in Cinzel with dimmed secondary text. It has a faint moonlit scene behind it and tone-coded action plates: Zones in teal and Next zone in ember. Other buttons in the bar become text links.
Why	Makes the player's location and the zone controls distinct from general navigation.
Where	<code>UIFoundation</code> : <code>zoneBarCandidatePresent</code> (a cheap probe for "zones" plus "next zone" buttons); <code>classifyZoneBar</code> (the innermost label matching <code>^zone\s*\d+\s*</code>: after stripping leading icons; host at most 5 levels up and not inside <code>header / nav</code>); - role writes <code>data-iw-ui="zone-bar zone-title zone-action"</code>, <code>data-iw-zone-action="zones prev next"</code>, <code>data-iw-zone-link="1"</code>; - cached <code>zoneBarResolutions</code>. <code>HeaderRenderer</code> adds <code>data-iw-header="zone-shell"</code> and the zone art variables. CSS: <code>header.css</code> §8, <code>ui-system.css</code> zone rail and <code>[data-iw-zone-link]</code> rules.
Depends on	English labels("Zones", "Next zone", "Zone N:"); the zone bar exists only on the Game route.
Runs	Each classify pass, with the probe first.
States & fallbacks	No zone bar (other routes) → nothing is tagged; header theming keeps the last zone (F-09).
Responsive	Wraps; the action group is pushed right (<code>margin-left: auto</code>). In merged layouts, see F-20.
Hover / focus / keyboard	Native buttons. Tone plates brighten on hover.
Accessibility	The scene is a masked <code>::before</code> at 0.34 opacity, decorative only.
Edge cases & findings	The tone falls back to <code>:first-of-type / :last-of-type</code> when <code>data-iw-zone-action</code> is missing (FUN-12).
Tests	<code>header-compact.test.mjs</code> , <code>route-swap-reclassify.test.mjs</code> , <code>zone-themes.test.mjs</code> .

F-20 Merged header chrome

What	The nav rail, the announcement and the zone bar render as one framed box under the header. Row 1: zone title (left) and announcement (right). Row 2: navigation tabs (left) and zone actions (right). The box has one frame and a divider.
Why	Three stacked boxes looked cluttered. The merge is done by CSS grid placement on existing nodes, without reparenting (rule 2) [Project record: docs/traps/header-chrome.md].
Where	<code>HeaderChrome.js</code> (<code>classifyHeaderChrome</code> , step 5 of the classify pass) writes <code>data-iw-chrome="shell nav notice zone-bar zone-text zone-actions"</code> . <code>ui-system.css</code> "Header chrome merge": the shell becomes a 2-column grid; <code>zone-bar</code> gets <code>display: contents</code> ; its children are placed in grid cells; a <code>::before</code> frame spans rows 2–3; an <code>::after</code> divider sits on row 3.
Depends on	Exact structural preconditions: - the rendered zone bar's parent has a direct <code><header></code> child; - the nav is the nearest preceding sibling containing nav tabs; - there are exactly zero or one non-<code>.panel</code> siblings (text ≤260 characters) between nav and zone bar; - the zone bar has exactly two children (text containing the zone title; actions containing the zone actions).
Runs	Each classify pass. Cached; re-resolved when the notice appears or disappears.
States & fallbacks	Any precondition fails → all marks are cleared and the three parts render separately, styled by F-16, F-18 and F-19.
Responsive	≤860 px: rows stack (zone text, notice, nav, actions).
Accessibility	<code>display: contents</code> on the zone bar removes its box but keeps its children. Chromium keeps them in the accessibility tree; historic <code>display: contents</code> accessibility bugs apply only to elements with their own role [Assumption: the bar is a plain <code>div</code>].

Edge cases & findings	Deliberately refuses rather than guesses, so a game layout change degrades to the un-merged look.
Tests	<code>header-compact.test.mjs</code> (34 checks), <code>menu-rows.test.mjs</code> .

2.4 ACTIVITY PANELS

F-21 Current Action panel and progress smoothing

What	The Current Action panel is framed. Its progress bar becomes an 18 px segmented track (10 segments) with a gradient fill and a slow shimmer. The fill moves continuously between the game's once-per-tick jumps. The queue ("Queued ..." with its button) becomes an inset row.
Why	The game steps the bar once per tick, so a CSS transition over the measured tick length makes it glide. It snaps back, rather than sliding backwards, when an action completes [Project record: docs/traps/current-action-progress.md].
Where	Classification: <code>UIFontFoundation.classifyActivityPanels</code> → <code>runActivityPanel</code> → <code>classifyCurrentAction</code> (L1085-1096) → <code>findCurrentActionProgress</code>. Smoothing: <code>smoothActionProgress</code> (L978-983) → <code>ProgressCadence.sampleProgress</code> (median of the last 5 tick intervals within 60-4000 ms, at least 3 samples, lead bias ×1.12, reset on drop) → inline <code>--iw-progress-duration</code>; <code>markProgressReset</code> adds <code>data-iw-progress-reset="1"</code> for one frame. CSS: <code>ui-system.css</code> L354-462 (text, track, segments, fill transition <code>width var(--iw-progress-duration, var(--iw-action-tick, 1s)) linear, ::before shimmer iw-control-meter-current 5.7s infinite, reset, reduced motion</code>), L464-500 (queue).
Depends on	Panel: a "Current Action" heading or leaf text, or <code>#current-action-panel</code>. Progress: <code>[role=progressbar] / [aria-valuenow][aria-valuemax]</code>, or a descendant with inline <code>width: N%</code>. Queue: a leaf with the text "queued".
Runs	Each classify pass (the host is cached; parts are re-run every pass). A tick changes the inline <code>width</code> , which produces a <code>style</code> mutation, a flush and a sample.
States & fallbacks	- No action running (no percentage): no sample, no duration; the CSS default is <code>--iw-action-tick (1s)</code>. - A completion reset (percentage drops) disables the transition for one frame and excludes the next interval from the median.
Responsive	≤640 px: tighter gaps.
Motion	The fill transition equals the measured tick; the shimmer loops forever while the panel exists (PERF-01). Reduced motion removes both.
Hover / focus / keyboard	The queue button is the game's own control.
Accessibility	A semantic <code>role=progressbar</code> from the game remains. The skin writes no ARIA.
Edge cases & findings	- The skin never computes or writes a progress value, only the transition length. - The duration write is an inline <code>style</code> change, visible to the observer; it re-enters once and returns because the percentage is unchanged.
Tests	<code>action-progress-theme.test.mjs</code> , <code>flush-quiescence.test.mjs</code> , <code>smoke.test.mjs</code> .

F-22 Action Log panel

What	The Action Log is framed. Its title (a game <code><button></code>) loses the plate look and reads as a heading link. "View All" becomes a small caps control. The log becomes a transparent, scrollable feed (max 300 px) whose rows have timestamps, a coloured first line and dim detail text. The XP/hr readout sits in the header row.
Why	Dense chronological text reads better as a ledger than as stacked cards.
Where	<code>classifyActionLog</code> (L1098-1103) → <code>classifyFeed</code> (L1059-1083): markers are leaves matching <code>^\d{1,2}:\d{2}:\d{2}\$</code> , else leaves reading <code>system</code> . The feed is the markers' common ancestor, and the rows are the direct children holding markers. <code>classifyPanelHeader</code> handles split headers. <code>ui-system.css</code> L246-350 and L503-611.
Depends on	"Action Log" heading; "View All" label; <code>hh:mm:ss</code> timestamps; the XP/hr pattern <code>[\d, .]+ xp / hr</code> .
Runs	Each classify pass; the feed and rows are re-tagged every pass.

States & fallbacks	No timestamps → the fallback picks the first child that is not a header, not skin-owned, not a form, and has no input or heading. An empty log → no rows.
Responsive	≥641 px: the header's first child becomes a flex row. ≤640 px: smaller row padding and font.
Hover / focus / keyboard	The title button keeps its native behaviour; hover changes colour.
Accessibility	No semantic change; the feed keeps its native element types.
Edge cases & findings	Row detection depends on the timestamp format; a localised clock would fall back to the unstyled feed [Assumption].
Tests	route-swap-reclassify.test.mjs , smoke.test.mjs .

F-23 World Chat panel

What	World Chat is framed. Messages sit in a dark #0B0C0A scroll box (max min(46vh, 430px)), with player names and bold text in theme colours. The composer is a two-column grid: a forged input with a gold focus outline, and an ember Send button (compact art from F-11).
Why	Readability, and consistency with the rest of the frame language.
Where	classifyWorldChat (L1105-1120): the input's placeholder matches /message\s+world\s+chat/i ; the composer is the nearest ancestor (≤3 levels) containing a button with the text send . Then classifyFeed . ui-system.css L607-689.
Depends on	The placeholder text "Message World Chat"; the "Send" label; hh:mm:ss or system markers; the game class .feed-panel .
Runs	Each classify pass.
States & fallbacks	Placeholder not found → no composer styling; the feed is still framed.
Responsive	≤430 px: the composer stacks and Send becomes full width; row header items wrap.
Hover / focus / keyboard	Input :focus outline; Send brightens on hover when enabled. Typing, submit and Enter-to-send are native.
Accessibility	Placeholder colour is restyled [Verify contrast]. Chat text is never rewritten, and player-controlled text is never parsed as HTML (NameScanner reads text nodes only).
Edge cases & findings	Item names inside chat messages can open item hovercards through NameScanner (F-45), which has no visible affordance (FUN-02).
Tests	route-swap-reclassify.test.mjs .

F-24 Daily XP Boost chip

What	The "Daily XP Boost ... +N% XP" line in the Skill Actions heading row becomes a small green, met-style chip.
Why	Signals an active bonus with the same green "met" language used for requirements.
Where	UIFoundation.classifyDailyBoost (L1336-1360; pattern DAILY_BOOST = /^daily\s+xp\s+boost\b.*\s*d+(?:\.\d+)?\s*%\s*xp/i) writes data-iw-skill-boost="1" on the innermost matching element in a heading row reading "Skill Actions" or "Actions". ui-system.css L192-208.
Depends on	English copy and heading position.
Runs	Each classify pass; the mark is cleared when the text no longer matches.
Edge cases & findings	When the Skill Actions frame is collapsed, the chip is hidden with the rest of the head row except the spine (F-13).
Tests	collapsible-frames.test.mjs (hidden with the head row), smoke.test.mjs .

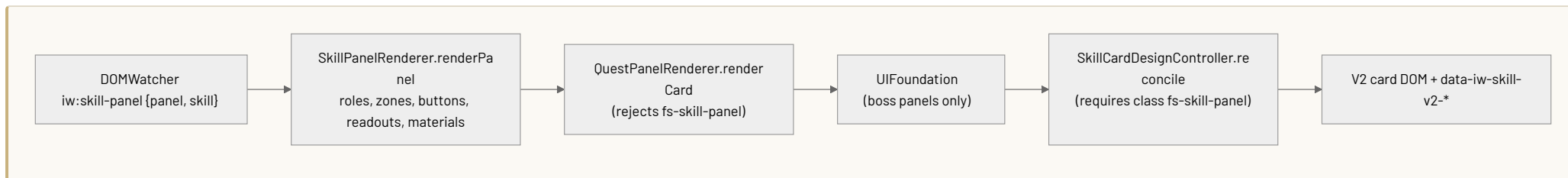
2.5 SKILL ACTIONS CARDS

The skill card is built in two layers, and both are always active:

- **Layer 1: SkillPanelRenderer** . Recognises the card, assigns semantic roles and zones, styles the command buttons, and neutralises the game's readout and material chrome.

- **Layer 2: SkillCardDesignController, the "V2" card.** Adds the compact medallion, the square themed action button with glyph and fitted label, the smoothed fill, the long-action countdown, and the material body.

The old choice between a "New" and a "Current" design was removed. `html[data-iw-skill-card-design="new"]` is always set while the skin is active (D-03). The CSS for both layers is concatenated into the single `skillpanel` style element (F-03).



F-25 Skill card detection and discipline identity

What	Each skill card is recognised as one of 12 disciplines, or as locked. The rest of the treatment (accent colour, medallion art, button glyph, labels) follows from that.
Why	<code>.compact-panel</code> is shared by skill, quest, boss, village and shop cards, so a skill card is identified positively , never from arbitrary body text [Source : <code>DOMWatcher.js</code> comments].
Where	<code>DOMWatcher.detectSkillType</code> (L143-203), <code>skillIdentitySignals</code> , <code>SKILL_IDENTITY_ALIASES</code> (L71-85), <code>skillSignature</code> , and the <code>skillTypeCache</code> <code>WeakMap</code> . The type is delivered in <code>iw:skill-panel.detail.skill</code> .
Depends on	English button labels and identity labels.
Runs	For each dirty <code>.compact-panel</code> in a flush. Cached per panel by a content signature: button texts, identity signals (leaf text ≤ 32 characters outside buttons and links, plus headings/ <code>[class*=skill-name]</code>), and locked copy.
States & fallbacks	Decision order: 1. Quest rejection. A <code>Turn In</code> or <code>Skip</code> (n) button plus a leaf starting <code>Reward:</code> $\rightarrow$ <code>unknown</code> . 2. Exact verbs. <code>fight</code> $\rightarrow$ <code>combat</code> , <code>mine</code> $\rightarrow$ <code>mining</code> , <code>prospect / cut</code> $\rightarrow$ <code>jewelcrafting</code> , <code>smelt / forge</code> $\rightarrow$ <code>smithing</code> , <code>brew</code> $\rightarrow$ <code>alchemy</code> , <code>enchant</code> $\rightarrow$ <code>spellcrafting</code> , <code>tailor / sew / weave</code> $\rightarrow$ <code>tailoring</code> , <code>fish</code> $\rightarrow$ <code>fishing</code> , <code>chop</code> $\rightarrow$ <code>woodcutting</code> , <code>craft parts / build</code> $\rightarrow$ <code>construction</code> . 3. Locked. A disabled control plus "coming soon"/"upcoming skill" copy $\rightarrow$ <code>locked</code> . 4. Identity label via the aliases. 5. Generic verbs. <code>craft</code> $\rightarrow$ <code>crafting</code> , then <code>gather / harvest</code> $\rightarrow$ <code>gathering</code> . 6. Anchored label fallbacks. 7. Otherwise <code>unknown</code> , which means "not a skill card".
Edge cases & findings	<code>gather / harvest</code> are deliberately deferred: Spellcrafting reuses them for mana harvesting. <code>craft</code> alone cannot identify a skill because Spellcrafting and Tailoring reuse it. - An equal-length label change (Mine $\rightarrow$ Fish) invalidates the cache, because the key is content. - A new discipline or renamed verb falls through to <code>unknown</code>; see <code>docs/traps/classification.md</code> "adding a new skill".
Tests	<code>smoke.test.mjs</code> (Mine $\rightarrow$ Fish invalidation), <code>quest-card-detection.test.mjs</code> , <code>skill-structure-signature.test.mjs</code> , <code>skill-card-system.test.mjs</code> .

F-26 Skill card structure and three-zone layout

What	Every skill card is laid out as three zones: identity (left column): medallion and discipline name; content (centre): action title, base-XP chip and materials; commands (right band): action button and pager. Cards get a discipline accent line and the forged ground.
Why	The game's card is a vertical stack of loosely structured text. A consistent grid makes 12+ cards scannable.
Where	<code>SkillPanelRenderer.annotateStructure</code> (L959-1248), <code>structureSignature</code> (L941-957), <code>applyPanelChrome</code> , <code>renderPanel</code> . The card gets the classes <code>fs-skill-panel</code> and <code>fs-skill--<type></code> , plus <code>data-fs-skill</code> , <code>data-iw-ui="skill-panel"</code> , <code>data-iw-skill</code> and <code>data-iw-skill-glyph</code> . Roles (<code>data-iw-skill-role</code>): <code>identity</code> , <code>identity-level</code> , <code>identity-icon</code> , <code>action-title</code> , <code>level-progress</code> , <code>action-button</code> , <code>nav-button</code> , <code>nav-group</code> , <code>xp-gain</code> , <code>requirement</code> , <code>reward</code> , <code>action-detail</code> , <code>progress-track</code> , <code>progress-fill</code> , <code>ingredient</code> . Zones (<code>data-iw-skill-zone</code>): <code>identity</code> , <code>content</code> , <code>commands</code> . Layout : <code>data-iw-skill-layout-shell="1"</code> on the common shell; <code>data-iw-skill-layout="three-zone"</code> when no unexpected flow child exists. CSS : <code>skillpanel.css</code> , sections from L81 onwards; the governing composition is L3369-3640, "full-height identity, right-hand command band".
Depends on	Text shapes: - level readout matching <code>LEVEL_PROGRESS_PATTERN</code> ("Lv N - X% • ... to go"); - verbs and <code>SKILL_META</code> title actions; - requirement text colour classes (<code>red</code>, <code>rose</code>, <code>orange</code>, <code>amber</code>, <code>yellow</code>, <code>danger</code>, <code>warning</code> $\rightarrow$ <code>unmet</code>); "Base reward:" lines; <code>[role=progressbar]</code> or width heuristics.

Runs	On <code>iw:skill-panel</code> . The expensive structural walk is gated by <code>structureSignature</code> : type, child count, and button labels with digits blanked . <code>disabled</code> is included only for locked cards, because the game disables every button during any request (busy toggle); it once caused scroll-anchoring jumps [Project record].
States & fallbacks	Non-skill cards are cleared only if this renderer owns them. Without the ownership check, there were once about 400 apply/clear cycles per second fighting the quest renderer [Source comment]. If the zones cannot be resolved, <code>three-zone</code> is not written and the earlier generic card styling applies.
Responsive	Governing composition: <code>--fs-skill-command-w: 240px</code> , <code>--fs-skill-command-h: 64px</code> , <code>--fs-skill-identity-w: 150px</code> . <code>≤640 px: 116 px rail</code> . <code>≤430 px: 78 px rail</code> . <code>≤384 px: the command band spans both columns</code> . The V2 container queries (F-31–F-35) refine this further.
Motion	Hover plate transitions; none under reduced motion.
Accessibility	No roles or ARIA are added to game nodes; hidden native text is covered in F-29 and A11Y-01.
Edge cases & findings	- The candidate memo can go stale (FUN-08). <code>skillpanel.css</code> holds 20+ successive passes; later passes override earlier ones (MAINT-01). - Locked cards are desaturated.
Tests	<code>skill-card-system.test.mjs</code> (124 checks across all sheets), <code>skill-structure-signature.test.mjs</code> , <code>smoke.test.mjs</code> , <code>flush-quiescence.test.mjs</code> .

F-27 Skill command button styling

What	Buttons on skill cards look like forged metal plates in three variants: primary (ember), secondary, and disabled (dimmed). When the Skills UI atlas is ready they use its action frame art. Icon-only buttons (<code>≤2</code> characters) become small plates.
Why	The game's buttons use utility colours that clash with the theme. Inline <code>!important</code> is used because Tailwind utilities and the game's own inline styles would otherwise win.
Where	<code>SkillPanelRenderer.styleButton</code> (L236–351), <code>classifyButton</code> , <code>FORGE</code> (L101–119), <code>ACTION_ART</code> (L145–161), <code>BUTTON_STYLES</code> (L166–225). All writes go through the <code>buttonStyleOwner</code> <code>InlineStyleOwner</code> (reversible), with <code>data-iw-btn-state="primary secondary disabled icon"</code> . The CSS fallbacks in <code>skillpanel.css</code> L339–417 mirror the inline values.
Depends on	The game's primary colour classes (<code>bg-ember</code> , <code>orange</code> , <code>primary</code> , <code>accent</code>); the <code>disabled</code> attribute; <code>SkillsArtService</code> atlas readiness (<code>[data-iw-skills-ui-ready="1"]</code>).
Runs	Per card render. A snapshot key (state, role, art, compact, current <code>style</code> attribute) skips unchanged buttons.
States & fallbacks	Before the atlas loads: a gradient plate. After: <code>background: var(--fs-button-background, <atlas window>)</code>, which lets the V2 layer and the zone theme steer the art through CSS variables [Project record: <code>sprites-and-buttons</code>]. - The level-progress button is skipped, and any previously owned styles on it are restored. V2 command buttons use V2 geometry tokens (<code>--iw-skill-v2-btn-w/h/pad/font/tracking</code>, <code>--iw-skill-v2-nav-w/h/radius</code>) instead of fixed <code>155×44 / 26×44 px</code>.
Responsive	Driven by the tokens above.
Motion	<code>transition: var(--fs-button-transition, ...)</code> .
Hover / focus / keyboard	Native. The <code>base.css</code> focus ring; V2 adds a 2 px accent outline at 5 px offset.
Accessibility	Label text is preserved. V2 hides it visually (<code>--iw-skill-v2-btn-font: 0px</code>) and shows an <code>aria-hidden</code> mirror (F-32), so the accessible name remains the game's.
Edge cases & findings	- Never put a shorthand and one of its own longhands in one inline map; that once caused about 120 flushes per second (<code>READOUT_STYLES</code> comment). - Inline <code>!important</code> beats every stylesheet; see the cascade rule in <code>CLAUDE.md</code>.
Tests	<code>revised-button-atlas.test.mjs</code> , <code>button-art-states.test.mjs</code> , <code>skill-card-system.test.mjs</code> , <code>smoke.test.mjs</code> (native inline styles restored).

F-28 Skill medallion artwork and identity text

What	A circular medallion with the discipline's illustrated icon, the discipline name in small caps (a cleaned label, e.g. "Herbalism"), and the action title without its leading emoji.
Why	Identity at a glance; the game's text carries emoji and inconsistent casing.

Where	<code>SkillPanelRenderer.ensureSkillArtwork</code> appends <code>span.fs-skill-medallion-art (aria-hidden , data-iw-skill-art=<type>)</code> into the identity zone. <code>SkillsArtService.decoratePanel(panel)</code> sets <code>--fs-*</code> custom properties and <code>data-iw-skills-ui-ready="1"</code>; <code>paintIcon</code> paints from <code>skills_icons_atlas.webp</code> using <code>skills_icons_index.json</code>. <code>ensureSkillPresentation</code> writes <code>data-iw-clean-text</code> on the identity and title nodes, and appends <code>span.fs-skill-identity-percent</code> and <code>span.fs-skill-identity-progress > span</code> (fill width = the game's percentage). - CSS: <code>skillpanel.css</code> L924-1251 (medallion, studs, atlas integration) and L1325-1533 (identity/title <code>font-size: 0</code> with <code>::before { content: attr(data-iw-clean-text) }</code>).
Depends on	<code>SKILL_META</code> labels and glyphs; the icon index aliases (<code>woodcutting</code> → <code>gathering</code> cell, <code>construction</code> → <code>crafting</code> cell; <code>unknown</code> → <code>generic</code>).
Runs	Per card render. The art is fetched once (the index JSON), and the sprite is painted inline on the skin's own span.
States & fallbacks	Index not loaded → <code>data-iw-skill-art-pending</code> , and the native identity icon stays visible. <code>:has()</code> hides the native icon only once the art is ready.
Accessibility	Art is <code>aria-hidden</code> . The visible label is CSS generated content over zero-size native text, so assistive technology may read both (A11Y-01).
Edge cases & findings	In V2 the identity zone hides all descendants except the medallion art, the level readout and the identity role chain (<code>skillcard-v2.css:122</code>).
Tests	<code>skill-card-system.test.mjs</code> , <code>skill-card-v2-render.test.mjs</code> .

F-29 Level/XP readout presentation

What	In the three-zone layout, the game's "Lv N - X% • N to go" line is shown as a centred plaque with the cleaned text. In the always-on V2 card, that native readout is hidden and replaced by the medallion badge (F-31).
Why	Visual consistency. The game's readout is a styled <code><button></code> whose chrome clashed with the card.
Where	<code>SkillPanelRenderer.neutraliseReadouts</code>, <code>sameTextShellChain</code>, <code>readoutBranch</code>, <code>READOUT_STYLES</code> (shorthand only), <code>readoutCursor</code> (L380-399). The readout branch is marked <code>data-iw-readout="1"</code>; the display copy goes in <code>data-iw-progress-display</code>. - CSS: <code>skillpanel.css</code> L1563-1580 (three-zone: <code>font-size: 0</code>, <code>pointer-events: none !important</code>, <code>cursor: default !important</code>, <code>::before</code> content) and <code>skillcard-v2.css:164</code> (V2: <code>display: none !important</code>).
Depends on	<code>LEVEL_PROGRESS_PATTERN</code> ; the game's <code>title="Click to cycle XP display"</code> on that button [Source comment].
Runs	Per card render; nodes that leave the readout branch are restored.
States & fallbacks	When the pattern does not match, no readout role is assigned and the native text shows.
Hover / focus / keyboard	None. The native control cannot be clicked or focused while the skin is active.
Accessibility	With <code>display: none</code> the control leaves the accessibility tree.
Edge cases & findings	FUN-01 (Medium): the game's "cycle XP display" button is unreachable. The JavaScript <code>readoutCursor</code> explicitly tries to keep its pointer affordance, but the later CSS disables and then hides it.
Tests	<code>skill-card-system.test.mjs</code> . No test asserts that the control stays operable; see §7.6.

F-30 Material (ingredient) lists and met highlighting

What	The game's single line of material requirements ("  Oak Log 3/5  Iron Ore 10/10") becomes a two-column grid of material chips. Satisfied materials are shown in green. Where the native text remains, the text of satisfied materials is highlighted green without changing the DOM.
Why	Scannability, and making met/unmet (game state, rule 5) obvious.

Where	SkillPanelRenderer : - INGR_PATTERN (/ ^ (? ! . * \ b x p \ b) . { 0 , 80 } \ b \ d + \ s * \ / \ s * \ d + \ b \ i) , elementTextStarts / entryStart , ingredientEntries ({ text , state met unmet }); - updateIngredientLists : the skin-owned div.fs-skill-ingredient-grid (role = "list" , aria-label = "Required materials") with span.fs-skill-ingredient-item (role = "listitem" , data-iw-ingredient-state) , inserted with source.after(list) ; - neutraliseIngredients (ingredientStyleOwner); - completedIngredientRanges → CSS highlight iw-skill-ingredient-met . CSS: skillpanel.css L419-460 (:: highlight(iw-skill-ingredient-met)) and L3933-4001(the native source line hidden with [data-iw-ingredient-list-source="1"] { display: none } ; the grid layout).
Depends on	"N/M" count format; entry boundaries at emoji or element edges. Text that <code>textContent</code> joins without separators is anchored on element boundaries[Project record: CLAUDE.md textContent rule].
Runs	Per card render, signature-guarded(data-iw-skill-ingredient-list).
States & fallbacks	Without the CSS Highlight API: no highlight, but the grid still shows state.
Responsive	≤430 px: the grid becomes one column.
Accessibility	The grid is an ARIA list with a label. The native line is <code>display: none</code> , so there are no duplicates.
Edge cases & findings	In V2 the material body(F-35)is the primary presentation; see <code>skillcard-v2-runtime-safe.css</code> .
Tests	<code>ingredient-entries.test.mjs</code> , <code>skill-card-v2.test.mjs</code> .

F-31 V2 card: medallion XP ring and level badge

What	A 43 px discipline medallion encircled by a conic XP ring filled to the skill's percentage, with a badge beneath reading "Lv N" and "X%". A card with no readable level shows "Lv —" and "—".
Why	The single, consistently placed level and percent for every card shape. Before this, some cards showed the game's own "LV 57" and "77.2%" at different heights[Source comment L60-65].
Where	SkillCardDesignController.levelReadout (L66-86): inline <code>--iw-skill-v2-progress: N%</code> on the card, and the appended <code>span[data-iw-skill-v2-level-readout].iw-skill-v2-level-readout > span.iw-skill-v2-level + span.iw-skill-v2-percent</code> . CSS: <code>skillcard-v2.css</code> medallion and <code>::after</code> conic gradient with a radial mask (L130-160).
Depends on	The percentage from <code>.fs-skill-identity-percent</code> , else the level-progress text, else the identity zone text; the level from <code>Lv N</code> , <code>Lv N + M</code> or <code>Lv -</code> .
Runs	Per V2 reconcile. Text is written only when changed.
States & fallbacks	No percentage → 0% ring and "—".
Responsive	Container <code>iw-skill-card</code> ≤580 px and ≤480 px scale the tokens down.
Motion	None. The ring updates in steps as the percentage changes.
Accessibility	The readout spans are not <code>aria-hidden</code> . With the native readout button hidden (F-29), this is the only level text assistive technology can read.
Tests	<code>skill-card-v2.test.mjs</code> , <code>skill-card-v2-render.test.mjs</code> .

F-32 V2 card: themed square action button, glyph and fitted label

What	The action button (Mine, Fight, Craft Parts, ...) is a square themed plate from the zone's card-v6 art. It has three states (idle, hover, pressed crossfade)and shows the discipline's icon in the centre, with the uppercase label drawn beneath, fitted to the button width. Crafting and Fishing, which have no icon, show the label only.
Why	Recognisable one-click action, consistent sizing across card shapes, and art that follows the zone theme.

Where	<code>ensureActionGlyph</code> appends <code>span[data-iw-skill-v2-action-glyph]</code> (<code>aria-hidden</code>) to the button's parent, never inside the control; it handles the case where the command zone is the button. <code>ensureActionLabel</code> adds the <code>aria-hidden</code> mirror <code>span[data-iw-skill-v2-action-label]</code>. <code>fitActionLabel</code> measures the label text with a <code>Range</code> and <code>getComputedStyle</code>, then writes <code>--iw-skill-v2-action-label-font</code> (4 px-9.5 px) on the card. It is keyed by label, layout, inline width and layout epoch in <code>data-iw-skill-v2-label-fit</code>, and refits after <code>document.fonts.ready</code>. - CSS: <code>skillcard-v2.css</code> (command group, per-type glyph <code>url('../assets/skills-ui/action-icons/<type>.svg')</code> at L376-385, label); <code>card-buttons.css</code> (plate from <code>--iw-card-action-*</code>); <code>card-button-atlas.css</code> (generated windows per theme); <code>skillpanel.css</code> L3780-3885 (revised-v5 hover/pressed crossfade on <code>::before</code> / <code>::after</code>).
Depends on	Button role from F-26; theme from F-09; card-v6 atlases.
Runs	Per V2 reconcile; the fit only when its key changes.
States & fallbacks	Unthemed (<code>data-iw-zone-theme="default"</code>): a gradient plate from <code>--fs-motion-*</code> tokens. Disabled: grayscale and 0.55-0.6 opacity; hover and pressed layers suppressed. Long action: the label is hidden and the glyph shows a countdown (F-34).
Responsive	Container queries resize the tokens. (<code>pointer: coarse</code>) enlarges only the nav height (F-36).
Motion	Crossfade 180 ms (hover) and 70 ms (pressed); <code>transition: none</code> under reduced motion.
Hover / focus / keyboard	Native <code><button></code> . Hover brightens; <code>:focus-visible</code> has a 2 px accent outline at 5 px offset; <code>:active</code> shows the pressed layer.
Accessibility	The native label stays the accessible name (its visible text is transparent/0 px). The glyph and label mirrors are <code>aria-hidden</code> . The fitted label can shrink to 4 px (A11Y-03).
Edge cases & findings	- A containing block created by a game wrapper once offset the icon by about 12 px; that is why the icon is a sibling of the button [Source comment]. - Inline <code>!important</code> from <code>styleButton</code> is steered through <code>var(--fs-button-background, ...)</code>.
Tests	<code>card-button-art.test.mjs</code> , <code>skill-card-v2-render.test.mjs</code> , <code>skill-card-system.test.mjs</code> , <code>revised-button-atlas.test.mjs</code> .

F-33 V2 card: activity fill smoothing

What	While an action runs, an accent band fills the action button continuously, instead of in 250 ms steps. The band snaps back when the next repetition starts, and the running card's button gets a darker track and accent border.
Why	The game rewrites the fill <code>width: max(8, elapsed%)</code> from a 250 ms <code>setInterval</code> . A linear transition of one measured tick turns the steps into motion without computing progress (rule 5) [Source comment L205-219].
Where	<code>smoothActionFill</code> (L223-233): reads the button's <code>:scope > span[style*="width"]</code> , samples it with <code>ProgressCadence.sampleProgress</code> keyed by card, writes <code>--iw-skill-v2-fill-duration</code> , and marks <code>data-iw-skill-v2-fill-reset="1"</code> for one frame on reset. CSS: <code>skillcard-v2.css</code> fill band and the <code>:has(> span[style*="width"])</code> running state; <code>card-buttons.css</code> clips the fill inside the art rail with <code>clip-path</code> and removes the game's <code>animate-pulse</code> .
Depends on	The game's fill span shape (a direct child span with inline width).
Runs	Once per tick (each width write is a <code>style</code> mutation that re-queues the card).
States & fallbacks	No fill span → idle. A non-percentage width → ignored.
Motion	Linear width transition over the measured tick; default 0.28 s. Removed under reduced motion.
Edge cases & findings	Removing <code>animate-pulse</code> from the fill is a presentation change to a game activity indicator; the band is still clearly visible as running.
Tests	<code>action-progress-theme.test.mjs</code> , <code>skill-card-v2-render.test.mjs</code> .

F-34 V2 card: long-action countdown

What	For an action lasting more than 11 s (for example, a long craft), the running card's button shows the remaining time (e.g. "0:42") in place of the icon and label.
Why	A long action gives little visual feedback. The countdown mirrors the game's own Current Action timer , so modifiers (housing, boosts) remain the game's decision [Source comment L241-243].

Where	<code>LONG_ACTION_SECONDS = 11</code> ; <code>durationSeconds (h:mm:ss , m:ss , 1h 2m 3s)</code> ; <code>currentActionRemaining</code> (the first leaf in the rendered Current Action panel, outside the progress bar, that parses as a duration); <code>syncLongActionTimer</code> ; <code>syncTimersOnTick</code> (on <code>iw:dom-flush</code> and <code>iw:name-scan-flush</code>). The glyph gets <code>data-iw-skill-v2-action-timer="1"</code> and the countdown text; the card gets <code>data-iw-skill-v2-long-action="1"</code> . CSS: the <code>skillcard-v2.css</code> <code>long-action</code> rules (button text transparent, label hidden, 15 px glyph text).
Depends on	A rendered Current Action panel (<code>[data-iw-panel="current-action"]</code> or <code>#current-action-panel</code>) and its timer text format.
Runs	Only on flushes whose roots touch the Current Action host. A containment test runs first; the rendered-host pick (<code>checkVisibility</code>) runs only then. This was measured as about 30 visibility reads per second before and about 1 after [Project record] .
States & fallbacks	Once started, the countdown stays while the action runs, even if the remaining time drops to 11 s or less. The action stopping (fill span gone) → timer cleared. No readable timer → no countdown.
Accessibility	The countdown text is inside the <code>aria-hidden</code> glyph, so it is not announced. The game's own panel remains the accessible source.
Edge cases & findings	Every running V2 card mirrors the same Current Action timer. The game runs one action at a time [Assumption] , so this is correct as long as only one card shows a fill.
Tests	<code>skill-action-countdown.test.mjs</code> .

F-35 V2 card: detail body and requirement note

What	Under the title, a compact auto-fit grid (up to 3 columns) lists the card's materials, sources and details. Materials show name and count, with a ✓ when met (green) and amber when unmet. When a requirement is unmet, a <code>△</code> amber note row appears at the foot.
Why	Dense cards; the V2 layout needs the information in one predictable place.
Where	<code>markSections</code> writes <code>data-iw-skill-v2-section="requirements materials details rewards queue sources"</code> on native nodes. <code>ensureDetailBody</code> builds <code>div[data-iw-skill-v2-body]</code> with rows <code>span.iw-skill-v2-body-row</code>, using <code>createElement / textContent</code> only, signature-guarded by <code>data-iw-skill-v2-body-signature</code>. <code>ensureFootRow / syncRequirementNote</code> build <code>div[data-iw-skill-v2-controls]</code> and <code>span[data-iw-skill-v2-req-note]</code> (<code>role="note"</code>, text joined with <code>"."</code>). <code>removeRetiredNodes</code>. CSS: <code>skillcard-v2.css</code> <code>body/controls</code>; <code>skillcard-v2-runtime-safe.css</code> visually hides native sections with a clip (not <code>display: none</code>, because <code>SkillPanelRenderer</code> reads computed visibility).
Depends on	Roles from F-26; the queue and sources text patterns in <code>markSections</code> .
Runs	Per V2 reconcile.
States & fallbacks	No materials → no body rows. Requirement met → no foot row.
Responsive	Container <code>iw-skill-body</code> auto-fit cells with a minimum of 96 px; <code>iw-skill-card</code> ≤480 px phone layout.
Accessibility	Native sections stay in the accessibility tree (clip-hidden) and the mirror body is not <code>aria-hidden</code> , so content can be announced twice (A11Y-01).
Edge cases & findings	CSS for the retired summary, expand and tab nodes remains (MAINT-01).
Tests	<code>skill-card-v2.test.mjs</code> (24), <code>skill-card-v2-render.test.mjs</code> (32), <code>skill-card-v2-wiring.test.mjs</code> .

F-36 Recipe pager (previous/next)

What	The game's ◀▶ recipe buttons become thin, cool-toned chevron plates. In V2 they sit centred below the square action button, with an octagonal card-v6 nav plate and a CSS-drawn chevron.
Why	The ornate framed arrows were dropped by the owner (2026-09) as too heavy.

Where	- Roles: <code>nav-button</code> (≤ 2 characters or aria prev/next), <code>data-iw-nav-direction="prev next"</code>, <code>nav-group</code>. <code>styleButton</code> nav geometry. - CSS: <code>skillpanel.css</code> L2389-2589 ("thin cool-toned arrows"; shape A <code>display: contents + order</code>; shape B absolute over the command column at ≥ 641 px); <code>skillcard-v2.css</code> nav group; <code>card-buttons.css</code> nav plate (<code>clip-path</code> octagon, hover/focus/active swaps).
Depends on	Button text or aria labels.
Runs	Per card render.
States & fallbacks	Disabled $\rightarrow$ 0.45 opacity. The native SVG children are hidden and replaced by the <code>::before</code> chevron.
Responsive	(<code>pointer: coarse</code>) $\rightarrow$ <code>--iw-skill-v2-nav-h: 26px</code> (target size, WCAG 2.2 2.5.8 minimum 24 px).
Hover / focus / keyboard	Native buttons. <code>:focus-visible</code> inset outline (-3 px).
Accessibility	The native SVG is hidden. The accessible name depends on the game's <code>aria-label</code> [Verify: if the game gives these icon buttons no <code>aria-label</code> , they are unnamed with or without the skin].
Edge cases & findings	- Two page shapes exist live: A, where the nav group is a sibling row, and B, where it sits over the command column. Both are pinned in CSS (L2389-2589). - A new shape would fall back to the base nav styling.
Tests	<code>skill-card-system.test.mjs</code> , <code>button-art-states.test.mjs</code> , <code>card-button-art.test.mjs</code> .

F-37 Skill Actions section frame

What	The "Skill Actions" panel wears the shared forged frame (F-12) with the Skills texture and atlas variables. Its collapse control folds all cards (F-13).
Why	The Skill Actions panel is found by heading rather than by <code>.panel</code> , and needs the atlas variables for its children.
Where	<code>SkillPanelRenderer.ensureSkillActionsFrame</code> : the nearest ancestor (≤ 8 levels) with a "Skill Actions" heading gets the class <code>fs-skills-section-frame</code> and <code>SkillsArtService.decoratePanel(frame)</code> variables. CSS: <code>ui-system.css</code> frame selector <code>.fs-skills-section-frame[data-iw-skills-ui-ready="1"]</code> ; <code>skillpanel.css</code> L1535-1691 frame polish; <code>skillcard-v2.css</code> :109 (folded frame hides V2 cards).
Runs	Per card render.
Edge cases & findings	A class is added to a game node (one of four class writes in the skin); <code>clearSkillPanels</code> removes it.
Tests	<code>collapsible-frames.test.mjs</code> , <code>panel-frame-nesting.test.mjs</code> .

2.6 QUESTS

F-38 Quest cards

What	Each quest card has three zones: Sigil medallion (left): the objective item's sprite, or a discipline glyph, with a percentage ring and badge. Text (centre): kicker "Work Order", title, brief in the flavour style, objective with a $\blacklozenge$ bullet, and a reward chip. Command rail (right): Turn In and Skip as three-slice atlas plates whose frames follow the label width. A ready-to-turn-in card uses the ember accent. The card frame art is chosen per zone theme from <code>approved-frames.png</code>.
Why	Quests were visually indistinguishable from skill cards. The three-slice art keeps the frame ends undistorted when labels change ("Turn In" vs "Turn In All (38)") [Source: <code>SkillsArtService.js</code> comment L37-59].

Where	QuestPanelRenderer.js : - isQuestCard, structureSignature, annotateStructure; - ensureDecoration: span.fs-quest-sigil is inserted as the first child of the body zone, containing span.fs-quest-sigil-icon and span.fs-quest-sigil-pct; - ensureAtlas, measureCommandBlock (--fs-quest-cmd-w), questState, renderCard, clearQuestPanels. Attributes: data-fs-quest, data-iw-quest-role (kicker, title, brief, objective, reward, progress-label, progress-track, progress-fill, turn-in, skip), data-iw-quest-zone, data-iw-quest-state="ready work-order active", data-iw-quest-reward, data-iw-quest-percent, data-iw-quest-glyph; class fs-quest-panel; inline --fs-quest-accent. CSS: skillpanel.css L2592-3367.
Depends on	Text: a Reward: leaf; Turn In / Skip (n) buttons or N% complete; the "work order" kicker; the discipline regex on the reward text(DISCIPLINE_STYLE). Data: ItemDatabase for the objective item; AtlasService for its sprite. Art: SkillsArtService action frame windows; quest-frames/approved-frames.png.
Runs	On iw:skill-panel for every .compact-panel (skill cards are rejected by class). The structural walk is gated by a signature: leaf count, labels with "(n)" stripped, track presence; disabled is excluded.
States & fallbacks	State: ready (Turn In enabled), work-order (Skip present) or active. Sigil icon: painted detached and attached only on success. While the atlas loads, it is marked pending and re-rendered when ready. Unknown item → discipline glyph (♠ generic). Command width: measured from the rendered rail; 0 means the hidden mirror, so it is not written.
Responsive	≤640 px: sigil 44 px, commands absolute top-right, text padded by --fs-quest-cmd-w. ≤400 px: further tightening.
Motion	A ready sigil icon plays a one-time 0.85 s settle animation (iw-quest-ready-settle , ui-system.css:1466-1473), only under prefers-reduced-motion: no-preference. Hover and disabled transitions.
Hover / focus / keyboard	Turn In and Skip are native buttons. Hover lifts the art; a disabled Turn In re-aims all three bands to the disabled cell (rule 5).
Accessibility	The sigil is aria-hidden. The reward chip text is CSS generated content over zero-size native text (A11Y-01).
Edge cases & findings	- var(--iw-font-flavour) is undefined, so the brief falls back to the UI font (FUN-05). - textContent joins elements without separators("XPTurn In"), so no \b is used before "Turn"[Source comment]. - A quest titled <Discipline> Work Order is rejected by detectSkillType early.
Tests	quest-card-detection.test.mjs, quest-structure-signature.test.mjs, quest-command-width.test.mjs, smoke.test.mjs.

F-39 Quest objective item hovercard

What	Hovering, focusing or tapping a quest's objective line (e.g. " ♦ Night Claw 22/100") opens the item hovercard (F-44) for that item.
Why	Players need to know where to get the objective item.
Where	QuestPanelRenderer.objectiveItemRef (strips leading non-letters and the trailing "n/m", then ItemDatabase.getBy_name) and ensureObjectiveTrigger. On the game's <p> it writes data-iw-item or data-iw-item-name, data-iw-tooltip-trigger="1", tabindex="0", aria-haspopup="dialog", aria-controls="iw-tip" and aria-expanded="false". clearObjectiveTrigger removes them. CSS: the skillpanel.css objective hover affordance.
Depends on	The name resolving in the item catalogue.
Runs	Per quest render.
States & fallbacks	Name not found → no trigger attributes; the line stays plain.
Hover / focus / keyboard	Tab reaches the objective. Enter or Space opens the card and moves focus into it; Escape closes it and returns focus.
Accessibility	A focusable <p> without a role (A11Y-02). NameScanner skips [data-iw-tooltip-trigger], so there is no double highlight.
Edge cases & findings	Teardown removes tabindex and the ARIA attributes unconditionally (FUN-09).
Tests	smoke.test.mjs, tooltip-virtual-anchor.test.mjs (shared engine).

2.7 INVENTORY

F-40 Inventory row overlay

What	Each inventory row is drawn by a skin overlay. From left to right it shows: • a 52 px icon socket with the item's sprite and a "+N" upgrade badge; • the item name in its tier colour (a hovercard trigger), with "+N" and "Lv N" suffixes; • up to five stat chips (fewer on small screens); • owned-item details: loadout, sockets, rolled stats, set and status; • requirement lines; • the quantity "×N". A 3 px category-colour rail marks the row type: gear, consumable, processed, resource, trade, potion, enchant, XP scroll, cache, orb, unknown. The game's own action buttons (Equip, Equipped, Set, List, icon actions) stay live and are restyled as small forged plates. Equipped rows keep a warm edge.
Why	The game's rows are plain text. Item data from <code>items.json</code> makes them informative without changing the actions (rule 1). Category and equipped state are separate visual axes, so neither destroys the other (rule 5).
Where	<code>InventoryRenderer.renderRow</code> (L611–716): • data extraction: <code>extractRowData</code>, <code>leafTexts</code>, <code>detailTexts</code>, <code>guessQuantity</code>; • <code>InventoryModel.buildInventoryDetails</code> and <code>itemDisplay</code> (<code>categoryClass</code>, <code>statChips</code>); • overlay creation: <code>div.fs-inv-row</code> via <code>innerHTML</code> with escaped values (L699); • <code>hideOriginalChildren</code> / <code>preserveInteractiveTree</code>, <code>syncRowState</code>, <code>paintIcon</code> (<code>AtlasService.paint</code>, fallback "<code>?</code>"), <code>configureIconTooltip</code>. CSS: <code>inventory.css</code> row shell, grid, categories, icon, name, stats, details, action kinds, media queries.
Depends on	• Rows: <code>.compact-row</code> or <code>[class*="item-row"]</code> inside the Inventory root (F-41), or rows containing Equip/Equipped/Unequip/List buttons. • Names: <code>ItemDatabase.getByName</code>; <code>AtlasService</code>. • Details: detail text shapes such as "<code>🔮 +N · stats</code>", "Not upgradable", loadout and socket lines.
Runs	On <code>iw:inventory-row</code> for each dirty row, and for all rows on <code>iw:item-db-updated</code> and <code>iw:atlas-updated</code>. Work is gated in two stages: 1. a cheap signature (child count, full text, DB and atlas revisions); 2. a row signature stored in <code>data-fs-inv</code> (id/name, enhancement, quantity, details, revisions). An unchanged row only re-hides new native children.
States & fallbacks	• Item not in the catalogue: plain name, category <code>unknown</code>, "<code>?</code>" icon, no chips. • Empty name (row still rendering): retried on the next frame. • Row leaves the inventory: overlay removed and native children restored. • Atlas not ready: icon painted after <code>AtlasService.ready()</code>.
Responsive	≤1024 px: smaller grid and icon, chips beyond 5 hidden, details capped at 26 px height. ≤768 px: rows wrap, 42 px icon, chips beyond 3 hidden.
Motion	Row hover lift; transition: <code>none</code> under reduced motion.
Hover / focus / keyboard	• The icon (<code>.fs-inv-icon</code>) and the item-name reference (<code>.iw-item-ref</code>, <code>role="button"</code>, <code>tabindex="0"</code>) open the hovercard on hover, focus + Enter/Space, or tap. • Native action buttons keep their behaviour; they are raised above the overlay (<code>z-index: 2</code>). • Disabled or <code>aria-disabled</code> actions drop to 0.45 opacity.
Accessibility	Non-interactive native children are <code>display: none</code> (removed from the accessibility tree); the overlay text replaces them. The icon has <code>aria-label</code> and <code>aria-haspopup</code>, and its <code>role</code> is removed on purpose (a <code>role=button</code> attracted a global <code>background-image !important</code>). Native controls stay in tab order.
Edge cases & findings	• <code>display: none</code> is written inline without <code>!important</code> through <code>InlineStyleOwner</code>, and restored on teardown. • The overlay is appended into the React row, which is allowed (rule 2 permits appending). A React re-render that replaces children leaves the overlay in place; the cheap signature detects the change. • Cloak alias table is temporary (<code>aliases.js</code>).
Tests	<code>inventory-model.test.mjs</code>, <code>item-display.test.mjs</code>, <code>inventory-root.test.mjs</code>, <code>smoke.test.mjs</code>, <code>data-services.test.mjs</code>.

F-41 Inventory panel chrome and filters

What	The Inventory panel gets the forged frame (F-12), a Cinzel title with a themed separator ornament, and a recessed list frame. The filter row (All, Gear, Materials, Consumables, Drops) and pager (Prev, Next, "n / m") become 26 px forged plates in one line. The active filter is lit ember. Tool icons become 30×31 px atlas plates, and bare SVG glyphs stay ornaments.
Why	A dense tool row on phones, and preserving the game's active-filter state (rule 5).
Where	<code>InventoryRenderer</code> : <code>findInventoryRoot</code> : an <code>aria-label="inventory"</code> ancestor, or an "Inventory" heading whose panel contains this row, which handles the hidden duplicate column; <code>classifyInventoryChromeOnce</code> (one sweep per flush per root) and <code>classifyInventoryChrome</code> : <code>data-iw-inventory-root</code>, <code>-title</code>, <code>-control="filter page icon glyph page-count"</code>, <code>-filter-state="active"</code>, <code>-filters</code>, <code>-list</code>; <code>ensureInventoryRule</code> : appends <code>div.fs-inv-rule</code> (<code>aria-hidden</code>) after the header row. CSS: <code>inventory.css</code> root, list, title, rule, controls, container queries, icon tools.
Depends on	English labels; active state from <code>aria-selected</code> , <code>aria-current="page"</code> , <code>data-state="active"</code> or an active-looking class.
Runs	Once per flush per inventory root.
States & fallbacks	No root found → rows can still render when their own actions identify them (F-40); the chrome stays native.
Responsive	≤1279 px: the filter row becomes container <code>iw-inventory-filters</code> , <code>nowrap</code> , font <code>clamp(7.5px, (100cqi - fixed)/22.5, 10.5px)</code> . ≤767 px: <code>k = 20.9</code> , tighter tracking, flexible widths. ≤560 px: the separator is scaled and lowered.
Motion	Hover transitions only.
Hover / focus / keyboard	Native controls; hover plates; the disabled pager is dimmed.
Accessibility	The ornament rule is <code>aria-hidden</code> . Glyph icons keep the game's colour.
Edge cases & findings	Icon tool windows use pixel atlas geometry (<code>background-size: 322.5px 173.63px</code>), audited by <code>audit-sprite-windows.mjs</code> .
Tests	<code>inventory-root.test.mjs</code> , <code>menu-rows.test.mjs</code> , <code>separator-symmetry.test.mjs</code> , <code>sprite-window</code> audit.

2.8 ITEM INFORMATION

F-42 Item catalogue service

What	Nothing visible directly. It powers item names, icons, stats, tooltips and the Village ledger. The tooltip footer says whether data is "live data", "cached data" or "stale cached data".
Why	The DOM does not carry item stats; the public <code>items.json</code> does.
Where	<code>ItemDatabase.js</code> : <code>ready</code> , <code>_load</code> (L132-158), <code>_fetchFresh</code> (L160-203), <code>_index</code> (L205-233), <code>startAutoRefresh</code> / <code>stopAutoRefresh</code> / <code>_scheduleRefresh</code> , <code>getById</code> , <code>getByName</code> , <code>find</code> , <code>all</code> , <code>source</code> , <code>revision</code> , <code>isReady</code> .
Depends on	<code>https://idleworlds.com/items.json</code> ; <code>chrome.storage.local</code> .
Runs	At boot. A fresh cache (<6 h) is indexed immediately, and a background refresh is scheduled. Stale cache. Indexed as <code>stale-cache</code>, then a network refresh is awaited. No cache. A network fetch is awaited. Refresh. A <code>setTimeout</code> chain runs every 6 h (15 min after a failure). Conditional request. Sends the stored validator (ETag/Last-Modified). A match cancels the body and records <code>iw-item-db-cache-checked</code>.
States & fallbacks	Fetch fails: stale data stays usable and a warning is logged. Nothing at all: features that need items degrade (plain names, ? icons, no tooltips). Legacy cleanup: a page <code>localStorage</code> key <code>iw-item-db-cache</code> from ≤1.5.x is removed once per page.
Edge cases & findings	- Absolute apex URL: on <code>www.</code> it is cross-origin (SEC-03). - Records are trusted field by field (SEC-02). - Duplicate names: the first record wins.

Tests	<code>data-services.test.mjs</code> (34 checks: cache freshness, validators, legacy eviction, failure paths). <code>npm run audit:items</code> validates the live contract manually.
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F-43 Sprite icons and upgrade badges

What	Item and gear icons come from two large sprite atlases. Upgraded gear ("Iron Sword +3") shows a "+3" badge, which is a sprite for known levels or text otherwise.
Why	The game's DOM uses emoji or no icons. Self-hosted atlases are immune to page CSP changes [Source : <code>vendor-assets.mjs</code>].
Where	<code>AtlasService</code> : <code>ready / _loadMissing</code> (<code>Promise.allSettled</code> over the gear manifest JSON and item CSV; backoff of 5 s, or 15 s when partially loaded), <code>_loadGearAtlas</code> , <code>_loadItemAtlas</code> (column and bounds validation), <code>resolve</code> (L332–380), <code>paint</code> (L382–395), <code>_paintBadge</code> (<code>span.iw-icon-badge[data-iw-badge]</code> , <code>aria-hidden</code>), <code>positionBadgeHost</code> (batches <code>getComputedStyle</code> in a microtask). CSS: <code>base.css</code> <code>.iw-icon-badge(--sprite)</code> ; sizing in <code>inventory.css</code> and <code>tooltip-engine.css</code> .
Depends on	The item id, else the normalised name. For gear, name fallbacks are tried in order: 1. the exact name; 2. the name without "+N"; 3. the cloak alias; 4. without "Lv N"; 5. without an "of X" suffix; 6. without a leading affix (16 words).
Runs	Per icon paint. The atlases load once per page, with retries.
States & fallbacks	Unresolved → <code>false</code> (the caller shows " ? ", a glyph or ♦) plus one console warning per name.
Edge cases & findings	Icons are painted with plain inline <code>background-*</code> on skin-owned hosts only (never on game nodes).
Tests	<code>data-services.test.mjs</code> , <code>smoke.test.mjs</code> .

F-44 Item hovercard (tooltip)

What	A 330 px forged card opens next to an item trigger. It shows: • 64 px art, the name in tier colour, and badges (tier, category, slot, requirement); • an effect line and a stat grid; • "How to get it" (crafted / zone or boss drop / gathered / upgrade / cache / shop / unknown); • a footer with a Wiki ↗ link, the data-source label and a close button.
Why	The single place for item knowledge across inventory, quests, bosses, village and prose.
Where	<code>TooltipEngine.js</code> : <code>initTooltipEngine</code> (creates <code>div#iw-tip</code>), <code>renderCard</code> , <code>renderAcquisition</code> , <code>renderStats</code> , <code>renderBadges</code> , <code>position</code> (L234–265), <code>show</code> (L338–377), <code>hide</code> , <code>scheduleShow</code> (120 ms), delegated listeners (L432–607), <code>showForItem</code> (virtual anchor), <code>hideTooltip(source)</code> , <code>itemRef</code> , <code>registerTooltipItem</code> . CSS: <code>tooltip-engine.css</code> .
Depends on	Triggers : <code>.iw-item-ref[data-iw-item data-iw-item-name]</code> or <code>[data-iw-tooltip-trigger="1"][data-iw-item data-iw-item-name]</code> . Data : <code>ItemDatabase</code> , with a fallback item from <code>registerTooltipItem</code> (boss rewards). Art : <code>AtlasService</code> .
Runs	• Desktop hover: capture-phase <code>mouseover</code> schedules show after 120 ms. Moving to another trigger hides the current card immediately. <code>mouseout</code> hides unless the pointer moves onto the card, which sits flush (<code>EDGE_GAP = 0</code>). • Pointer tracking: <code>mousemove</code> hides the card when the pointer is on neither the anchor nor the card. • Click: shows the card (desktop), or toggles it (touch: (<code>hover: none</code>), (<code>pointer: coarse</code>)). Clicks elsewhere hide it. The click is never cancelled, so game rows still receive it. • Keyboard: Enter/Space on a trigger toggles the card with <code>preventDefault</code> and focuses the first interactive element inside it. Escape hides it and restores focus. • Scroll/resize: the card repositions, or hides if the pointer left. • Window blur: hides. • <code>iw:dom-flush</code>: an orphan check. • Data updates (<code>iw:atlas-updated</code>, <code>iw:item-db-updated</code>): refresh an open card unless keyboard focus is inside it.

States & fallbacks	- Unknown item → no card. - Missing art → a category glyph. - Unknown source → "Source not documented..." with a skin-written hint (CONT-01). - Stale data → the source label is styled <code>is-stale</code>.
Responsive	Clamped to the viewport with a 10 px margin, flipped above when there is no room below, <code>max-height: calc(100dvh - 20px)</code> with a scrolling body. <code>≤420 px: width 100vw - 20px</code> , smaller art, one-column stats.
Motion	None: the card is shown and hidden by class and display, with <code>transition: none</code> .
Hover / focus / keyboard	See Runs . Keyboard-opened cards are "keyboard-owned": mouse movement does not close them. <code>focusout</code> outside closes them.
Accessibility	<code>role="dialog"</code> , <code>aria-modal="false"</code> , <code>tabindex="-1"</code> , <code>aria-label="<name> item details"</code> . Triggers get <code>aria-controls="iw-tip"</code> , <code>aria-haspopup="dialog"</code> and <code>aria-expanded</code> . Content on hover stays dismissible (Escape), hoverable (flush) and persistent, which matches the intent of WCAG 1.4.13 [Assumption : no formal audit performed].
Edge cases & findings	<code>craft_level</code> and <code>tier</code> reach <code>innerHTML</code> unescaped (SEC-01). - The wiki slug is not URL-encoded (SEC-08). <code>#iw-tip</code> persists after teardown (FUN-10). <code>cleanupAnchor</code> removes <code>aria-describedby</code> from any anchor (FUN-09).
Tests	<code>tooltip-virtual-anchor.test.mjs</code> , <code>smoke.test.mjs</code> (no duplicate tooltip after re-enable), <code>world-boss-panels.test.mjs</code> (keyboard opening via boss rewards).

F-45 Item-name detection in game text

What	Item names that appear in the game's own text (log lines, chat, quest copy, descriptions) become hover targets. Resting the pointer on the exact word, or tapping it on touch devices, opens the hovercard. No visible marking is drawn (FUN-02).
Why	Item knowledge everywhere without wrapping React text in elements, which would break React's reconciliation. The CSS Custom Highlight API and <code>Range</code> geometry are used instead [Source : module header].
Where	<code>NameScanner.js</code> : <code>buildTrie</code> (per DB revision), <code>findMatches</code> (longest match at ASCII word starts, <code>≥3</code> characters), <code>scanForItemNames</code> (TreeWalker with skip containers), <code>rebuildHighlight</code> (rAF, re-registers only on change), <code>hitTest</code> (<code>elementFromPoint</code> → <code>caretPositionFromPoint</code> / <code>caretRangeFromPoint</code> → range rects; fallback subtree walk), <code>bindHover</code> , <code>clearItemNameScan</code> .
Depends on	<code>ItemDatabase</code> names; skip containers <code>.iw-item-ref</code> , <code>[data-iw-tooltip-trigger]</code> , <code>.fs-inv-row</code> , <code>.fs-skill-header</code> , <code>.iw-tip</code> , <code>[role="tooltip"]</code> , <code>input</code> , <code>textarea</code> , <code>select</code> ; skip tags <code>SCRIPT</code> , <code>STYLE</code> , <code>INPUT</code> , <code>TEXTAREA</code> , <code>SELECT</code> , <code>OPTION</code> , <code>CANVAS</code> .
Runs	On <code>iw:name-scan-flush</code> for changed roots, and on <code>iw:item-db-updated</code> for the whole body. The pointer is hit-tested once per animation frame while the mouse moves. On touch devices a click hit-test runs, and the game's click continues.
States & fallbacks	Without the Highlight API: matches are still indexed and hover still works. Scroll hides a prose-anchored card.
Accessibility	Keyboard users cannot reach prose matches, because no element is focusable. The same items are reachable where explicit triggers exist (inventory, quests, rewards, village).
Edge cases & findings	FUN-02: the highlight is invisible. PERF-06: the fallback walk runs on every pointer frame that misses. <code>SKIP_TAGS</code> also lists <code>SVG</code> and <code>IW-TIP</code>, which are ineffective (MAINT-04). - Ranges are live and are rebuilt when React rewrites a text node.
Tests	<code>smoke.test.mjs</code> ; <code>tooltip-virtual-anchor.test.mjs</code> .

2.9 WORLD BOSSES AND ZONE CONTROL

F-46 World Boss encounter cards

What	Each boss card (Ancient Treant, Abyssal Behemoth, World Eater) becomes an illustrated encounter card: • a 104 px portrait from a 2×2 art atlas; • a per-boss accent colour; • a grid for difficulty, buff, timer, participation, status and progress; • a themed 132×38 px action plate labelled PREJOIN or QUEUED. A green notice at the top of the World Bosses panel says there are no minimum requirements and no risk in joining (skin-authored text, CONT-01).
Why	Makes world events feel like encounters, and keeps the join state legible.
Where	<code>UIFoundation.classifyBossCards</code> ("World Bosses" heading → root → leaf <code>.compact-panel</code> gets <code>data-iw-boss="card"</code>) and the <code>iw:skill-panel</code> refresh for cards in the resolved root. <code>WorldBossPanels.decorateWorldBossPanel</code>: • <code>data-iw-encounter=<bossKey zone></code>; • <code>data-iw-boss-role</code> values <code>header</code>, <code>title</code>, <code>action</code>, <code>participation</code>, <code>difficulty</code>, <code>buff</code>, <code>timer</code>, <code>status</code>, <code>progress</code>, <code>details</code>; • appended <code>div.iw-boss-art</code> (<code>aria-hidden</code>); • <code>data-iw-boss-action-state="active idle"</code> and <code>data-iw-boss-action-label</code> ("Queued" when the button reads "Prejoined", "Prejoin" when it reads "Prejoin"); • <code>SkillsArtService.decoratePanel(action)</code> (inline variables on the game button) and <code>data-iw-boss-art-ready</code>; • <code>p.iw-boss-notice</code>. CSS: <code>ui-system.css</code> boss sections (card ground, encounter grid, art atlas positions, action plate, <code>font-size: 0</code> label with <code>::after { content: attr(data-iw-boss-action-label) }</code>).
Depends on	Boss names in the card title; the button text "Prejoin"/"Prejoined"; the "World Bosses" heading.
Runs	Each classify pass and on <code>iw:skill-panel</code> for boss cards.
States & fallbacks	Unknown boss name → no portrait or accent, but roles still apply. Atlas not ready → the plate falls back to a gradient.
Responsive	≤760 px and ≤380 px layouts (40 and 8 rules).
Hover / focus / keyboard	The native action button keeps its behaviour. Hover and disabled plate states.
Accessibility	The visible label comes from CSS content, while the accessible name remains the game's text ("Prejoined"). Sighted users see "QUEUED" and screen-reader users hear "Prejoined" (A11Y-01, CONT-02).
Edge cases & findings	• The notice text is a gameplay claim written by the skin (CONT-01). • Participation link hit area and row placement fixes are pinned by tests.
Tests	<code>world-boss-panels.test.mjs</code> (43), <code>boss-action-label.test.mjs</code> , <code>boss-participation-hit.test.mjs</code> .

F-47 World Boss "Possible Rewards"

What	A "Possible Rewards" disclosure under each boss card shows a grid of reward tiles (icon plus name). Each tile opens the item hovercard. The open/closed choice is remembered per boss.
Why	The DOM does not list boss drops; the item catalogue does.
Where	<code>WorldBossPanels.rewards(card, boss)</code>: • signature <code>boss dbRevision atlasReady</code>; • builds <code>details.iw-boss-rewards</code> with a summary and <code>button.iw-boss-reward[data-iw-tooltip-trigger="1"] [data-iw-item][data-iw-item-name]</code> tiles (<code>aria-label</code>), <code>registerTooltipItem</code>, <code>AtlasService.paint</code> (the orb uses the copper orb art), fallback "◆". Item ids come from: the <code>BOSSES</code> table ids; <code>items.json</code> records with <code>acquisition_type</code> <code>BossDrop</code> whose acquisition text names the boss; plus <code>trader_token</code> and <code>boss_upgrade_orb</code>. Fallback records come from bundled <code>assets/world-bosses/rewards.json</code> and two inline records (L15-16). <code>bindRewardDisclosure</code> stores <code>iw-boss-rewards-collapsed</code> <code>:<bossKey></code> on <code>toggle</code>.
Depends on	<code>ItemDatabase</code> ; boss key.
Runs	Per boss decoration; rebuilt only when the signature changes.

States & fallbacks	DB not loaded → fallback records (clearly marked as documented fallbacks in code).
Responsive	Up to 9 columns; tiles 44×70 px.
Motion	Icon lift on hover under <code>(hover: hover)</code> and <code>(pointer: fine)</code> and <code>(prefers-reduced-motion: no-preference)</code> .
Hover / focus / keyboard	Tiles are real <code><button></code> s: focusable, and Enter/Space opens the card. The <code>summary</code> is natively keyboard operable.
Accessibility	Native <code><details></code> / <code><summary></code> semantics; tile <code>aria-label</code> . Reward names use <code>var(--iw-font-body, sans-serif)</code> , which is undefined (FUN-05).
Edge cases & findings	Drop rates and orb rules shown in fallback copy are skin-authored (CONT-01).
Tests	<code>world-boss-panels.test.mjs</code> .

F-48 Zone Control "Dominion Ward"

What	The Zone Control card becomes a "Dominion Ward" visual: • a crystal-and-crossed-swords crest that changes with control (red, blue or contested); • a readout with a kicker and state ("Crimson Dominion", "Azure Dominion", "Ward contested", "Under siege", "Fortified"); • a 10-segment meter showing each side's strength, with a split marker; • faction labels ("Crimson Oath", "Azure Covenant"). A short energy "impact" animation plays when strength changes.
Why	Turns a numeric race into a readable territorial state, without changing the numbers.
Where	<code>WorldBossPanels.ensureDominion (section.iw-control-dominion with aria-label="Zone dominion status", crest layers aria-hidden, meter role="progressbar" with aria-valuemin / max / now / valuetext), updateDominion, controlProgress, strengthTeam, controlStrengths, playStrengthImpact</code> (Web Animations API, 1000/1250 ms, blue delayed 170 ms; checks <code>prefers-reduced-motion</code> and listens for its change during the animation). The native control-strength, HP and progress nodes are marked with <code>data-iw-boss-role</code> and hidden by CSS (the mirror replaces them). CSS: <code>ui-system.css zone control</code> section and 8 keyframes.
Depends on	Text "controls Zone N" / "Race to capture" / "contested"; red/blue class or colour words (<code>red</code> , <code>rose</code> , <code>pink</code> , <code>crimson</code> vs <code>blue</code> , <code>sky</code> , <code>cyan</code> , <code>azure</code>); numeric leaves.
Runs	Each classify pass for the zone card; the impact animation only when a strength value changes.
States & fallbacks	<code>data-iw-control="red blue contested"</code> . If team colours cannot be read and exactly two numbers exist, the order is assumed red then blue (FUN-12).
Responsive	≤760 px and ≤380 px variants; a mobile impact variant at ≤760 px.
Motion	Continuous loops: • captured aura, 7.4 s; • resting light, 7.7 s; • sword filaments, 7.3 / 10.9 / 13.7 / 17.3 s, running only while contested; • crystal flows, 8.3 / 11.7 s; • meter currents, 5.7 / 8.1 s; • a 1.2 s crest crossfade. All are removed under reduced motion, and the impact is skipped (PERF-01).
Accessibility	The meter exposes values through ARIA. The decorative layers are hidden. The native numbers are hidden with <code>display: none</code> , so the mirror is the only accessible source.
Edge cases & findings	Invented faction and state names (CONT-02).
Tests	<code>zone-control-motion.test.mjs</code> (39), <code>world-boss-panels.test.mjs</code> .

2.10 MARKET

F-49 Market listing rows

What	Market listings become smoked-glass rows. Empty slots (the game's <code>opacity-30</code>) stay visibly empty, and item-name buttons read as plain text links.
Why	Consistency with the frame language. Keeps the game's own red/green stepper colours (rule 5).
Where	<code>UIFoundation.classifyMarket</code> : the "Market" section title → frame <code>data-iw-market="root"</code> ; leaf <code>.compact-row</code> outside inventory overlays and feeds → <code>data-iw-market="row"</code> ; name buttons(<code>button.text-left</code> , <code>button[class*=flex-1]</code>) → <code>data-iw-market="namebtn"</code> . CSS: <code>ui-system.css</code> market section.
Depends on	The "Market" heading; Tailwind classes <code>text-left</code> , <code>flex-1</code> , <code>opacity-30</code> .
Runs	Each classify pass (cached, revalidated).
Edge cases & findings	Rows with Equip/List buttons may also be claimed by the inventory renderer (F-40). The market classifier skips rows that already carry an inventory overlay.
Tests	<code>route-swap-reclassify.test.mjs</code> .

2.11 VILLAGE

F-50 Village route panels

What	On the Village route: • Housing hero ("Village"): the current house illustration, name and tier with five diamond pips (held pips gold), salvage and next-upgrade lines, and an ember Upgrade button. • Add-on slots ("Village Add-ons"): each slot is a card with the installed building's illustration (or a dashed empty footprint), its name (hovercard trigger), effects, and actions (Install / Upgrade ember CTA, Uninstall / Destroy with danger hover). • Install picker: each option shows small building art.
Why	The route is text-heavy; the game's own building art makes it legible [Project record : <code>CLAUDE.md</code> VillagePanels].
Where	<code>UIFoundation.classifyVillagePanels</code> (headings "Village" → housing, "Village Add-ons" → addons; root <code>.panel</code>). <code>VillagePanels.js</code> : • <code>decorateVillagePanel</code>, <code>decorateHousing</code> (tier from "Current tier: N", else name), <code>decorateSlot</code> (positional parse; <code>slotSignature</code>), <code>ensureArt</code> (<code>div.iw-village-art</code>, inline <code>--iw-village-sprite</code>), <code>ensureOptionArt</code>, <code>ensureTrigger</code> / <code>clearTrigger</code>, <code>clearVillagePanel</code>; • namespace <code>data-iw-village*</code>, skin nodes <code>data-iw-village-owned</code>. CSS: <code>ui-system.css</code> village section.
Depends on	English headings and verbs; the <code>villageBuildings.js</code> name map; <code>assets/village/*</code> ; <code>ItemDatabase</code> (building items <code>construction_building_tier_N</code>).
Runs	Each classify pass; per-slot signature gating.
States & fallbacks	Unknown building → generic art. Vacant slot → dashed footprint and muted text. Picker closed → no options.
Responsive	≤760 px: the head uses <code>display: contents</code> and actions form a row. ≤380 px: tighter.
Motion	Hover transitions; none under reduced motion.
Hover / focus / keyboard	Native buttons. The building name <code><p></code> becomes focusable with Enter/Space → hovercard (A11Y-02).
Accessibility	Art is <code>aria-hidden</code> . The destroy action adds a danger hover colour: a distinction the game did not draw.
Edge cases & findings	• Picker option art flip-flops between re-decorations (FUN-06). • Trigger teardown removes <code>tabindex</code> /ARIA unconditionally (FUN-09). • <code>clearVillagePanel(document.body)</code> at teardown removes every <code>[data-iw-tooltip-trigger]</code> in the body, including quest triggers; this is harmless at full teardown.
Tests	<code>village-panels.test.mjs</code> (53; also pins the generated <code>villageBuildings.js</code> against <code>buildings.json</code>).

F-51 Dashboard Village scene

What	A new Village frame inserted directly after the Skill Actions panel on the dashboard. It shows a cobblestone plot composition: the house in the middle, four corner plots and one bottom-centre plot, each with its building art or an empty plot. A count line reads "N installed / M slots", and the ledger (F-52) sits beside the composition.
Why	The village is not in the dashboard DOM at all, so the skin reads it from the game's own API or reuses what the Village route rendered [Project record : docs/traps/village.md].
Where	VillageScene.js : - reconcileVillageScene (L256–332), findAnchor (an h2 "Skill Actions"/"Actions" → closest .panel or .fs-skills-section-frame), readRenderedVillage (from F-50 tags), readHousingPerks , normaliseVillage (strict validation: tier and slots are integers 0–5, installed is an array, duplicate slot → invalid), render , own() (data-iw-village-scene-owned), art() (), clearVillageScene . - The frame is section.iw-village-scene[data-iw-ui="section-frame"][data-iw-panel="village-scene"][aria-label="Village"] , inserted with target.after(frame) . CSS: village-scene.css .
Depends on	GET /api/player?section=<route>&scope=core with session cookies and x-idleworlds-league (ssf if the path starts with /ssf , else standard)(SEC-04); villageBuildings.js ; ItemDatabase .
Runs	Every classify pass (async), and on iw:item-db-updated . Network reads happen only when: - no request is in flight; - Date.now() ≥ nextRead , which is 0 initially, then +5 min, or +30 s after a failure, or at most +10 s after a route change; - document.visibilityState is not hidden . Each request is abortable, with a 10 s timeout. A generation counter drops late responses.
States & fallbacks	"Loading your village..." (initial). Loaded (API or the rendered Village route). "Showing the last loaded village. Refresh unavailable; retrying shortly." "Village could not be loaded. Open the Village tab to view your buildings." League change (path /ssf... ↔ standard): the snapshot is dropped. Tab change within the league: the snapshot survives. No anchor (Skill Actions not rendered): the frame is removed and the request aborted.
Responsive	The composition is clamp(290px, 23vw, 320px) high, max 680 px wide, in container iw-village (≤470 px smaller plots). The scene and ledger wrap at a 592 px container width (iw-village-frame). ≤480 px viewport: 12 px frame padding.
Motion	None, apart from the ledger chevron rotation (F-52).
Accessibility	A section with aria-label="Village" ; the heading is role="heading" aria-level="2" ; sprites use alt="" .
Edge cases & findings	SEC-04: an authenticated API read that needs owner approval. - Collapsible like other frames (F-13): a re-render removes only the scene's own children and keeps the collapse toggle. - Distinct namespace from F-50, to avoid collisions.
Tests	village-scene.test.mjs (30), village-scene-layout.test.mjs (19).

F-52 Village ledger

What	Beside the scene is a ledger. It lists the home (slot count plus the tier perks printed on the Village route, e.g. "Base actions take 6s") and each installed building as a collapsible heading over its benefits (stats, skill levels). An Overall stats block sums installed buildings only . The whole "Buildings" list can be folded, and open/closed choices persist.
Why	The benefit numbers exist only in the item catalogue: each add-on is the item construction_building_tier_<N> . Housing is not an item, so only its printed text is shown, not guessed [Source : CLAUDE.md].
Where	VillageLedger.js : buildVillageLedger , entry , buildingItem (ItemDatabase.find({id, name})), buildingBenefits , totalBenefits (15 stat specs plus per-skill levels), wire (click → toggle preference and persist iw-village-ledger), applyEntry (data-iw-vs-open , aria-expanded , title), clearVillageLedger . Nodes are created with createElement / textContent only. CSS: village-scene.css ledger rules.
Depends on	The F-51 snapshot; ItemDatabase ; itemDisplay.deriveDisplayStats .
Runs	On each scene render (signature: snapshot, status, perks, DB revision).
States & fallbacks	Building not in the catalogue → entry with name only. No buildings → vacant lines.

Responsive	Ledger max width 310 px beside the scene at ≥592 px container width; stretches when folded or stacked.
Motion	Chevron rotation; none under reduced motion.
Hover / focus / keyboard	Entry heads are native <code><button></code> s (plated by the generic control rule), with <code>aria-controls</code> pointing to generated ids <code>iw-vs-entry-N</code> .
Accessibility	Buttons with <code>aria-expanded</code> ; stats in a <code>dl</code> ; "Overall stats" is <code>role="heading" aria-level="3"</code> .
Edge cases & findings	Uses its own <code>data-iw-vs-toggle</code> / <code>data-iw-vs-open</code> namespace, so <code>CollapsibleFrames</code> never adopts it.
Tests	<code>village-ledger.test.mjs</code> (41).

2.12 OVERLAYS AND CROSS-CUTTING BEHAVIOUR

F-53 Pop-up and modal framing

What	Any game dialog that covers the viewport (for example "Players Online" or confirmation modals) gets a dark blurred scrim, and its content card gets the forged frame: border, corner filigree through <code>border-image</code> , and textured ground.
Why	Portalled dialogs are not route panels, so they reached the page bare before this module existed. The frame uses only the border box, because these cards scroll and pseudo-element ornaments would scroll away [Source : CLAUDE.md].
Where	<code>OverlayFramer.js</code> : <code>frameOverlays</code> (every <code>iw:dom-flush</code>), <code>isScrim</code> , <code>pickCard</code> , <code>tagScrim</code> , <code>visible</code> , <code>clearOverlayFramer</code> . <code>data-iw-overlay="scrim panel"</code> . CSS: <code>overlay.css</code> (injected before <code>ui-system</code>).
Depends on	• Candidates: <code>[class*="fixed"]</code> or inline <code>position: fixed</code>. • A scrim must be: computed <code>position: fixed</code>; <code>z-index ≥ 20</code>; covering ≥85% of the viewport width and height, starting within 15%; visible; non-empty; and either blurred or with a dark, sufficiently opaque background (<code>alpha ≥ 0.15</code>, <code>luminance < 90</code>). • The card is a <code>.panel</code> (see FUN-07), or else the largest visible child that is not itself full-screen (up to 3 levels deep).
Runs	Every flush. Tagged scrim are revalidated first; the candidate query then walks the document (PERF-03).
States & fallbacks	A card inside skin-owned UI, or already framed (section frame, inventory root, Skill Actions frame, <code>.compact-panel</code>), is not re-framed. A layer that stops qualifying is untagged.
Accessibility	No semantic change; the game's dialog semantics are untouched.
Edge cases & findings	Detection is by rendered state, not class names, so toasts and tooltips that do not cover the viewport are ignored.
Tests	<code>overlay-framer.test.mjs</code> (detection, negative controls, teardown).

F-54 Reduced-motion support

What	With the operating-system "reduce motion" preference, the skin shows no transitions or animations.
Where	14 <code>@media (prefers-reduced-motion: reduce)</code> blocks across <code>base.css</code> , <code>inventory.css</code> , <code>skillpanel.css</code> , <code>skillcard-v2.css</code> , <code>header.css</code> , <code>ui-system.css</code> , <code>compact-buttons.css</code> , <code>village-scene.css</code> and <code>collapsible.css</code> . 4 rules in <code>ui-system.css</code> that animate only under <code>prefers-reduced-motion: no-preference</code> : 3 in plain <code>no-preference</code> blocks, and 1 combined with <code>(hover: hover)</code> and <code>(pointer: fine)</code> . <code>WorldBossPanels.playStrengthImpact</code> checks <code>matchMedia('(prefers-reduced-motion: reduce)')</code> .
Edge cases & findings	• <code>base.css</code> uses a universal <code>*</code> rule that also stops the game's animations and transitions (FUN-03). • Pseudo-element animations need explicit rules, because <code>*</code> does not match <code>::before</code> / <code>::after</code>. The skin includes those rules for its own shimmer and dominion layers.
Tests	<code>zone-control-motion.test.mjs</code> ; <code>static-invariants.test.mjs</code> . The project records that an earlier reduced-motion test could not fail (CLAUDE.md "negative control").

F-55 Responsive and touch behaviour

What	The skin adapts from 320 px phones to wide desktops, and enlarges touch targets on coarse pointers.
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Why	Phones and tablets are first-class. Breakpoints are aligned to Tailwind's so skin and game layouts change together (V1.6.0 mobile audit).
Where	66 @media blocks [Generated: Appendix C.7]. Style-rule counts per width condition: • max-width: 640px — 76 rules; • 760px — 40; • 768px — 18; • 860px — 17; • 1024px — 14; • 430px — 14; • 420px — 12; • 400px — 8; • 380px — 8; • 384px — 7; • 1024px and min-width: 641px — 6; • 560px — 5; • 1280px — 5; • 1279px / 767px — 4 each; • min-width: 861px — 4; • 520px — 3; • min-width: 641px — 2. Container queries: iw-skill-card (≤580, ≤480), iw-village (≤470), iw-village-frame (≥592); cqi-based font fitting in iw-inventory-filters and iw-nav-row. (pointer: coarse): V2 nav height 26 px; collapse toggle hit area 32 px. (hover: none), (pointer: coarse) in JavaScript switches tooltips to tap-to-toggle. JavaScript breakpoint epoch: F-05.
Edge cases & findings	• Breakpoints at 760/860/420/400/384/380/520 px are skin-specific and do not align with Tailwind [Generated]. They were chosen from measured layouts [Project record: docs/traps/mobile.md]. • The mobile audit harness is node claude/audit-mobile.mjs.
Tests	menu-rows.test.mjs, header-compact.test.mjs, village-scene-layout.test.mjs, quest-command-width.test.mjs, viewport-swap.test.mjs; fixture renders at 320/360/390/430/600/768 px(npm run fixtures).

3. CSS ARCHITECTURE

This chapter explains how the skin's CSS is delivered, how it wins (and sometimes loses) in the cascade, what every stylesheet contains section by section, and how to integrate it more cleanly.

It is paired with two generated references, so that **every selector and rule group** is documented without transcribing 1,373 rules by hand:

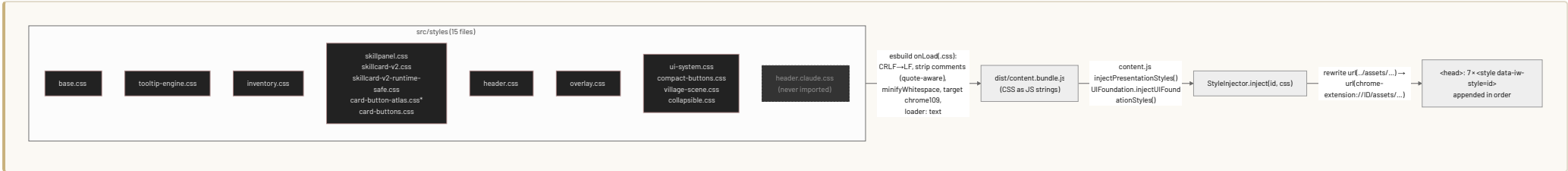
- **Appendix B – CSS rule index.** One row per rule in every stylesheet, with:
 - line, `@media / @container` context, selectors and specificity;
 - declaration count, `!important` count and every property set;
 - game (host) hooks versus skin hooks, fragility flags, and the nearest authored comment.
- **Appendix C – CSS dependency surface.** Host hooks, skin hooks, fragile patterns, `:has()` rules, `!important` by property, custom properties (defined, used, set by JS), media and container queries, animations and transitions, pseudo-elements, pseudo-classes.
- `tools/css-superseded.md`. Declarations that can never apply because a later rule with the identical selector overrides them.

All three are produced by scripts in `tools/` and are reproducible (§3.14).

Verification of the generator [Generated].

- `tools/css-inventory.mjs` counts **1,373 rules** in the 14 shipped sheets. An independent parse with `rrweb-cssom` (the CSS parser `jsdom` uses) produced the same count.
- A comment-stripped regex found **3,879 !important** declarations. The CSSOM reports 3,878, because `inventory.css` repeats one property inside a single rule and the CSSOM merges it.
- The inventory refuses to run if it meets an unaccounted stylesheet, or if it parses zero rules.

3.1 DELIVERY PIPELINE



* `card-button-atlas.css` is **generated** by `build-tools/build-card-buttons.mjs`; do not hand-edit it.

3.1.1 The seven style elements

The concatenation order inside each element is fixed by string concatenation in the importing module: `content.js` for `skillpanel`, and `UIFoundation.injectUIFoundationStyles` for `ui-system` [Source; Generated: Appendix B].

Order	data-iw-style	Files, in concatenation order	Rules	Selectors	Declarations	!important	@media blocks	:has() rules	Rules touching game hooks	Highest specificity
1	base	base.css	58	92	388	66	1	0	24	(0,12,0)
2	tooltip-engine	tooltip-engine.css	51	54	260	24	1	1	9	(0,3,0)
3	inventory	inventory.css	108	142	437	178	6	6	21	(0,4,1)
4	skillpanel	skillpanel.css + skillcard-v2.css + skillcard-v2-runtime-safe.css + card-button-atlas.css + card-buttons.css	609	702	2,593	2,351	30	12	558	(0,13,2)
5	header	header.css	81	98	471	400	6	6	3	(0,3,0)
6	overlay	overlay.css	2	2	11	11	0	0	0	(0,1,0)
7	ui-system	ui-system.css + compact-buttons.css + village-scene.css + collapsible.css	464	534	1,728	849	22	12	53	(0,30,3)
	Total	14 files	1,373	1,624	5,888	3,879	66	37	668	

Invariants(asserted by `tests/static-invariants.test.mjs` and `tests/smoke.test.mjs`)[**Test run**]:

- exactly 7 style elements;
- `ui-system` is injected last;
- all 7 are removed on disable.

3.1.2 The asset URL contract

`StyleInjector.rewriteAssetUrls` rewrites only the form `url(.. /assets/<path>)`, with optional `'` or `"` quotes. It uses the regex `/url\\(\\s*(['"])?\\.\\.\\.\\/assets\\/([\\^'"\\s]+)\\1\\s*)\\/g` ([StyleInjector.js:17](#)).

Warning. Any other spelling(`url(/assets/...)`, `url(assets/...)`, `url("../.. /assets/...")`) is left unchanged. The injected `<style>` has no base URL of its own, so the path resolves against the **game page** and 404s on `idleworlds.com`.

Assets must also be covered by `web_accessible_resources . assets/*` covers everything.

3.1.3 Not shipped

`src/styles/header.claude.css` (560 lines; 53 rules; 270 `!important`) is imported by no module, so esbuild never bundles it. Two further archived drafts live in `claude/`: `header.chatgpt.css` and `inventory.pre-forge.css`. All three are historical; they are candidates for deletion (MAINT-01).

3.2 CASCADE MODEL

3.2.1 Where the skin's rules sit

Competitor	How the skin relates to it	Consequence
Game stylesheets (Tailwind utilities and components, served by Next.js)	The skin's <code><style></code> elements are appended to the end of <code><head></code> at <code>document_idle</code> , after server-rendered <code><link rel="stylesheet"></code> elements. [Assumption] : CSS chunks that the game loads later during client-side navigation may be inserted after the skin's elements.	At equal specificity, the skin wins only if its element comes later. <code>!important</code> makes load order irrelevant against the game's normal declarations.
Game inline styles (<code>style="width: 40%"</code> , colours set by components)	Inline normal declarations beat every author rule except <code>!important</code> ones.	Properties the game sets inline (fill widths, some colours) can only be overridden with <code>!important</code> . The skin deliberately does not override the progress <code>width</code> ; it only adds a transition.
Skin inline <code>!important</code> (<code>BackgroundPainter</code> , <code>SkillPanelRenderer</code> button, readout and ingredient owners)	Inline <code>!important</code> beats every stylesheet declaration, <code>!important</code> or not.	To make such a value steerable from CSS, the renderer writes <code>var(--token, fallback)</code> inline, and CSS sets the token (<code>--fs-button-background</code> , <code>--fs-button-border</code> , <code>--fs-button-shadow</code> , <code>--fs-button-transition</code>) [Project record : docs/traps/sprites-and-buttons.md].
Skin inline normal (custom properties on game nodes and <code><html></code> , <code>display:none</code> from <code>InventoryRenderer</code> , geometry on skin-owned nodes)	Custom properties never conflict with game rules. <code>display: none</code> from <code>InventoryRenderer</code> is inline and not <code>!important</code> , so a game <code>!important display</code> would beat it [Source : <code>InventoryRenderer.js:456</code>].	—
User agent	The skin resets UA defaults it relies on (<code>dd margin</code> , <code>summary</code> marker, button padding).	—

3.2.2 Specificity strategy

- Attribute hooks.** Almost every selector keys on a skin-owned attribute (`[data-iw-ui="..."]`, `[data-iw-skill-role="..."]`, ...), which gives specificity (0,1,0) per hook. Game hooks are mostly `.compact-panel`, `button` and utility-class substrings (Appendix C.1).
- Compound scoping for the dense card systems.** The skill card rules chain the card hooks:
 - `.compact-panel.fs-skill-panel[data-iw-skill-layout="three-zone"]` is (0,3,0) before any descendant;
 - V2 rules prefix `html[data-iw-skill-card-design="new"] .compact-panel.fs-skill-panel[data-iw-skill-v2="1"][data-iw-skill-layout]`, which is (0,5,1).

The `html` type selector plus attribute is used as a cheap tie-breaker: 80 rule starts in `skillpanel.css` and 96 in `skillcard-v2.css` begin with `html` [**Generated**].

- Opt-out `:not()` chains instead of overrides.** The generic control rules in `base.css` and `ui-system.css` list every control that paints its own art. Each `:not(X)` adds X's specificity (Selectors Level 4), so the chain in `ui-system.css:1019` is **(0,30,3)**. `CLAUDE.md` calls it (0,4,1) (drift D-01).

Rule of thumb from the project. Never out-specify the generic rule. **Add your hook to its `:not()` chain**, in both `ui-system.css` and the matching `base.css` chains.

- `:where()` zeroes specificity in two `base.css` rules (L366, L452).** This lets the player-name heading keep its colour without raising the rule.
- `:is()` / `:has()` take their most specific argument.** The shared frame rule `:is(section-frame, inventory-root, .fs-skills-section-frame[ready]):not(:has(:is(...)))` is (0,4,0) to (0,6,1) depending on the pseudo-element. `collapsible.css` therefore opts **out** of the frame rule (`:not([data-iw-collapsed="1"])`) rather than trying to beat it.

6. **Retired hooks kept for specificity.** `:not([data-iw-order-handle])` appears 10 times in `base.css` and 6 times in `ui-system.css`, although the Rearrange feature was removed on 2026-09-15. The exclusions match nothing, but other rules may be tuned against the specificity they add [Source: `CLAUDE.md`].

Highest-specificity rules [Generated]:

Rule	Specificity	Why so high
ui-system.css:1019 generic button:not(...) rest state	(0,30,3)	30 :not() exclusions
ui-system.css:1033 hover	(0,30,2)	same chain
ui-system.css:1065 disabled	(0,29,3)	same chain
card-buttons.css:46 V2 nav plate hover/focus	(0,13,2)	V2 prefix + :is() states
base.css control chains	up to (0,12,0)	exclusion chains

3.2.3 `!important`: where, why, and how to remove it safely

Census [Generated: Appendix C.5]:

- 3,879 of 5,888 declarations (66%) are `!important`.
- The most common properties: `color` 199, `font-size` 185, `padding` 165, `display` 163, `background` 160, `width` 146, `box-shadow` 142, `border` 116, `height` 114, `margin` 113, `min-width` 110, `border-radius` 106, `min-height` 105, `position` 92.
- `!important` declarations by file:

File	<code>!important</code>
skillpanel.css	1,807
ui-system.css	720
skillcard-v2.css	500
header.css	400
inventory.css	178
collapsible.css	75
base.css	66
compact-buttons.css	54
card-buttons.css	30
tooltip-engine.css	24
skillcard-v2-runtime-safe.css	14
overlay.css	11
village-scene.css	0
card-button-atlas.css	0

Why it is used. The source comments and `docs/traps` give four reasons, labelled [Source/Project record] with inference where noted:

1. **Against the game's utilities, regardless of load order.** The skin cannot control where later game CSS is inserted (§3.2.1).
2. **Against game inline styles.** Nothing else can override them from a stylesheet.
3. **Against the skin's own earlier passes.** `skillpanel.css` grew as successive "passes" (§3.4.4), each re-asserting a composition over the previous one with equal or higher specificity and `!important`. This is the main source of volume. At least **341 `!important` declarations are provably dead**: a later identical selector overrides them (`tools/css-superseded.md`). 302 of the 350 dead declarations are in `skillpanel.css` and 46 in `header.css`.
4. **To defeat the generic control rules** in `base.css` and `ui-system.css`, which are themselves `!important` at high specificity.

Where it is *not* used. Three places deliberately avoid it:

- `village-scene.css` (0 `!important`): the Village scene is entirely skin-owned DOM, so there is nothing to beat.
- `card-button-atlas.css`: it only defines custom properties.
- A handful of `transition` / `animation` declarations, so that the reduced-motion `!important` rules can win.

Safe removal procedure. Apply it one rule group at a time; each step has a check:

1. **Delete dead declarations first.** Work through `tools/css-superseded.md`. Each row is overridden by an identical later selector, so removing it cannot change the computed style. Check: `npm test` (the cascade suites load all sheets in order) and `npm run fixtures`.
2. **Remove superseded passes.** Where a whole section of `skillpanel.css` is overridden by a later "final" section (§3.4.4 marks them), delete the earlier section. Check: `skill-card-system.test.mjs`, `skill-card-v2-render.test.mjs` and the fixture screenshots at 320-1536 px.
3. **Find the actual competitor for the remaining `!important` declarations.** In DevTools → Elements → Computed, expand the property to see which declarations it overrides:
 - **only skin rules** → resolve the ordering inside the skin (merge rules), then drop `!important` from both;
 - **a game utility class** → keep `!important`, or move the skin into a cascade layer strategy (§3.13);
 - **a game inline style** → keep `!important`, or, in an official integration, stop the component from writing that inline style.
4. **Never drop `!important` from a rule that competes with a renderer's inline `!important`.** It loses either way. Instead, make the renderer write `var(--token, fallback)` inline and set the token from CSS.
5. **Keep the reduced-motion `!important` rules** (`transition: none !important`, `animation: none !important`). They must beat every other motion declaration.

3.2.4 Inheritance and custom properties

The skin's theming is carried almost entirely by **custom properties**. They inherit, so a value set on `<html>` or on a card reaches every descendant rule without extra selectors [[Generated](#): Appendix C.6 lists every custom property with where it is defined, used and set by JS].

Scope	Set by	Examples	Notes
<code>:root</code> defaults	<code>base.css</code> , <code>vi-system.css</code>	<code>--iw-ink-*</code> , <code>--iw-th-*</code> , <code>--iw-font-*</code> , <code>--iw-r-*</code> , <code>--iw-space-*</code> , <code>--iw-control-h</code> , <code>--iw-frame-title-size</code> , <code>--iw-zone-atlas</code> / <code>--iw-corner-filigree</code> / <code>--iw-zone-separator</code> (defaults)	The skin's design tokens (§3.3)
<code>:root</code> game token overrides	<code>base.css</code> (<code>!important</code>)	<code>--background</code> , <code>--foreground</code> , <code>--card</code> , <code>--popover</code> , <code>--primary</code> , <code>--accent</code> , <code>--ring</code> , <code>--border</code> , <code>--input</code> , <code>--muted</code> , <code>--sidebar*</code> , <code>--panel-bg</code> , <code>--panel-border</code> , <code>--surface-*</code>	Re-themes game components the skin never classifies. This is the clearest official integration point.
<code>:root[data-iw-zone-theme="<theme>"]</code>	<code>base.css</code> (9 palettes), <code>skillpanel.css</code> (quest frame windows), <code>card-button-atlas.css</code> (V2 plate windows)	<code>--iw-th-accent</code> , <code>--iw-th-edge</code> , <code>--fs-quest-frame-position</code> , <code>--iw-card-action-idle</code>	Attribute set by <code>HeaderRenderer</code>
Inline on <code><html></code>	<code>HeaderRenderer.applyZoneTheme</code> , <code>SkillsArtService.applyThemeVariables</code>	<code>--iw-zone-atlas: url(...)</code> , <code>--iw-corner-filigree</code> , <code>--iw-zone-separator</code> , <code>--iw-compact-atlas</code> , <code>--iw-action-idle</code> , <code>--iw-action-idle-paint</code> , ... (12 kinds × 2)	Outside the observed <code><body></code> , so writes never trigger a flush
Inline on game containers	<code>HeaderRenderer</code> (header, nav, announcement, zone bar), <code>SkillsArtService.decoratePanel</code> (skill/quest cards, Skill Actions frame, boss action buttons), <code>QuestPanelRenderer</code> , <code>SkillCardDesignController</code> , <code>UIFoundation</code>	<code>--iw-header-surface</code> , <code>--fs-skills-ui-atlas</code> , <code>--fs-ui-action-idle-size</code> , <code>--fs-quest-accent</code> , <code>--fs-quest-cmd-w</code> , <code>--iw-skill-v2-progress</code> , <code>--iw-skill-v2-fill-duration</code> , <code>--iw-skill-v2-action-label-font</code> , <code>--iw-progress-duration</code>	Removed on teardown

Scope	Set by	Examples	Notes
Rule-local tokens	Component rules	<code>--fs-skill-accent</code> (per discipline), <code>--fs-forge-*</code> , <code>--fs-chev-*</code> , <code>--iw-skill-v2-*</code> (≈ 30 geometry tokens), <code>--hd-*</code> (header), <code>--iw-frame-pad-*</code> , <code>--iw-nav-k</code> , <code>--fs-tier</code> , <code>--fs-frame</code>	Container-query blocks redefine the tokens rather than the properties

Undefined custom properties (used with `var()` but never defined anywhere) resolve to their fallback (FUN-05):

- `--iw-font-flavour` $\rightarrow$ `var(--iw-font-ui)`. The intended token is `--iw-font-flav`.
- `--iw-font-body` $\rightarrow$ `sans-serif`.
- `--iw-font-display` $\rightarrow$ `'Cinzel', serif`. The fallback is what the author intended.

Inheritance hazards. Several font and colour declarations are applied to containers and inherited by native text. For example, `[data-iw-panel]` sets `font-family` and `color !important`. A game element that relies on *inheriting* its colour inherits the skin's instead. Elements with their own utility colour class keep it, which is how rule 5 state colours survive.

3.3 DESIGN TOKENS

Family	Defined	Purpose	Values / examples
Ink scale	<code>base.css</code> <code>:root</code> L74+	Warm near-black grounds	<code>--iw-ink-950</code> #070806 ... <code>--iw-ink-600</code>
Theme accent layer <code>--iw-th-*</code>	<code>base.css</code> <code>:root</code> (default "Ashen Iron"), $9 \times$ <code>:root[data-iw-zone-theme]</code> (L216–318), derived edges (L320–368, <code>color-mix</code>)	Every edge, hairline, bracket, brass, rule, plate, call-to-action, ground and glow colour	<code>--iw-th-accent</code> , <code>-accent-dim</code> , <code>-edge</code> , <code>-hairline(-hi)</code> , <code>-bracket(-dim)</code> , <code>-brass</code> , <code>-rule(-soft)</code> , <code>-plate</code> , <code>-cta(-hi)</code> , <code>-ground-a/-b/-wash</code> , <code>-glow</code> , <code>-edge-mid/-soft/-faint</code>
Lines	<code>base.css</code>	Rules and borders	<code>--iw-line</code> , <code>--iw-line-hi</code> , <code>--iw-line-hot</code>
Legacy aliases	<code>base.css</code>	Older rules	<code>--iw-gold</code> , <code>--iw-gold-dim</code> , <code>--iw-ember</code> , <code>--iw-ember-hi</code>
Text	<code>base.css</code>	Text tiers	<code>--iw-text</code> , <code>--iw-text-hi</code> , <code>--iw-dim</code> , <code>--iw-faint</code>
Semantic	<code>base.css</code> L137–139	Good/bad/info	<code>--iw-good</code> #82B88A, <code>--iw-bad</code> #D27171, <code>--iw-info</code> #75A8C8
Tier colours	<code>base.css</code> L142+ (<code>--iw-t-common</code> ... <code>--iw-t-mythic</code>) and <code>.tier-*</code> classes (L609–616)	Item tier naming colours	<code>--iw-t-common</code> #B9B4A8, <code>--iw-t-uncommon</code> #6FBF73, ... (bands by tier number in <code>itemDisplay.js</code>)
Fonts	<code>base.css</code>	Typefaces	<code>--iw-font-head</code> (Cinzel), <code>--iw-font-ui</code> (Barlow), <code>--iw-font-flav</code> (Crimson Text italic)
Radii	<code>base.css</code>	Flattened corners	<code>--iw-r-panel</code> 3px, <code>--iw-r-control</code> 2px, <code>--iw-r-slot</code> 2px, plus the aliases <code>--iw-r</code> , <code>--iw-r-sm</code>
Motion	<code>base.css</code>	Default tick	<code>--iw-action-tick</code> : 1s
Spacing and controls	<code>ui-system.css</code> L8–25	Rhythm	<code>--iw-space-1...5</code> (4/6/9/12/16 px), <code>--iw-control-h</code> 32px, <code>--iw-control-h-lg</code> 36px, <code>--iw-frame-title-size</code> 18px, <code>--iw-frame-edge</code> , <code>--iw-frame-inner</code> , <code>--iw-steel</code>
Frame art	<code>base.css</code> L195–214 + inline on <code><html></code>	Per-zone art	<code>--iw-zone-atlas</code> , <code>--iw-corner-filigree</code> , <code>--iw-zone-separator</code>
Discipline accents	<code>skillpanel.css</code> L1–21	Per-skill accent	<code>--fs-skill-accent</code> : combat #B84A20, mining #84919B, smithing #B28A2A, gathering #579A5D, alchemy #9271B2, jewelcrafting #4E9FB8, spellcrafting #8B6FC3, tailoring #A56E86, woodcutting #8B6A3A, construction #5F7F72, crafting #5E8FB7, fishing #478FA8, locked #6A6257
Inventory categories	<code>inventory.css</code>	Row identity	<code>--fs-tier</code> by <code>.fs-cat-*</code> : gear #6E9BC4, consumable #68B96A, processed #A6AEB2, resource #9E7B54, trade #D4AD63, unknown #8C8578, potion #CE5C6E, enchant #8E7BDB, xpscroll #3FB2AC, cache #B366C4, orb #E0913E

Family	Defined	Purpose	Values / examples
Header-local	header.css §0	Header plates	--hd-bracket(-dim), --hd-plate, --hd-rule(-soft), --hd-teal #4FC7D8, --hd-teal-dim #1E5C68, --hd-gap
V2 card geometry	skillcard-v2.css L14+	Card layout	--iw-skill-v2-icon 43px, -identity-w 144px, -command-w 96px, -btn-w 64px, -btn-h 66px, -nav-w 30px, -nav-h 22px, -action-label-font 9.5px (ceiling; JS writes the fitted value), ...

Note — a JavaScript mirror of CSS tokens. Some token values are duplicated in JavaScript and must be changed together:

- the forged plate tokens exist in skillpanel.css L22-79 **and** in SkillPanelRenderer.FORGE (L101-119), and in build-tools/render-fixtures.mjs;
- the compact button geometry block in compact-buttons.css (/* COMPACT_GEOMETRY */) is generated by build-tools/preview-compact-buttons.mjs.

3.4 STYLESHEET-BY-STYLESHEET REFERENCE

Each table maps a line range to what it targets and its load-bearing declarations. Rule-level detail (every selector, specificity, property and fragility flag) is in the linked Appendix B page.

3.4.1 base.css — 713 lines, 58 rules, 66 !important (rule index)

Lines	Section	Targets	Key declarations and notes
1-9	Header	—	—
10-72	Typefaces	@font-face ×6	Cinzel 600-700 (variable), Barlow 400/500/600/700, Crimson Text italic; font-display: swap; url('../assets/fonts/...')
74-193	Root tokens	:root	Ink, --iw-th-*, lines, text, semantic, tier, fonts, radii, --iw-action-tick. Game token overrides (--background ... --sidebar-accent) are all !important. --iw-th-edge is declared twice (harmless)
195-214	Per-zone frame theme	:root	Defaults: --iw-zone-atlas: url('../assets/skills_ui_atlas.webp'), --iw-corner-filigree: url('../assets/inventory/panel_corners.webp'), --iw-zone-separator
216-318	Per-zone accent palettes	:root[data-iw-zone-theme="glacial infernal verdant forged-metal celestial voidborn runic-arcane lunar-spectral tempest-oceanic"]	Redefines the --iw-th-* layer per theme (0,1,1)
320-368	Derived inner frames	:root[data-iw-zone-theme]	--iw-th-edge-mid/-soft/-faint, --iw-line/-hi via color-mix(in srgb, ...). Requires Chrome 111
370-431	Core surfaces	html, body, h1-h3, .compact-panel, [class*="compact-panel"], .compact-row, [class*="item-row"], [class*="rounded-3xl 2xl xl lg md"]	html, body background, colour and font !important. Body gets radial + linear gradients with background-attachment: fixed. Headings use Cinzel. The forged ground (texture + gradients, soft-light blend) goes on every game compact panel and row. Tailwind radii are flattened
433-559	Controls	button, [role=button], input, select, textarea, nav/header/tab buttons	- Base control fill (0,0,1), not !important, overriding the UA buttonface. [class*="hover:underline"] text-link buttons: background, border, shadow and padding cleared (longhands only, so the gradient-clipped player name survives). - Font on controls; button shadow and transition, with a long :not() exclusion chain; hover brightness(1.08). :focus-visible 1px gold outline, !important. - Forged inputs; placeholder colour; nav/tab radius and weight.

Lines	Section	Targets	Key declarations and notes
561–607	Shared icon slot	<code>.iw-ico (unused), .iw-icon-badge, .iw-icon-badge--sprite</code>	+N badge placement for <code>AtlasService</code>
609–616	Tier colours	<code>.tier-commontier-mythic</code>	Used by the tooltip name (<code>tierClass</code>)
618–660	Extension-owned button primitive	<code>.iw-btn, --primary, --disabled</code>	Unused by any module (MAINT-01)
662–683	Progress/XP primitive	<code>.iw-xp*</code>	Unused (MAINT-01)
685–691	Scrollbars	<code>::-webkit-scrollbar, -track, -thumb</code>	Page-wide 7 px themed scrollbar (Chromium-only pseudo-elements)
693–697	Reduced motion	<code>*</code>	<code>transition: none !important; animation: none !important</code> – page-wide, including the game (FUN-03)
698–713	Pills and bordered rounded boxes	<code>[class*="rounded-full"][class*="px-"]</code> , <code>button[class*="rounded-full"]</code> , <code>[class*="border"][class*="rounded-3xl 2xl xl"]</code>	Radius flattening; border colour and shadow for bordered rounded containers

3.4.2 tooltip-engine.css – 411 lines, 51 rules, 24 !important (rule index)

Lines	Section	Targets	Key declarations and notes
1–25	Header	—	Records the design contract: same frame language as panels
27–56	Inline item reference	<code>.iw-item-ref, :hover, :focus-visible, .iw-item-info</code>	Inherits font, gold colour, <code>cursor: help</code> , bottom rule on hover/focus; the "i" glyph is hidden
58–97	Card frame	<code>.iw-tip</code>	Tokens <code>--fs-tip-w 330px, -pad-x 12, -art 64, -corner-h 20, -corner-w 18; position: fixed; z-index: 99999; max-width: calc(100vw - 20px); max-height: calc(100dvh - 20px); forged ground; opacity: 0; pointer-events: none; transition: none</code>
98–134	Ornament and open state	<code>.iw-tip::before (hairline), ::after (corner filigree border-image: var(--iw-corner-filigree) 50% / h w / 0 stretch), .iw-tip > *, .is-open, :focus-visible</code>	<code>.is-open → opacity: 1; pointer-events: auto</code>
135–226	Head, art slot, name	<code>.iw-tip-head, .iw-tip-art, -art-host, -art-fallback, .iw-tip-name, .iw-tip-close</code>	64 px slot built like the inventory icon; 22 px close button
227–270	Badges	<code>.iw-tip-badge, .t (tier), .req (requirement)</code>	Rectangular chips (house style: no pills)
271–292	Sections	<code>.iw-tip-body (scroll, overscroll-behavior: contain, scrollbar-gutter: stable), .iw-tip-sec, .iw-tip-sec-title</code>	Cinzel eyebrow in tracked caps
293–357	Stats	<code>.iw-tip-stats (2 columns; :not(:has(> :nth-child(2))) → 1 column), .iw-tip-stat .k/.v, .v.amber, .v.good (unused), .iw-tip-stat-note, .iw-tip-effect</code>	Only <code>:has()</code> rule in the file
358–399	Acquisition, flavour, foot	<code>.iw-tip-acq(.is-unknown), .iw-tip-flavour (unused), .iw-tip-foot:empty, .iw-tip-link</code>	—
400–411	Source label and phone	<code>.iw-tip-source.is-stale; @media (max-width: 420px)</code>	Phone: width <code>100vw - 20px</code> , smaller art and corners, one-column stats

3.4.3 inventory.css — 884 lines, 108 rules, 178 !important (rule index)

Lines	Section	Targets	Key declarations and notes
1-25	Header	—	"Forged pass, 2026-08-25" rationale
26-388	Inventory chrome	[data-iw-inventory-root="1"] (overflow hidden), [data-iw-inventory-list="1"] (recessed card with ::before / ::after), [data-iw-inventory-title="1"], .fs-inv-rule (::before line, ::after separator ornament var(--iw-zone-separator) 186×62), [data-iw-inventory-control="filter page"] (26 px plates, top: 3px optical nudge), [data-iw-inventory-filters="1"] (container iw-inventory-filters, cqi font fit), [data-iw-inventory-filter-state="active"] (ember, rule 5), [data-iw-inventory-control="icon"] (30×31 px, display: grid, pixel atlas window background-size: 322.5px 173.63px), _="glyph" (15 px, drop shadow, colour untouched), _="page-count"	Media: ≤1279 px k=22.5, ≤767 px k=20.9, ≤560 px ornament. --fs-inv-command-w: 54px appears unused [Generated: C.6]
389-502	Row shell	.compact-row:has(> .fs-inv-row) / [class*="item-row"]:has(...) (flex, min-height 62, top border, hover), .fs-inv-row (grid 52 px / 1fr / 38 px), .fs-cat-* → --fs-tier, ::before 3 px rail, ::after hover tint, .has-details, .has-requirements	6 :has() rules key the row shell on the overlay's presence
503-551	Icon slot	.fs-inv-icon (52 px socket, overflow: visible for the badge, cursor: help), :hover, :focus-visible, .is-equipped (warm edge #8A6A2C)	Equipped is a separate axis from category (rule 5)
552-702	Identity and information rails	.fs-inv-body, .fs-inv-name (2-line clamp, Cinzel 14, --fs-tier), .fs-inv-sub, .fs-inv-plus, .fs-inv-name-plain, .fs-inv-stats, .fs-stat(--pos --tier), .fs-inv-details, .fs-inv-detail--loadout socket effect status set, .fs-inv-requirement(s), .fs-inv-qty	.fs-inv-requirement colour set twice: L657 is superseded by L680 [Generated: css-superseded.md]
703-763	React-owned action rail	[data-fs-action-host="1"] (display: contents; font-size: 0; color: transparent), [data-fs-suppressed="1"] (display: none !important : hides native row branches), [data-fs-preserved-action="control"] (order 2, z-index 2) and its button, a, [role=button] plates	The rules that make native row children disappear while controls stay
764-823	Action kinds	[data-fs-action-kind="equipped equip set secondary icon"], hover re-assertions, :disabled / [aria-disabled="true"] 0.45	Equipped is green and Equip brass; hover keeps their meaning (rule 5)
824-884	Responsive collapse	@media (max-width: 1024px), (max-width: 768px), (prefers-reduced-motion)	Chip limits by :nth-child (5 and 3), icon 42 px, wrapping; the root ::after border depends on ui-system.css frame art

3.4.4 skillpanel.css — 4,001 lines, 483 rules, 1,807 !important (rule index)

This file is **append-only history**. Each banner section is a design pass that re-asserts parts of the card over earlier passes. The table marks which passes still govern the live card, based on the later sections that override them and on `tools/css-superseded.md` (302 superseded declarations in this file, 97 of them in L200-999 and 146 in L1000-1999).

Lines	Section (banner)	Targets	Status	Notes
1-21	Discipline accents	.fs-skill--<type>	Live	--fs-skill-accent per discipline
22-79	Forged plate tokens	.compact-panel.fs-skill-panel	Live	--fs-forge-*, --fs-chev-*. Mirrored in SkillPanelRenderer.FORGE
81-143	Base card	card, ::before accent line, .fs-skill-wrapper (contents), .fs-skill-header (none), three-zone grid	Partly superseded	The grid is redefined at L529 and later
145-287	Role typography	identity, action-title, level-progress, xp-gain, requirement ([data-iw-req-state="unmet"] #D58282), reward, action-detail, ingredient, .iw-item-ref, progress rail	Partly live	—

Lines	Section (banner)	Targets	Status	Notes
289–337	Readout and ingredient neutralisation	<code>[data-iw-readout]</code> , <code>[data-iw-ingr]</code> + pseudo-elements	Live	Strips game chrome from readouts
339–417	Command rail and CSS fallbacks	action/nav buttons by <code>data-iw-btn-state</code>	Live as fallback	Mirrors inline <code>BUTTON_STYLES</code>
419–460	Zone flex and highlight	zones; <code>::highlight(iw-skill-ingredient-met)</code>	Live	The only <code>::highlight()</code> rule in the skin
462–520	Early responsive	<code>@media (max-width: 768px)</code> , <code>(max-width: 520px)</code>	Largely superseded	Documented dead 88 px action width rule
523–922	"Premium three-zone action card"	identity medallion, content, commands; media 1024/640/420	Largely superseded	—
924–1150	"Option 3 art-direction fidelity"	<code>.fs-skill-medallion-art</code> , studs, <code>:has(> .fs-skill-medallion-art[data-iw-skill-art-ready="1"])</code> hides the native icon	Partly live	—
1153–1251	Atlas integration	<code>[data-iw-skills-ui-ready="1"]</code> texture, medallion, corner filigree, separator, nav, XP plaque	Partly live	—
1253–1324	Live wrapper compatibility	<code>[data-iw-skill-layout-shell="1"]</code> grid	Partly live	<code>:has(> [data-iw-skill-layout-shell="1"])</code>
1325–1533	"Final approved Skills composition"	identity and title <code>font-size: 0 + ::before { content: attr(data-iw-clean-text) }</code> ; identity progress mirror; native track/fill/xp-gain/reward visually hidden; base-exp plaque; locked desaturation	Live (text replacement)	A11Y-01
1535–1691	"Skills polish"	identity percent; level-progress <code>font-size: 0</code> ; pointer-events: none !important ; cursor: default !important (L1563–1580); nav SVG backgrounds; <code>.fs-skills-section-frame</code>	Live	FUN-01
1693–1908	"Compact responsive Skills composition"	tighter sizes; media 1024–641 and 640	Partly superseded	—
1911–2083	"Alignment pass": real command stacking, true artwork ratios	155×44 action (ratio 3.52), studs, quest-rail artwork command, base EXP 150×50	Partly superseded by V2	—
2085–2349	"Recipe navigation"	frameless nav, zeroed hidden rows, rhythm, narrow layouts	Partly live	—
2351–2386	"Base readout: simplified"	plain chip	Live	Sprite dropped by the owner
2389–2589	"Recipe pager — thin cool-toned arrows"	shape A (<code>nav-group display: contents + order</code>), shape B (absolute over command column at ≥641 px), 26×44 plate, <code>::before</code> chevron with derived <code>--fs-chev-ink</code>	Live for the three-zone layout; V2 repositions it	—
2592–3367	Quest cards	<code>.compact-panel.fs-quest-panel</code> frame; <code>:root[data-iw-zone-theme] --fs-quest-frame-position</code> (L2706–2733); sigil (glyph <code>::before</code> , ring sprite <code>::after</code> with radial mask, <code>approved-frames.png</code>); roles; reward chip (<code>::before attr</code>); three-slice command rail; ready state; media 640/400	Live	Only quest CSS in the skin
3369–3640	"Full-height identity, right-hand command band"	<code>--fs-skill-command-w 240px</code> , <code>--fs-skill-command-h 64px</code> , <code>--fs-skill-identity-w 150px</code> ; shell grid; media 640 (116 px), 430 (78 px)	Live: supersedes earlier layouts	—
3642–3778	"Exact themed button states"	<code>html[data-iw-zone-theme]:not([...="default"])</code> action, nav and quest backgrounds from <code>--iw-action-*</code> / <code>--iw-chevron-*</code> ; hover/active/disabled swap variables	Live (values now come from revised-v5)	—
3780–3885	revised-v5 crossfade	<code>html[data-iw-button-atlas="revised-v5"] ::before</code> hover (180 ms) and <code>::after</code> pressed (70 ms) layers	Live	—

Lines	Section (banner)	Targets	Status	Notes
3887-3931	384 px fallback	command band spans both columns	Live	—
3933-4001	Material grid	[data-iw-ingredient-list-source="1"] display: none; .fs-skill-ingredient-grid 2 columns; met state; ≤430 px one column	Live	—

Status labels are an analysis aid, not a proof. "Superseded" means later sections redefine the same targets. Only the declarations listed in `tools/css-superseded.md` are *proven* dead. Before deleting a whole section, run `npm test` and `npm run fixtures` and compare screenshots.

3.4.5 skillcard-v2.css — 543 lines, 108 rules, 500 !important (rule index)

All selectors are scoped to `html[data-iw-skill-card-design="new"] .compact-panel.fs-skill-panel[data-iw-skill-v2="1"][data-iw-skill-layout]`.

Lines	Section	Key declarations and notes
1-13	Header	V2 is the only design; the old/new switch was removed
14-108	Shared composition	V2 tokens on the card; container: iw-skill-card / inline-size; radius 7; radial + gradient + texture ground; ::before hairline; ::after corner filigree (17/16 px, z-index 3, opacity 0.85); shell grid (areas identity/content/body/tabs/command)
109-156	Fold, zones and identity	L109: cards hidden inside a collapsed Skill Actions frame. L122: identity zone hides every descendant except the medallion art, level readout and identity chain (display: none). Medallion 43 px; ::after conic XP ring from --iw-skill-v2-progress with a radial mask; level badge
157-165	Native text and readout	Native identity text font-size: 0. L164: [data-iw-skill-role="level-progress"] display: none !important (FUN-01)
166-230	Title row	Title ::before clamp(17px, 2cqi, 23px); .fs-skill-base-exp chip; retired .iw-skill-v2-summary / -expand rules (dead)
231-280	Command group	Action button absolutely centred; --fs-motion-*, --fs-button-border, --fs-button-shadow feeding the inline var() s; color: transparent; hover brightness; :focus-visible 2 px accent outline (offset 5 px); disabled grayscale 0.55
281-318	Activity fill	> span[style*="width"] accent band; transition: width var(--iw-skill-v2-fill-duration, .28s) linear; reset rule; reduced motion (L311); running-state track via :has()
319-348	Shell-child button case	Glyph and label placed in grid-area: commands when the button is the zone
349-370	Long action	Countdown in glyph (15 px), label hidden
371-387	Discipline icons	10 rules, background-image: url('../assets/skills-ui/action-icons/<type>.svg')
388-432	Pager	Nav group below the button; wrappers normalised (position: static; transform: none; filter: none; contain: none; will-change: auto) so the containing block is the command column; ::before 7 px chevron; Lucide <svg> hidden
433-468	Body and foot	.iw-skill-v2-controls shown only with a requirement note; .iw-skill-v2-body container iw-skill-body, auto-fit up to 3 columns (min 96 px); material cells (✓ when met)
469-514	Container queries	@container iw-skill-card (max-width: 580px) tokens; (max-width: 480px) phone layout
515-519	Reduced motion; identity wrappers	* { transition: none !important }; identity ancestors display: contents
520-530	Touch	@media (pointer: coarse) → --iw-skill-v2-nav-h: 26px
531-543	Oldest layout	@container iw-skill-card (max-width: 480px) explicit grid lines for cards whose zones are direct children

3.4.6 skillcard-v2-runtime-safe.css – 29 lines, 1 rule, 14 !important

[data-iw-skill-v2-section] is visually hidden with position: absolute; width/height: 1px; clip-path: inset(50%); opacity: 0; pointer-events: none. It deliberately does not use display: none, because SkillPanelRenderer reads computed visibility to choose live nodes, and display: none would make those nodes unresolvable. The nodes stay in the accessibility tree (A11Y-01).

3.4.7 card-button-atlas.css – 73 lines, 9 rules, 0 !important (generated)

One rule per theme, html[data-iw-zone-theme="<theme>"], defines --iw-card-action-idle|hover|clicked and --iw-card-nav-idle|hover|clicked as transparent url(card-v6/action-atlas.png|nav-atlas.png) x% y% / size no-repeat. Regenerate with node build-tools/build-card-buttons.mjs.

3.4.8 card-buttons.css – 55 lines, 8 rules, 30 !important

Lines	Section	Key declarations and notes
1-22	V2 action plate	--fs-motion-* = --iw-card-action-*; --fs-button-background; transparent border; background-origin/clip: padding-box; radius 2; transition: none; ::before / ::after inset: 0 crossfade layers
23-34	Fill clip	> span[style*="width"]: clip-path: inset(5px max(0px, calc(100% - var(--iw-skill-v2-btn-w) + 2*border + 5px)) 5px 5px round 2px); accent gradient; animation: none (removes the game's animate-pulse); disabled grayscale 0.6
35-55	Nav plate	--iw-card-nav-*; octagon clip-path: polygon(...); (not !important); hover/focus/active swaps; :focus-visible outline inset -3 px. Highest specificity in the group: (0,13,2) at L46

3.4.9 header.css – 1,045 lines, 81 rules, 400 !important (rule index)

Lines	Section	Targets	Key declarations and notes
1-31	Header	—	Explains why the frame art is used through border-image corners rather than the stretched header_frame.webp
32-47	§0 Local tokens	[data-iw-header="root"], [data-iw-ui="main-nav"], [data-iw-header="announcement"], [data-iw-header="zone-shell"]	--hd-* (bracket, plate, rule, teal #4FC7D8 / #1E5C68, gap)
48-79	§1 Corner-bracket plate	status cards, utility buttons, zone actions	8 linear-gradient corner brackets. Partly superseded by §4/§4b and ui-system.css:833 (header.css:59 border is dead)
80-165	§2 Header plate	[data-iw-header="root"]	width: 100%; contain: layout inline-size; padding 16/20; overflow: hidden; background: var(--iw-header-surface) ... / cover; > * z-index 1; ::after corner filigree 34/32 at inset: 0; layout grid 62fr/38fr
166-370	§3 Identity region	identity-region (grid: crest/profile/utilities; max-width: min(560px, 46vw)), ::before smoked glass (rgba(..., .72) + backdrop-filter: blur(4px)), .fs-header-crest (104×106, var(--iw-header-crest)), profile, brand (caps with rules), profile-name (typography only; colour left to the game), profile-name button reset, profile-title, profile-meta, profile-online (real button reduced to text + teal dot ::before)	Rule 5: the player-name colour is never set
371-438	§4 Utility buttons	[data-iw-header="utilities"] (flex, space-between, margin -8 px), utility-button (flex: 1 1 0, 42 px high), svg 18 px, hover	Partly superseded by §4b
439-531	§4b Purpose-made plate art	utility buttons and status cards: smoked glass	Historical sprite-crop notes; the specificity note at L498

Lines	Section	Targets	Key declarations and notes
532–658	§5/§5a Status tiles	<code>status-grid</code> (2 × 4); gold spans; timer/other in column 2; <code>@media (min-width: 861px)</code> combat/boost plaque with <code>:has([data-iw-header-stat="boost"])</code> ; status-card typography (Georgia serif 12 px, white, wrap)	6 <code>:has()</code> rules in this file (L581–601, L790, L837)
659–726	§6 Navigation rail	<code>main-nav-shell</code> , <code>main-nav</code> , <code>nav-tab</code> , <code>:hover</code> , <code>[data-iw-state="active"]</code> , <code>:first-child</code>	30 declarations superseded by <code>ui-system.css</code> L699–780 (identical selectors, later element)—conflict X-01
727–764	§7 Announcement bar	<code>[data-iw-header="announcement"]</code>	Teal gradient; <code>::before</code> four-point star via <code>clip-path</code>
765–931	§8 Zone bar	<code>zone-shell</code> (flex wrap), title branch <code>> :has([data-iw-ui="zone-title"])</code> , <code>zone-title ~ *</code> dimmed with <code>::before</code> divider (except <code>zone-link</code>), action group <code>margin-left: auto</code> , <code>::before</code> moonlit scene <code>var(--iw-zone-scene)</code> at opacity .34 with <code>-webkit-mask-composite: source-in / mask-composite: intersect</code> , <code>zone-title</code> Cinzel 16, <code>zone-action</code> sizing, tones by <code>[data-iw-zone-action]</code> with <code>:first-of-type / :last-of-type</code> fallbacks	Position fallbacks: FUN-12. Several <code>zone-action</code> declarations superseded by <code>ui-system.css:833</code>
932–1038	§9 Responsive	<code>@media (max-width: 1280px)</code> one column; <code>(max-width: 860px)</code> crest 108×116, utilities flex, 2 auto columns, <code>nav-tab</code> 14 px, <code>zone-action</code> flex; <code>(max-width: 768px)</code> <code>var(--iw-header-surface-mobile)</code> ; <code>(max-width: 560px)</code> <code>main-nav</code> horizontal scroll, hidden scrollbar	—
1039–1045	§10 Reduced motion	utility button, nav tab, zone action	<code>transition: none</code>

3.4.10 overlay.css — 55 lines, 2 rules, 11 !important (rule index)

Selector	Declarations
<code>[data-iw-overlay="scrim"]</code>	<code>background-color: rgba(7,7,11,.72)</code> , <code>backdrop-filter: blur(3px)</code>
<code>[data-iw-overlay="panel"]</code>	<code>isolation: isolate</code> ; <code>border</code> ; <code>border-image: var(--iw-corner-filigree) 50% / 25px 23px / 0 stretch</code> ; <code>radius</code> ; forged ground with <code>skills_panel_texture.webp</code> ; <code>background-attachment: scroll</code> ; shadows. No pseudo-elements , because the card is its own scroll container

3.4.11 ui-system.css — 2,460 lines, 350 rules, 720 !important (rule index)

Lines	Section	Targets	Key declarations and notes
1–25	Header and tokens	<code>:root</code>	<code>--iw-space-*</code> , <code>--iw-control-h(-lg)</code> , <code>--iw-frame-title-size</code> , <code>--iw-frame-edge/-inner</code> , <code>--iw-steel</code>
27–205	Shared forged frame	<code>:is(section-frame, inventory-root, .fs-skills-section-frame[ready]):not(:has(:is(...))) + ::before</code> hairline + <code>::after</code> corners (<code>inset: -3px -1px</code> , <code>border-width: 34px 32px</code> , <code>border-image: var(--iw-corner-filigree) 50% / 34px 32px / 0 stretch</code>) + <code>> * z-index 1</code> ; <code>@media (max-width: 1024px)</code> padding and 27/25 corners; nav-rail padding tokens; nested layout columns stripped (L164–190); <code>[data-iw-ui="section-title"]</code> ; <code>[data-iw-skill-boost="1"]</code>	Collapsed frames and the nav rail opt out through <code>:not()</code>
206–352	Activity surfaces	<code>[data-iw-panel]</code> (flex column, font, colour), <code>[data-iw-panel] * box-sizing</code> , split header grid, header parts, header/tools buttons (excluding the collapse toggle), <code>panel-control</code> , Action Log title un-plated, <code>@media (min-width: 641px)</code>	—

Lines	Section	Targets	Key declarations and notes
353–501	Current action	title/strong 15 px, p faint; progress 18 px track, ::before 10 segments, ::after highlight; fill > * gradient + transition: width var(--iw-progress-duration, var(--iw-action-tick, 1s)) linear + will-change: width; > *::before shimmer iw-control-meter-current 5.7s infinite; reset; reduced motion; queue grid + 32 px button	PERF-01
502–691	Feed language (log and chat)	feed (scroll box; scrollbar-color / -width), action-log max 300 px, world-chat min(46vh, 430px), transparent wrappers(*:has([data-iw-panel-part="feed-row"])), feed-row, time, action-log row typography, game .feed-panel background, chat strong / b, composer, chat-input (+ ::placeholder, :focus), send; @media (max-width: 640px), (max-width: 430px)	Depends on the game class .feed-panel
692–815	Main navigation rail	main-nav-shell, main-nav (flex, max-content), nav-tab (32 px plate, Barlow 11.5 px caps), :last-of-type, :hover:not(:disabled), [data-iw-state="active"] (+ ::after ember underline), [data-iw-nav-link="toolkit"] (+ ::after ↗, hover)	Overrides header.css §6(X-01)
816–844	Zone command rail	zone-bar, zone-title, zone-action plate(L833)	Overrides parts of header.css §1/§8
845–1007	Header chrome merge	[data-iw-chrome="shell"] 2-column grid; header row margin; zone-bar display: contents (pseudo-elements none); zone-text row 2 column 1; notice row 2 column 2; nav row 3 column 1(plate reset); zone-actions row 3 column 2; shell ::before frame(rows 2–3), ::after divider; @media (max-width: 860px) stacked rows 2–5; [data-iw-zone-link] text-link reset	—
1008–1095	Generic control language	L1019 button:not(...×30), [role="button"]:not(...)(0,30,3)rest; L1033 hover; L1057–1062 min-height floors (classified controls only); L1065 disabled saturate(.58) brightness(.84); input[type=search], input[placeholder*="Search" i]; @media (max-width: 768px) full-width nav, compact Toolkit	Opt-out chain (§3.2.2)
1096–1148	One-row rail on phones and tablets	@media (max-width: 1279px) :is(main-nav, section-frame):has(> nav-tab) container iw-nav-row, nowrap, --iw-nav-k: 28.1; nav-tab font clamp(7.5px, (100cqi - fixed)/k, 11.5px); @media (max-width: 767px) Toolkit display: none, k = 23.25	Measured constants [Project record: docs/traps/mobile.md]
1149–1482	World Boss cards and rewards	[data-iw-boss="card"] ground, hairline, inset; transparent wrappers; decoration-dotted buttons as links; .iw-boss-notice; [data-iw-encounter] grid(104 px art auto 1fr); per-boss accents; role placements; .iw-boss-art from encounters.png (background-size: 200%, per-boss positions); boss action 132×38 plate; [data-iw-boss-art-ready] atlas action window; [data-iw-boss-action-label] font-size: 0 + ::after { content: attr(...) } (–.07em optical fix); details.iw-boss-rewards, summary marker, 9-column grid, 44×70 tiles, 34 px icons; (prefers-reduced-motion: no-preference) icon lift; quest-ready settle animation (L1463–1481); [data-iw-control="red blue contested"] colours	--iw-font-body undefined (FUN-05)
1483–1921	Zone Control "Dominion Ward"	[data-iw-encounter="zone"]; .iw-control-dominion grid; crest ::before aura(7.4 s); crest art layers(zone-control-contested-v3.png)1.2 s crossfade; captured tint ::after (clip-path, mix-blend-mode: color, 7.7 s); sword energy filaments(masks; 7.3/10.9/13.7/17.3 s; contested only); crystal core flows(8.3/11.7 s); readout; meter(10 segments; red fill with blue ::before sized by control split); currents(5.7/8.1 s); split marker; factions; 8 @keyframes (L1806–1863); native control-hp/progress/strength display: none; timer ⚡; @media (max-width: 760px), (max-width: 380px); reduced motion(.iw-control-dominion *, *::before, *::after → none)	PERF-01
1922–1976	Market	[data-iw-market="root row namebtn"]; [class*="opacity-30"] empty slots	Keeps game stepper colours
1977–2460	Village route	[data-iw-village*] intro/note; slot card grid (art head) with ground and hairline; [data-iw-village-art] plot + ::before sprite var(--iw-village-sprite) 88% auto; generic/vacant ::after dashed footprint; actions(install/upgrade ember call-to-action, destroy danger hover); picker grid; option and option-owned; .iw-village-option-art 34 px; housing hero (art 132 px spanning 4 rows); house name/tier/note/salvage/next(keeps the game's emerald)/max; tier pips (diamonds; held gold); @media (max-width: 760px), (max-width: 380px); reduced motion	—

3.4.12 compact-buttons.css – 127 lines, 27 rules, 54 !important

Lines	Section	Key declarations and notes
1-19	Base	<code>[data-iw-compact-layer]</code> <code>display: none</code> unless <code>html[data-iw-compact-atlas="compact-ghost-v3"]</code> . The button gets <code>position: relative; isolation: isolate</code> with background, border, shadow, filter and transform cleared, a colour transition of 180 ms, and a 10 px cap (nav tab 12 px, send 14 px)
20-81	<code>/* COMPACT_GEOMETRY */</code> (generated by <code>preview-compact-buttons.mjs</code>)	Mid/cap/icon sizes and x positions; per-state layer y (idle 2.94%, hover 50%, clicked 97.06%); layers absolutely positioned at <code>inset: 0</code> with a three-slice layout (mid band content-box, <code>::before</code> left cap, <code>::after</code> right cap, z-index -3/-2/-1); single-window icon kind
82-94	States	<code>:hover</code> / <code>:focus-visible</code> / <code>:active</code> → hover layer opacity 1 (180 ms); <code>:active</code> → clicked layer (70 ms)
95-107	Selection legibility	Inset bottom accent for the active filter, <code>[data-iw-compact-selected]</code> and equipped (green)
108-118	Active nav label	Forced light text on the active route tab, including native children
119-127	Disabled and reduced motion	Hover/clicked layers hidden, 0.45 opacity; <code>transition: none</code>

3.4.13 village-scene.css – 192 lines, 72 rules, 0 !important

Lines	Section	Key declarations and notes
1-15	Frame and title	<code>.iw-village-scene; .iw-vs-title</code> <code>var(--iw-font-display, 'Cinzel', serif)</code> (undefined token; the fallback is intended)
16-23	Scene container	<code>.iw-vs-scene</code> { <code>position: relative; width: 100%; container: iw-village / inline-size; height: clamp(290px, 23vw, 320px); max-width: 680px; overflow: hidden; background: cobblestone.png</code> }. <code>width: 100%</code> is load-bearing
24-70	Plots	Captions overlaid; house absolute top-centre 40%×52%; plots 27%×130 px by <code>[data-slot="1..5"]</code> ; <code>object-fit: contain</code> ; placeholder glyph; <code>[data-plot-state="locked"]</code> opacity
71-82	Narrow	<code>@container iw-village (max-width: 470px); @media (max-width: 480px)</code> frame padding 12 px
83-192	Ledger	<code>.iw-vs-body</code> flex wrap in container <code>iw-village-frame</code> (wraps at 592 px = 330 + 250 + 12); <code>@container</code> (min-width: 592px) ledger max-width 310; titles and toggles; <code>[data-iw-vs-open="0"]</code> hides list/body; entry head (generic plate from <code>ui-system</code>), icon, label, chevron rotation (0.16 s); body rule; notes; <code>d1 stats (dt / dd)</code> ; skill-level rows in mythic amber; vacant dashed lines; totals card; folded stretch; reduced motion

3.4.14 collapsible.css – 202 lines, 15 rules, 75 !important

Lines	Section	Key declarations and notes
1-46	Head gutter and toggle	<code>[data-iw-collapse-head]</code> { <code>position: relative; padding-right: 30px</code> }; <code>[data-iw-collapse]</code> 22 px square, absolutely positioned right and centred, with plate and accent colour
47-64	Chevron	<code>::after</code> rotated border corner: -135° expanded, 45° collapsed (<code>[aria-expanded="false"]</code>); 0.16 s
65-87	Touch, hover, focus	<code>@media (pointer: coarse)</code> <code>::before</code> { <code>inset: -5px</code> } (32 px target); hover; <code>:focus-visible</code> outline
88-93	The fold	<code>[data-iw-collapsed="1"] > *:not([data-iw-collapse]):not([data-iw-collapse-head])</code> <code>display: none !important</code>
94-134	The collapsed bar	Tokens <code>--iw-frame-pad-y: 7px</code> , <code>--iw-frame-pad-x: 22px/16px</code> ; head reset; <code>padding-left: 22px</code>
135-151	The spine	<code>[data-iw-collapsed="1"] [data-iw-collapse-head] *:not([data-iw-collapse-spine]):not([data-iw-collapse])</code> <code>display: none !important</code> ; spine <code>display: block</code> . Depends on JS spine marks (FUN-04)
152-168	Title treatment	<code>[data-iw-collapse-title]</code> 14 px / 20 px Cinzel, single line, ellipsis
169-200	One flourish	Collapsed <code>::after</code> top-left corner only (<code>border-width: 34px 0 0 32px</code>)
201-202	Reduced motion	Chevron <code>transition: none</code>

3.4.15 header.claude.css – inactive

An alternative header sheet (560 lines, 53 rules, 270 !important) that no module imports, so it never ships. Its rule index is generated for completeness (rule index). Recommendation: delete it, or move it to an archive folder outside src/ .

3.5 SELECTORS, PSEUDO-CLASSES AND PSEUDO-ELEMENTS

Kind	Count (rules)	Notes
:not()	130	Mostly opt-out chains (§3.2.2) and exclusions of skin-owned nodes
:hover	62	Surface-only changes; the skin avoids changing game text colours on hover (rule 5)
:disabled (+ [aria-disabled="true"] : 22)	57	Disabled plates, pagers, command buttons
:is()	39	Grouping inside the frame, V2 and compact rules
:has()	37	See the list in Appendix C.4. Used to key styles on the presence of skin marks (overlay present, art ready, nav group present, fill running), and for leaf-frame detection
:root	23	Tokens and per-theme palettes
:focus-visible	20	Keyboard focus rings on controls, tooltip, triggers
:first-child / :last-child / :nth-child / :first-of-type / :last-of-type / :only-child / :empty	14/3/3/2/2/1/2	Position-based: fragile (§3.8)
:active	9	Pressed art layers
:where()	2	Specificity-neutral exclusions in base.css
:focus	1	Chat input
::before / ::after	140 / 100	Ornaments (hairlines, corners, studs, chevrons), three-slice caps, crossfade layers, generated label text (content: attr(...) : A11Y-01)
::-webkit-scrollbar*	5	Chromium/WebKit-only scrollbars
::placeholder	2	Inputs and chat
::highlight(iw-skill-ingredient-met)	1	Met material text. There is no ::highlight(iw-item-name) (FUN-02)
::marker	1	Rewards summary

Attribute substring selectors on game classes (Appendix C.3): [class*="hover:underline"] (11 rules), [class*="chat-name-"] (10), [class*="item-row"] (7), [class*="decoration-dotted"] (6), [class*="rounded-*"] (several), [class*="opacity-30"] (2), [class*="border"] , [class*="px-"] , [class*="tab"] (1 each), [style*="width"] (9: the game's inline progress widths).

3.6 MEDIA QUERIES AND CONTAINER QUERIES

Breakpoint system. The skin uses Tailwind's breakpoints where it must change together with the game's own layout: 640, 768, 1024 and 1280 px. The -1 px forms (max-width: 767px , 1279px) match Tailwind's min-width boundaries exactly. Other breakpoints are component-specific and were chosen from measured layouts:

Condition	Rules	Files	Purpose
max-width: 640px	76	skillpanel.css , ui-system.css	Phone skill and quest card layouts, activity panels
max-width: 760px	40	ui-system.css	Boss, dominion and village phone layouts
max-width: 768px	18	inventory.css , skillpanel.css , header.css , ui-system.css	Tablet portrait; mobile header art; full-width nav

Condition	Rules	Files	Purpose
max-width: 860px	17	header.css , ui-system.css	Header stacking; merged chrome stacking
max-width: 1024px	14	inventory.css , skillpanel.css , ui-system.css	Frame padding and corners; inventory density
max-width: 430px	14	skillpanel.css , ui-system.css	Small phones
max-width: 420px	12	tooltip-engine.css , skillpanel.css	Tooltip full width
max-width: 400px / 384px / 380px	8 / 7 / 8	skillpanel.css , ui-system.css	Very small phones
max-width: 1024px and min-width: 641px	6	skillpanel.css	Tablet skill card band
max-width: 560px / 520px	5 / 3	inventory.css , header.css , skillpanel.css	Ornament and nav scroll; legacy
max-width: 1280px / 1279px / 767px	5 / 4 / 4	header.css , inventory.css , ui-system.css	Header single column; cqi font fitting
min-width: 861px / 641px	4 / 2	header.css , skillpanel.css , ui-system.css	Header plaque; pager shape B; log header
prefers-reduced-motion: reduce / no-preference	14 / 3 (+1 combined)	9 files	Motion (F-54)
pointer: coarse	2	skillcard-v2.css , collapsible.css	Touch targets
@container iw-skill-card (max-width: 580px / 480px)	4 / 11	skillcard-v2.css	V2 card density by card width
@container iw-village (max-width: 470px)	4	village-scene.css	Narrow scene
@container iw-village-frame (min-width: 592px)	3	village-scene.css	Scene and ledger side by side

JavaScript counterpart. Viewport.js watches (min-width: 640/768/1024/1280/1536px) and bumps a layout epoch, so caches can re-resolve the visible duplicate column (F-05).

Why container queries. A panel's width is not a function of viewport width: panels sit in 1–3 column grids. The V2 card, the inventory filter row, the nav row and the Village scene therefore size by their container (cqi units and @container) [**Project record:** docs/traps/mobile.md].

3.7 ANIMATIONS AND TRANSITIONS

Keyframes (all in ui-system.css) [**Generated:** Appendix C.8]:

Keyframes	Defined	Used by	Duration and iteration	Runs when
iw-quest-ready-settle	L1473	Ready quest sigil icon (L1466)	0.85 s, once	Quest becomes ready; no-preference only
iw-control-meter-current	L1838	Current Action fill shimmer (L435); dominion currents (L1759)	5.7 s, infinite	Always while visible (paused under reduced motion)
iw-control-captured-aura	L1810	Crest ::before (L1530)	7.4 s, infinite	Captured state visible
iw-control-resting-light	L1806	Crest art ::after (L1572)	7.7 s, infinite	Not contested
iw-control-filament-red / -blue	L1814 / L1821	Sword energy ::before / ::after (L1630–1633)	7.3 / 10.9 / 13.7 / 17.3 s, infinite	Contested (opacity-gated)
iw-control-crystal-flow	L1828	Crystal core (L1644)	8.3 s, infinite	Contested

Keyframes	Defined	Used by	Duration and iteration	Runs when
<code>iw-control-crystal-undertow</code>	L1833	Crystal core <code>::after</code> (L1654)	11.7 s, infinite	Contested

Transitions. 55 rules declare `transition`. The policy recorded in the sources:

- Hover and focus changes are short (70–240 ms).
- Progress fills use `linear` over the **measured tick**, so they move continuously (F-21, F-33).
- The tooltip uses `transition: none`, so hiding is immediate and cannot linger.
- Width and cadence belong to the game; the skin never animates to a computed value.

Web Animations API. `WorldBossPanels.playStrengthImpact` (1000 / 1250 ms, once per strength change) is the only JavaScript animation, and it checks reduced motion.

Reduced motion. See F-54.

- `base.css:693` applies `*` (the whole page) and does not match pseudo-elements.
- Pseudo-element loops therefore have explicit rules: `ui-system.css:461`, `:1884`, `:1913`.

3.8 DOM-STRUCTURE DEPENDENCIES AND FRAGILE RULES

The complete list of game hooks with rule counts is Appendix C.1. Risk summary:

Dependency	Rules	What breaks if the game changes it	Detection
<code>.compact-panel</code>	561	Every skill, quest, boss and village card style, and the forged ground on all compact panels	<code>smoke</code> , <code>skill-card-system</code> , <code>fixtures</code>
<code>button</code> (element)	170	Command and pager styling; generic control language	<code>cascade suites</code>
<code>[aria-disabled="true"]</code> , <code>:disabled</code>	22 + 57	Disabled appearance	<code>cascade suites</code>
<code>[role="button"]</code>	21	Generic control rules	—
Tailwind class substrings (<code>hover:underline</code> , <code>chat-name-</code> , <code>decoration-dotted</code> , <code>rounded-*</code> , <code>opacity-30</code> , <code>px-</code> , <code>border</code> , <code>tab</code>)	≈45	Text-link exclusions (player names could gain plates), radius flattening, market empty slots	Visual review; <code>claude/audit-mobile.mjs</code>
<code>[style*="width"]</code>	9	V2 fill band and clip (F-33)	<code>action-progress-theme</code> , <code>skill-card-v2-render</code>
<code>.compact-row</code> , <code>[class*="item-row"]</code>	7 + 7	Inventory row shell	<code>inventory-root</code> , <code>smoke</code>
<code>.feed-panel</code>	2	Chat and log background	—
Element types (<code>p</code> , <code>strong</code> , <code>span</code> , <code>summary</code> , <code>dd</code> , <code>dt</code> , <code>time</code> , <code>b</code> , <code>svg</code> , <code>input</code> , <code>select</code> , <code>textarea</code> , <code>hr</code> , <code>nav</code> , <code>header</code>)	small	Placement and typography inside classified parts	—
Position (<code>:first-child</code> ... <code>:first-of-type</code>)	25	Zone action tone fallback (FUN-12); first-row borders; chip limits; tooltip stat layout	—
Sibling combinators (<code>~</code> , <code>+</code>)	6	Zone title dimming (<code>zone-title ~ *</code>); feed row spacing	—

Structural contracts expressed in JavaScript, not CSS. Most fragility is moved into the classifiers, which write stable `data-iw-*` roles. Chapter 6 §6.5 lists those assumptions, and Appendix D lists every role attribute. When the game's markup changes, the usual failure is that a classifier stops writing a role, so the CSS silently stops applying and the surface renders native. That is the designed fail-safe.

3.9 CONFLICTS AND OVERRIDDEN RULES

ID	Conflict	Evidence	Effect today	Resolution
X-01	header.css §6 navigation (L659-726) vs ui-system.css main navigation rail (L692-815): identical selectors, both !important; ui-system is later	30 superseded declarations in header.css L660-729 [Generated: css-superseded.md]	ui-system.css governs the rail. For nav-tab, header.css:684 still contributes the properties ui-system.css:725 does not set: flex, display: flex, align-items, justify-content, gap, white-space, text-shadow and transition	Merge into one rule set; delete the dead declarations
X-02	header.css §1 corner-bracket plate and §4 vs §4b and ui-system.css:833	e.g. header.css:59 border, :415 border-radius, :419 box-shadow superseded	Later plate art governs	Delete superseded declarations
X-03	skillpanel.css successive passes	302 superseded declarations; §3.4.4 status column	Later passes govern	Remove dead declarations first, then superseded sections, with fixtures
X-04	SkillPanelRenderer.readoutCurs or keeps a pointer on the XP-cycle button vs skillpanel.css:1578 pointer-events: none !important and skillcard-v2.css:164 display: none !important	Source	The control is unreachable (FUN-01)	Decide product intent; see Chapter 5 hardening/behaviour split
X-05	Renderer inline !important plates vs stylesheet button art	docs/traps/sprites-and-buttons.md	Resolved: inline values are var(--fs-button-*, fallback) and CSS sets the tokens	Keep the pattern for any new inline !important
X-06	Shared frame corner rule (0,4,0) !important vs collapsed bar	ui-system.css:100 :not([data-iw-collapsed="1"])	Resolved by opt-out	—
X-07	Generic control box-shadow / background-image !important vs component plates	Opt-out chains; docs/traps/sprites-and-buttons.md L116	Resolved by exclusions; every new self-painting control must be added	—
X-08	base.css universal reduced-motion vs the game's own animations	base.css:693	Game animations frozen for reduced-motion users (FUN-03)	Scope to skin selectors
X-09	inventory.css .fs-inv-requirement colour set twice	L657 vs L680	L680 governs	Delete L657 declaration
X-10	InventoryRenderer inline display: none (not !important) vs any game !important display on row children	InventoryRenderer.js:456	None observed [Assumption: the game uses no !important display utilities on these children]	If needed, write 'important' priority
X-11	V2 identity zone hides every descendant except skin marks (skillcard-v2.css:122) vs the game adding a new meaningful element there (e.g. a boost badge)	Source	A new game element in the identity branch would be invisible	Review when the game adds identity content (rule 5)

3.10 BROWSER-SPECIFIC BEHAVIOUR

Feature	Where	Chromium behaviour	Notes
::-webkit-scrollbar*	base.css L685-691 (page-wide), header.css L1035	Themed 7 px scrollbars	Chromium/WebKit only; scrollbar-color / scrollbar-width used for feeds and header nav

Feature	Where	Chromium behaviour	Notes
<code>color-mix(in srgb, ...)</code>	145 uses	Chrome 111+	Older engines: invalid at computed-value time, so many edges lose colour (§1.10)
<code>:has()</code>	37 rules	Chrome 105+	Rules dropped on older engines
Container queries and <code>cqi</code>	6 blocks; 7 units	Chrome 105+	—
<code>::highlight()</code> + <code>CSS.highlights</code>	1 rule	Chrome 105+	—
<code>backdrop-filter: blur()</code>	14	Composited; GPU	Glass plates and scrim; cost on low-end GPUs
<code>-webkit-mask + mask</code> (and <code>-webkit-mask-composite: source-in / mask-composite: intersect</code>)	<code>header.css</code> L856-863, <code>skillcard-v2.css</code> L136-137, <code>skillpanel.css</code> L2781-2782, <code>ui-system.css</code> L1613-1614	Prefixed forms for older Chromium; unprefixed from Chrome 120 [Assumption : version from general platform knowledge]	Both are provided
<code>border-image</code> with <code>var()</code> sources	frames, tooltip, overlay	Works; slices use percentages so 1× and 2× sheets align	—
<code>display: contents</code>	zone bar, action host, V2 identity wrappers, nav group	Removes the box; children keep semantics	Test with screen readers [Verify]
<code>dvh</code>	tooltip <code>max-height</code>	Chrome 108+	—
<code>font-display: swap</code> , variable Cinzel	<code>base.css</code>	Flash of fallback text until fonts load	V2 label fitting re-measures after <code>document.fonts.ready</code>
<code>background-attachment: fixed</code>	<code>body</code> ground	May force repaint on scroll on some devices [Verify with the perf harness]	—

3.11 PERFORMANCE CHARACTERISTICS

Concern	Evidence	Impact	Recommendation
Sprite atlas decode size (PERF-07)	<code>gear_icons_atlas.png</code> 1280×19,584 px ≈ 95.6 MiB decoded RGBA; <code>item_icons_atlas.png</code> 1280×6,144 ≈ 30 MiB; theme atlases 1720×926 ≈ 6 MiB; boss, quest, cobblestone and card sheets 6 MiB each [Generated : PNG/WebP header dimensions]	One icon forces a whole-sheet decode. The gear sheet is taller than the 16,384 px maximum texture size common on GPUs [Assumption], so it may need tiled or CPU rasterisation. High memory on phones [Verify]	Split atlases, use lossless WebP/AVIF, or per-item images (the game's own CDN in an official integration)
<code>:has()</code> invalidation	37 rules; the frame rule <code>:not(:has(:is(section-frame...)))</code> is evaluated for frames on attribute changes	Style recalculation when classified attributes change inside frames	The renderers already compare before writing; keep that discipline
Long <code>:not()</code> chains on <code>button</code>	3 rules at (0,29-30,x)	Matching cost per button per recalc (≈800 buttons on a 200-row inventory [Project record])	Replace with a positive opt-in hook in an official integration
Substring attribute selectors	≈60 rules	Evaluated on class changes	Official integration: component variants instead
Universal descendants	<code>[data-iw-panel] * (box-sizing), V2 * (reduced motion), .iw-control-dominion *, base.css *</code>	Broad matching sets	Acceptable; scope where cheap
Infinite animations (PERF-01)	9 loops (§3.7)	Continuous paint or composite while visible	Pause when off-screen or when the tab is hidden; limit to <code>transform / opacity</code>
<code>backdrop-filter</code> , blend modes	14 + 3	GPU memory and compositing	Consider pre-baked translucent textures on mobile

Concern	Evidence	Impact	Recommendation
Duplicate and superseded rules	350 dead declarations; about 1,800 <code>!important</code> in <code>skillpanel.css</code>	Parse and cascade cost. The bundle carries 324,572 bytes of minified CSS text (48.6% of 668,309 bytes) ; <code>skillpanel.css</code> alone is 123,490 bytes [Generated : the build's CSS pipeline replicated by the inspection script]	§3.2.3 removal procedure

Measured page-level costs (flush quiescence, boot cost, scroll stalls) are in `docs/traps/performance.md` [**Project record**] and can be reproduced with `node claude/perf-harness.mjs --runs 3` (Chapter 7).

3.12 KNOWN CSS DEFECTS AND GAPS

Finding	Summary	Location
FUN-01	Game "cycle XP display" button disabled, then hidden	<code>skillpanel.css:1563-1580</code> , <code>skillcard-v2.css:164</code>
FUN-02	No <code>::highlight(iw-item-name)</code> style, so item names in prose have no affordance	(missing)
FUN-03	Page-wide reduced-motion rule also freezes game animations	<code>base.css:693</code>
FUN-04	The collapsed title depends on JavaScript spine marks that can go missing after a kill-switch round trip	<code>collapsible.css:139</code> + <code>CollapsibleFrames.js:91-117</code>
FUN-05	Undefined font tokens (<code>--iw-font-flavour</code> , <code>--iw-font-body</code>)	<code>skillpanel.css:2840</code> , <code>ui-system.css:1478</code> , <code>1701</code>
FUN-12	Position-based zone action tone fallback	<code>header.css:902</code> , <code>915</code>
A11Y-01	Text replaced by CSS generated content; duplicate or altered announcements	<code>skillpanel.css</code> L1325+, quest reward chip, boss action label
A11Y-03	Very small text (4 px fitted label; 9 px names)	<code>skillcard-v2.css</code> , <code>ui-system.css:1478</code>
MAINT-01	Unused rules (<code>.iw-ico</code> , <code>.iw-btn*</code> , <code>.iw-xp*</code> , retired V2 summary/expand/collapsed, <code>.iw-tip-flavour</code> , <code>.v.good</code> , <code>--fs-inv-command-w</code>), inactive <code>header.claude.css</code> , 350 superseded declarations	various
PERF-07	Very large sprite atlases	<code>assets/*_atlas.png</code>

3.13 RECOMMENDATIONS FOR A CLEANER INTEGRATION

- These are ordered from "no visual change" to "requires IdleWorlds source access". None was applied in this handover.
- Delete dead CSS** (no visual change): the 350 rows in `tools/css-superseded.md`, the unused primitives listed in MAINT-01, `header.css` §6 duplicates (X-01), and `header.claude.css`. Verify with `npm test` and `npm run fixtures`.
 - Fix the token typos** (small visual change): `--iw-font-flavour` → `--iw-font-flav`, and `--iw-font-body` → `--iw-font-ui`. Decide intentionally, because the quest brief and reward names would change font (FUN-05).
 - Scope reduced motion to the skin** (behaviour change for reduced-motion users): replace `base.css:693` * with selectors under skin hooks (FUN-03).
 - Collapse `skillpanel.css` into one composition**. Keep the live sections marked in §3.4.4, rewrite them as one block per component, and drop `!important` where the only competitor is another skin rule (§3.2.3 steps 2–3).
 - Adopt cascade layers** once the game's CSS strategy is known [**Verify: Tailwind version and whether the game ships native `@layer`**]:
 - The game already uses native layers** (Tailwind v4 emits `@layer` `theme`, `base`, `components`, `utilities`). Unlayered skin rules beat every layered game rule at normal importance, so most skin `!important` can be removed without specificity games.

- **The game's CSS is unlayered** (Tailwind v3 compiles its directives away). Wrap the game's CSS in a layer in the official build, or keep `!important` only for properties the game sets inline.
 - **Caution.** `!important` in layers reverses order: important declarations in *earlier* layers win. Do not move `!important` skin rules into a layer without re-testing.
6. **Move tokens into the game's theme** (official integration). `base.css` already overrides the game's design tokens (`--background`, `--card`, `--primary`, ...). An official integration defines the Ashen Iron and per-zone palettes as the game's theme values (Tailwind theme / CSS variables), and sets `data-zone-theme` from game state instead of text scanning.
 7. **Replace classifiers with component variants** (official integration). Where the skin writes `data-iw-skill-role`, `data-iw-quest-role`, `data-iw-panel` and so on, IdleWorlds components can render the same attributes (or class variants) directly. The CSS can then keep its selectors while the JavaScript classifiers are deleted (Chapter 6 §6.7).
 8. **Replace text-replacement tricks with real markup** (accessibility). Render the cleaned identity and title text, the reward chip and the "Queued" label in the component, instead of `font-size: 0 + content: attr()` (A11Y-01).
 9. **Split or replace the sprite atlases** (PERF-07), and remove `exact-v3` from the package (SIZE-01).
 10. **Guard the cascade in CI.** Add a stylelint configuration with `declaration-no-important` (warning), `selector-max-specificity` (e.g. `0,6,1`), and `selector-max-compound-selectors`. Keep `tools/css-superseded.mjs` at zero as a gate, and keep `build-tools/audit-sprite-windows.mjs`.

3.14 REGENERATING THE CSS REFERENCES

```
node handoff/reskin-technical-handover/tools/css-inventory.mjs
```

```
node handoff/reskin-technical-handover/tools/css-superseded.mjs
```

```
node handoff/reskin-technical-handover/tools/hook-census.mjs
```

- The first writes Appendix B (per-file pages), Appendix C and `tools/css-inventory.json`. It fails on an unaccounted stylesheet or zero parsed rules.
- The second writes `tools/css-superseded.md` from the JSON.
- The third writes the attribute census used by Appendix D.

The scripts have no dependencies beyond Node.js and read `src/` only. When a stylesheet is added, update the `INJECTION` map in `css-inventory.mjs` and the `FILE_ORDER` list in `css-superseded.mjs`, or both scripts fail.

4. DEVELOPMENT CONSTRAINTS AND LIMITATIONS

The reskin was built as a **third-party browser extension, without access to the IdleWorlds source code, build, backend or art pipeline** [Source: CLAUDE.md, module headers]. Almost every unusual technique in the codebase traces back to one of the constraints below.

For each constraint this chapter records:

- the **evidence** for it;
- **how it shaped** the implementation;
- whether the resulting code is **deliberate design** (it would still be right with full access) or a **workaround** (it exists only because access was missing);
- how a developer **with full access** would replace it;
- whether and **how the constraint can be removed**.

4.1 SUMMARY

ID	Constraint	Classification	Removable with full access?	Replacement in one line
C1	No access to the game's source or build	Workaround driver	Yes	Implement the theme inside the IdleWorlds codebase
C2	React owns and continuously rewrites the DOM	Design, given C1	Yes (obsolete in-app)	Render the design in React components
C3	Appending during hydration breaks React (#418)	Workaround	Yes	Nothing to gate: components render in React's lifecycle
C4	The isolated content-script world cannot see page JavaScript	Workaround	Yes	Read state from app stores; use the app's own fetch client
C5	Almost no stable hooks; English copy is the only identifier	Workaround	Yes	Components expose roles and variants; i18n keys
C6	The game renders its panel stack twice (hidden duplicate column)	Workaround	Yes	Components know which layout is active
C7	Tailwind utilities and inline styles win the cascade	Workaround	Yes	Theme tokens, cascade layers, component variants
C8	Unknown CSS load order (late chunks)	Workaround	Yes	One controlled CSS pipeline
C9	One observer on a constantly mutating 15k-node page	Design, given C1/C2	Yes (obsolete in-app)	React renders; no observer
C10	Page CSP and third-party origins can blank assets	Design	Partly	Serve art from the game's own CDN and origin
C11	Everything must be reversible at runtime (kill switch)	Design	Replace	A feature flag or theme setting
C12	Presentation only: never change behaviour or destroy state	Design (keep)	Keep as a principle	Code review rule in the app
C13	Needed data is not in the DOM (item stats, village on dashboard)	Workaround	Yes	Props and stores already hold the data
C14	Data contracts(items.json , /api/player)are undocumented	Workaround	Yes	Typed internal APIs
C15	MV3 extension model: no UI, chrome.storage only	Design (minimal permissions)	Replace	Game settings UI
C16	Modern CSS/JS platform features, Chrome only	Design	Partly	Follow the game's browser support policy

ID	Constraint	Classification	Removable with full access?	Replacement in one line
C17	No access to the art pipeline; atlases generated or imported externally	Workaround	Yes	Game asset pipeline, per-item assets, CDN
C18	Committed, unminified bundle for auditability	Design	Replace	Normal app build with source maps
C19	Tests cannot use the real app	Workaround	Yes	Component, integration and visual tests in the app repo
C20	Performance budget on an idle game that ticks constantly	Design	Changes shape	Profiling inside React (memoisation, virtualisation)
C21	Responsive behaviour must fit the game's existing breakpoints	Workaround	Yes	Shared breakpoint tokens; container queries in components
C22	Accessibility is limited to what can be layered on	Workaround	Yes	Real semantics and markup in components
C23	Legal and ownership boundaries (game art, outbound link, authored copy)	Policy	Decision needed	Owner approval, or content from the game
C24	Tooling depends on Windows/Python/sibling repositories not in the repo	Workaround	Yes	Reproducible asset pipeline in CI

4.2 CONSTRAINTS IN DETAIL

C1. No access to the IdleWorlds source code or build

Evidence	CLAUDE.md : "React keeps ownership of every gameplay node ... this extension adds decoration, semantic role attributes and its own overlay elements. It never rewrites the game's DOM." The project ships as an MV3 extension (manifest.json).
Shaped	The entire architecture: classifiers that recognise surfaces from rendered DOM (<code>UIFoundation</code> , <code>DOMWatcher.detectSkillType</code> , <code>QuestPanelRenderer.isQuestCard</code> , ...); role attributes as the contract between JavaScript and CSS; a runtime style injector; a kill switch.
Classification	Workaround driver. This is the root constraint; C2–C9, C13, C14, C17, C19, C21 and C22 follow from it.
With full access	Implement the theme as part of the IdleWorlds front end: - design tokens in the theme configuration; - component variants for panels, cards and controls; - data already in props. The CSS in <code>src/styles</code> can be ported selector by selector, because it keys on semantic roles that components can render directly (Chapter 6 §6.7).
Removal	Follow the migration plan in §6.7. The extension then becomes unnecessary.

C2. React owns and continuously rewrites the DOM

Evidence	Rule 2 ("No reparenting"), rule 3 (reversible), the <code>docs/traps/performance.md</code> measurements, and module comments on node replacement (e.g. <code>TooltipEngine</code> "React can replace the trigger without a conventional mouseleave").
Shaped	Append-only decoration: overlays, medallions and toggles are appended; <code>display: contents</code> and grid placement are used instead of moving nodes. Idempotent renderers: signatures such as <code>data-fs-inv</code> and <code>structureSignature</code> , and compare-before-write helpers such as <code>setOwnText</code> and <code>setData</code> . Caches revalidated by <code>isConnected</code> , <code>contains</code> and the layout epoch. Skin-owned nodes excluded from classifier sweeps (<code>:not([data-iw-...])</code> , <code>closest()</code> guards). Game text never rewritten; clean labels come from CSS <code>content: attr()</code> (A11Y trade-off).
Classification	Design, given C1. In an extension these are exactly the right rules. They are unnecessary inside React.
With full access	Render the decorations as part of the components. React's reconciliation handles updates; no signatures or observers are needed.
Removal	Obsolete once components own the markup. Keep the rule-5 discipline (never repaint state uniformly) as a design-review rule.

C3. Appending during hydration breaks React

Evidence	CLAUDE.md and the HydrationGate.js header: appending skin nodes mid-hydration made React throw #418 and re-render the tree.
Shaped	- A second content script in the MAIN world (src/page/hydration-signal.js) reads React's private root fields and writes <code>html[data-iw-page-hydrated]</code>. - HydrationGate.waitForPageHydration polls for it, with a 4 s cap. - start() awaits the gate before the first boot.
Classification	Workaround. It reads undocumented React internals (__reactContainer\$... , memoizedState.isDehydrated).
With full access	Not needed: code inside the app runs after hydration by construction (useEffect , or rendering during hydration with matching server output).
Removal	- Delete src/page/hydration-signal.js , src/modules/HydrationGate.js and tests/hydration-gate.test.mjs . - Remove the second content_scripts entry from the manifest. - Remove the waitForPageHydration call in content.js . While still an extension, the safer alternative would be for the game to dispatch a documented, public event or attribute when hydration completes, replacing the private-field probe.

C4. The isolated content-script world cannot see page JavaScript

Evidence	- The hydration signal must run in the MAIN world (C3). - VillageScene replicates the x-idleworlds-league header because "the page monkey-patches fetch to add it; the isolated world does not see the patch" [Source: VillageScene.js comments L43-66; Project record: docs/traps/village.md].
Shaped	- The league is derived from location.pathname (a /ssf prefix → ssf), and the header is added manually. - No access to React state, so everything is read from DOM text (C5).
Classification	Workaround.
With full access	Read the league and the player's village from the app's state and API client, which already carries auth and league context.
Removal	Replace VillageScene 's fetch with a store selector. Delete leagueOf and sectionOf .

C5. Almost no stable hooks; English copy is the identifier

Evidence	CLAUDE.md : "The live app has almost no stable hooks (#current-action-panel , .panel , data-skin)". Classifier regexes throughout, listed in Chapter 6 §6.5.
Shaped	- Heading and label regexes: "Inventory", "Skill Actions", "World Bosses", "Village Add-ons", "Zone N:", "Turn In", "Reward:", "Message World Chat", ... - Verb tables for skills; leading emoji runs stripped before anchored regexes. - Element-boundary anchoring, because textContent joins elements without separators. Positive identification rules so shared .compact-panel cards are not misclassified. - Separate attribute namespaces per module.
Classification	Workaround. It is also the main maintenance risk: any copy change, localisation, or emoji change in the game can silently disable a surface (it falls back to native).
With full access	Components render explicit roles (data-role="skill-card" , data-skill="mining") or variant props. Localisation uses message keys, not rendered strings.
Removal	Add stable data-* hooks to the game components first (even before a full migration). Then switch each classifier to the hook, keeping the text fallback until the hook ships everywhere. This alone removes most fragility (§6.7 phase 1).

C6. The panel stack is rendered twice

Evidence	Viewport.js header: a hidden xl:grid copy and an xl:hidden copy, swapping visibility at 1280 px; a breakpoint crossing mutates nothing.
----------	---

Shaped	<code>isRendered</code> / <code>pickRendered</code> / <code>preferRendered</code> (<code>checkVisibility</code>), the <code>matchMedia</code> layout epoch, and every classifier cache keyed on the epoch. Inventory root resolution checks that the root contains this row. <code>QuestPanelRenderer.measureCommandBlock</code> skips 0-width measurements.
Classification	Workaround.
With full access	Render one responsive layout, or let the components know which layout is active, so the theme never has to choose.
Removal	Delete <code>Viewport.js</code> and the epoch checks once classifiers are gone.

C7. Tailwind utilities and inline styles win the cascade

Evidence	3,879 <code>!important</code> declarations (66%); substring selectors on utility classes; inline <code>!important</code> from renderers (Chapter 3 §3.2).
Shaped	<code>!important</code> as the default. - Long <code>:not()</code> opt-out chains, (0,30,3) specificity. - The <code>var(--token, fallback)</code> inline pattern. - Radius and colour flattening through <code>[class*="rounded-..."]</code>. - Overriding the game's own design tokens on <code>:root</code>.
Classification	Workaround.
With full access	- Put the palette into the game's theme tokens. - Use cascade layers or the Tailwind theme so component styles win without <code>!important</code>. - Stop components from writing inline colours that the theme must override.
Removal	Chapter 3 §3.13, steps 1–7.

C8. Unknown CSS load order

Evidence	The skin's <code><style></code> elements are appended at <code>document_idle</code> . Whether the game inserts CSS later (client-side navigation chunks) is not observable from the repository [Assumption].
Shaped	<code>!important</code> on declarations that must win regardless of order. <code>ui-system</code> is injected last so that ties among the skin's own sheets resolve predictably.
Classification	Workaround.
With full access	A single CSS pipeline with a defined order (or layers).
Removal	Part of C7.

C9. One observer on a constantly mutating page

Evidence	Rule 4 (one <code>MutationObserver</code>); <code>DOMWatcher</code> flush budget of 60; <code>docs/traps/performance.md</code> (for example: <code>annotateStructure</code> took 72% of cost before the structure signature; ~10 s boot from per-row style reads; self-driving flush loops); <code>tests/flush-quiescence.test.mjs</code> .
Shaped	- A central queue with round-robin budgeting. - Event fan-out to renderers. - An attribute filter limited to eight attributes. - Compare-before-write everywhere. - Batching reads before writes. - No shorthand plus longhand in inline maps. - Caches with coverage checks (the <code>[]</code>.<code>every()</code> trap).
Classification	Design, given C1/C2. It is excellent engineering for an extension, and unnecessary in-app.
With full access	Components re-render only when their props change. There is no observer.
Removal	Obsolete in-app. While the extension exists, keep all these disciplines: they are pinned by tests.

C10. Page CSP and third-party origins

Evidence	<code>vendor-assets.mjs</code> header: a previous build loaded atlases from <code>raw.githubusercontent.com</code> and fonts from Google, so a CSP change on the game "could blank every icon".
Shaped	- All art and fonts are vendored into <code>assets/</code>, pinned to a sprite repository commit, and served from <code>chrome-extension://</code> through <code>web_accessible_resources</code>. <code>StyleInjector</code> rewrites URLs. - The build refuses to run without the required assets (unless <code>--allow-missing-assets</code>).
Classification	Deliberate design for an extension: no runtime third-party dependencies.
With full access	Serve assets from the game's own origin or CDN under the game's CSP, with cache headers and per-asset URLs.
Removal	Move <code>assets/</code> into the game's static pipeline. Delete <code>vendor-assets.mjs</code> and the URL rewrite. Keep "no third-party runtime origins" as policy.

C11. Everything must be reversible at runtime

Evidence	Rule 3; README "Kill switch"; <code>StyleInjector</code> , <code>InlineStyleOwner</code> , every <code>clear*()</code> ; smoke-test round trips.
Shaped	- CSS injected as removable elements. - Inline styles owned property by property, with native values restored. - Each module removes exactly what it added (own namespaces). - Page-lifetime listeners inert while disabled. - The storage listener kept alive.
Classification	Deliberate design . Reversibility is also how the skin guarantees it can never strand a player with broken UI.
With full access	A theme setting or feature flag chooses between the stock and fantasy themes at render time. There is no runtime teardown.
Removal	Replace with a flag. Keep the principle for any remaining runtime enhancement.

C12. Presentation only; never destroy state information

Evidence	Rules 1 and 5; examples throughout (active tab, equipped item, met/unmet, disabled, team colour, player-name colour left to the game).
Shaped	- Restyling native controls instead of replacing them. - Mirrors are <code>aria-hidden</code> while the native label stays the accessible name. - State is read from the game's classes and attributes. - Colour rules avoid changing text colour on controls whose colour is state.
Classification	Deliberate design (keep) .
Known exceptions	FUN-01 (a game control hidden); FUN-03 (game animations frozen for reduced-motion users); CONT-01/CONT-02 (skin-authored copy and relabelled "Prejoined" → "Queued").
With full access	The same principle becomes a UI review checklist item.
Removal	Not applicable: retain.

C13. Needed data is not in the DOM

Evidence	<code>ItemDatabase</code> exists because stats are not rendered. <code>VillageScene</code> exists because the village "simply is not in the dashboard's DOM" [Source: <code>CLAUDE.md</code>]. <code>VillageLedger</code> joins building items from <code>items.json</code>. - World Boss rewards are derived from item acquisition text plus a bundled fallback.
Shaped	- Network reads (<code>items.json</code>, <code>/api/player</code>), with a storage cache and a refresh chain. - Name normalisation and alias tables to join DOM text to records. - Fallback data (<code>rewards.json</code>, inline records) and skin-authored explanatory copy.
Classification	Workaround .

With full access	Components already hold item and player objects. Rewards, benefits and acquisition come from the same server data the game uses.
Removal	Delete <code>ItemDatabase</code> 's fetch and cache, <code>aliases.js</code> , name joins, and <code>VillageScene</code> 's API read. The tooltip renders from props.

C14. Undocumented data contracts

Evidence	<code>build-tools/audit-items-contract.mjs</code> (manual, not in CI); <code>normaliseVillage</code> strict validation; D-07 size ambiguity; the <code>items.json</code> field names assumed in <code>itemDisplay.js</code> / <code>TooltipEngine.js</code> .
Shaped	- Defensive parsing (e.g. tier and slots must be integers 0-5; duplicate slots rejected). - Graceful degradation when data is missing. - Validation of CSV columns and bounds for atlases. - Conditional requests (ETag/Last-Modified). But: item fields are trusted per field (SEC-01, SEC-02).
Classification	Workaround.
With full access	Typed API clients shared with the game. Contract tests run in the game's CI.
Removal	Replace with typed selectors. Until then, add field-type validation (Chapter 5 hardening H-01/H-02).

C15. MV3 extension model with minimal permissions

Evidence	Manifest: <code>permissions: ["storage"]</code> , two host permissions, no service worker, popup, options or icons.
Shaped	- The kill switch is only reachable through DevTools (F-02). - Preferences (collapse, ledger, boss rewards) live in <code>chrome.storage.local</code>. - No background fetching or messaging. - No extension UI.
Classification	Deliberate design: the smallest reasonable permission surface.
With full access	A settings toggle in the game ("Theme: Classic / Fantasy") and per-account preferences stored server-side or in the app's local storage.
Removal	Replace storage keys with app settings. Migrate <code>iw-collapsed-frames</code> , <code>iw-village-ledger</code> and <code>iw-boss-rewards-collapsed:*</code> only if continuity matters.

C16. Modern platform features, Chrome only

Evidence	Chapter 1§1.10: <code>color-mix</code> (111), <code>world: "MAIN"</code> (111), <code>:has</code> , container queries, CSS Highlights, <code>checkVisibility</code> ; <code>chrome.*</code> APIs.
Shaped	Terse CSS (derived colour scales, <code>:has()</code> keyed composition, container-query layouts), with no fallback styles for older engines.
Classification	Deliberate design for a Chrome extension audience. Gap: the manifest does not declare <code>minimum_chrome_version</code> , and the build target says 109 (PKG-01).
With full access	Follow the game's supported-browser matrix. Most features used are Baseline in current Chrome, Edge, Firefox and Safari [Assumption: verify against current compatibility data]. <code>::highlight()</code> and <code>caretPositionFromPoint</code> are the ones to check.
Removal	Not a removal, but align targets: set <code>minimum_chrome_version: "111"</code> and the esbuild target to <code>chrome111</code> (non-behavioural).

C17. No access to the art pipeline

Evidence	§1.8 provenance: - atlases vendored from a personal sprite repository; - village and zone art imported from sibling repositories; - many families AI-generated; - generator inputs not tracked (<code>sources/</code>, <code>themes-v2/</code>, <code>source-v4/</code>); - generated CSS (<code>card-button-atlas.css</code>, <code>COMPACT_GEOMETRY</code>); - the <code>audit-sprite-windows.mjs</code> gate.
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Shaped	- Giant sprite atlases with percentage windows. - Three-slice bands to keep frame ends undistorted. - Exact-alpha registration for state art. - A 142.8 MB release with 60.8 MB unreachable (SIZE-01). 95.6 MiB decoded gear atlas (PERF-07).
Classification	Workaround.
With full access	Use the game's own asset pipeline: per-item icons (or smaller atlases) from the CDN, SVG/CSS for frames where possible, and art generated and reviewed under the game's content policy.
Removal	- Drop <code>exact-v3</code> from packaging (\$6.8). - Replace the atlases with game assets. - Retire the import scripts and generated CSS.

C18. Committed, unminified bundle

Evidence	CLAUDE.md ("dist/content.bundle.js is committed, and CI fails if it is stale"; "deliberately un-minified for readability"); the build's soft budget warning.
Shaped	Every <code>src/</code> change requires a rebuild and a commit of <code>dist/</code> . Reviewers can audit exactly what ships. The bundle is 668 KB (48.6% CSS).
Classification	Deliberate design for auditability of an extension (store reviewers and users can read the code).
With full access	A normal app build: minified, code-split, with source maps.
Removal	Obsolete in-app. For the extension, keep it; optionally add a minified release build alongside the readable one.

C19. Tests cannot use the real app

Evidence	CLAUDE.md "Verification discipline": fixtures model the DOM; "No session can verify live; only the user can"; DevTools capture snippets in <code>cclaude/</code> .
Shaped	47 standalone suites: jsdom for logic, Playwright for layout and cascade. - Synthetic DOM shapes copied from live captures. - Negative controls ("a check reporting zero is broken"). - A mobile audit harness and a perf harness against synthetic pages.
Classification	Workaround. It is also a disciplined test strategy given the constraint.
With full access	Component tests against real components; visual regression (Playwright screenshots) on the real app in CI; contract tests for APIs.
Removal	Port the assertions that describe behaviour (flush quiescence becomes render counts; teardown becomes flag switching; rule-5 state checks). Drop fixture-specific DOM modelling.

C20. Performance budget on an always-ticking idle game

Evidence	Perf harness on a "ticking 15k-node page"; measured costs in <code>docs/traps/performance.md</code> [Project record]; PERF-01...07.
Shaped	Flush and DOM work: flush budget; caches; no layout reads in loops; rAF coalescing; one sweep per flush; the rendered-host check deferred behind a containment test (V2 timer). Progress smoothing: CSS transitions instead of JavaScript animation. Open cost: infinite decorative loops remain (PERF-01).
Classification	Design.
With full access	The budget stays, but the tools change: React profiling, memoisation, list virtualisation (inventory), and pausing animations when panels are hidden.
Removal	Not removable; re-baseline in-app.

C21. Responsive behaviour within the game's layout

Evidence	V1.6.0_MOBILE_LAYOUT_AUDIT.md ; docs/traps/mobile.md ; Tailwind-aligned breakpoints plus component breakpoints; cqi font fitting with measured constants (k = 28.1 , 23.25 , 22.5 , 20.9).
Shaped	Container queries where panel width is independent of viewport width. Measured font-fitting constants for single-row menus. The Toolkit link hidden below 768 px.
Classification	Workaround (the game's grid could not be changed).
With full access	Adjust the game's grid and menus directly, share breakpoint tokens between theme and layout, and avoid measured magic constants.
Removal	Replace cqi formulas with layout changes (e.g. overflow menus).

C22. Accessibility can only be layered on

Evidence	font-size: 0 + content: attr() label replacement; clip-hidden native sections plus visible mirrors; aria-hidden mirrors; focusable <p> triggers (A11Y-01/02/03).
Shaped	Native semantics were preserved wherever possible: controls, labels, ARIA state untouched. Where visual copy had to differ, mirrors or generated content were used, with trade-offs.
Classification	Workaround.
With full access	Render the cleaned text as real text, use proper roles (button for triggers), and use a single source for each piece of content.
Removal	Part of the component migration (§6.7 phase 3).

C23. Legal and ownership boundaries

Evidence	- Game art re-hosted (§1.8). - An outbound Toolkit link in the game's navigation (SEC-09). - Skin-authored gameplay statements (CONT-01) and invented terminology (CONT-02). - Fonts without bundled licence text. - AI-generated art.
Shaped	The content choices listed above.
Classification	Policy constraint. It needs owner decisions, not code.
With full access	IdleWorlds decides which art, copy, links and terminology are official.
Removal	Decision checklist in Chapter 6 §6.9.

C24. Tooling depends on local environments and sibling repositories

Evidence	- Windows PowerShell scripts (make-exact-button-states.ps1 , make-source-preview-button-states.ps1). - Python builders (build-compact-buttons.py , build-revised-buttons.py). - Import scripts that default to ../idleWorlds-game-sprites-BC and ../idleWorlds-art-source . - A real browser profile for the live header audit (build-tools/header-live-profile/ , SEC-07).
Shaped	Art cannot be rebuilt from a clean clone. Live audits depend on a logged-in local profile.
Classification	Workaround.
With full access	A reproducible asset pipeline in CI; test accounts for live audits instead of personal profiles.
Removal	- Vendor the source art into an asset repository. - Port the PowerShell to Node. - Document the Python dependencies. - Delete personal profiles from working copies.

4.3 WHICH WORKAROUNDS ARE SAFE TO REMOVE TODAY (STILL AS AN EXTENSION)

These need no IdleWorlds source access:

Item	Change	Risk	Verify with
Unreachable exact-v3 art (SIZE-01)	Exclude <code>^assets/skills-ui/buttons/exact-v3/</code> in <code>package-release.mjs</code> . Optionally delete the <code>else</code> branch in <code>SkillsArtService.applyThemeVariables</code> , or keep it as a guarded fallback	None while all 9 themes are in <code>revised-v5/index.json</code>	<code>npm test (zone-themes, revised-button-atlas)</code> , <code>npm run package</code> , a load-unpacked smoke check
Metadata JSON in the zip	Add <code>registration.json</code> , <code>verification.json</code> , <code>index.json</code> (non-runtime ones), <code>zone-theme-map.json</code> and <code>buildings.json</code> to <code>EXCLUDE</code> . Keep <code>skills_*_index.json</code> , <code>gear_icons_manifest.json</code> and <code>item_icons_index.csv</code> (fetched at runtime)	Excluding a runtime-fetched index breaks icons	<code>extension-e2e.test.mjs</code>
Dead CSS	§3.13 step 1	None (proven dead)	<code>npm test</code> , fixtures
Browser targets	Add <code>minimum_chrome_version: "111"</code> ; esbuild <code>chrome111</code>	None	<code>npm run build</code> (the bundle may change slightly; commit dist)
Font licence	Add <code>assets/fonts/OFL.txt</code> and credits	None	—
Hydration probe (C3)	Keep until the game provides a public signal	—	—

5. JAVASCRIPT DOCUMENTATION AND SAFETY REVIEW

This chapter documents every JavaScript feature of the skin and reviews its safety **with evidence**. Every "the code does not do X" statement is backed in one of two ways:

- by the **lexical census** in `tools/js-api-census.md`, which lists every call site of every security-relevant API, and "none" where an API never appears;
- by a **cited line** that shows the guard.

Where a risk or an uncertainty exists, it is disclosed as a finding with a proportionate mitigation (§5.6).

Scope. All shipped JavaScript:

- `src/content.js`;
- `src/page/hydration-signal.js`;
- the 33 files in `src/modules/`.

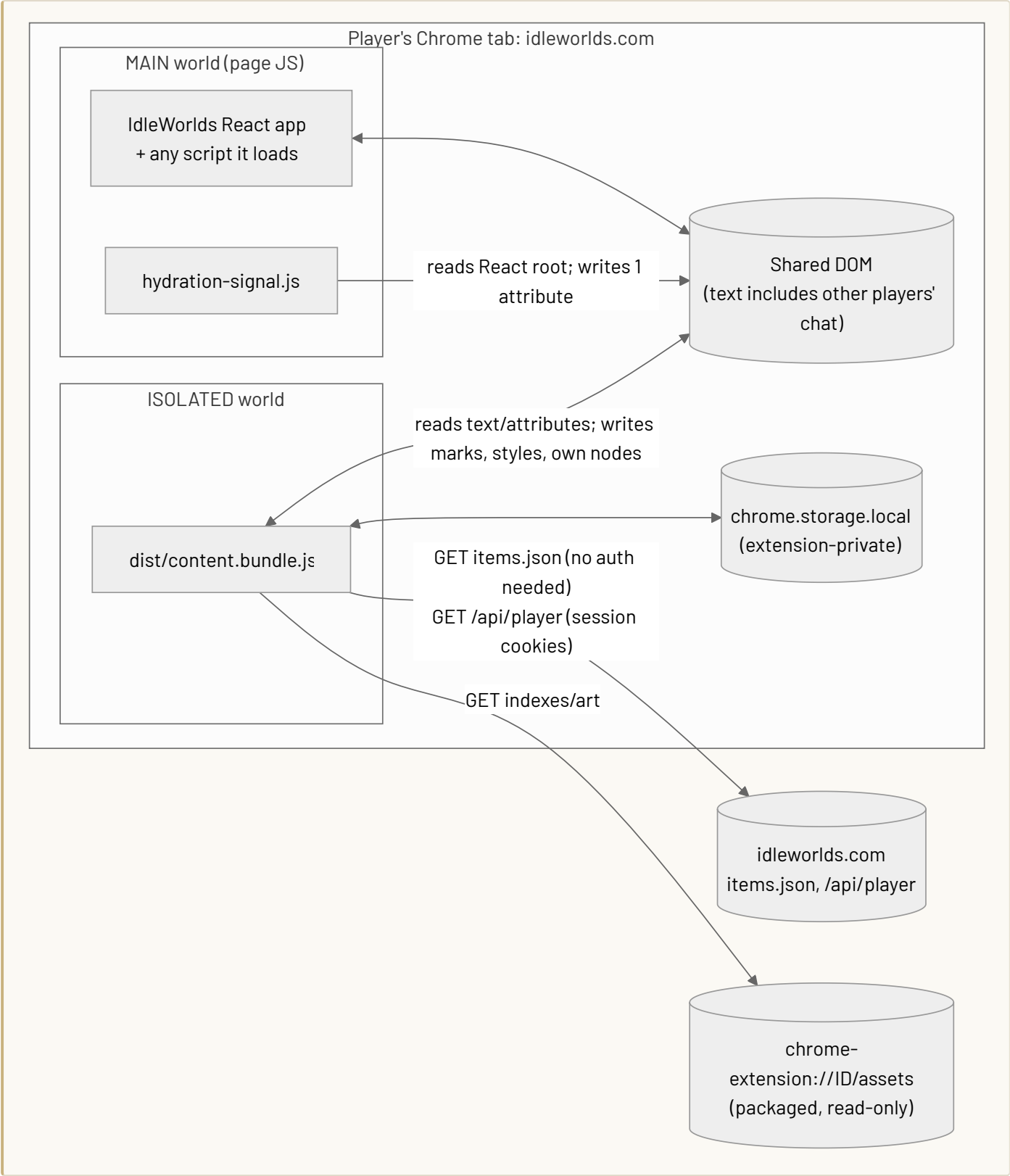
Together these are 35 files and 11,869 lines, bundled into `dist/content.bundle.js`, except the page script. Build tools, tests and `cLaude/` probes do not ship. They are reviewed only where they touch secrets or live accounts (SEC-07).

5.1 THREAT MODEL AND METHOD

5.1.1 What must be protected

Asset	Why it matters
The player's authenticated session (cookies, league context)	Account takeover or unwanted actions
Game integrity (actions, purchases, trades, chat)	The skin must never act for the player (rule 1)
The game page's integrity	Script injection into <code>idleworlds.com</code> would be an XSS in the game's origin
Player privacy	No tracking, no exfiltration of player data
Game state legibility	Wrong or hidden state misleads the player (rule 5)
Performance and availability	A skin that stalls the tab degrades the game
The extension's own integrity	Update and supply chain; packaging hygiene

5.1.2 Adversaries and trust boundaries



Adversary or input	Can it reach the skin?	What the skin does with it
Other players (chat messages, player names, market listings)	Yes: the text is in the DOM	Read as text only : <code>textContent</code> , <code>nodeValue</code> , <code>Range</code> . Never parsed as HTML (§5.2.4). Chat text is only indexed for item names (<code>NameScanner</code>)
The game's own scripts (or anything the page loads)	Yes: shared DOM and <code>document</code> events	They could forge <code>iw:*</code> events or skin attributes. Effect: extra reconciliation work, or a tooltip for a <i>real</i> catalogue item (§5.3.1). They cannot read extension storage or call extension APIs

Adversary or input	Can it reach the skin?	What the skin does with it
The game server (<code>items.json</code> , <code>/api/player</code>)	Yes	Parsed as JSON. Item fields reach one HTML string builder where two fields are unescaped (SEC-01). Village data is strictly validated
Network attacker	No, while HTTPS holds	All runtime requests are <code>https://</code> same-site, or <code>chrome-extension://</code>
Other extensions	Shared DOM only	Same capabilities as page scripts
Supply chain	Build time only	No runtime dependencies; dev dependencies are locked (<code>package-lock.json</code>); the bundle is committed and unminified for review (C18)
The developer's own workstation	Yes (working copy)	A real browser profile with saved credentials sits in the tree (SEC-07)

5.1.3 Method

1. **Full read** of every shipped JavaScript file, with notes per function.
2. **Lexical API census** (`tools/js-api-census.mjs`): network, storage, extension APIs, messaging, HTML sinks, dynamic code, navigation, synthetic interaction, observers and timers, listeners, globals, diagnostics, layout reads. The tool fails if it finds zero `fetch` , `MutationObserver` or `addEventListener` , because a census reporting nothing is broken.
3. **Data-flow tracing** from each external input (DOM text, `items.json` , `/api/player` , storage, events) to each sink (`innerHTML` , attributes, URLs, CSS custom properties, storage, network).
4. **Test execution**: `npm test` passed on 2026-09-17 (exit 0; 3 min 26 s; [evidence log](#)) [**Test run**].
5. **Classification** of every issue as Confirmed (C), Plausible, needs a test (P), or Verify live/with owner (V), with severity High, Medium, Low or Info.

Limits.

- This is a source review, not a penetration test. No live page, server or real account was exercised.
- Chrome-behaviour statements marked [**Assumption**] come from general platform knowledge, not from this repository: CSP handling of isolated worlds, cross-world `CustomEvent.detail` , storage quotas.

5.2 GLOBAL SECURITY POSTURE

5.2.1 Manifest surface

Item	Value	Assessment
<code>permissions</code>	<code>["storage"]</code>	Minimal. No <code>tabs</code> , <code>cookies</code> , <code>webRequest</code> , <code>scripting</code> , <code>activeTab</code> , <code><all_urls></code>
<code>host_permissions</code>	<code>https://idleworlds.com/*</code> , <code>https://www.idleworlds.com/*</code>	Scoped to the game
Content scripts	Bundle (ISOLATED, <code>document_idle</code> , top frame); hydration signal (MAIN, <code>document_start</code> , top frame)	<code>all_frames: false</code> : no injection into iframes
Background / service worker	none	No persistent process, no cross-tab messaging
Popup / options / devtools pages	none	No extension UI surface
<code>externally_connectable</code>	absent	Web pages cannot message the extension
<code>content_security_policy</code>	absent, so the MV3 default applies	No remote code possible for extension pages (none exist)
<code>web_accessible_resources</code>	<code>assets/*</code> (+ 4 PNG globs), matches both game hosts	Game pages can load any packaged asset, and can therefore detect the extension (SEC-05)

Item	Value	Assessment
minimum_chrome_version	absent	Effective minimum is 111 (PKG-01)
update_url, key	absent	Distributed unpacked or through the store [Verify distribution channel]

5.2.2 Network destinations (complete)

Source: `fetch()` call sites in `tools/js-api-census.md`. There are **no** `XMLHttpRequest`, `WebSocket`, `EventSource`, `sendBeacon`, `<script>` creation, or `window.open` call sites.

#	Call site	URL	Method, credentials, cache	Data sent	Response use
N1	ItemDatabase.js:165	<code>https://idleworlds.com/items.json</code> (constant, L13)	GET; credentials: default <code>same-origin</code> ; cache: <code>'no-cache'</code> (conditional revalidation)	Nothing beyond the URL and standard headers. Cookies are sent on the apex host (same origin); on <code>www</code> the request is cross-origin, so no cookies are sent and CORS applies (SEC-03)	JSON → <code>items</code> array → in-memory Maps + <code>chrome.storage.local</code> cache
N2	VillageScene.js:312	<code>/api/player?section=<route>&scope=core</code> (relative, so the current origin)	GET; credentials: <code>'same-origin'</code> ; cache: <code>'no-store'</code> ; <code>AbortController</code> , 10 s timeout	Session cookies ; header <code>x-idleworlds-league</code> : <code>standard ssf</code> ; <code>section</code> from a fixed whitelist (<code>dashboard</code> , <code>housing</code> , <code>market</code> , <code>leaderboards</code> , <code>dungeon</code>) via <code>encodeURIComponent</code>	<code>normaliseVillage()</code> keeps <code>housing.tier</code> , <code>villageAddons.totalSlots</code> , <code>installed[]</code> .{ <code>slot</code> , <code>itemKey</code> , <code>name</code> }; everything else is dropped with the response object (SEC-04)
N3	AtlasService.js:256	<code>chrome-extension://<id>/assets/gear_icons_manifest.json</code>	GET, <code>no-store</code>	—	Validated JSON (icons array, numeric geometry)
N4	AtlasService.js:286	<code>chrome-extension://<id>/assets/item_icons_index.csv</code>	GET, <code>no-store</code>	—	CSV parsed; required columns and finite bounds validated
N5	SkillsArtService.js:93	<code>chrome-extension://<id>/assets/skills_icons_index.json</code> , <code>skills_ui_index.json</code>	GET, <code>no-store</code>	—	<code>validateIndex</code>
—	<code>CSS url()</code> , <code></code>	<code>chrome-extension://<id>/assets/...</code> only (after <code>StyleInjector</code> rewrite, or via <code>assetUrl</code> with constant paths)	Browser-initiated	—	Images and fonts
—	Links	<code>https://idleworlds.com/wiki/items/<slug></code> (tooltip), <code>https://idleworldstorage.com</code> (nav)	Navigation only on click , <code>target="_blank"</code> , <code>rel="noopener noreferrer"</code>	Referrer suppressed	—

No request goes to any third-party origin at runtime. No data read from the page, storage or the API is ever placed in a request URL, header or body. The only request inputs are the fixed `section` whitelist and the league derived from `location.pathname` [Source: N2 construction].

5.2.3 Storage (complete)

Key	Area	Writer (file:line)	Value	Contains personal data?	Retention
<code>iw-skin-enabled</code>	<code>chrome.storage.local</code>	Developer via DevTools (no code writes it)	<code>true / false</code>	No	Until changed
<code>iw-item-db-cache</code>	<code>chrome.storage.local</code>	<code>ItemDatabase._writeCache</code> (L252-257)	<code>{ items, generatedAt, validator, cachedAt }</code> : the public item catalogue (≈5.5 MB per the source comment [Verify])	No (public game data)	Overwritten on change; never deleted
<code>iw-item-db-cache-checked</code>	<code>chrome.storage.local</code>	<code>ItemDatabase._markCacheChecked</code> (L259-262)	<code>{ generatedAt, checkedAt }</code>	No	Overwritten
<code>iw-collapsed-frames</code>	<code>chrome.storage.local</code>	<code>CollapsibleFrames.persist</code> (L174-178)	<code>{ "<panel:slug> <title:lowercase heading>": true }</code>	No. Keys are panel or heading names	Until changed
<code>iw-village-ledger</code>	<code>chrome.storage.local</code>	<code>VillageLedger.persist</code> (L119-123)	<code>{ "<entry key>": false }</code> for closed entries	No	Until changed
<code>iw-boss-rewards-collapsed</code> <code>:<bossKey></code>	<code>chrome.storage.local</code>	<code>WorldBossPanels.bindRewardDisclosure</code> (read L25, write L35)	<code>true / false</code>	No	Until changed
<code>iw-item-db-cache (legacy)</code>	page <code>localStorage</code>	<code>ItemDatabase.evictLegacyCache</code> (L38-49): <code>getItem</code> then <code>removeItem</code> only	Removes a ≤v1.5.x cache	No	Deleted once per page session if present

- No other page storage is touched: `sessionStorage`, `indexedDB` and `document.cookie` are **none** in the census.
- `chrome.storage.local` is readable only by the extension. Content scripts cannot read other extensions' storage, and pages cannot read it [\[Platform\]](#).
- Nothing written is secret. The only player-specific data the skin *holds* is the village snapshot (in memory) and panel preferences (storage).

Note. The storage cache is roughly 5.5 MB. The default `chrome.storage.local` quota is 10 MB in current Chrome [\[Assumption: quota raised from 5 MB in Chrome 114\]](#). On older Chrome, or if the catalogue grows past the quota, the write fails. `storageSet` catches the failure and warns, and the skin still works from the network copy.

5.2.4 HTML sinks and dynamic code

Dynamic code: none. The census finds no `eval`, `new Function`, string-argument timers, dynamic `import()`, `<script>` creation, `DOMParser`, `createContextualFragment`, `document.write`, `insertAdjacentHTML` or `outerHTML` writes. The only code that runs is the packaged bundle and the page script. **Remote code: none.**

innerHTML writes: exactly three [\[Generated: census\]](#).

Sink	What is interpolated	Escaping	Verdict
<code>SkillPanelRenderer.js:1421</code>	Nothing: constant <code>''</code>	n/a	Safe (constant)
<code>InventorYRenderer.js:699</code>	See the list below this table	<code>esc()</code> (L83-87) escapes <code>&</code> <code><</code> <code>></code> , not <code>'</code> . Every attribute in the template is double-quoted, so <code>'</code> cannot break out	Safe. Every non-constant value is escaped or numeric

Sink	What is interpolated	Escaping	Verdict
TooltipEngine.js:353	Item record fields from <code>items.json</code> , the cache, or the bundled fallback: <code>name</code> , <code>category</code> , <code>subcategory</code> , <code>req_text</code> , <code>req_level</code> , <code>req_skill</code> , <code>effects_raw</code> , stat labels and values, <code>acquisition_*</code> , <code>source_zone</code> , <code>drop_rate</code> , <code>drop_boosted_by</code> , <code>wiki_slug</code> , <code>craft_level</code> , <code>tier</code> ; internal constants	<code>esc()</code> (L49-53) escapes <code>&</code> & <code><</code> & <code>></code> <code>"</code> <code>'</code> for all fields except <code>craft_level</code> (L101) and <code>tier</code> in the recipe line (L102)	SEC-01. Two fields are unescaped

Values interpolated into the inventory overlay:

- **Item name:** from DOM text or the catalogue, via `itemRef()`, which escapes the name, id and label (TooltipEngine L660-680).
- **Level and enhancement:** from DOM text.
- **Stat chips:** from `itemDisplay.statChips`.
- **Details and requirements:** DOM text via `InventoryModel`.
- **Quantity:** DOM text.
- **detail.count:** a number produced by `InventoryModel` aggregation (L236-240).
- **Class names:** constants from `categoryClass`, `detail.kind` (escaped) and `c.kind` (compared to constants).

SEC-01 in detail.

```
// TooltipEngine.js:99-102
main = 'Crafted &middot; ' + esc(skill);
const bits = [];
if (item.craft_level !== null && item.craft_level !== '') bits.push(esc(skill) + ' Lv ' + item.craft_level);
if (item.tier !== null && item.tier !== '') bits.push('Tier ' + item.tier + ' recipe');
```

- **Precondition.** A record with `acquisition_type === 'Crafted'` whose `craft_level` or `tier` is a string containing markup. The record would have to come from the `items.json` response or the extension's own storage cache. The bundled `rewards.json` contains no such values [Source].
- **Impact.** The markup is parsed into the page's DOM inside `#iw-tip` when a player opens that item's card:
 - an element such as `<img src=x onerror=...>` would run its handler in the **page's** JavaScript world, under the page's CSP;
 - that is script execution in `idleworlds.com`'s origin, with the player's session.
- **Likelihood.** Low. The data source is the game's own server; an attacker able to change `items.json` already controls content served by that origin. The finding matters for defence in depth, for a compromised CDN or cache, and because the same pattern could be copied elsewhere.
- **Mitigation.**
 - **H-01**(no visual change): `esc(item.craft_level)` and `esc(item.tier)`.
 - **H-02:** type-validate records.
 - **H-03:** build the card with DOM APIs.

5.2.5 Listeners, observers and timers (complete)

Resource	Registered at	Target and options	Lifetime	Active while disabled?
MutationObserver (the only one)	DOMWatcher.startWatcher (L358)	<code>document.body</code> : <code>childList</code> , <code>subtree</code> , <code>characterData</code> , <code>attributes</code> with an 8-name filter	Boot → teardown (stopWatcher L459, disconnect L461)	No
MediaQueryList change ×5	Viewport.startLayoutWatch (L85-101)	(min-width: 640/768/1024/1280/1536px)	Boot → teardown (removeEventListener, L104)	No

Resource	Registered at	Target and options	Lifetime	Active while disabled?
chrome.storage. onChanged	Runtime.onStorageChanged (L108-122) via content.js L216	iw-skin-enabled	Page lifetime (by design: the re-enable path)	Yes, required
document iw:dom-flush ×5	content.js:97; UIFoundation.js:1457; HeaderRenderer.js:463; TooltipEngine.js:583; SkillCardDesignController.js:523	—	Page lifetime	Registered; work is gated by guard (inactive) or finds nothing
document iw:name-scan- flush ×2	content.js:109; SkillCardDesignController.js:523	—	Page lifetime	Same
document iw:skill-panel ×4	SkillPanelRenderer.js:1576; QuestPanelRenderer.js:465; UIFoundation.js:1460; SkillCardDesignController.js:520	—	Page lifetime	Same
document iw:inventory- row	InventoryRenderer.js:757	—	Page lifetime	Same
document iw:item-db- updated ×4	content.js:114; InventoryRenderer.js:758; TooltipEngine.js:607; VillageScene.js:251	—	Page lifetime	Same (VillageScene checks isRuntimeActive)
document iw:atlas- updated ×2	InventoryRenderer.js:759; TooltipEngine.js:606	—	Page lifetime	Same
document mouseover / mous eout / mousemove (capture)	TooltipEngine.js:467, 481, 501	—	Page lifetime	Registered; with no triggers, return immediately
document click, keydown	TooltipEngine.js:513, 537	—	Page lifetime	Same
window scroll (capture), resize, blur	TooltipEngine.js:577-579	—	Page lifetime	Same
#iw-tip mouseenter / mou seleave / focusou t / click	TooltipEngine.js:434-459	Skin element	Page lifetime	Element hidden
document mousemove (passive, capture), click	NameScanner.js:357, 368	—	Page lifetime, bound on first scan	rAF callback gated by guard
window scroll (passive, capture)	NameScanner.js:380	—	Page lifetime	Trivial
click on each collapse toggle	CollapsibleFrames.js:217	Skin <button>	Until the button is removed	Buttons removed
click on each ledger toggle	VillageLedger.js:189	Skin <button>	Until the node is removed	Nodes removed
toggle on each rewards <details>	WorldBossPanels.js:31	Skin element	Until the node is removed	Nodes removed
MediaQueryList change (reduced motion)	WorldBossPanels.js:147-148	During an impact animation only	Removed on finish or cancel	No

Resource	Registered at	Target and options	Lifetime	Active while disabled?
<code>setTimeout</code> chains	<code>hydration-signal.js:58</code> (50 ms, ≤ 15 s); <code>HydrationGate.js:47</code> (50 ms, ≤ 4 s); <code>ItemDatabase.js:114</code> (6 h / 15 min); <code>TooltipEngine.js:402</code> (120 ms show delay); <code>TooltipEngine.js:585</code> (rAF fallback); <code>VillageScene.js:310</code> (10 s abort); <code>Runtime.js:177</code> (rAF fallback)	—	Cleared: <code>stopAutoRefresh</code> (refresh), <code>clearShowTimer</code> , <code>clearTimeout</code> in <code>finally</code> (village)	Refresh timer cleared on teardown
<code>requestAnimationFrame</code> (one-shot)	12 sites (census)	—	One frame	Callbacks guarded
<code>queueMicrotask</code>	<code>AtlasService.js:88</code> ; <code>InventoryRenderer.js:319</code>	—	One microtask	—
<code>setInterval</code>	none			
<code>IntersectionObserver</code> / <code>ResizeObserver</code>	none			
Web Animations	<code>WorldBossPanels.js:137</code>	Skin elements	≤ 1.25 s; cancelled on stop	—

5.2.6 Globals and page-world exposure

- **No globals.** The bundle is an esbuild IIFE (`format: 'iife'`, `build.mjs:99`). The census finds **no** `window.*`, `globalThis.*` or `self.*` property writes. In any case, content-script globals live in the isolated world and are invisible to the page.
- **Page script scope.** `hydration-signal.js` is wrapped in an IIFE (L24–64). It defines no globals, adds no listeners, and makes one attribute write (L38).
- **Page-visible artefacts.** Everything the skin changes in the shared DOM is visible to the page, and is non-sensitive:
 - `data-iw-*` and `data-fs-*` attributes;
 - four classes on game nodes;
 - inline custom properties, and `chrome-extension://<id>` URLs in custom properties and `<img>`;
 - the seven `<style>` elements;
 - skin-owned elements;
 - `CustomEvents` on `document`.

The extension ID in those URLs identifies the installed extension.

5.2.7 Authentication state and sensitive information

Question	Answer	Evidence
Does the skin read cookies, tokens or credentials?	No	Census: no <code>document.cookie</code> , no <code>chrome.cookies</code> , no reads of storage beyond its own keys and the legacy key check
Does it send authenticated requests?	Yes, one (N2, <code>/api/player</code>), plus N1 on the apex host, where the default <code>same-origin</code> credentials attach cookies to a public file	<code>VillageScene.js:312–316</code> ; <code>ItemDatabase.js:165</code>
Does it change authentication state?	No	No login, logout or session endpoints; no writes to cookies or auth storage
Does it read form fields or chat input?	No. Inputs, textareas and selects are excluded from name scanning (<code>NameScanner SKIP_TAGS / SKIP_CONTAINERS</code>). The chat input is only matched by its <code>placeholder</code> attribute	<code>UIFoundation.js:1107–1108</code> ; <code>NameScanner.js:45–48</code>
Does it log sensitive data?	No. Console output is lifecycle messages, error messages, item counts and item names without icons (<code>AtlasService.js:377</code>)	Census "console output"

Question	Answer	Evidence
Does it transmit page data anywhere?	No. No request carries page-derived data (§5.2.2)	—

5.2.8 Does it modify game data or act for the player?

Question	Answer	Evidence
Synthetic clicks or submits?	None	Census: no <code>.click()</code> , no <code>requestSubmit / submit</code>
Calls to game action endpoints?	None. Only N1 and N2, both GET reads	§5.2.2
Dispatches events to game elements?	No. <code>dispatchEvent</code> is used only for the skin's own <code>iw:*</code> events on <code>document</code> (<code>AtlasService.js:225</code> , <code>DOMWatcher.js:45</code> , <code>ItemDatabase.js:224</code>)	Census
Cancels game input?	Only on skin-owned controls and item triggers: <code>CollapsibleFrames.js:218–219</code> — <code>preventDefault</code> + <code>stopPropagation</code> on the skin's own toggle button; <code>VillageLedger.js:190</code> — <code>preventDefault</code> on its own button; <code>TooltipEngine.js:553</code> — <code>preventDefault</code> for Enter/Space when focus is on an item trigger. Triggers are skin elements, or game <code><p></code> elements the skin made focusable (quest objective, village building name), which have no default Enter/Space action. Tooltip clicks are never cancelled (<code>TooltipEngine.js:513–535</code>)	Census; source
Moves focus?	Only for the tooltip: into the card on keyboard open, back to the trigger on Escape or close (<code>TooltipEngine.js:374–375</code> , <code>464</code> , <code>542</code> , <code>559</code>)	Census
Changes game values or text?	No. No write to game text nodes (smoke test: "game text nodes are never detached"). Values are only <i>read</i>	[Test run] <code>smoke.test.mjs</code>
Changes game control state (<code>disabled</code> , <code>aria-*</code>)?	Partly: <code>aria-expanded</code>, <code>aria-controls</code> and <code>aria-haspopup</code> on item triggers (skin elements, and the two game <code><p></code> types above); <code>tabindex="0"</code> on those <code><p></code> s; - no <code>disabled</code> writes. Teardown removes them unconditionally (FUN-09)	<code>QuestPanelRenderer.js:183–186</code> ; <code>VillagePanels.js:155–158</code>
Hides or disables game controls?	Yes, one: the "cycle XP display" button (FUN-01). Other native elements hidden by CSS are non-interactive text branches replaced by mirrors	§3.12

5.2.9 Content Security Policy compatibility

Mechanism	Subject to the page's CSP?	Assessment
Content script execution (bundle)	No: extension content scripts are not governed by the page CSP [Platform]	Safe from page CSP changes
MAIN-world script (<code>hydration-signal.js</code>)	Injected by the browser as an extension content script, not as an inline <code><script></code> . [Assumption]: not blocked by <code>script-src</code>	If blocked, the gate falls back to its 4 s timeout
<code><style></code> elements inserted from the isolated world	[Assumption]: Chromium exempts DOM insertions made from an extension's isolated world from the page's <code>style-src</code> . A strict <code>style-src</code> without <code>'unsafe-inline'</code> should be tested live	[Verify] before IdleWorlds deploys a strict style CSP
Inline <code>style</code> attribute writes (<code>element.style.setProperty</code>)	CSSOM writes are not blocked by <code>style-src</code> [Platform]	Safe
<code>chrome-extension://</code> images and fonts	[Assumption]: Chromium registers the extension scheme as bypassing page CSP for web-accessible resources. This premise underlies <code>vendor-assets.mjs</code>	[Verify] with the live CSP
<code>fetch</code> from the isolated world (N1, N2)	[Assumption]: not subject to the page's <code>connect-src</code> ; subject to CORS with the page origin	N1 on <code>www</code> needs CORS (SEC-03)

Mechanism	Subject to the page's CSP?	Assessment
Trusted Types	The page could enforce <code>require-trusted-types-for 'script'</code> . Whether that applies to <code>innerHTML</code> writes from an isolated world is [Verify]	H-03 removes the three <code>innerHTML</code> sinks, which makes this moot

5.3 CROSS-CUTTING ANALYSIS

5.3.1 Forged events and attributes (page scripts)

- `iw:*` **events** are dispatched on the shared `document`, so page scripts can listen and dispatch.
 - Under Chromium's world isolation, an object `detail` created in one world is not readable from the other (it appears as `null`) [\[Assumption\]](#). Forged events would then reach consumers with `detail === null`.
 - Consumers either use optional chaining (`e.detail?.roots || []` → whole-document work: `content.js:98`, `SkillCardDesignController.js:521-524`) or dereference inside `guard` (`InventoryRenderer.js:757`: `e.detail.row` throws → caught → one `warnOnce`).
 - Even if `detail` were readable, every consumer treats `detail` as **element references to re-validate** (`isConnected`, classification), never as data to trust.
 - **Worst case**: a page script dispatching events in a loop forces repeated whole-document reconciliation. That is a self-inflicted slowdown in its own tab, not a data or integrity risk.
 - Hardening **H-23**: ignore events whose `detail` is not an object the skin created (a module-private `WeakSet` of detail objects).
- **Skin attributes** are read back as inputs in these places:
 - `data-iw-item / data-iw-item-name` on triggers are **lookup keys** into the catalogue (`TooltipEngine.show`, L341-346). A forged trigger can only open a card for a *real* catalogue item, rendered from catalogue data.
 - `data-iw-*` role marks drive CSS and later classifier passes. A page could spoof them to restyle its own elements.
 - `data-iw-page-hydrated` can be spoofed to boot early (SEC-06).
 - None of these grant access to storage, network or extension APIs.

5.3.2 Injection through game text (other players)

- **Chat, player names and listings** reach the skin as DOM text.
- **Paths that read text**: `NameScanner` (`nodeValue` → `trie match` → `Range`); `classifiers` (`textContent` → `regex`); `InventoryRenderer` (`textContent` → `esc()` → `template`); `SkillCardDesignController / VillageLedger / WorldBossPanels` (`textContent` → `textContent` writes).
- **None** assigns page text to `innerHTML`, a URL, `style` or storage.
- **Regular expressions** applied to page text contain no nested quantifiers [\[Source: heuristic scan\]](#). Patterns such as `/atk\s*\d+.*def\s*\d+.*hp\s*\d+/i` backtrack at worst polynomially, over bounded header text, which is not a practical ReDoS vector.

5.3.3 Prototype pollution and JSON handling

- **Records**. `items.json`, `/api/player` and the index JSONs are parsed with `Response.json()`. Item records are stored in `Maps` keyed by strings (`ItemDatabase.js:212-221`), so a `__proto__` item name or id becomes a normal key, not a prototype write.
- **Village data**. `normaliseVillage` builds new objects field by field.
- **Preferences**. They are read with `Object.entries` and copied into `Maps` (`CollapsibleFrames.js:165-169`, `VillageLedger.js:107-111`).
- **Merging**. No recursive merge of untrusted objects exists.

5.3.4 URL construction

URL	Built from	Injection possible?
chrome-extension://.../assets/<file>	assetUrl() with constant paths, generated tables (villageBuildings.js , zoneThemes.js), or packaged index fields(index.atlas)	No: no page or network data
/api/player?section=...	Whitelisted section via encodeURIComponent	No
Wiki link	https://idleworlds.com/wiki/items/ + esc(wiki_slug)	Only the path on idleworlds.com can vary. No scheme or host change is possible, because the prefix is a constant https:// URL. Not URL-encoded(SEC-08)
Toolkit link	Constant	No

5.3.5 Race conditions and stale async work

Hazard	Guard	Evidence
A village API response arriving after a route or league change or after teardown	Generation counter, AbortController , isRuntimeActive() check before applying	VillageScene.js:276-279, 305-322, 334-336
Catalogue refresh after teardown	stopAutoRefresh clears the timer. An in-flight fetch can still resolve and dispatch iw:item-db-updated ; consumers are inert while inactive	ItemDatabase.js:91-95 ; content.js:74-80
Atlas load failure loops	Backoff (5 s, or 15 s when partial); rejection when nothing loaded	AtlasService.js:176-253
Kill-switch change during the hydration wait	Not guarded: the value is read before the ≤ 4 s wait and the listener registered after it	content.js:196-216 (H-07)
Tooltip open when React replaces the trigger	Orphan check on iw:dom-flush and mousemove	TooltipEngine.js:583-590, 501-511
Label fit measured before web fonts load	Refit after document.fonts.ready	SkillCardDesignController.js:531

5.3.6 Memory

- **Per-element caches** use WeakMap (for example: skill type, structure signatures, fill samples, tooltip items, spines, strength history). InlineStyleOwner tracks elements through WeakRef + FinalizationRegistry .
- **Module-level strong references:**
 - SkillPanelRenderer.lastCandidatePanel / lastCandidates keep one panel's candidate list (FUN-08).
 - NameScanner.matchIndex is a Map of text nodes → matches. It is pruned when nodes disconnect (rebuildHighlight) and cleared on teardown.
 - Runtime._warned and AtlasService._missingWarned are unbounded Set s of strings (small).
 - CompactButtons.decorated is a Set of elements, rebuilt each pass.
- **Assets:** see PERF-07 (≈126 MiB of decoded sprite atlases).

5.3.7 Error isolation

Every consumer is wrapped in guard / guardEach (Runtime.js:145-177), and exceptions are reported once per label (warnOnce). One malformed element therefore leaves only itself un-skinned (smoke.test.mjs). Async paths catch their own rejections: content.js:136-143, VillageScene.js:323-331, and the ItemDatabase refresh chain.

One gap: hydration-signal.js has no try/catch . A throwing getter on React internals would stop its polling with an uncaught error in the page console, and the gate would then time out (H-06).

5.3.8 Supply chain and release integrity

- **Runtime dependencies:** none (`package.json` has `devDependencies` only). The bundle contains only project source and three JSON files.
- **Dev dependencies** are locked by `package-lock.json` ; CI uses `npm ci` . Build tools run on the developer machine and in CI only.
- **CI** has `permissions: contents: read` , never writes, and fails when the committed bundle differs from a fresh build. The shipped JavaScript is therefore reproducible from source at a commit.
- **Release** zip: built from an allowlist (`package-release.mjs`), with version agreement and a source-map check. The zip is not signed by the build; Chrome Web Store signing, if used, is out of the repository's scope [Verify distribution].
- **Vendored art** is pinned to a commit of an external repository (`vendor-assets.mjs:25`). The download has no content hash check; integrity relies on HTTPS plus the pinned commit, and the files are committed after download.

5.4 PER-FEATURE JAVASCRIPT REVIEWS

Every subsection uses the same headings, so the checklist can be audited feature by feature.

The **Safety statement** table always answers the same five questions:

1. Does it handle **external data**?
2. Does it touch **sensitive information**?
3. Does it execute **dynamic code**?
4. Does it read or change **authentication state**?
5. Does it **modify application or game data**?

Hardening IDs (**H-xx**) preserve functionality, and behaviour-change IDs (**B-xx**) alter the feature. Both are listed with priorities in §5.7.

ID	Feature	Files
J-01	Activation lifecycle and kill switch	<code>content.js</code>
J-02	Page-world hydration signal	<code>src/page/hydration-signal.js</code>
J-03	Hydration gate	<code>HydrationGate.js</code>
J-04	Runtime utilities	<code>Runtime.js</code>
J-05	Style injection	<code>StyleInjector.js</code>
J-06	Reversible inline styles	<code>InlineStyleOwner.js</code>
J-07	Viewport and breakpoint epoch	<code>Viewport.js</code>
J-08	DOM watcher and event bus	<code>DOMWatcher.js</code>
J-09	Item catalogue	<code>ItemDatabase.js</code> , <code>normaliseItemName.js</code>
J-10	Sprite atlas service	<code>AtlasService.js</code> , <code>aliases.js</code>
J-11	Skills art service	<code>SkillsArtService.js</code>
J-12	Item hovercard	<code>TooltipEngine.js</code>
J-13	Item-name scanning	<code>NameScanner.js</code>
J-14	Inventory overlay	<code>InventoryRenderer.js</code> , <code>InventoryModel.js</code> , <code>itemDisplay.js</code>
J-15	Surface colour repainting	<code>BackgroundPainter.js</code>
J-16	Skill card renderer	<code>SkillPanelRenderer.js</code>
J-17	V2 skill card controller	<code>SkillCardDesignController.js</code> , <code>ProgressCadence.js</code>
J-18	Quest cards	<code>QuestPanelRenderer.js</code>
J-19	Classifier hub (frames, nav, zone bar, activity panels, market, boost)	<code>UIFoundation.js</code>
J-20	Toolkit link	<code>UIFoundation.ensureToolkitLink</code>

ID	Feature	Files
J-21	Header renderer and zone theming	HeaderRenderer.js, zoneThemes.js
J-22	Merged header chrome	HeaderChrome.js
J-23	Collapsible frames	CollapsibleFrames.js
J-24	Compact buttons	CompactButtons.js
J-25	World Boss and Zone Control panels	WorldBossPanels.js
J-26	Village route panels	VillagePanels.js, villageBuildings.js
J-27	Dashboard Village scene (API read)	VillageScene.js
J-28	Village ledger	VillageLedger.js
J-29	Overlay framing	OverlayFramer.js

J-01 Activation lifecycle and kill switch

Purpose; why JavaScript. Decide whether the skin should be active, wait for the page to be ready, build every surface in dependency order, and tear everything down on request without a reload. Stylesheets alone cannot wait for hydration, read a stored preference, or remove themselves.

Where. `src/content.js`:

Symbol	Lines	Role
<code>injectPresentationStyles</code>	57–72	Inject the 7 style elements
<code>scanRoots</code>	74–81	Rescan item names
<code>bindConsumersOnce</code>	83–117	Register page-lifetime consumers
<code>boot</code>	119–153	Build the skin
<code>teardown</code>	163–189	Remove the skin
<code>applyEnabled</code>	191–194	Boot or tear down
<code>start IIFE</code>	196–217	Entry point

Flow.

- `start()` calls `setRuntimeActive(false)`, so nothing can reconcile yet.
- `await storageGet('iw-skin-enabled')`. On failure → `null`.
- `await waitForPageHydration()` (J-03), then log the outcome.
- `applyEnabled(stored === null ? true : stored !== false)`.
- `boot()`, only if not already booted:
 - `setRuntimeActive(true)`.
 - Inject styles.
 - `bindConsumersOnce()`: init tooltip, inventory, skill panel, quest panel, UI foundation and header; subscribe to `iw:dom-flush`, `iw:name-scan-flush` and `iw:item-db-updated`.
 - `initSkillCardDesignController()`.
 - `AtlasService.ready()`, `ItemDatabase.startAutoRefresh()` and `ready()` (rejections are logged).
 - `paintBackground()`.
 - `startWatcher()` last.
 - Log `v1.6.0` booted.
- `onStorageChanged('iw-skin-enabled', v ⇒ applyEnabled(v !== false))` for the page lifetime.
- `teardown()`, only if booted: stop the watcher and refresh timer; hide the tooltip; clear the name scan, inventory, V2, skills, quests, UI foundation, header, background paint and overlays; remove styles and the hydration latch. **Then** `setRuntimeActive(false)`, then log.

Reads and writes.

Kind	Detail
Storage	Reads <code>iw-skin-enabled</code> . Never writes it
DOM	Indirect, through modules
Events	Listens to <code>iw:dom-flush</code> (→ <code>paintBackground</code> , <code>frameOverlays</code>), <code>iw:name-scan-flush</code> (→ <code>scanForItemNames</code>), <code>iw:item-db-updated</code> (→ <code>rescan</code> body)
Console	3 <code>log</code> lines; 2 <code>warn</code> lines on data-service failure

Runs; initialisation and cleanup. Once per page load, plus on each storage change. Consumers are bound once per page. Teardown reverses boot. `onStorageChanged` is never removed, by design.

Resources held. One `chrome.storage.onChange` listener; three `document` listeners (page lifetime). No timers, observers, network, globals or injected elements of its own.

Lifecycle, performance, failure modes, edge cases, compatibility.

- Boot cost is dominated by the initial discovery flush [**Project record:** `docs/traps/performance.md`].
- If storage is unavailable, the skin boots, with `enabled` as the default.
- A module throwing during init is contained by `guard` , and the rest boots.
- **Edge case (H-07):** the stored value is read before the up-to-4 s hydration wait, and the change listener is attached after it. A kill-switch change during that window is missed until the next change.
- `teardown()` before `boot()` only marks the runtime inactive.
- Compatibility: MV3 `chrome.storage` promises (Chrome 88+ [**Assumption**]).

Safety statement.

Question	Answer and evidence
External data	No network or page data is used directly; only the stored boolean
Sensitive information	No
Dynamic code	No
Authentication state	No
Modifies app or game data	No. Orchestration only

Concerning APIs and existing safeguards. `chrome.storage.onChange` is a page-lifetime listener. It is keyed to one key (`Runtime.js:111`), and handler errors are caught (L114–118).

Hardening (preserves functionality). H-07: register the storage listener before the hydration wait (or re-read the key after it), so a toggle during the wait is honoured.

Changes that alter the feature. A user-facing toggle (popup or keyboard shortcut) would change the product surface (C15).

Rewrite with internal components. Replace the whole lifecycle with a theme flag read at render time, for example `<ThemeProvider theme={settings.fantasy ? 'fantasy' : 'classic'}>` . No teardown is needed, because React unmounts.

J-02 Page-world hydration signal

Purpose; why JavaScript. Detect when React has finished hydrating, so the skin does not append nodes during hydration (C3). Only page-world JavaScript can see React's root object.

Where. `src/page/hydration-signal.js` (64 lines, IIFE); manifest `content_scripts[1]` (`world: "MAIN"` , `run_at: document_start`).

Flow.

1. `check()` builds the candidate containers: `document` , `<html>` , `<body>` , `#__next` , and each child of `<body>` (L45–46).

2. For each container, find its own property whose name starts with `__reactContainer$` (L30–34).

3. Read `container[key]?.stateNode?.current?.memoizedState` (L53).

4. Decide:

- not an object, or `isDehydrated` is not boolean → `signal('unknown')`;
- `isDehydrated` `=== false` → `signal('1')`;
- `true` → stop scanning containers (still hydrating).

5. If no signal was written and less than 15 s have passed → `setTimeout(check, 50)`.

6. `signal(state)` writes `data-iw-page-hydrated` on `<html>` only if absent (L36–39).

Reads and writes. Reads own-property **keys** of up to about $4 + n$ DOM nodes, and one nested React object path. Writes **one attribute** on `<html>`. Nothing else.

Runs. At `document_start`. Polls every 50 ms, for up to 15 s (≤ 300 checks). Stops at the first signal.

Resources held. A `setTimeout` chain. No listeners, observers, network, storage or globals (IIFE scope).

Lifecycle, performance, failure modes, edge cases, compatibility.

- **Cost:** each check is `Object.keys` on a handful of nodes.
- **React changes shape** → "unknown" → the skin boots immediately.
- **No React root found** → polling until 15 s; the gate has already timed out at 4 s.
- **A getter throws** → uncaught exception in the page console, polling stops (**H-06**).
- `world: "MAIN"` requires Chrome 111.

Safety statement.

Question	Answer and evidence
External data	Reads page JavaScript objects (React internals) read-only ; values are compared to booleans and never stored or forwarded
Sensitive information	No . It reads only <code>memoizedState.isDehydrated</code> ; React state contents are not traversed
Dynamic code	No
Authentication state	No
Modifies app or game data	No . One attribute on <code><html></code> , which React does not manage

Concerning APIs and existing safeguards. Access to React private fields is fragile, but read-only. It runs in the page world, so page scripts could tamper with it. That gives them nothing they did not already have. The latch can be spoofed to boot early (SEC-06), with no security impact.

Hardening (preserves functionality).

- **H-06:** wrap `check()` in `try/catch`; on error, `signal('unknown')` and stop.
- Optionally cap the scan to `document`, `#__next` and the body's first element.

Changes that alter the feature. Replacing the probe with a public event from IdleWorlds (for example `document.dispatchEvent(new Event('idleworlds:hydrated'))`) needs a game change.

Rewrite with internal components. Delete it (C3).

J-03 Hydration gate

Purpose; why JavaScript. Wait (bounded) for J-02's latch before the first boot.

Where. `HydrationGate.js`: `waitForPageHydration` (L30–51), `clearHydrationLatch` (L54–56). Constants: `HYDRATION_GATE_ENABLED = true`, `TIMEOUT_MS = 4000`, `POLL_MS = 50`.

Flow.

1. If disabled → resolve `{state: 'disabled'}`.
2. Otherwise poll the `data-iw-page-hydrated` attribute every 50 ms.
3. On a value, or on timeout, **remove the attribute** and resolve `{state, waitedMs}`.

Reads and writes. Reads and removes one attribute on `<html>`.

Runs. Once per page (first boot). `clearHydrationLatch` runs on teardown.

Resources held. A `setTimeout` chain of at most 4 s. It polls rather than observes (rule 4 allows only one observer).

Failure modes and edge cases. A latch written after the timeout persists until teardown (FUN-13). A spoofed latch releases the boot early (SEC-06).

Safety statement.

Question	Answer
External data	Reads one attribute value; only its truthiness and string are used, and only for logging
Sensitive information	No
Dynamic code	No
Authentication state	No
Modifies app or game data	No

Hardening. Also remove a late latch after the timeout, for example in the first `iw:dom-flush` (FUN-13). **Rewrite.** Delete it (C3).

J-04 Runtime utilities

Purpose; why JavaScript. Shared services: the activation flag; error isolation (`guard` , `guardEach` , `warnOnce`); `chrome.storage.local` wrappers with a test fallback; `assetUrl` ; `raf` .

Where. `Runtime.js` :

Symbol	Lines
<code>hasChrome</code>	20
<code>setRuntimeActive</code> / <code>isRuntimeActive</code>	28–34
<code>assetUrl</code>	42–49
<code>storageGet</code> / <code>storageSet</code> / <code>storageRemove</code>	59–102
<code>onStorageChanged</code>	108–122
<code>warnOnce</code>	132–138
<code>guard</code>	145
<code>guardEach</code>	160
<code>raf</code>	175–177

Flow.

- `guard(label, fn)` : returns `false` without calling `fn` when inactive; otherwise `try { fn() } catch { warnOnce }`.
- Storage functions `await chrome.storage.local.*` inside `try/catch` . On failure they warn once and return `null / false` .
- `assetUrl` strips leading slashes and calls `chrome.runtime.getURL` .

Reads and writes. `chrome.storage.local` (the keys in §5.2.3); console.

Resources held. A module `Set` of warned labels; an in-memory `Map` fallback (tests only).

Safety statement.

Question	Answer and evidence
External data	No
Sensitive information	No. Warnings include error messages and labels only (<code>Runtime.js:136</code>)
Dynamic code	No
Authentication state	No
Modifies app or game data	No

Concerning APIs. `chrome.storage.local` is extension-private. `chrome.runtime.getURL` only builds URLs.

Hardening. None needed.

Rewrite. An app-level error boundary and logger; app settings storage.

J-05 Style injection

Purpose; why JavaScript. Make all presentation CSS removable at runtime (C11), and point `url(../assets/...)` at packaged files (C10).

Where. `StyleInjector.js`: `rewriteAssetUrls` (L17), `inject` (L28), `removeAll` (L39).

Flow.

1. `inject(id, css)`: if the `id` is new, create `<style data-iw-style=id>`, set `textContent = rewriteAssetUrls(css)`, and append it to `head`, or to `<html>` if there is no head.
2. `removeAll()`: `document.querySelectorAll('style[data-iw-style]')`.`forEach`(`el => el.remove()`), then clear the set.

Reads and writes. Creates and removes 7 `<style>` elements.

Runs. Boot and teardown.

Safety statement.

Question	Answer and evidence
External data	No. The CSS text is compiled into the bundle
Sensitive information	No
Dynamic code	No. <code>textContent</code> is used (not <code>innerHTML</code>), so no markup is parsed; the CSS comes from the package
Authentication state	No
Modifies app or game data	No

Concerning APIs and safeguards. Inserting `<style>` into a page with a strict CSP (§5.2.9 [Verify]). `removeAll` would also remove any foreign `style[data-iw-style]`, which is harmless.

Hardening. Track the created elements and remove only those, instead of querying by attribute.

Rewrite. The app's CSS pipeline (C7/C8).

J-06 Reversible inline styles

Purpose; why JavaScript. Some visuals need inline declarations (to beat inline game styles or utilities), and teardown must restore the game's original inline values exactly (rule 3).

Where. `InlineStyleOwner.js`: `createInlineStyleOwner()` returns `{ set, restoreElement, restoreWithin, restoreAll }`.

Flow.

1. `set(e1, prop, value, priority = 'important')`:
 1. Look up this element's state. If the current inline value differs from what the owner last applied, treat the current value as the new *native* value (the game changed it).
 2. `style.setProperty(prop, value, priority)`.
 3. Record the post-write serialised value and priority.
 4. Return whether anything changed.
2. `restoreElement(e1)`: for each owned property still equal to what was applied, restore the native value or remove the property. A property the game changed since is left alone.
3. Tracking uses a `WeakMap` plus `WeakRef` + `FinalizationRegistry`, so detached elements are collected.

Reads and writes. Inline style properties on the elements callers pass.

Callers. `BackgroundPainter` (important); `SkillPanelRenderer` × 3 owners (important); `InventoryRenderer` (`display: none`, `normal` priority).

Safety statement.

Question	Answer
External data	No. Values come from callers' constants or computed colours
Sensitive information	No
Dynamic code	No
Authentication state	No
Modifies app or game data	Inline presentation only. Native inline values are preserved and restored (smoke test: "native inline skill/background styles survive teardown exactly" [Test run])

Hardening. None needed.

Rewrite. Component styles; no inline ownership.

J-07 Viewport and breakpoint epoch

Purpose; why JavaScript. Pick the rendered copy of duplicated panels, and invalidate caches when a Tailwind breakpoint is crossed (C6). CSS cannot tell JavaScript which copy is visible.

Where. `Viewport.js`: `isRendered` (L36, `checkVisibility()`, `else rect`), `pickRendered` (L45), `preferRendered` (L55), `getLayoutEpoch` (L73), `startLayoutWatch` (L85–101), `stopLayoutWatch` (L103–107).

Flow.

1. `startLayoutWatch(onCross)` creates five `matchMedia('(min-width: Npx)')` lists and adds a `change` listener to each.
2. Each crossing increments the epoch and calls `onCross` (`DOMWatcher` → `discover(body)`).

Reads and writes. Layout visibility (read); no DOM writes.

Runs. Visibility checks are called by classifiers; listeners live from boot to teardown.

Resources held. 5 `MediaQueryList` listeners (removed on stop).

Performance. `checkVisibility()` forces style calculation; callers limit it (for example the V2 timer's containment pre-check).

Safety statement. No external data, sensitive information, dynamic code or authentication. **No modification** of anything.

Hardening. None.

Rewrite. Components know their layout; delete.

J-08 DOM watcher and event bus

Purpose; why JavaScript. Keep every decoration current as React mutates the page, within a per-frame budget (C9). Also identify skill cards (F-25).

Where. `DOMWatcher.js`:

Symbol	Lines / role
Constants	<code>SEL_INV_ROW</code> , <code>SEL_SKILL_PANEL</code> , <code>FLUSH_BUDGET = 60</code>
<code>emit</code>	<code>L45: document.dispatchEvent(new CustomEvent(type, {detail, bubbles: false}))</code>
Skill detection	<code>detectSkillType (L143-203)</code> + signature cache
Queueing	<code>queueContext</code> , <code>discover</code> , <code>scheduleFlush</code>
Draining	<code>takeConnected</code> , <code>takeNameRoot</code> , <code>drainGlobalBudget</code> , <code>flushPending</code>
Lifecycle	<code>startWatcher (L353-434)</code> , <code>stopWatcher (L459)</code> , <code>getScanRoots</code> , <code>on / off</code>

Flow.

- `startWatcher` creates the observer on `document.body` (§5.2.5 options).
- For each mutation record:
 - childList**: queue the target's nearest row/panel context (repaint only for compact buttons); `discover()` each added element; queue parents of added text.
 - characterData**: queue the parent context, with no repaint.
 - attributes**: `style` → skills and background; `disabled / aria-disabled` → skills only; others → the node's own context.
- `discover(body)` runs once initially, and again on breakpoint crossings.
- `flushPending` (rAF) drains at most 60 elements round-robin, then emits:
 - `iw:inventory-row` per row;
 - `iw:skill-panel` per panel, with its detected type;
 - one `iw:dom-flush {roots}`;
 - one `iw:name-scan-flush {roots}`.
- If work remains, it reschedules.

Reads and writes. Reads the DOM structure and text of mutated regions. **Writes nothing to the DOM.** Dispatches four event types on `document`.

Runs. On every mutation batch; one rAF flush per frame at most.

Resources held. 1 `MutationObserver`; rAF callbacks; four pending Set s; a `WeakMap` skill-type cache.

Lifecycle, performance, failure modes, edge cases, compatibility.

- Budget**: large bursts spread over frames.
- Isolation**: consumers are guarded individually, so one throwing consumer cannot strand the queue (the comment at L336-339).
- Invisible changes**: changes carried only by `data-iw-*` or by `<html>` are invisible, by design.
- Pending frame after stop**: a queued rAF may still fire after `stopWatcher`, and drains empty sets.
- Test**: `flush-quiescence.test.mjs` asserts zero flushes on a quiet page [[Test run](#)].

Safety statement.

Question	Answer and evidence
External data	Page DOM text, used for skill identity only through anchored regexes. It never leaves the process
Sensitive information	No . Observes the whole body, including chat, but only element references and short label strings are kept (<code>skillSignature</code>); no text is stored beyond the per-panel signature
Dynamic code	No
Authentication state	No

Question	Answer and evidence
Modifies app or game data	No. No DOM writes

Concerning APIs and existing safeguards.

- A subtree `MutationObserver` over the whole body is powerful. It is bounded by an attribute filter, a flush budget and cheap per-record handling.
- The events are page-visible and forgeable (§5.3.1). Consumers re-validate element references and are guarded.

Hardening (preserves functionality).

- **H-23**: mark genuine event details (module-private `WeakSet`) and ignore others.
- Optionally dispatch on a private `EventTarget` instead of `document`, which also removes page visibility of the bus. This is behaviour-neutral for the skin, but tests that listen on `document` would need updating.

Rewrite with internal components. Obsolete: React props and state drive renders.

J-09 Item catalogue

Purpose; why JavaScript. The DOM does not carry item stats, tiers or acquisition data. The public `items.json` does. JavaScript fetches, caches, indexes and refreshes it for inventory, tooltips, name scanning, quests, bosses and the Village ledger.

Where. `ItemDatabase.js` (singleton `ItemDatabase`) and `normaliseItemName.js`.

Symbol	Lines
Constants <code>API_URL</code> , <code>CACHE_KEY</code> , <code>CHECKED_KEY</code> , <code>CACHE_MAX_AGE_MS = 6 h</code> , <code>REFRESH_RETRY_MS = 15 min</code>	13–21
<code>evictLegacyCache</code>	38–49
<code>responseValidator</code>	54–58
<code>ready</code>	75–86
<code>startAutoRefresh</code> , <code>stopAutoRefresh</code>	87–95
<code>_scheduleRefresh</code>	111–124
<code>_refreshOnce</code>	126–131
<code>_load</code>	132–158
<code>_fetchFresh</code>	160–203
<code>_index</code>	205–233
<code>_readCache</code>	235–250
<code>_writeCache</code> , <code>_markCacheChecked</code>	252–262
Lookups <code>getById</code> , <code>getByName</code> , <code>find</code> , <code>all</code>	264–284

Flow.

1. `ready()` → `_load()`, memoised. A rejection clears the memo, so a transient failure is retried later.
2. `_load()` evicts the legacy page- `localStorage` key, then reads the cache from `chrome.storage.local`:
 - **Fresh** (<6 h since cached or confirmed): `_index(..., 'cache')`, then a background `_refreshOnce()`.
 - **Stale**: `_index(..., 'stale-cache')`, then `await _refreshOnce()`. On failure it keeps the stale data and warns.
 - **None**: `await _refreshOnce()`.
3. `_fetchFresh()` runs `fetch(API_URL, {cache: 'no-cache'})`:
 - a non-OK status throws;
 - if the ETag/Last-Modified validator matches the indexed table, it cancels the body and records a confirmation;
 - otherwise `res.json()`; a missing or empty `items` array throws;

- the same `generatedAt` updates only the validator or confirmation;
 - otherwise `_index(..., 'network')` and `_writeCache(...)` (not awaited).
4. `_index()` builds `_byId (String(item_id))` and `_byName` (normalised name, first record wins), increments `_revision`, and dispatches `iw:item-db-updated`.
 5. `startAutoRefresh()` chains `setTimeout(_refreshOnce, 6 h)`. After a failure the chain retries in 15 min. `stopAutoRefresh()` clears the timer.

Reads and writes.

Kind	Detail
Network	N1 (§5.2.2)
Storage	<code>iw-item-db-cache</code> , <code>iw-item-db-cache-checked</code> (read and write); page <code>localStorage['iw-item-db-cache']</code> (<code>getItem</code> and <code>removeItem</code> only)
Events	Dispatches <code>iw:item-db-updated</code> { <code>revision</code> , <code>generatedAt</code> , <code>source</code> , <code>count</code> }
Console	Load counts; failure warnings

Runs; initialisation and cleanup. Loads at boot. Refreshes every 6 h while active. Cleanup is `stopAutoRefresh()` on teardown; indexes stay in memory, harmlessly, for the next boot.

Resources held. One `setTimeout` (6 h or 15 min). The in-memory catalogue: all records from `items.json`, about 5.5 MB decoded per the source comment [Verify]. No listeners.

Lifecycle, performance, failure modes, edge cases, compatibility.

- **Cache writes.** A cache write of several MB is deliberately not awaited. A failed write (quota) is warned and the skin continues.
- **Confirmation writes.** Recording a confirmation avoids rewriting the whole table (L15–18 comment).
- **Name collisions.** First record wins. Collisions are not reported.
- **Malformed records.** A record that is not an object makes `_index` throw. The caller then rejects and the skin logs; there is no partial index. On `www.`, CORS applies (SEC-03).
- **Legacy key check.** `localStorage.getItem` of the legacy key reads the entire legacy value (≤ 5 MB) once, just to test for its presence. `key()` iteration or a `length` check would avoid that read, but it happens at most once per page.

Safety statement.

Question	Answer and evidence
External data	Yes: the <code>items.json</code> response and the storage cache. Validation covers the envelope only (a non-empty <code>items</code> array); per-field types are not validated (SEC-02). Records are stored in <code>Map</code> s, so prototype pollution is not possible (§5.3.3)
Sensitive information	No. The public catalogue. The request carries cookies on the apex host because of default credentials (SEC-03); no response data is personal
Dynamic code	No
Authentication state	Not read or changed. Cookies may be attached automatically (above)
Modifies app or game data	No. Its one page-storage effect is removing its own legacy key (L43)

Concerning APIs and existing safeguards. `fetch` (N1): the URL is constant, the request method is GET, the status is checked, and the envelope is validated. Page `localStorage` access is limited to the extension's own legacy key (census: L42–44 only).

Hardening (preserves functionality).

- **H-02:** validate each record before indexing: `item_id` string or number; `name` non-empty string; `tier / craft_level / req_level` finite numbers or absent; text fields strings. Skip invalid records and count them in one warning.
- **H-05:** `credentials: 'omit'`. Consider a same-origin relative URL (`/items.json`) so `www` works without CORS [Verify that `www` serves it].

- Optionally check `Content-Type: application/json` and cap the response size before parsing (for example, reject above 20 MB).

Changes that alter the feature. None needed.

Rewrite with internal components. Item objects come from the game's own state or API client. Remove the fetch, cache, name index and aliases (C13).

J-10 Sprite atlas service

Purpose; why JavaScript. Map an item (id or name, including "+N" upgrades) to a window in the two sprite atlases, and paint it on a host element. Mapping a name to sprite coordinates requires data lookup; CSS cannot do it.

Where. `AtlasService.js` (singleton), `aliases.js`, `normaliseItemName.js`.

Symbol	Lines / role
URLs	L45–48, via <code>assetUrl</code>
<code>REQUIRED_ITEM_COLUMNS</code>	L54
<code>setAttr</code>	L65 (compare first)
<code>positionBadgeHost</code>	L79–97 (batched reads in a microtask)
<code>parseCSV</code>	L106–140
<code>ready</code>	L176–197 (backoff)
<code>_loadMissing</code>	L199–253
<code>_loadGearAtlas</code>	L255–283
<code>_loadItemAtlas</code>	L285–330
<code>resolve</code>	L332–380
<code>paint</code>	L382–395
<code>_applySprite</code> , <code>_paintBadge</code> , <code>_clearBadge</code>	L397–466

Flow.

- `ready()` : if not complete and not backing off, `_loadMissing()` fetches the missing metadata with `Promise.allSettled` :
 - the gear manifest JSON: the `icons` array must be non-empty; a schema warning if entries lack numeric geometry;
 - the item CSV: required columns and finite bounds are validated, and atlas dimensions are derived from the extents.
- On any success it bumps the revision and dispatches `iw:atlas-updated` . On failures it backs off: 5 s when nothing loaded, 15 s when partially loaded. It throws if nothing is usable.
- `resolve({id, name})` tries, in order:
 - item by id;
 - item by normalised name;
 - gear by name;
 - gear without "+N" (the badge level is recorded);
 - cloak alias;
 - without "Lv N";
 - without "of X";
 - without a leading affix word.

Otherwise it warns once for that name and returns `null` .

- `paint(host, ref)` :

- writes inline `background-image` (constant packaged atlas URL), `background-size`, `background-position` and `background-repeat` on **the host passed in**;
- sets `data-iw-atlas`;
- paints or clears the "+N" badge: `span.iw-icon-badge[data-iw-badge]`, `aria-hidden`, a sprite from the gear "+N" entry or the text "+N".

Hosts. Always skin-owned elements:

- `.fs-inv-icon` (InventoryRenderer);
- `.iw-tip-art-host` (TooltipEngine);
- `.fs-quest-sigil-icon`, painted detached (QuestPanelRenderer);
- boss reward icons (WorldBossPanels).

Reads and writes. Packaged JSON and CSV (N3, N4). Inline styles on skin hosts. The `iw:atlas-updated` event. Console warnings.

Runs. Metadata loads once, with retries. Paint runs per icon render.

Resources held. In-memory indexes (gear and item maps); a `Set` of warned names; the microtask queue for badge hosts.

Lifecycle, performance, failure modes, edge cases, compatibility.

- **Backoff.** It prevents fetch storms when each row calls `ready()` (the Audit S3.2 comment at L182–189).
- **Batched style reads.** Badge-host position checks are batched; they once cost ~10 s at boot [Project record, comment L67–78].
- **Atlas size.** The atlases are very large to decode (**PERF-07**).
- **Unresolved names.** An item without an icon falls back to the caller's glyph.
- **No teardown.** Hosts are removed by their owners. The badge span lives inside the host.

Safety statement.

Question	Answer and evidence
External data	Packaged files only (integrity is provided by the extension package). Names and ids come from DOM text or the catalogue and are used only as lookup keys
Sensitive information	No. Console warnings include item names (non-sensitive)
Dynamic code	No
Authentication state	No
Modifies app or game data	No. It writes only to skin-owned hosts

Concerning APIs. `fetch` of `chrome-extension://` URLs; `queueMicrotask`; `getComputedStyle` (batched).

Hardening (preserves functionality). Limit warnings: cap `_missingWarned` at a few hundred entries, or aggregate them into one summary warning per minute.

Changes that alter the feature. H-24 (split atlases) changes files, not visuals.

Rewrite with internal components. Item components render `<img src={item.iconUrl}>`, or sprites from the game's asset pipeline, with no name resolution.

J-11 Skills art service

Purpose; why JavaScript. Turn the packaged Skills UI and icon indexes into CSS custom properties (sprite windows, three-slice bands, textures), and point button-art variables at the current zone theme. The window geometry is data-driven, which pure CSS cannot compute from JSON.

Where. `SkillsArtService.js`:

Symbol	Lines / role
revisedButtonAtlas import	Bundled JSON, L10
validateIndex	L79-90
fetchIndex	L92-96
spriteGeometry , bandGeometry	L97-118
paint	L120-130
setVar , setData	L132-141
clearThemeVariables	L143-152
applyThemeVariables	L154-184
applyUiVariables	L185-219
clearUiVariables	L221-235
load	L236-246
Public object	ready , isReady , paintIcon , decoratePanel , clearPanel , applyThemeVariables , clearThemeVariables , iconEntry

Flow.

1. `ready()` → `load()` fetches `skills_icons_index.json` and `skills_ui_index.json` (N5). Each index must have finite dimensions and entries with a key and finite geometry. The service then builds maps. A failure clears the memo and warns once. There is **no backoff**, but callers use pending flags and the files are packaged.
2. `decoratePanel(panel)` writes inline custom properties on the panel:
 - `--fs-skills-panel-texture` , `--fs-skills-ui-atlas: var(--iw-zone-atlas, url(...))` , `--fs-skills-nav-prev/-next` ;
 - `--fs-ui-<token>-size/-position` for 9 tokens, plus band variables for 2 sliced tokens;
 - `--fs-ui-action-cap-ratio` ;
 - `data-iw-skills-ui-ready="1"` .

The panels are **game nodes**: skill cards, quest cards, the Skill Actions frame, and boss action buttons.

3. `paintIcon(host, key)` → `paint()` : inline background on the skin-owned medallion span; `data-iw-skills-atlas` , `data-iw-skills-atlas-index` .
4. `applyThemeVariables(html, theme)` :
 - writes `data-iw-compact-atlas="compact-ghost-v3"` and `--iw-compact-atlas` ;
 - writes `data-iw-button-atlas="revised-v5"` ;
 - for 12 button kinds, writes `--iw-<kind> (image)` and `--iw-<kind>-paint` (a full background shorthand with a percentage window).

The `exact-v3` branch is unreachable (SIZE-01).

5. `clearPanel` / `clearThemeVariables` remove exactly those properties and attributes.

Reads and writes. Packaged JSON (N5). Inline custom properties and `data-iw-*` on game panels and `<html>` . Inline background on skin spans.

Runs. Per card, frame or boss render (the functions compare first where it matters); theme variables on zone changes.

Resources held. Module indexes and maps.

Safety statement.

Question	Answer and evidence
External data	Packaged indexes only. <code>index.atlas</code> (a filename from packaged JSON) is placed into <code>assetUrl('assets/' + ...)</code>

Question	Answer and evidence
Sensitive information	No
Dynamic code	No
Authentication state	No
Modifies app or game data	No. Presentation custom properties on game containers; removed on teardown (<code>cClearPanel</code>)

Concerning APIs. `style.setProperty` on game nodes (custom properties only, under the skin's own prefixes). A CSS injection risk would exist only if an index value contained `)` or `;`; packaged, validated numeric geometry prevents it.

Hardening (preserves functionality).

- Validate `index.atlas` against `/^[\\w.-]+\\. (webp|png)$ / .`
- Add a retry backoff to `ready()` , mirroring `AtlasService`.

Changes that alter the feature. Remove the unreachable `exact-v3` branch (no visual change today; removes a fallback).

Rewrite with internal components. Theme tokens and component CSS modules, with no runtime geometry. The three-slice logic becomes a CSS `border-image` on standalone images.

J-12 Item hovercard (`TooltipEngine`)

Purpose; why JavaScript. Show a rich item card for item triggers across the page, with correct positioning, hover handoff, keyboard access and data refresh. This needs delegated event handling, data lookup and geometry.

Where. `TooltipEngine.js` (688 lines):

Symbol	Lines / role
<code>esc</code>	L49-53
<code>isMobile</code>	L55-57: <code>(hover: none)</code> , <code>(pointer: coarse)</code>
<code>findItem</code>	L61
<code>registerTooltipItem</code>	L68
<code>categoryGlyph</code> , <code>cleanEffectText</code>	L72-88
<code>renderAcquisition</code>	L91-146
<code>renderStats</code>	L148-162
<code>renderBadges</code>	L164-184
<code>renderCard</code>	L186-229
<code>position</code>	L234-265
<code>clearShowTimer</code> , <code>isOpen</code> , <code>nodeInside</code>	L269-282
<code>TRIGGER_SELECTOR</code> , <code>findTrigger</code>	L284-293
<code>isVirtual</code> , <code>anchorContainsPointer</code> , <code>activeSurfaceForNode</code> , <code>pointerNode</code> , <code>pointerIsOnActiveSurface</code>	L300-330
<code>cleanupAnchor</code>	L332-336
<code>show</code>	L338-377
<code>hide</code>	L379-398
<code>scheduleShow</code>	L400-411
<code>initTooltipEngine</code>	L415-608
Virtual anchor, <code>showForItem</code> , <code>hideTooltip</code> , <code>isTooltipOpen</code>	L610-652
<code>itemRef</code> , <code>itemRefIfKnown</code>	L660-688

Flow.

1. `initTooltipEngine()` , once per page:
 1. inject the `tooltip-engine` CSS (a no-op after `content.js`);
 2. create `div#iw-tip.iw-tip` with `role="dialog"` , `aria-modal="false"` , `aria-label` , `tabindex="-1"` , `display:none` , and append it to `<body>` ;
 3. register listeners (§5.2.5).
2. **Hover.** Capture-phase `mouseover` on a trigger → `scheduleShow` (120 ms). The show fires only if the anchor still `:hover` s, or on touch devices. `mouseout` → hide, unless moving into the card. `mousemove` keeps the pointer invariant (anchor or card).
3. **Click.** On a trigger: show (desktop) or toggle (touch). The click is **not** cancelled. Elsewhere: hide.
4. **Keyboard.** Enter/Space on a trigger → `preventDefault` , toggle, move focus to the first interactive element in the card. Escape → hide and restore focus. `focusout` from a keyboard-owned card → hide.
5. **show(anchor) :**
 1. look up the record: `findItem({id, name})` from `data-iw-item` / `data-iw-item-name` , else the registered fallback;
 2. `renderCard(item)` returns `{html, artHost, painted}` ;
 3. `STATE.el.innerHTML = html` (**sink**);
 4. insert the painted art host;
 5. set `aria-label` , and set `aria-controls` / `aria-haspopup` / `aria-expanded="true"` on the anchor;
 6. `position()` : measure while hidden, place below or flip above, clamp to a 10 px margin;
 7. add `.is-open` ; focus the card if the open came from the keyboard.
6. **hide()** : remove `.is-open` (guarded); `display: none` ; `visibility: hidden` ; `cleanupAnchor` (remove `aria-describedby` , set `aria-expanded="false"`).
7. **Scroll/resize** → reposition, or hide if the pointer left. **Window blur** → hide. `iw:dom-flush` → orphan check in the next frame. **Data events** → re-render an open card unless it is keyboard-owned.
8. **showForItem(item, rectProvider, source)** opens the card on a virtual anchor (NameScanner). `hideTooltip(source)` closes only a card opened by that source.

Reads and writes.

Kind	Detail
DOM reads	Trigger attributes; <code>:hover</code> ; geometry; <code>elementFromPoint</code>
DOM writes	<code>#iw-tip</code> (content, inline position/visibility/display, class, <code>aria-label</code>); trigger <code>aria-controls</code> , <code>aria-haspopup</code> , <code>aria-expanded</code> ; <code>aria-describedby</code> removal on anchors
Focus	Moves focus into the card and back
Events	Consumes <code>iw:dom-flush</code> , <code>iw:atlas-updated</code> , <code>iw:item-db-updated</code>

Runs. Listeners live for the page lifetime. A render happens per open or refresh.

Resources held. 15 listeners (§5.2.5); one element; one `setTimeout` (the show delay); rAF callbacks; a `WeakMap` of fallback items.

Lifecycle, performance, failure modes, edge cases, compatibility.

- **Teardown.** Only calls `hideTooltip()` . The element and listeners persist (FUN-10); with no triggers left they do nothing.
- **Anchors.** React replacing a trigger is caught by the orphan checks. A virtual-anchor containment bug was fixed (L438–442 comment).
- **Very short viewports.** The card clamps and its body scrolls.

Safety statement.

Question	Answer and evidence
External data	Yes: catalogue records (J-09) and fallback records. All text fields go through <code>esc()</code> except <code>craft_level</code> and <code>tier</code> on L101-102 (SEC-01) . The wiki slug is escaped but not URL-encoded (SEC-08). DOM attributes are used only as lookup keys (L341-346); a forged trigger can only show a real catalogue record
Sensitive information	No
Dynamic code	No script is created. The <code>innerHTML</code> sink could parse injected markup under SEC-01's precondition, and inline event-handler attributes in that markup would run in the page world
Authentication state	No
Modifies app or game data	No. It does not cancel game clicks (L513-535). It cancels Enter/Space only on item triggers (L553). It moves focus only for its own card

Concerning APIs and existing safeguards.

- `innerHTML` (L353). `esc()` on nearly every field; constants for classes and glyphs.
- `focus()`. Only for keyboard-initiated opens and closes.
- **Document-level capture listeners.** They return early when the event target is not a trigger.
- `elementFromPoint`. Only while a card is open.

Hardening (preserves functionality).

- **H-01:** `esc(item.craft_level)`, `esc(item.tier)`.
- **H-03:** build the card with `createElement` / `textContent` (or a small tagged-template helper that escapes by default), which removes the `innerHTML` sink.
- **H-04:** `encodeURIComponent(item.wiki_slug)`, and validate the slug with `/^[a-z0-9-]+$/.`
- **H-11** (shared with FUN-09): remove `aria-describedby` only where the skin added it.
- **H-12:** remove `#iw-tip` on teardown and recreate it on boot, keeping listeners bound once.
- **H-22:** early-return listeners when `!isRuntimeActive()`.

Changes that alter the feature.

- **B-11:** none required here.
- Making the card persistent on desktop click, or adding a hover grace period, would change interaction design.

Rewrite with internal components. An `<ItemTooltip item={item}>` component using the app's popover primitive (with focus management and portal). Data comes from props, and markup is JSX, which escapes by construction.

J-13 Item-name scanning (`NameScanner`)

Purpose; why JavaScript. Detect catalogue item names inside arbitrary game text (log, chat, descriptions) without adding elements, and open the hovercard for the word under the pointer. Text matching and geometry hit-testing need JavaScript.

Where. `NameScanner.js`:

Symbol	Lines / role
<code>SKIP_TAGS</code>	L45-48
<code>SKIP_CONTAINERS</code>	L61
Trie build and match	<code>buildTrie</code> (L68), <code>longestMatchAt</code> (L95), <code>findMatches</code> (L138)
<code>queueHighlightRebuild</code> , <code>rangesIntact</code> , <code>buildRanges</code> , <code>rebuildHighlight</code>	L168-234
<code>pointTextNode</code> , <code>hitInTextNode</code> , <code>hitTest</code>	L254-316
<code>handlePointer</code>	L327-352
<code>bindHover</code>	L353-386

Symbol	Lines / role
scanForItemNames	L394-443
clearItemNameScan	L446-453

Flow.

- Scan.** `scanForItemNames(root)` (on `iw:name-scan-flush` roots, and on `iw:item-db-updated` for the body):
 - build or reuse a trie of normalised catalogue names (per DB revision);
 - walk text nodes, pruning skip containers and skip tags;
 - find the longest matches at word starts (≥ 3 characters);
 - update `matchIndex` (text node $\rightarrow$ matches);
 - queue a highlight rebuild.
- Highlight.** `rebuildHighlight` (rAF) drops disconnected nodes, rebuilds only broken `Range`s, and re-registers `CSS.highlights.set('iw-item-name', new Highlight(...ranges))` only when the range set changed.
- Hover.** `bindHover` (once) listens to:
 - `mousemove` (passive, capture) $\rightarrow$ rAF $\rightarrow$ `handlePointer` (guarded);
 - `click` (touch-like only) $\rightarrow$ hit-test $\rightarrow$ `showForItem`;
 - `scroll` (passive, capture) $\rightarrow$ hide the name-scan card.
- Hit test.** `handlePointer` $\rightarrow$ `hitTest(x, y)`:
 - `elementFromPoint`;
 - the caret API gives the text node; its match ranges' client rects are checked;
 - otherwise the **fallback** walks the element's text subtree plus the direct children of 5 ancestors (**PERF-06**).

On a hit it calls `showForItem(item, rectProvider, 'name-scan')`; when the pointer leaves, `hideTooltip('name-scan')`.

Reads and writes.

Kind	Detail
DOM reads	Text node values (including chat); geometry
DOM writes	None. Painting uses the CSS Highlight registry
Events	Consumes the pointer, click and scroll events above

Runs. Per name-scan flush; per animation frame while the pointer moves.

Resources held. 3 page-lifetime listeners; `matchIndex` (`Map` of text node $\rightarrow$ matches and ranges); the trie; one registered highlight.

Lifecycle, performance, failure modes, edge cases, compatibility.

- No visible marking.** There is no `::highlight(iw-item-name)` style (**FUN-02**).
- Fallback cost.** The walk runs on misses (**PERF-06**).
- Stale offsets.** Stale ranges are dropped silently (L193-203).
- Compatibility.** Without the Highlight API, painting is skipped but hover still works. `caretPositionFromPoint` (Chrome 128) falls back to `caretRangeFromPoint`.
- Keyboard users** cannot reach prose matches.

Safety statement.

Question	Answer and evidence
External data	Yes: page text, including other players' chat. It is only compared against catalogue names; no page text is written anywhere, stored, or used in HTML or URLs

Question	Answer and evidence
Sensitive information	Chat text is read transiently in memory for matching. <code>matchIndex</code> stores only node references and offsets, not text copies (node values stay in the DOM)
Dynamic code	No
Authentication state	No
Modifies app or game data	No. Pure reads plus highlight registration; clicks are never cancelled

Concerning APIs and existing safeguards. Page-lifetime `mousemove` : rAF-throttled, passive, and gated by `guard` while inactive. `elementFromPoint` and caret APIs are read-only.

Hardening (preserves functionality).

- **H-15:** run the fallback walk only when the caret APIs are unavailable, or cap it to elements with fewer than N text nodes.
- Skip scanning chat feed rows if chat parsing is judged undesirable (a product decision).
- Fix `SKIP_TAGS` to use lowercase `svg` checks (MAINT-04).

Changes that alter the feature. B-02: add a visible highlight style, **or** remove prose scanning entirely (FUN-02).

Rewrite with internal components. Render item mentions as `<ItemLink>` where the game composes log and quest messages from structured data. Chat could keep a text matcher if wanted.

J-14 Inventory overlay

Purpose; why JavaScript. Replace each inventory row's plain text with a structured overlay built from DOM text plus catalogue data, hide the redundant native text branches, keep native action controls live, and classify the panel chrome. This requires parsing, data joins and element creation.

Where. `InventoryRenderer.js`, `InventoryModel.js` (pure) and `itemDisplay.js` (pure).

Symbol	Lines
<code>esc</code>	L83-87
Context resolution	<code>findInventoryRoot</code> , <code>resolveInventoryContext</code>
<code>classifyInventoryChromeOnce</code>	L317 (microtask reset at L319)
<code>classifyInventoryChrome</code>	L324
<code>suppressDisplayBranch</code> , <code>preserveInteractiveTree</code> , <code>hideOriginalChildren</code>	L452, L469, L493
<code>displayStyleOwner.set(eL, 'display', 'none', '')</code>	L456
<code>scheduleEmptyRetry</code>	L529 (rAF at L532)
<code>configureIconTooltip</code>	L541
<code>paintIcon</code>	L558-586
<code>detailHTML</code>	L588-601
<code>syncRowState</code>	L603-609
<code>renderRow</code>	L611-716
<code>reconcileAll</code>	L718-722
<code>clearInventoryRenderer</code>	L725+
<code>initInventoryRenderer</code>	L757-759

Flow.

1. On `iw:inventory-row {row}` → `renderRow(row)` (guarded).
2. **Cheap signature:** child count, full text, DB revision, atlas revision. If unchanged and an overlay exists, re-hide new native children, sync state, and return.

3. **Context:** the row must be inside the Inventory root, or contain Equip/Equipped/Unequip/List buttons. Otherwise clear the row and return.
4. **Chrome:** classify the panel once per flush (title, rule, filters with active state, pager, icons, list).
5. **Extraction:** name, quantity, detail texts and enhancement from leaf texts. An empty name retries next frame.
6. **Data:** `item = ItemDatabase.getByName(name); buildInventoryDetails; rowSignature`. If it equals `data-fs-inv`, re-hide and return.
7. **Build:** construct `div.fs-inv-row` by `innerHTML` with escaped values (§5.2.4), **append it into the row**, then `hideOriginalChildren`:
 - interactive hosts are preserved and tagged `data-fs-preserved-action` with `data-fs-action-kind`;
 - non-interactive native children are hidden via inline `display: none` (owner-restorable) and `data-fs-suppressed`.
8. `syncRowState` sets `is-equipped` / `has-set-action` from preserved control kinds. `paintIcon` paints the atlas sprite (fallback " ? "). `configureIconTooltip` makes the skin icon a trigger.
9. `reconcileAll` runs on `iw:item-db-updated` and `iw:atlas-updated`.
10. `clearInventoryRenderer` removes overlays, rules and all attributes, and restores `display`.

Reads and writes.

Kind	Detail
DOM reads	Row text and structure; button labels; ARIA and <code>data-state</code> for the active filter
DOM writes (game nodes)	<code>data-fs-inv</code> , <code>data-fs-preserved-action</code> , <code>data-fs-action-host</code> , <code>data-fs-suppressed</code> , <code>data-fs-action-kind</code> , <code>data-iw-inventory-*</code> ; inline <code>display: none</code> on native text branches; appended <code>div.fs-inv-row</code> and <code>div.fs-inv-rule</code>
DOM writes (skin nodes)	Overlay content; icon trigger attributes

Runs. Per dirty row; all rows on data events; one chrome sweep per flush per root.

Resources held. Two page-lifetime `document` listeners (L758–759) plus the `iw:inventory-row` consumer; `WeakMap` caches; a `Set` for per-flush sweeps; an inline style owner.

Lifecycle, performance, failure modes, edge cases, compatibility.

- **Cost.** Signature gating keeps ticks cheap. A full-document heading query runs per root resolution but is cached.
- **Unknown items.** Rendered plainly.
- **Rows leaving the inventory.** Cleared and restored.
- **Row identity.** A React re-render that replaces the row element produces a new row, which is decorated afresh; the old overlay leaves with the old row.

Safety statement.

Question	Answer and evidence
External data	Yes: DOM text (item names can be player-influenced only through the game's item system) and catalogue records. All interpolations are escaped with <code>esc()</code> inside double-quoted attributes, or are constants or numbers (§5.2.4). <code>itemRef</code> escapes name, id and label (TooltipEngine L660–680)
Sensitive information	No (inventory contents are the player's game data, displayed locally only)
Dynamic code	No
Authentication state	No
Modifies app or game data	No. Native action controls keep their handlers and positions (they are not moved, only restyled and raised). Only presentation attributes, inline <code>display</code> , and appended overlay elements are written, and all are removed on teardown (smoke test [<code>Test run</code>])

Concerning APIs and existing safeguards. `innerHTML` (L699), with escaping evidence in §5.2.4. `esc()` omits `'`, which is safe only because all attributes are double-quoted; a future single-quoted attribute would break this.

Hardening (preserves functionality).

- **H-03:** use DOM construction, or make `esc()` also escape `'` (cheap defence in depth).
- Assert (in a test) that every attribute in the template uses double quotes.

Changes that alter the feature. None needed.

Rewrite with internal components. An `InventoryRow` variant in the game renders the rich layout from the item object, with the existing action buttons. No hiding or overlay is needed.

J-15 Surface colour repainting (`BackgroundPainter`)

Purpose; why JavaScript. Some navy backgrounds are set inline or through arbitrary Tailwind values, so no generic stylesheet selector can target them. JavaScript reads computed colours and repaints known navy values.

Where. `BackgroundPainter.js` :

Symbol	Lines
<code>GAME_NAVIES</code>	L19-24
<code>SURFACE_SELECTOR</code>	L26-39
<code>ourColour</code>	L41-50
<code>normHex</code>	L53-59
<code>shouldSkip</code>	L65-71
<code>isSurfaceCandidate</code>	L73-84
<code>paintElement</code>	L90-116
<code>paintBackground</code>	L118-134
<code>clearBackgroundPaint</code>	L137-142

Flow.

1. `paintBackground(root)` walks all elements under `root` (`TreeWalker`).
2. For each element that is not skipped and is a surface candidate: `getComputedStyle`. A background whose RGB (alpha ignored) is in the 20-colour navy set → inline `background-color: <warm ink> !important`. Each border side in the set → `border-*-color: #342D20 !important`.
3. Mark `data-iw-painted="1"`.
4. On teardown, `restoreAll()` restores the native inline values and removes the marks.

Reads and writes. Computed styles (read). Inline colour properties and `data-iw-painted` on game nodes (write).

Runs. At boot on `<body>`; per `iw:dom-flush` for changed roots (or the whole document when there are none).

Resources held. One `InlineStyleOwner`.

Lifecycle, performance, failure modes, edge cases, compatibility.

- **Cost.** PERF-04.
- **Translucency.** Alpha ignored (FUN-11).
- **Game rewrites.** React writing a stock colour back is repainted on the next flush.
- **Error containment.** Exceptions are caught for the whole walk (the `try/catch` in `paintBackground`, L122-132).

Safety statement.

Question	Answer
External data	Computed colours only
Sensitive information	No
Dynamic code	No
Authentication state	No

Question	Answer
Modifies app or game data	No. Presentation-only inline colours, restored exactly

Hardening (preserves functionality).

- **H-13:** skip elements whose computed colour has alpha < 1. This preserves translucency, so it is listed as **B-05** because it changes appearance.
- Limit the walk to candidates with `querySelectorAll(SURFACE_SELECTOR)` plus two body levels, instead of visiting every element.

Rewrite with internal components. Delete it. The palette lives in theme tokens.

J-16 Skill card renderer (SkillPanelRenderer)

Purpose; why JavaScript. Skill cards have no stable internal hooks. JavaScript identifies each semantic part of a card (identity, title, level readout, action and nav buttons, requirements, rewards, materials, progress), assigns roles and zones, applies inline button plates that must beat inline game styles, neutralises readout and material chrome, and adds medallion art, a progress mirror and a material grid.

Where. `SkillPanelRenderer.js` (1,577 lines):

Area	Functions (line)
Buttons	<code>setOwnedStyle</code> (56), <code>classifyButton</code> (68), <code>compactCommandButton</code> (231), <code>styleButton</code> (236)
Readouts	<code>readoutCursor</code> (395), <code>sameTextShellChain</code> (401), <code>readoutBranch</code> (426), <code>neutraliseReadouts</code> (459)
Materials	<code>elementTextStarts</code> (545), <code>entryStart</code> (581), <code>completedIngredientRanges</code> (592), <code>ingredientEntries</code> (627), <code>rebuildIngredientHighlight</code> (652), <code>updateIngredientHighlights</code> (668), <code>updateIngredientLists</code> (690), <code>clearIngredientLists</code> (738), <code>neutraliseIngredients</code> (751)
Structure	<code>setRole</code> (793), <code>clearStructureRoles</code> (798), <code>commonAncestorWithin</code> (816), <code>isPresentationHidden</code> (827), <code>visibleFirst</code> (841), <code>textCandidates</code> (855; memo 852–853), <code>findBestText</code> (872), <code>findProgress</code> (879), <code>structureSignature</code> (941), <code>annotateStructure</code> (959)
Presentation	<code>ensureSkillArtwork</code> (1250), <code>ensureSkillActionsFrame</code> (1341), <code>setOwnText</code> (1369), <code>setOwnData</code> (1377), <code>ensureSkillPresentation</code> (1381)
Lifecycle	<code>applyPanelTreatment</code> (1447), <code>clearPanelInlineTreatment</code> (1459), <code>ownsPanel</code> (1479), <code>clearPanelChrome</code> (1483), <code>applyPanelChrome</code> (1510), <code>renderPanel</code> (1524), <code>clearSkillPanels</code> (1552), <code>initSkillPanelRenderer</code> (1572)

Flow.

1. On `iw:skill-panel {panel, skill} → renderPanel`:
 - **not a skill** (`unknown /quest`): clear only if this module owns the card (`ownsPanel`);
 - **a skill**: `applyPanelChrome` (classes `fs-skill-panel` and `fs-skill--<type>`, `data-fs-skill`, `data-iw-ui`, `data-iw-skill`, `glyph`), then `applyPanelTreatment`.
2. `applyPanelTreatment` runs these in order:
 1. `annotateStructure`, gated by `structureSignature` (type, child count, button labels with digits blanked);
 2. `ensureSkillArtwork` (medallion span; `SkillsArtService.decoratePanel / paintIcon`; pending if the art is not ready);
 3. `ensureSkillPresentation` (`data-iw-clean-text`, `data-iw-progress-display`, percent and progress mirror spans, base-EXP chip; the Skill Actions frame);
 4. `styleButton` for every button (`InlineStyleOwner`, important; skipped when the state snapshot is unchanged);
 5. `neutraliseReadouts` (inline chrome removal on readout branches; `cursor` kept as `pointer` for real controls);
 6. `neutraliseIngredients` (inline chrome removal; `data-iw-ingr`; material grid list; CSS highlight ranges for met materials).
3. `clearSkillPanels` restores all owned inline styles and removes classes, attributes, skin spans, grid lists and the highlight.

Reads and writes.

Kind	Detail
DOM reads	Card text, structure, button labels, <code>disabled</code> , requirement text classes, computed visibility(<code>getComputedStyle : visibleFirst</code> L830, unexpected flow child L1238), <code>getBoundingClientRect</code> (in <code>findProgress</code>)
DOM writes on game nodes	4 class writes(L1512-1516, L1349); ~25 <code>data-iw-*</code> / <code>data-fs-*</code> attributes(Appendix D); inline important styles through three owners; inline custom properties via <code>SkillsArtService</code>
Skin nodes	<code>span.fs-skill-medallion-art</code> , <code>span.fs-skill-identity-percent</code> , <code>span.fs-skill-identity-progress</code> > <code>span</code> (constant <code>innerHTML</code> , L1421), <code>span.fs-skill-base-exp</code> , <code>div.fs-skill-ingredient-grid</code> (<code>role="list"</code>)inserted after the native material line
Highlight	<code>CSS.highlights['iw-skill-ingredient-met']</code>

Runs. Per dirty card, gated by signatures. Cleanup at teardown.

Resources held. One page-lifetime event consumer; `WeakMap` s (structure signatures, button snapshots); three inline style owners; one highlight registration; the module-level memo (FUN-08).

Lifecycle, performance, failure modes, edge cases, compatibility.

- **Structural walk cost.** `annotateStructure` once took 72% of the skin's cost before digit-blanked signatures [Project record, comment L912-925].
- **Busy toggles.** The game disables all buttons during requests, so `disabled` is excluded from signatures (scroll-anchoring jumps)[**Project record**].
- **Shorthands.** Shorthand and longhand in one inline map caused ~120 flushes/s (`READOUT_STYLES` comment).
- **Card shapes.** Unexpected card shapes skip `three-zone`.
- **Memo.** Stale candidate memo (FUN-08).

Safety statement.

Question	Answer and evidence
External data	DOM text only, used for regex classification and copied into skin nodes with <code>textContent</code> (<code>setOwnText</code> , L1369-1375) or into <code>data-*</code> attributes via <code>setAttribute</code> . It is never parsed as HTML (the only <code>innerHTML</code> is a constant, L1421)
Sensitive information	No
Dynamic code	No
Authentication state	No
Modifies app or game data	No values or handlers: presentation attributes, owned inline styles and appended spans only. Exception in effect: the CSS applied to the <code>level-progress</code> role makes the game's XP-cycle button unreachable (FUN-01 , CSS-caused; this module itself preserves its pointer cursor, L380-399)

Concerning APIs and existing safeguards. Inline `!important` on game buttons: owned and restorable, and verified by the smoke test (native inline styles restored exactly). Layout reads: batched inside signature-gated passes.

Hardening (preserves functionality).

- **H-09:** clear the `textCandidates` memo at the end of `annotateStructure` (or key it by a pass counter).
- Add a test that the level-progress button keeps `pointer-events` and visibility, once B-01 is decided.

Changes that alter the feature. B-01: restore access to the XP-cycle control (a design decision on how it should look in the V2 medallion).

Rewrite with internal components. A `SkillCard` component with explicit slots(`identity` , `title` , `commands` , `materials`). Roles become component structure; no classification and no inline plates.

J-17 V2 skill card controller (SkillCardDesignController , ProgressCadence)

Purpose; why JavaScript. Adds the V2 card presentation on top of J-16:

- level readout and XP ring value;
- section marks;
- action glyph and fitted label;
- smoothed fill duration;
- long-action countdown mirrored from Current Action;
- material body and requirement note.

Measuring label width, sampling tick intervals and mirroring timer text all need JavaScript.

Where. SkillCardDesignController.js (548 lines):

Area	Functions (line)
Utilities and sections	setData (21), setText (26), markSections (29), levelReadout (66), removeRetiredNodes (92)
Glyph and label	ensureActionGlyph (117), ensureActionLabel (150), labelTextBox (191), fitActionLabel (317)
Fill and countdown	smoothActionFill (223), actionButtonOf (235), durationSeconds (245), currentActionRemaining (255), clearLongActionTimer (266), syncLongActionTimer (273), touchesHost (299), syncTimersOnTick (302)
Body and note	controlsHost (361), ensureDetailBody (372), ensureFootRow (440), syncRequirementNote (459)
Lifecycle	enhanceSkillCardV2 (478), clearSkillCardV2 (497), reconcile (512), bindOnce (517), initSkillCardDesignController (527), clearSkillCardDesignController (543)

ProgressCadence.js: nowMs (42), median (46), sampleProgress (60-111), formatDuration (113).

Flow.

1. initSkillCardDesignController() (each boot): active = true; bindOnce() (consumers for iw:skill-panel, iw:dom-flush, iw:name-scan-flush, all gated by active && isRuntimeActive()); document.fonts.ready → refit labels; set html[data-iw-skill-card-design="new"]; remove the legacy toggle node.
2. reconcile(panel, skill) requires the fs-skill-panel class (added by J-16), then enhanceSkillCardV2:
 1. data-iw-skill-v2="1", type and state;
 2. markSections;
 3. levelReadout (--iw-skill-v2-progress inline; readout spans);
 4. removeRetiredNodes;
 5. ensureActionGlyph (sibling of the button), ensureActionLabel (aria-hidden mirror);
 6. fitActionLabel (Range text rect + getComputedStyle; writes --iw-skill-v2-action-label-font between 4 px and 9.5 px; keyed in data-iw-skill-v2-label-fit);
 7. smoothActionFill (samples the button's span[style*=width] percentage → --iw-skill-v2-fill-duration; the reset flag for one frame);
 8. syncLongActionTimer;
 9. ensureDetailBody and syncRequirementNote.
3. sampleProgress(samples, key, pct):
 - keeps the last 5 inter-tick intervals within 60-4000 ms and needs at least 3;
 - returns {reset, durationMs}, where durationMs = median × 1.12;
 - a drop in percentage marks a reset and excludes the next interval;
 - the returned duration is null when it is within 15 ms of the previous one.
4. On a flush whose roots touch the Current Action host → syncTimersOnTick reads the remaining time once and updates the running cards' glyph text.

Reads and writes.

Kind	Detail
DOM reads	Card roles and text; the Current Action panel's timer text (<code>currentActionRemaining</code>); geometry for label fit; <code>checkVisibility</code> via <code>pickRendered</code>
DOM writes	<code>html[data-iw-skill-card-design]</code> ; on cards: <code>data-iw-skill-v2*</code> attributes and 4 inline custom properties; <code>data-iw-skill-v2-section</code> on native nodes; skin nodes (readout, glyph, label, body, controls, note) created with <code>createElement</code> / <code>textContent</code> only

Runs. Per `iw:skill-panel` ; on flushes touching Current Action; once after fonts load.

Resources held. Consumers for three event types (page lifetime, gated); `WeakMap` fill samples; one rAF per fill reset.

Lifecycle, performance, failure modes, edge cases, compatibility.

- **Label measurement.** It forces layout only when its key changes. The layout epoch is in the key, so hidden-mirror cards (0 px) retry after a breakpoint crossing.
- **Countdown.** It mirrors the game's timer (it never computes remaining time). If the timer text format changes, no countdown appears.
- **Teardown.** `clearSkillCardDesignController` removes the `<html>` attribute, legacy nodes and all V2 additions.

Safety statement.

Question	Answer and evidence
External data	DOM text (card labels, timer text), copied via <code>textContent</code> only; no <code>innerHTML</code> in this module (census)
Sensitive information	No
Dynamic code	No
Authentication state	No
Modifies app or game data	No. It never changes progress or timer values (rule 5): it measures cadence and mirrors text. Mirrors are <code>aria-hidden</code> where they duplicate a control label. Glyph and label nodes are siblings of the button, never inside it (L117–149)

Concerning APIs and existing safeguards. `document.fonts.ready` (read-only promise). `getComputedStyle` inside the fit, gated by its key.

Hardening (preserves functionality).

- Guard `fitActionLabel` against 0-width containers (already present via the epoch key).
- Make the V2 readout and material body `aria-hidden` **only if** the native equivalents stay exposed. Otherwise leave as is (see A11Y-01 and B-10).

Changes that alter the feature.

- **B-11:** raise `LABEL_FLOOR_PX` (for example to 7 px), and allow ellipsis or wrapping instead of 4 px text.
- **B-10:** decide one accessible source for level and material text.

Rewrite with internal components. Props provide level, percentage, timer and materials; CSS transitions keep the smoothing. The countdown reads the game's action state directly.

J-18 Quest cards (`QuestPanelRenderer`)

Purpose; why JavaScript. Positively identify quest cards (which share `.compact-panel`), assign roles and zones, add the sigil medallion with the objective item's sprite, measure the command block width, and make the objective line an item hovercard trigger.

Where. `QuestPanelRenderer.js` (466 lines): `textLeaves` (57), `questButtons` (66), `isQuestCard` (81), `findProgress` (106), `setRole` / `setZone` / `clearRoles` (120–147), `structureSignature` (149), `disciplineStyle` (157), `objectiveItemRef` (168), `clearObjectiveTrigger` (183), `ensureObjectiveTrigger` (194), `annotateStructure` (213), `commonAncestor` (282), `ensureDecoration` (291), `ensureAtlas` (353), `measureCommandBlock` (386), `questState` (400), `renderCard` (408), `clearCard` (436), `clearQuestPanels` (457), `initQuestPanelRenderer` (461).

Flow.

- 1. On `iw:skill-panel` → `renderCard(card)`. `isQuestCard` requires, with no word-boundary assumptions (`textContent` joins elements):
 - a `.compact-panel` that is not `fs-skill-panel`;
 - a "turn in" or "% complete" text prefilter;
 - a leaf starting `Reward:` ;
 - and a Turn In/Skip button or a `% complete` leaf.
- 2. Not a quest → clear if previously decorated (`fs-quest-panel` class).
- 3. `annotateStructure` , gated by signature (leaf count, labels with "(n)" stripped, track presence).
- 4. Style roles; add the `fs-quest-panel` class; set `--fs-quest-accent` from the discipline in the reward text; set `data-iw-quest-state` (`ready` , `work-order` or `active`)and `data-fs-quest` .
- 5. `objectiveItemRef` : strip the objective text's leading glyphs and trailing `n/m` , then `ItemDatabase.getByName` . If found → `ensureObjectiveTrigger` on the game `<p>` .
- 6. `ensureDecoration` : insert `span.fs-quest-sigil` as the first child of the body zone. Paint the icon detached, then attach it if painted. If the atlas is not ready, mark pending and re-render on ready. Add the percentage ring text.
- 7. `ensureAtlas` (SkillsArtService variables on the card); `measureCommandBlock` (`getBoundingClientRect` → `--fs-quest-cmd-w` ; skip when 0).

Reads and writes.

Kind	Detail
DOM reads	Card text and structure; button labels and <code>disabled</code> ; geometry of the command block
Game nodes	Class <code>fs-quest-panel</code> ; <code>data-fs-quest</code> , <code>data-iw-quest-*</code> ; inline <code>--fs-quest-accent</code> , <code>--fs-quest-cmd-w</code> , <code>SkillsArtService</code> variables; on the objective <code><p></code> : <code>data-iw-item</code> / <code>data-iw-item-name</code> , <code>data-iw-tooltip-trigger</code> , <code>tabindex="0"</code> , <code>aria-haspopup="dialog"</code> , <code>aria-controls="iw-tip"</code> , <code>aria-expanded="false"</code>
Skin nodes	Sigil, icon and percentage spans (<code>aria-hidden</code> sigil)

Runs. Per dirty `.compact-panel` ; one extra pass when atlas data arrives.

Safety statement.

Question	Answer and evidence
External data	DOM text (quest copy) for classification and for an item-name lookup key; catalogue record (J-09) for the icon. Text reaches skin nodes via <code>textContent</code> and attributes via <code>setAttribute</code> . No HTML sinks (census)
Sensitive information	No
Dynamic code	No
Authentication state	No
Modifies app or game data	No values or handlers. It adds focusability and ARIA to one game <code><p></code> per quest. Turn In/Skip remain native buttons (restyled only)

Concerning APIs and existing safeguards. Focusability added to game text (A11Y-02). Teardown removes the attributes unconditionally (FUN-09).

Hardening (preserves functionality). H-11: record whether `tabindex` / `aria-*` existed before writing, and restore instead of removing.

Changes that alter the feature. B-12: give triggers `role="button"` , which changes semantics for assistive technology.

Rewrite with internal components. A `QuestCard` component with `<ItemLink>` for the objective.

J-19 Classifier hub (UIFoundation)

Purpose; why JavaScript. Recognise and tag the page's major surfaces so the shared CSS can style them:

- navigation, zone bar and section frames;
- activity panels(Current Action with progress smoothing, Action Log, World Chat);
- the Daily XP Boost line, World Boss cards, Market rows and Village panels.

It then runs the dependent decorators in a fixed order.

Where. UIFoundation.js (1,468 lines):

Area	Functions (line)
Utilities	normText (49), setRole (53), setPanel (58), setPanelPart (63), commonAncestor (97), sameTextShell (108), matchingLeaves (122), directChildUnder (127)
Navigation	warmBackground (133), deriveTabActive (145), resolveMainNav (199), applyMainNavState (240), ensureToolkitLink (291), classifyMainNav (306)
Zone bar	zoneBarCandidatePresent (340), classifyZoneBar (352)
Frames	ownedElsewhere (468), inOverlay (472), beginClassifyPass (506), headingIndex (510), classifySectionFrames (591)
Bosses and market	classifyBossCards (724), classifyMarket (804)
Village	classifyVillagePanels (878)
Current Action	findCurrentActionProgress (902), progressPercent (946), markProgressReset (962), smoothActionProgress (978)
Activity panels	findActivityPanelHost (1012), classifyPanelHeader (1030), classifyFeed (1059), classifyCurrentAction (1085), classifyActionLog (1098), classifyWorldChat (1105), runActivityPanel (1144), activityPanelLabelNodes (1167), classifyActivityPanels (1245)
Daily boost	classifyDailyBoost (1336)
Lifecycle	queueClassify (1362), clearUIFoundation (1404), injectUIFoundationStyles (1451), initUIFoundation (1455)

Flow.

1. `initUIFoundation`: inject the `ui-system` group; `on('iw:dom-flush', queueClassify); on('iw:skill-panel', ...)` for boss cards in the resolved root; queue one pass.
2. `queueClassify` (rAF, at most once per frame) runs, in order:
 1. `beginClassifyPass`;
 2. `classifyMainNav` (+ Toolkit link);
 3. `classifyZoneBar`;
 4. `classifyActivityPanels`;
 5. `classifySectionFrames`;
 6. `classifyHeaderChrome` (J-22);
 7. `classifyDailyBoost`;
 8. `classifyBossCards` (→ J-25);
 9. `classifyMarket`;
 10. `classifyVillagePanels` (→ J-26);
 11. `reconcileVillageScene()` (J-27, async; `.catch(warnOnce)`);
 12. `decorateCompactButtons` (J-24);
 13. `decorateCollapsibleFrames` (J-23), last.
3. **Resolution caching.** Each classifier caches its resolution (host elements, roles) and revalidates it cheaply (connectivity, roles, layout epoch, coverage) before rescanning.
4. **Activity panels.** Hosts are resolved from headings or leaf text plus structural matchers. Per pass, the parts (progress, queue, feed rows, composer) are re-tagged. Current Action progress is sampled into `--iw-progress-duration` , with a

one-frame reset flag.

5. `clearUIFoundation` clears, in order: header chrome, collapsible frames, Village scene, compact buttons, boost, boss cards, Village panels, caches, the Toolkit link and progress properties. It then removes every `data-iw-ui`, `-tab`, `-state`, `-panel`, `-panel-part`, `-panel-header`, `-zone-action`, `-zone-link`, `-boss` and `-market` attribute document-wide.

Reads and writes.

Kind	Detail
DOM reads	Headings, labels, button text, ARIA and class state, computed background colour (<code>warmBackground</code> , L135; <code>inOverlay</code> , L482), <code>location.pathname</code> / <code>hash</code> (read only)
Game nodes	The <code>data-iw-*</code> role attributes listed; inline <code>--iw-progress-duration</code> on the Current Action track
Skin nodes	The Toolkit <code><a></code> (J-20)

Runs. One rAF pass after each flush (coalesced).

Resources held. Two page-lifetime consumers; module caches of resolutions; `WeakMap` progress samples.

Lifecycle, performance, failure modes, edge cases, compatibility.

- **Whole-document scans.** Avoided through cheap probes and caches (the zone-bar probe; the heading index per pass).
- **Classifier misses.** Any miss leaves that surface native (fail-safe).
- **Stale comment.** The attribute filter description in the code comment (L929-933) is out of date (D-04).

Safety statement.

Question	Answer and evidence
External data	DOM text and attributes for classification only. <code>location</code> is read to infer the active route, and never written (census: no <code>location</code> writes)
Sensitive information	No. Chat text is not read. World Chat is found through the input's <code>placeholder</code> attribute (L1106-1107); only the feed's timestamp leaves are matched
Dynamic code	No
Authentication state	No
Modifies app or game data	No. Roles and one inline custom property. Game controls are untouched

Concerning APIs and existing safeguards. Teardown removes `data-iw-ui` document-wide, including marks written by other modules. That is acceptable, because teardown clears everything.

Hardening (preserves functionality). Scope `clearUIFoundation`'s sweeps to the attributes this module wrote. This aligns with the ownership rule and prevents a future module from losing marks it still needs during a partial teardown.

Changes that alter the feature. None.

Rewrite with internal components. Components render roles and variants directly (§6.7). The Current Action smoothing becomes a CSS transition driven by the known tick interval from game state (no sampling).

J-20 Toolkit link

Purpose; why JavaScript. Add an outbound navigation link into the game's nav rail. JavaScript is required to insert an element.

Where. `UIFoundation.ensureToolkitLink` (L291-304); `TOOLKIT_URL = 'https://idleworldstoolkit.com'` (L289); removal in `clearUIFoundation`.

Flow. If the nav track has no `:scope > [data-iw-nav-link="toolkit"]`, create `<a data-iw-nav-link="toolkit" data-iw-ui="nav-tab" href=TOOLKIT_URL target="_blank" rel="noopener noreferrer" title="IdleWorlds Toolkit (opens in a new tab)">Toolkit</a>` with every attribute set before insertion, then `track.append(link)`.

Reads and writes. Appends one element to the game's nav track.

Runs. Each classify pass (a no-op once present).

Safety statement.

Question	Answer and evidence
External data	No. The URL is a constant
Sensitive information	No. <code>rel="noreferrer"</code> prevents sending the game URL as a referrer; <code>noopener</code> prevents the new tab from accessing <code>window.opener</code>
Dynamic code	No
Authentication state	No
Modifies app or game data	No. It adds a link, and does not alter game nav items or their handlers

Concerning aspects. A third-party destination (SEC-09): it is controlled by the skin author [**Assumption**], and its content and privacy practices are outside this review.

Hardening (preserves functionality). None needed technically.

Changes that alter the feature. B-07: remove the link, move it to a settings or about area, or keep it with owner approval.

Rewrite with internal components. If kept, a normal nav item in the game's router config with an external-link icon.

J-21 Header renderer and zone theming (`HeaderRenderer` , `zoneThemes`)

Purpose; why JavaScript. Identify the live header's parts (profile, utilities, status tiles), prepend the crest, point art variables at packaged images, read the current zone number, and apply the per-zone header painting and page theme. The zone comes from page text, so JavaScript must read it.

Where. `HeaderRenderer.js` (465 lines):

Area	Functions (line)
Zone	<code>currentZoneNumber</code> (70), <code>zoneSurfaceUrl</code> (83), <code>applyZoneSurface</code> (90), <code>applyZoneTheme</code> (118)
Variables	<code>setRole</code> (139), <code>setAssetVar</code> (144), <code>applyHeaderVars</code> (149), <code>applyNavVars</code> (159), <code>applyAnnouncementVars</code> (165), <code>applyZoneVars</code> (169)
Header parts	<code>isVisible</code> (185), <code>pickVisible</code> (190), <code>findLiveHeader</code> (194), <code>ensureCrest</code> (211), <code>classifyProfile</code> (223), <code>statKind</code> (294), <code>classifyStatus</code> (308), <code>tagStatusCards</code> (338)
Resolution	<code>headerResolutionValid</code> (357), <code>classifyHeader</code> (366), <code>findAnnouncement</code> (390), <code>classifyAdjacent</code> (407)
Lifecycle	<code>reconcile</code> (429), <code>queueReconcile</code> (436), <code>clearHeaderRenderer</code> (445), <code>initHeaderRenderer</code> (462)

`zoneThemes.js` is **generated**: `ZONE_THEMES` , `THEME_NAMES` , `zoneTheme()` .

Flow.

- `initHeaderRenderer` injects the header CSS, subscribes `queueReconcile` to `iw:dom-flush` , and queues one pass.
- `reconcile` (rAF) → `classifyHeader` (cached):
 - `findLiveHeader` (text probes) → layout;
 - `classifyProfile` (roles, crest prepended into the identity region);
 - `classifyStatus` (roles, stat kinds) and `applyHeaderVars` (URL variables for packaged art).
- `classifyAdjacent` : nav rail variables; announcement role and variables; zone shell role and variables.
- `applyZoneSurface(root)` : `currentZoneNumber()` returns `[data-iw-ui="zone-title"]` , else a text scan (PERF-02); remember the last zone; write the surface variables only when the key changes.
- `applyZoneTheme()` : `zoneTheme(zone)` gives the theme, then on `<html>` : `data-iw-zone-theme` , three URL variables, and `SkillsArtService.applyThemeVariables` . With no theme, remove them.

6. `clearHeaderRenderer` : reset caches; remove the `<html>` theme attribute and variables; remove header roles and card, stat and zone attributes; remove the crest; remove the header variables from every element (`querySelectorAll('*')`).

Reads and writes.

Kind	Detail
DOM reads	Header text, classes, visibility rects; the zone label text
Game nodes	<code>data-iw-header*</code> , <code>data-iw-zone</code> ; inline <code>--iw-header-*</code> , <code>--iw-nav-*</code> , <code>--iw-announcement-frame</code> , <code>--iw-zone-*</code> custom properties
<code><html></code>	<code>data-iw-zone-theme</code> ; <code>--iw-zone-atlas</code> , <code>--iw-corner-filigree</code> , <code>--iw-zone-separator</code> ; button-art variables and two attributes (via J-11)
Skin nodes	<code>span.fs-header-crest</code> (<code>aria-hidden</code>), prepended into a game container

Runs. One rAF pass after each flush.

Resources held. One page-lifetime consumer; resolution cache; `lastZoneNumber` .

Safety statement.

Question	Answer and evidence
External data	DOM text (header copy, zone label). The zone number is parsed as an integer and used only as a table index (<code>ZONE_THEMES[zone]</code>) and in a constant-pattern filename (<code>zone_\${zone}.webp</code> , validated as an integer between 1 and 34, L84-86)
Sensitive information	The header contains the player's name and stats. They are read for structure detection only (regex tests), never stored or transmitted, and the name colour is left untouched
Dynamic code	No
Authentication state	No
Modifies app or game data	No. Roles, custom properties and one decorative prepended span

Concerning APIs and existing safeguards. URL construction from page-derived data is limited to an integer in a fixed range (L84). A crafted zone label could only select another packaged zone image.

Hardening (preserves functionality).

- **H-16:** skip the full-document zone text scan when the zone bar is absent and a last zone is known. Or cache the scan result per flush.
- Remove the dead `--iw-header-frame-bar` mapping (MAINT-04), and fix the joined-statement formatting (D-10).

Changes that alter the feature. None.

Rewrite with internal components. The header component receives `zone` and `theme` from game state, and sets `data-zone-theme` on the app root.

J-22 Merged header chrome (`HeaderChrome`)

Purpose; why JavaScript. Detect the exact sibling structure (nav rail, optional announcement, zone bar) so CSS grid can present them as one framed box without moving nodes.

Where. `HeaderChrome.js` (144 lines): `setRole` (27), `clearMarks` (31), `norm` (35), `looksLikeShell` (42), `resolutionValid` (46), `elementChildren` (61), `resolve` (65-124), `classifyHeaderChrome` (126), `clearHeaderChrome` (141).

Flow.

1. `resolve()` : the rendered zone bar (`pickRendered`) whose parent has a direct `<header>` child becomes the shell.
2. Nav = the nearest preceding sibling containing nav tabs (stopping at `HEADER`).
3. Notice = zero or one qualifying sibling in between.

4. The zone bar must have exactly two children: text with the title, and actions.
5. Any mismatch → `clearMarks()` and return. Otherwise write `data-iw-chrome` roles.

Reads and writes. Reads DOM structure and short text lengths. Writes `data-iw-chrome` on game nodes.

Runs. Step 6 of each classify pass; cached, and re-resolved when the notice appears or disappears.

Safety statement. No external data beyond structure; no sensitive information; no dynamic code; no authentication. **No modification** except role attributes (CSS `display: contents` and grid placement do the visual merge).

Hardening. None needed.

Rewrite with internal components. Render the merged header layout directly.

J-23 Collapsible frames (`CollapsibleFrames`)

Purpose; why JavaScript. Add a collapse control to each frame, remember the choice per panel, and mark the heading "spine" so CSS can hide everything except the title. Toggle state, persistence and the per-panel title depth need JavaScript.

Where. `CollapsibleFrames.js` (295 lines):

Symbol	Line
<code>FRAME, STORE_KEY = 'iw-collapsed-frames'</code>	top of file
<code>frameTitle</code>	54
<code>frameLandmarks</code>	67
<code>headOf</code>	83
<code>spines WeakMap</code>	91
<code>markSpine</code>	106
<code>collapseTarget</code>	136
<code>frameKey</code>	152
<code>loadPreferences</code>	162
<code>persist</code>	174
<code>applyState</code>	180
<code>ensureToggle</code>	205
<code>decorateCollapsibleFrames</code>	230
<code>clearCollapsibleFrames</code>	284

Flow.

1. `decorateCollapsibleFrames(document)` is the last step of each classify pass:
 1. `loadPreferences()` runs once per activation: `storageGet` → a `Map` of keys that are `true`, excluding keys touched meanwhile.
 2. Find the leaf frames, and for each one:
 - find the title, the head (the frame child containing the title) and the target (the closest leaf `.panel`, else the frame);
 - build the key (`panel:<slug>` or `title:<lowercase title>`);
 - `markSpine(head, title);`
 - `data-iw-collapse-head="1";`
 - `ensureToggle(head, frame, key, name);`
 - `applyState`.
 3. Remove orphaned toggles and their marks.

2. Toggle click:

1. `preventDefault()` + `stopPropagation()` ;
 2. flip the preference;
 3. `persist()` (`storageSet` of the whole bag);
 4. `applyState()` : `data-iw-collapsed` on the target, `aria-expanded` , `aria-label / title` `Collapse <name>` or `Expand <name>` .
3. `clearCollapsibleFrames` removes toggles and all five attributes, and drops the in-memory preferences. Stored preferences are kept. **The `spines` cache is not reset (FUN-04).**

Reads and writes.

Kind	Detail
DOM reads	Frame structure; title text
Game nodes	<code>data-iw-collapse-head</code> , <code>data-iw-collapse-title</code> , <code>data-iw-collapse-spine</code> , <code>data-iw-collapsed</code> ; appended <code><button type="button" data-iw-collapse></code> in the head
Storage	<code>iw-collapsed-frames</code>

Runs. Each classify pass; each toggle click.

Resources held. One `click` listener per toggle button, released with the button; the preference `Map` ; the `spines` `WeakMap` .

Safety statement.

Question	Answer and evidence
External data	Stored preference bag: only <code>value === true</code> entries are accepted (L168). Title text is used as a key
Sensitive information	No. Keys are panel slugs or heading names, stored locally
Dynamic code	No
Authentication state	No
Modifies app or game data	No. The click handler stops propagation only for its own button (L217–225), so a heading row's game click handler does not also fire

Concerning APIs and existing safeguards. `stopPropagation` on a skin control inside a game container. It is scoped to the skin's own button, which is the intended isolation.

Hardening (preserves functionality).

- **H-08:** reset `spines` in `clearCollapsibleFrames` (`let` `spines` = `new WeakMap()` reassigned), or early-return only when the title still carries `data-iw-collapse-title` .
- Add a kill-switch round-trip test for collapsed titles (§7.6).

Changes that alter the feature. None.

Rewrite with internal components. A `CollapsiblePanel` wrapper with persisted state in app settings.

J-24 Compact buttons (`CompactButtons`)

Purpose; why JavaScript. Mark small game controls with a shape kind and selection state, and append three art layer spans into each, so CSS can crossfade themed art. Classification by label and neighbours needs JavaScript.

Where. `CompactButtons.js` (77 lines): `CONTROL` (L2), `loadoutPair` (14), `kind` (20), `clear` (37), `decorateCompactButtons` (43–72), `clearCompactButtons` (74).

Flow. For every element matching `CONTROL` :

1. `kind()` classifies it:
 - skin collapse toggle → skip;
 - preserved inventory action → `icon / text` ;

- classified filter, pager, nav tab, zone action or send → `text` ;
- I/II loadout pair → `icon` ;
- "Change Zone" → `text` ;
- an empty or timer-glyph button beside "Change Zone" → `icon` .

2. Set `data-iw-compact-button` .

3. Selection is read from `aria-pressed` / `aria-selected` , else `data-state="active"` , else an active-looking class. Set or remove `data-iw-compact-selected` .

4. Append any missing `span[data-iw-compact-layer="idle|hover|clicked"][aria-hidden="true"]` .

5. Controls that no longer match are cleared.

Reads and writes. Reads labels, ARIA, `data-state` and classes. Writes two attributes, and **appends up to three spans into native buttons**.

Runs. Each classify pass.

Resources held. A `Set` of decorated elements (rebuilt each pass).

Safety statement.

Question	Answer
External data	DOM labels only
Sensitive information	No
Dynamic code	No
Authentication state	No
Modifies app or game data	No. Spans are <code>aria-hidden</code> and decorative. The button's label, handlers and <code>disabled</code> are untouched

Concerning aspects. Children appended inside React-owned buttons. React tolerates extra trailing children in most updates, and the observer re-adds layers if React replaces them [**Project record:** `compact-button-reconcile.test.mjs`].

Hardening. None needed.

Rewrite with internal components. A `Button variant="compact"` with the art in CSS (or pseudo-elements on the component's own markup).

J-25 World Boss and Zone Control panels (`WorldBossPanels`)

Purpose; why JavaScript. Assign roles on boss cards, add portrait art and a rewards disclosure derived from item data, relabel the join action visually, and build the Dominion Ward mirror for Zone Control, including an impact animation when strengths change.

Where. `WorldBossPanels.js` (364 lines):

Area	Symbols (line)
Data	<code>BOSSSES</code> (L9-13), inline fallback records (L15-16), bundled <code>rewards.json</code>
Rewards disclosure	<code>bindRewardDisclosure</code> (20), <code>rewards</code> (250)
Helpers	<code>mark</code> / <code>setText</code> / <code>owned</code> (45-56)
Zone Control	<code>controlProgress</code> (57), <code>strengthTeam</code> (67), <code>controlStrengths</code> (77), <code>ensureCrestLayers</code> (104), <code>stopStrengthImpact</code> (117), <code>playStrengthImpact</code> (124), <code>ensureDominion</code> (155), <code>updateDominion</code> (188)
Lifecycle	<code>decorateWorldBossPanel</code> (284), <code>clearWorldBossPanel</code> (356)

Flow.

1. `decorateWorldBossPanel({root, heading})` , called from `UIFoundation` :

1. insert the skin notice paragraph, once per root (L285–290);
 2. for each leaf `.compact-panel`, identify the boss by name, or Zone Control by text;
 3. write `data-iw-encounter` and `data-iw-boss-role` roles;
 4. append `div.iw-boss-art`;
 5. set the action state and label: "Queued" when the button text matches `prejoined`, "Prejoin" when it matches `prejoin` (L312–313);
 6. decorate the action button with atlas variables.
2. **Boss cards** → `rewards(card, boss)`, signature `key|dbRevision|atlasReady`:
1. build `details.iw-boss-rewards > summary + button.iw-boss-reward` tiles(`data-iw-tooltip-trigger`, `data-iw-item`, `data-iw-item-name`, `aria-label`);
 2. `registerTooltipItem(tile, record)`;
 3. paint the icon.
- Reward ids come from `BOSSSES`, plus catalogue `BossDrop` records whose acquisition text names the boss, plus two fixed ids. `bindRewardDisclosure` restores and persists the open state on `toggle`.
3. **Zone Control** → `ensureDominion` (the `section` structure, meter `role="progressbar"`) and `updateDominion`:
1. read strengths and progress from native numbers, and team from colour words and classes;
 2. mark the native nodes;
 3. set texts, fill inline `width` and `--iw-control-split`, and the meter's `aria-value*`;
 4. `playStrengthImpact` when a value changed.
4. **Impact animation**: skipped under reduced motion. `Element.animate` runs for 1000–1250 ms. A temporary `change` listener on the reduced-motion query is removed on finish.
5. `clearWorldBossPanel(root)`: stop animations, clear the atlas variables, remove owned nodes and attributes.

Reads and writes.

Kind	Detail
DOM reads	Card text, numbers, class and style colour words
Game nodes	<code>data-iw-encounter</code> , <code>data-iw-boss-role</code> , <code>data-iw-boss-action-state</code> , <code>data-iw-boss-action-label</code> , <code>data-iw-boss-art-ready</code> , <code>data-iw-control</code> , <code>data-iw-control-crest</code> ; inline atlas custom properties on the action button
Skin nodes	Notice <code>p</code> , art <code>div</code> , rewards <code>details</code> and tiles, Dominion <code>section</code> (all <code>data-iw-boss-owned</code>); inline <code>width</code> and <code>--iw-control-split</code> on the skin meter
Storage	<code>iw-boss-rewards-collapsed:<bossKey></code>
Tooltip	<code>registerTooltipItem</code>

Runs. Each classify pass (and on `iw:skill-panel` for boss cards); animations only on value changes.

Resources held. One `toggle` listener per rewards element; one temporary media-query listener per running impact; `WeakMap`s of history, animations and cleanups; the `Map` of reward preferences.

Safety statement.

Question	Answer and evidence
External data	DOM text and numbers (parsed to numbers); catalogue records; bundled fallbacks. Tile labels and texts use <code>textContent</code> and <code>setAttribute</code> (no HTML sinks, census)
Sensitive information	No
Dynamic code	No
Authentication state	No
Modifies app or game data	No values or handlers. It adds skin-authored gameplay statements to the page (CONT-01), and visually relabels "Prejoined" as "QUEUED" (CONT-02; the accessible name stays the game's)

Concerning aspects.

- **Content accuracy.** CONT-01/CONT-02.
- **Team colour inference.** The position fallback (FUN-12).
- **Web Animations.** Reduced-motion aware; temporary listeners removed.

Hardening (preserves functionality). When team colours cannot be read, prefer not rendering the Dominion mirror (leave the card native) over guessing red-then-blue. This changes a degraded case only, and is listed as **B-13**.

Changes that alter the feature.

- **B-06:** remove or replace the notice and fallback copy with game-provided text, and keep the game's "Prejoined" wording.
- **B-13:** as above.

Rewrite with internal components. `BossCard` and `ZoneControlCard` components using the game's own boss and zone-control state, and reward tables from the server.

J-26 Village route panels (`VillagePanels` , `villageBuildings`)

Purpose; why JavaScript. Parse the Village route's housing and add-on slot cards, attach the matching building or house art, mark actions, and make building names item hovercard triggers.

Where. `VillagePanels.js` (380 lines):

Symbol	Line
<code>mark / unmark / owned</code>	74–87
<code>nativeChildren</code>	88
<code>ensureArt</code>	102
<code>removeArt</code>	124
<code>resolveBuilding</code>	130
<code>ensureTrigger</code>	143
<code>clearTrigger</code>	155
<code>decorateSlot</code>	169
<code>ensureOptionArt</code>	239
<code>undecorateSlot</code>	250
<code>slotSignature</code>	265
<code>houseTier</code>	281
<code>ensureTierTrack</code>	293
<code>decorateHousing</code>	308
<code>decorateVillagePanel</code>	350
<code>clearVillagePanel</code>	370

`villageBuildings.js` is **generated**: 34 buildings and 5 houses → file names.

Flow.

1. For the `addons` kind, each slot card is gated by `slotSignature` . `decorateSlot` then:
 - parses the head row and copy positionally;
 - marks slot state, name and effects;
 - classifies action verbs;
 - decorates the picker options (option art prepended into the option name `<p>` : **FUN-06**);

- calls `ensureArt` (`div.iw-village-art` with inline `--iw-village-sprite: url(chrome-extension://.../assets/village/<file>)`);
 - calls `ensureTrigger` on the building name `<p>`, when the item resolves.
- For the `housing` kind, `decorateHousing` reads the tier from "Current tier: N" (or the house name), marks roles, and adds the tier pips and house art.
 - `clearVillagePanel(root)` undoes both.

Reads and writes.

Kind	Detail
DOM reads	Headings, lines and button labels
Game nodes	<code>data-iw-village*</code> attributes; trigger attributes including <code>tabindex</code> / <code>aria-*</code> on the building name <code><p></code>
Skin nodes	Art, option art, tier track (<code>data-iw-village-owned</code>)

Runs. Each classify pass on the Village route.

Safety statement.

Question	Answer and evidence
External data	DOM text; the generated table. Names are used as lookup keys. Art URLs come only from the generated table's file names (constants)
Sensitive information	No
Dynamic code	No
Authentication state	No
Modifies app or game data	No values or handlers. Install, Upgrade, Uninstall and Destroy remain native buttons (restyled; destroy gets a danger hover)

Concerning aspects. FUN-06 (option art flip-flop), FUN-09 (unconditional trigger teardown), A11Y-02.

Hardening (preserves functionality). H-10: read the option name from the first **text** node, or skip skin-owned children. H-11: attribute ownership for triggers.

Changes that alter the feature. None needed.

Rewrite with internal components. Village components with the building art from the game's asset pipeline.

J-27 Dashboard Village scene (`VillageScene`): authenticated API read

Purpose; why JavaScript. Show the player's village (house and installed buildings) on the dashboard, where the game renders no village markup. The data must come from somewhere: either the Village route's rendered DOM (when the player has visited it) or the game's own `/api/player` endpoint.

Where. `VillageScene.js` (346 lines):

Symbol	Line
Constants <code>REFRESH_MS = 300000</code> , <code>RETRY_MS = 30000</code> , <code>ROUTE_REFRESH_MS = 10000</code>	14–17
Module state	18–41
<code>leagueOf</code>	51–53
<code>sectionOf + SECTIONS</code> whitelist	61–66
<code>normaliseVillage</code>	68–94
<code>own</code>	104–110
<code>art</code>	112–119
<code>findAnchor</code>	121

Symbol	Line
<code>readHousingPerks</code>	146
<code>readRenderedVillage</code>	164
<code>render</code>	181
<code>bindOnce</code>	248–254
<code>reconcileVillageScene</code>	256–332
<code>clearVillageScene</code>	334–346

Flow.

- `reconcileVillageScene()` (each classify pass, and on `iw:item-db-updated`):
 - return immediately if the runtime is inactive;
 - if the route (`pathname + search`) changed: when the **league** changed (`/ssf` prefix), drop the snapshot, perks and message; increment `generation`; abort any in-flight request; pull `nextRead` forward to at most 10 s;
 - if the Village route's rendered panels are tagged (J-26), `readRenderedVillage()` becomes the snapshot (no network);
 - `findAnchor()`: the rendered `.panel` (or Skill Actions frame) holding an `h2` "Skill Actions"/"Actions". None → remove the frame, abort, return;
 - ensure `section.iw-village-scene` (all nodes via `own()`), inserted with `target.after(frame)`;
 - `render()`.
- Network read**, only if no request is in flight, `Date.now() ≥ nextRead`, and `document.visibilityState !== 'hidden'`:
 - `nextRead = now + 5 min`; create an `AbortController` with a 10 s timeout;
 - `fetch('/api/player?section=' + encodeURIComponent(sectionOf(pathname)) + '&scope=core', {method: 'GET', credentials: 'same-origin', cache: 'no-store', signal, headers: {'x-idleworlds-league': league}})`;
 - `!response.ok` → error; `normaliseVillage(await response.json())` → `null` → error;
 - if aborted, superseded (`generation`) or inactive → drop the result;
 - otherwise `snapshot = result`; `render('')`.
- On error**: `nextRead = now + 30 s`; render "Showing the last loaded village. Refresh unavailable; retrying shortly." or "Village could not be loaded. Open the Village tab to view your buildings."
- `normaliseVillage(data)`, **strict**:
 - `player.housing.tier` is an integer 0–5;
 - `player.villageAddons.totalSlots` is an integer 0–5;
 - `installed` is an array, and every row has an integer `slot` in `1...totalSlots`;
 - each row resolves by `itemKey` or name, or has a name;
 - no duplicate slot.

Otherwise `null`. It returns only `{tier, capacity, slots[5]: {slot, state, name, file, itemKey}}`.
- `render(status)`: signature `JSON [snapshot, status, perks, dbRevision]`. It removes only its own children, keeping the collapse toggle, and builds the heading, count, stage and ledger (J-28) with `createElement` / `textContent`; art uses `<img src=assetUrl(constant file) alt="" decoding="async">`.
- `clearVillageScene()`: increment `generation`, abort, remove the frame, clear the ledger, reset state.

Reads and writes.

Kind	Detail
Network	N2 (§5.2.2): authenticated GET with session cookies and the league header
DOM reads	The Skill Actions heading and panel; the Village route's tagged panels (text)
DOM writes	Inserts one skin section as the next sibling of the Skill Actions panel. All descendants are skin nodes

Kind	Detail
Memory	The snapshot (tier, capacity, slot names and keys), perks lines, status message

Runs. Every classify pass, which returns early unless the anchor exists. Network at most every 5 min per page (30 s after failures), plus up to one read ≤ 10 s after each route change, only while the tab is visible.

Resources held. One page-lifetime `iw:item-db-updated` listener; one `AbortController` and one `setTimeout` per request (both cleared in `finally`); module state.

Lifecycle, performance, failure modes, edge cases, compatibility.

- **Server-side changes.** An API shape change $\rightarrow$ `normaliseVillage` returns `null` $\rightarrow$ the error copy, no crash. Tier or slot values above 5 are rejected; the game adding a 6th tier would break the scene until the constants are updated.
- **Session expiry.** Expired session: the API likely returns 401 $\rightarrow$ an error message $\rightarrow$ retries every 30 s [**Verify server behaviour**].
- **Multiple tabs.** Each tab polls independently.

Safety statement.

Question	Answer and evidence
External data	Yes: the authenticated <code>/api/player</code> JSON. Parsed with <code>Response.json()</code> , validated strictly, reduced to a minimal snapshot. Strings(<code>name</code> , <code>itemKey</code>) are rendered with <code>textContent</code> or used as lookup keys, and never used in URLs . Image files come only from the generated table
Sensitive information	Yes, transiently: the <code>scope=core</code> response is the player's core profile payload [Verify contents with IdleWorlds]. It lives in memory only for the duration of <code>normaliseVillage</code> . Only tier, capacity and slot names and keys are retained (in memory). Nothing is stored, logged or sent anywhere
Dynamic code	No
Authentication state	Uses the session: cookies are sent automatically by <code>credentials: 'same-origin'</code> . Does not read or modify authentication state: no cookie access, no token handling, no login or logout requests
Modifies app or game data	No. GET only; no state-changing endpoints; the league header only selects which character's data is read (the page itself sends the same header [Project record: <code>docs/traps/village.md</code>])

Concerning APIs and existing safeguards.

- **Authenticated fetch.** Mitigated by: GET only; fixed path; whitelisted `section`; derived league; a 10 s timeout; a generation counter against stale writes; `isRuntimeActive` checks; visibility gating; the refresh interval and route floor; strict schema validation; the data-minimising snapshot.
- **Residual concerns (SEC-04).** Owner approval of the API use and polling rate; the full payload being parsed in the extension; no backoff for auth or rate-limit statuses.

Hardening (preserves functionality).

- **H-14:**
 - treat 401/403 as "stop polling until the next route change";
 - honour `429 / Retry-After` with exponential backoff;
 - skip the network read while the scene frame is collapsed;
 - add `Accept: application/json`;
 - reject responses whose `Content-Type` is not JSON.
- Keep the in-memory snapshot minimal (already true).
- Document the endpoint use for IdleWorlds.

Changes that alter the feature. B-08: remove the network read and use only the Village route's rendered data. The scene then shows "Open the Village tab..." until visited. Alternatively, get owner sign-off, or an official lightweight endpoint (for example `GET /api/player/village`).

Rewrite with internal components. A dashboard `VillageSummary` component fed by the game's existing player store. No extra request.

J-28 Village ledger (VillageLedger)

Purpose; why JavaScript. Compute each installed building's benefits and the totals from catalogue records, and render a two-level disclosure with persisted open and closed state.

Where. VillageLedger.js (415 lines):

Symbol	Line
STORE_KEY = 'iw-village-ledger' , BENEFITS (15 stat specs)	top of file
loadPreferences	104
persist	119
applyEntry	125
icon	143
statList	157
chevron	171
wire	187
entry	203
buildingItem	238
buildingBenefits	244
totalBenefits	274
buildVillageLedger	316
clearVillageLedger	410

Flow.

1. buildVillageLedger({snapshot, house, perks, own}) :
 1. load preferences (once);
 2. build the head with a section toggle, the home entry (slots and perks lines), vacant lines, building entries (ItemDatabase.find({id: itemKey, name}) → buildingBenefits), the totals block(totalBenefits over installed items only) and a note.
2. Each toggle button → wire() : on click , preventDefault , flip the preference, persist() , then applyEntry (data-iw-vs-open , aria-expanded , title).
3. clearVillageLedger resets caches; stored preferences are kept.

Reads and writes. Catalogue records(read); skin nodes only (through VillageScene's own()); storage iw-village-ledger (only false values accepted and stored, L110 and L121).

Runs. On each scene render (signature-gated).

Resources held. One click listener per toggle button; a live rows array (reset on clear).

Safety statement.

Question	Answer
External data	Catalogue fields (numbers formatted, labels as textContent); snapshot strings via textContent
Sensitive information	The player's own building list, displayed locally
Dynamic code	No
Authentication state	No
Modifies app or game data	No. preventDefault only on its own buttons

Hardening. Clamp or format numbers defensively (for example ignore non-finite catalogue values), which also covers H-02.

Rewrite with internal components. A ledger component using server-computed benefit totals, if IdleWorlds exposes them.

J-29 Overlay framing (OverlayFramer)

Purpose; why JavaScript. Detect the game's full-screen modal layers by **rendered state** (portalled dialogs have no stable classes), and mark the scrim and its content card for the frame CSS.

Where. OverlayFramer.js (188 lines): SKIN_OWNED , ALREADY_FRAMED ; visible (56), area (64), isScrim (70), pickCard (101), tagScrim (121), untag (132), frameOverlays (145), clearOverlayFramer (185).

Flow.

1. frameOverlays() runs on every iw:dom-flush .
2. Revalidate the tagged scrims: untag those that are disconnected or no longer qualify.
3. Query candidates: [class*="fixed"] , and [style*="position: fixed"] or [style*="position:fixed"] . Skip skin-owned ones.
4. isScrim(el) requires:
 - computed position: fixed ;
 - z-index ≥ 20 ;
 - a box covering ≥85% of the viewport on both axes, starting within 15%;
 - visibility and descendants;
 - a backdrop blur, **or** a dark translucent background.
5. tagScrim sets data-iw-overlay="scrim" . pickCard (FUN-07) chooses the card; data-iw-overlay="panel" is added unless the card is skin-owned or already framed.
6. clearOverlayFramer removes all data-iw-overlay attributes.

Reads and writes. Computed style and geometry (read). One attribute on game nodes (write).

Runs. Every flush (PERF-03).

Resources held. A Set of tagged elements.

Safety statement. No external data beyond rendered state; no sensitive information (dialog contents are not read); no dynamic code; no authentication. **No modification** beyond the attribute. Dialog behaviour, focus traps and ARIA are the game's.

Hardening (preserves functionality).

- Skip the candidate query when no childList mutation added a fixed-position element since the last pass (cache by flush roots).
- Correct the comment or the behaviour of pickCard (FUN-07; **B-09** if the behaviour changes).

Rewrite with internal components. A Modal component variant with the frame styling.

5.5 PERMISSION AND EXPOSURE MINIMISATION

The extension already requests the minimum it needs (§5.2.1). The remaining exposure can be reduced without changing features:

Surface	Today	Recommendation	Effect on features
web_accessible_resources	assets/* plus redundant PNG globs; every packaged file is fetchable by game pages	List only the directories referenced at runtime (fonts, header, header/zones, inventory, quest-frames art, skills-ui atlases and buttons (not exact-v3), village, world-bosses art, and the three runtime index files). Consider use_dynamic_url: true [Verify: behaviour with CSS url() in injected styles before adopting]	None if the list is complete; extension-e2e.test.mjs would catch a missing path

Surface	Today	Recommendation	Effect on features
Packaged files	60.8 MB unreachable exact-v3 art; metadata JSON	Exclude in package-release.mjs (§4.3)	None
Host permissions	apex + www	Keep. Confirm www is still served [Verify]; drop it if it redirects to the apex host	None
Manifest	No minimum_chrome_version	"minimum_chrome_version": "111"	Older browsers cannot install, instead of rendering broken CSS
Network credentials	N1 sends cookies on the apex host	credentials: 'omit' for items.json (H-05)	None [Verify: the endpoint is public]
Authenticated read	N2 polls /api/player	Owner decision (SEC-04); H-14 backoff	See B-08
Event bus	iw:* on the page document	Private EventTarget or detail token (H-23)	None
Development secrets	A real browser profile in the working tree	Remove it from the tree (H-20, SEC-07)	None

5.6 FINDINGS REGISTER

Severity: High / Medium / Low / Info.

Confidence:

- **C** — confirmed by source reading, with the cited line;
- **P** — plausible from the source; a test is recommended to demonstrate it;
- **V** — needs verification against the live game or with the IdleWorlds team.

Result: no High-severity issue was found. Every Medium item is a policy, performance, accessibility or product-behaviour decision rather than an exploitable flaw.

Security and privacy

ID	Sev.	Conf.	Finding	Evidence	Impact	Action
SEC-01	LOW	C	Two catalogue fields reach innerHTML unescaped	TooltipEngine.js:101-102 → :353	Markup injection into the page (a script via inline handlers) only if items.json or the extension cache were hostile	H-01; H-02; H-03
SEC-02	LOW	C	Catalogue records are not type-validated (network and cache)	ItemDatabase.js:180-182, 205-221, 235-246	Defence-in-depth gap for SEC-01 and for malformed data	H-02
SEC-03	INFO	V	items.json fetched from an absolute apex URL with default credentials	ItemDatabase.js:13, 165	Cookies sent needlessly on apex; on www depends on CORS; degraded skin if blocked	H-05
SEC-04	MEDIUM	C+V	Authenticated /api/player polling (session cookies, league header), full core payload parsed	VillageScene.js:305-331	API use and server load outside the game's own client; needs owner approval; no auth or rate-limit backoff	Owner decision; H-14; B-08
SEC-05	INFO	C	Page-visible surface: forgeable iw:* events, readable marks, fetchable packaged files, extension detectable	DOMWatcher.js:45; manifest web_accessible_resources	Page scripts can cause extra work or detect the extension; no data exposure	H-19; H-23
SEC-06	LOW	C	MAIN-world probe of React internals; spoofable latch	hydration-signal.js:30-58	Early boot only; brittle across React upgrades	H-06; replace with a public signal (C3)
SEC-07	MEDIUM	V	Real browser profile (saved logins, autofill, wallet per .gitignore) and business documents inside the working tree	.gitignore; build-tools/header-live-profile/; output/	Credential leak if the folder is zipped, synced or shared during handover	H-20; rotate credentials if shared
SEC-08	INFO	C	Wiki slug HTML-escaped but not URL-encoded	TooltipEngine.js:203	Path-only variation on idleworlds.com	H-04

ID	Sev.	Conf.	Finding	Evidence	Impact	Action
SEC-09	INFO	C	Outbound third-party link injected into game navigation	UIFoundation.js:289-304	Product and trust decision; noopener noreferrer present	B-07

Functional

ID	Sev.	Conf.	Finding	Evidence	Impact	Action
FUN-01	MEDIUM	C+V	The game's "cycle XP display" button is disabled and hidden by CSS	skillpanel.css:1563-1580 ; skillcard-v2.css:164 ; contrast SkillPanelRenderer.js:380-399	A game control is unreachable (rule 1)	B-01; add an operability test
FUN-02	LOW	C	Item-name highlight registered with no ::highlight(iw-item-name) style	NameScanner.js:227-230 ; no rule in any sheet	Hover cards on prose have no visual affordance; scanning cost for an invisible effect	B-02
FUN-03	LOW-MED	V	Page-wide reduced-motion rule also stops the game's animations and transitions	base.css:693	Game activity indicators may freeze for reduced-motion users (rule 5)	B-03
FUN-04	LOW	C/P	Collapse spine cache survives teardown, so collapsed titles can disappear after a kill-switch round trip	CollapsibleFrames.js:91, 106-117, 284-295 ; collapsible.css:139	A collapsed bar shows only the chevron	H-08 + test
FUN-05	LOW	C	Undefined font tokens(--iw-font-flavour , --iw-font-body)	skillpanel.css:2840 ; ui-system.css:1478, 1701	Quest brief and boss reward names use unintended fonts	B-04
FUN-06	LOW	C/P	Village picker option art flip-flops between re-decorations	VillagePanels.js:224, 239-247	Icons appear and disappear on alternate passes	H-10 + test
FUN-07	LOW	V	Overlay card selection picks the innermost .panel (comment says outermost)	OverlayFramer.js:104-105	Frame on a sub-panel in nested modals	Decide intent; B-09
FUN-08	LOW	P	Candidate memo never invalidated	SkillPanelRenderer.js:852-869	Stale role resolution when the same card re-annotates with no other card in between	H-09 + test
FUN-09	LOW	C	Trigger teardown removes tabindex / aria-* / aria-describedby it may not own	QuestPanelRenderer.js:183-186 ; VillagePanels.js:155-158 ; TooltipEngine.js:332-336	Would delete native values if the game adds them (rule 3)	H-11
FUN-10	INFO	C	#iw-tip and page-lifetime listeners persist after teardown	TooltipEngine.js:415-428	Deviation from rule 3 wording; inert	H-12 or document
FUN-11	LOW	V	Alpha ignored when repainting navy surfaces	BackgroundPainter.js:53-59, 93-97	Translucent layers become opaque	B-05
FUN-12	LOW	V	Position-based fallbacks for team colour and zone action tone	header.css:902, 915 ; WorldBossPanels.js:77-103	Wrong state colour if the game reorders (rule 5)	B-13; stable hooks
FUN-13	INFO	C	A late hydration latch persists until teardown	HydrationGate.js:39-47 ; hydration-signal.js:58	Stray attribute	J-03 hardening

Accessibility

ID	Sev.	Conf.	Finding	Evidence	Impact	Action
A11Y-01	MEDIUM	V	Visible text supplied by CSS generated content over zero-size native text; mirrors not aria-hidden while native sections stay exposed	skillpanel.css L1325+ and quest reward chip; ui-system.css boss label; skillcard-v2-runtime-safe.css ; SkillCardDesignController.js:66-86, 372-439	Duplicate or mismatched announcements (e.g. "QUEUED" shown, "Prejoined" read)	B-10; screen-reader test
A11Y-02	LOW	C	Focusable <p> triggers without a role	QuestPanelRenderer.js:194-211 ; VillagePanels.js:143-153	Unclear semantics for assistive technology	B-12
A11Y-03	LOW	C	Fitted labels down to 4 px; 9 px names	SkillCardDesignController.js:184-187, 351	Illegible text for many users (the native accessible name remains)	B-11

Performance

ID	Sev.	Conf.	Finding	Evidence	Impact	Action
PERF-01	MEDIUM	C	Infinite decorative CSS animations for the page lifetime	<code>ui-system.css:435, 1530, 1572, 1630-1654, 1759</code>	Continuous paint and composite work	H-17
PERF-02	LOW	P	Full-document zone text scan on routes without the zone bar, twice per header reconcile	<code>HeaderRenderer.js:70-81, 92, 121</code>	Per-flush cost on Market, Village and similar routes	H-16; perf harness on those routes
PERF-03	LOW	Project record	Overlay candidate query over the whole document every flush	<code>OverlayFramer.js:145-183</code>	~48 ms of 225 ms at 13k nodes (recorded)	J-29 hardening
PERF-04	LOW	C	Background walk visits every element of each root	<code>BackgroundPainter.js:118-134</code>	Boot and layout-change cost	J-15 hardening
PERF-05	INFO	C	668 KB unminified bundle (48.6% CSS); 3,879 !important	build output; Appendix B	Parse and cascade cost	§3.13
PERF-06	LOW	P	Name-scan pointer fallback walks the hovered element's text subtree on misses	<code>NameScanner.js:284-316</code>	Per-frame cost while moving over large containers	H-15; perf harness with pointer movement
PERF-07	MEDIUM	C/V	Sprite atlases decode to ~126 MiB (gear 1280×19,584 px)	<code>assets/gear_icons_atlas.png</code> , <code>assets/item_icons_atlas.png</code>	Memory and GPU pressure, especially on phones	H-24

Content, maintainability and packaging

ID	Sev.	Conf.	Finding	Evidence	Impact	Action
CONT-01	MEDIUM	V	Skin-authored gameplay statements shown as if they were game UI	<code>WorldBossPanels.js:15-16, 286</code> ; <code>rewards.json</code> ; <code>TooltipEngine.js:130</code>	Players may be misinformed if the rules differ	B-06
CONT-02	LOW	V	Invented terminology and a relabelled game state ("Prejoined" → "QUEUED")	<code>WorldBossPanels.js:312-313</code> ; Dominion labels	Confusion with official terms	B-06
MAINT-01	—	C	CSS history layers and dead rules (350 proven dead declarations)	<code>tools/css-superseded.md</code> ; §3.4.4	Maintenance cost	§3.13 step 1
MAINT-02	—	C	Specificity escalation, (0,30,3); retired <code>:not()</code> hooks	<code>ui-system.css:1019</code>	Fragile cascade	§3.13
MAINT-03	—	C	Detection by English copy and utility-class substrings	§6.5	Breaks on copy, localisation or markup changes	§6.7 phase 1
MAINT-04	—	C	Dead or odd code: <code>--iw-header-frame-bar</code> ; <code>exact-v3</code> branch; ineffective <code>SKIP_TAGS</code> ; joined statements	<code>HeaderRenderer.js:151, 155, 232, 462</code> ; <code>SkillsArtService.js:178-181</code> ; <code>NameScanner.js:45-48</code>	Confusion	Clean up
MAINT-05	—	C	Generated files and non-rebuildable art sources	§1.8; C17, C24	Reproducibility	C24 removal
MAINT-06	—	C	Committed build output with a CI staleness gate	<code>ci.yml</code>	Merge friction (by design)	Keep for the extension
MAINT-07	—	C	Documentation drift D-01...D-10	§1.13	Misleading guidance	Fix docs
TEST-01	LOW	C	A regression suite tests a stale copy of the module's logic instead of the shipping code. The copied <code>textLeaves</code> lacks the <code>.fs-quest-sigil</code> exclusion; the copied <code>structureSignature</code> still includes <code>disabled</code> , which the source deliberately removed	<code>tests/quest-structure-signature.test.mjs</code> ("logic copied verbatim") vs <code>QuestPanelRenderer.js:57-64, 149-155</code>	The suite can pass while the real signature regresses, and it asserts an outdated contract	Extract the real functions from the source, as <code>skill-structure-signature.test.mjs</code> does (§7.6 T-09)
SIZE-01	MEDIUM	C	60.8 MB unreachable art plus metadata in the release	§1.8	Download size; web-accessible clutter	H-18
PKG-01	LOW	C	No icons; no <code>minimum_chrome_version</code> ; build target 109 vs effective 111	<code>manifest.json</code> ; <code>build.mjs:89, 101</code>	Store requirement; misleading support floor	H-18

5.7 HARDENING AND BEHAVIOUR-CHANGE PLAN

5.7.1 Hardening that preserves functionality (H)

Effort: S < ½ day, M ≈ 1–2 days, L > 2 days.

ID	Change	Findings	Effort	Verification
H-01	<code>esc()</code> the <code>craft_level</code> and <code>tier</code> interpolations	SEC-01	S	Unit test: a record with <code>craft_level: ''</code> renders literal text; no <code>img</code> element inside <code>#iw-tip</code>
H-02	Validate catalogue records (types, required fields) at <code>_index</code> , for network and cache	SEC-02	S-M	<code>data-services.test.mjs</code> : malformed records skipped with a count; valid records unchanged
H-03	Replace the tooltip and inventory <code>innerHTML</code> templates with DOM construction, or an auto-escaping template helper; make inventory <code>esc()</code> also escape <code>'</code>	SEC-01	M	Snapshot the rendered DOM before and after on the fixtures; the census shows 1 constant <code>innerHTML</code> (or 0)
H-04	<code>encodeURIComponent</code> and validate the wiki slug	SEC-08	S	Unit test with <code>../</code> and <code>?</code> in a slug
H-05	<code>credentials: 'omit'</code> for <code>items.json</code> ; consider a relative URL	SEC-03	S	Live check on apex and <code>www</code> [Verify]
H-06	<code>try/catch</code> in <code>hydration-signal.js</code> ; <code>signal('unknown')</code> on error	SEC-06	S	<code>hydration-gate.test.mjs</code> : a throwing getter yields "unknown"
H-07	Register the kill-switch listener before the hydration wait, or re-read after it	J-01	S	Test: a toggle during the wait is honoured
H-08	Reset <code>spines</code> on collapse teardown	FUN-04	S	<code>collapsible-frames.test.mjs</code> : disable → enable → collapse keeps the title visible
H-09	Invalidate the <code>textCandidates</code> memo per annotate pass	FUN-08	S	<code>skill-structure-signature.test.mjs</code> : re-annotate the same card after a structural change
H-10	Read option names from text nodes, ignoring skin children	FUN-06	S	<code>village-panels.test.mjs</code> : two re-decorations keep the art
H-11	Attribute ownership for <code>tabindex</code> / <code>aria-*</code> written on game nodes (record native values, restore on teardown)	FUN-09	M	Test: a pre-existing <code>aria-describedby</code> survives open, close and teardown
H-12	Remove <code>#iw-tip</code> on teardown, recreate on boot (listeners stay bound once)	FUN-10	S	Smoke round trip: no <code>#iw-tip</code> while disabled; exactly one after re-enable
H-14	Village API backoff: stop on 401/403 until a route change; honour 429 and <code>Retry-After</code> ; skip while the frame is collapsed; check the JSON content type	SEC-04	M	<code>village-scene.test.mjs</code> with mocked statuses
H-15	Restrict the name-scan fallback walk (caret API present, or size cap)	PERF-06	S	Perf harness with synthetic pointer movement
H-16	Avoid the full-document zone scan when the zone bar is absent	PERF-02	S	Perf harness on a Market-shaped page
H-17	Pause decorative loops when off-screen or the tab is hidden (for example <code>animation-play-state: paused</code> via a <code>data-iw-offscreen</code> flag)	PERF-01	M	Perf harness; <code>zone-control-motion.test.mjs</code>
H-18	Packaging: exclude <code>exact-v3</code> and metadata; add icons, <code>minimum_chrome_version: "111"</code> , esbuild <code>chrome111</code>	SIZE-01, PKG-01	S	<code>npm run package</code> ; <code>extension-e2e.test.mjs</code>
H-19	Narrow <code>web_accessible_resources</code>	SEC-05	S	<code>extension-e2e.test.mjs</code> plus fixture screenshots
H-20	Remove the real browser profile from the working tree; use a test account for live audits	SEC-07	S	Manual
H-22	Early-return page-lifetime listeners while the runtime is inactive	FUN-10	S	Smoke: no listener work while disabled
H-23	Accept only skin-created event details (private <code>EventTarget</code> or <code>WeakSet</code> token)	SEC-05	S	Test: a forged <code>iw:skill-panel</code> from the page is ignored

ID	Change	Findings	Effort	Verification
H-24	Split or recompress the sprite atlases (no visual change)	PERF-07	M-L	Sprite-window audit; fixture screenshots; memory measurement

(H-13 was reclassified as B-05 because it changes appearance.)

5.7.2 Changes that alter the feature (B)

These need a product or design decision before implementation.

ID	Change	Findings	Decision owner
B-01	Make the XP-cycle control reachable again (style the native button inside the V2 medallion, or show it on hover/focus)	FUN-01	Design and IdleWorlds
B-02	Add a visible item-name highlight, or remove prose scanning	FUN-02	Design
B-03	Scope reduced-motion rules to skin-owned selectors	FUN-03	Accessibility
B-04	Fix the font tokens (the quest brief and reward names change font)	FUN-05	Design
B-05	Preserve translucency when repainting navy surfaces	FUN-11	Design
B-06	Replace skin-authored boss copy with game-provided text; keep the "Prejoined" wording	CONT-01, CONT-02	IdleWorlds
B-07	Remove, relocate or approve the Toolkit link	SEC-09	IdleWorlds
B-08	Remove the <code>/api/player</code> read (Village route data only), or replace it with an approved endpoint	SEC-04	IdleWorlds
B-09	Frame the outermost modal panel	FUN-07	Design [Verify live]
B-10	One accessible source for replaced text (real text or <code>aria-hidden</code> generated content)	A11Y-01	Accessibility
B-11	Raise the minimum fitted label size; allow wrapping or ellipsis	A11Y-03	Design
B-12	<code>role="button"</code> on focusable item triggers	A11Y-02	Accessibility
B-13	Leave Zone Control native when team colours cannot be read, instead of guessing	FUN-12	Design

5.7.3 Recommended order

- Before any wider distribution:** H-20 (secrets out of the tree), H-01, H-18, and owner decisions on SEC-04/B-08, CONT-01/B-06 and SEC-09/B-07.
- Next maintenance release:** H-02, H-04, H-05, H-06, H-07, H-08, H-09, H-10, H-12, H-14, H-16, H-22, H-23, with the tests listed.
- Performance pass:** H-15, H-17, H-24, plus the CSS cleanup (§3.13 steps 1–4).
- Design and accessibility pass:** B-01, B-02, B-03, B-04, B-10, B-11, B-12.
- Official integration:** Chapter 6 §6.7. Most findings disappear with the classifiers.

5.8 CONCLUSION OF THE SAFETY REVIEW

What the evidence shows the JavaScript does not do:

- no remote or dynamic code (census: no `eval`, `Function`, dynamic `import()`, script creation, or string timers);
- no third-party network requests at runtime (all five `fetch` sites listed);
- no access to cookies or other sites' data;
- no messaging or globals;
- no synthetic clicks, form submissions or game action requests;

- no persistence of personal data beyond panel preferences.

Game text, including other players' chat, is handled as text: never parsed as HTML, never used in URLs, never stored.

Genuine risks and uncertainties, disclosed above and none of them High severity:

- **Two unescaped catalogue fields in one HTML template (SEC-01).** Low likelihood, because the data source is the game's own server; trivial to fix.
- **An authenticated polling read of the game's player API (SEC-04).** Carefully constrained, but it needs the owner's approval.
- **Development-environment secret material in the working tree (SEC-07).**
- **Items that alter game behaviour, legibility or accessibility.** Content, behaviour and accessibility decisions: CONT-01, FUN-01, FUN-03, A11Y-01.

Verified live? No. Claims about live-page behaviour, CSP handling, server responses and assistive technology output are labelled [**Verify**] and must be confirmed on the live game (§7.8).

6. INTEGRATION AND MAINTENANCE GUIDE

This chapter covers:

- installing and running the skin;
- options for integrating it with IdleWorlds;
- start-up order and dependencies;
- every configuration constant and customisation procedure;
- **every assumption** the skin makes about the game;
- how game updates break it, and how to diagnose and fix that;
- troubleshooting;
- the long-term migration path;
- how to prove that a change preserves visuals and functionality.

6.1 INSTALLATION AND SETUP

6.1.1 Prerequisites

For	Requirement
Running the skin	Chrome (or Chromium) 111 or newer (§1.10). Other browsers are unsupported
Building and testing	Node.js (Cl uses 24); <code>npm ci</code> ; <code>npx playwright install chromium</code> once
Rebuilding art (optional)	Python 3 with imaging libraries [Verify packages] , Windows PowerShell, and the sibling art repositories (C24)

6.1.2 Load the extension

From the repository (developer):

1. `npm ci`
2. `npm run build` (or `node build-tools/build.mjs --allow-missing-assets` if the vendored atlases are absent)
3. Open `chrome://extensions`, enable **Developer mode**, then **Load unpacked** → select the repository root (the folder containing `manifest.json`).
4. Open `https://idleworlds.com` and log in.

From a release zip:

1. `npm run package` writes `release/idleworlds-fantasy-skin-v<version>.zip`.
2. Unzip it into a folder.
3. **Load unpacked** → that folder.

The Chrome Web Store needs a manifest icon set before submission (PKG-01).

6.1.3 Development loop

<code>npm ci</code>
<code>npm run build</code>
<code>npm test</code>
<code>npm run fixtures</code>

- After editing `src/`, run `npm run build`, **commit `dist/content.bundle.js` together with the change** (CI fails on a stale bundle), then click the reload icon for the extension in `chrome://extensions` and refresh the game tab.
- `npm test` rebuilds with `--allow-missing-assets`, syntax-checks the bundle, runs all 47 suites and the sprite-window audit. It takes about 3.5 minutes on the inspection machine [[Test run](#)].
- A single suite runs with `node tests/<name>.test.mjs`. Most suites load `dist/`, so **build first**.
- `npm run vendor` is needed only when `ASSET_REVISION` changes (`build-tools/vendor-assets.mjs:25`).

6.1.4 Kill switch

The skin can be removed from, and restored to, the open page without a reload. There is no UI for it (C15).

1. On the IdleWorlds tab, open **DevTools → Console**.
2. In the **JavaScript context selector** (top-left of the Console, normally "top"), choose the **IdleWorlds Fantasy Skin** content-script context.
3. Run one of:

```
chrome.storage.local.set({ 'iw-skin-enabled': false }) // full teardown
```

```
chrome.storage.local.set({ 'iw-skin-enabled': true }) // rebuild
```

Expected results:

After	Console shows	Page state
false	[IW Fantasy Skin] disabled – active presentation removed	Removed: all 7 <code>style[data-iw-style]</code> , every <code>data-iw-*</code> and <code>data-fs-*</code> mark, skin nodes, and owned inline styles (restored to native). <code>#iw-tip</code> remains, hidden (FUN-10)
true	[IW Fantasy Skin] v1.6.0 booted	Skin rebuilt

Notes:

- The value persists across reloads and tabs. Remember to set it back to `true`.
- `chrome.storage.local.remove('iw-skin-enabled')` also enables the skin (a missing key means enabled).
- A change made during the first ≤ 4 s of a page load may be missed (H-07).

6.1.5 Confirming the skin is active

Run in the **page** context (the default "top" context):

```
{
  styles: [...document.querySelectorAll('style[data-iw-style]')].map(s => s.dataset.iwStyle),
  theme: document.documentElement.dataset.iwZoneTheme,
  design: document.documentElement.dataset.iwSkillCardDesign,
  frames: document.querySelectorAll('[data-iw-ui="section-frame"]').length,
  skillCards: document.querySelectorAll('.fs-skill-panel').length,
  inventoryRows: document.querySelectorAll('.fs-inv-row').length,
}
```

Expected:

- `styles`: `["base", "tooltip-engine", "inventory", "skillpanel", "header", "overlay", "ui-system"]`;
- `theme`: a theme name, or `"default"`;
- `design`: `"new"`;
- non-zero counts on the matching routes.

The console also logs `[IW Fantasy Skin] boot after page hydration: 1 (...ms)`. A state of `timeout` or `unknown` means the hydration probe did not work (§6.6).

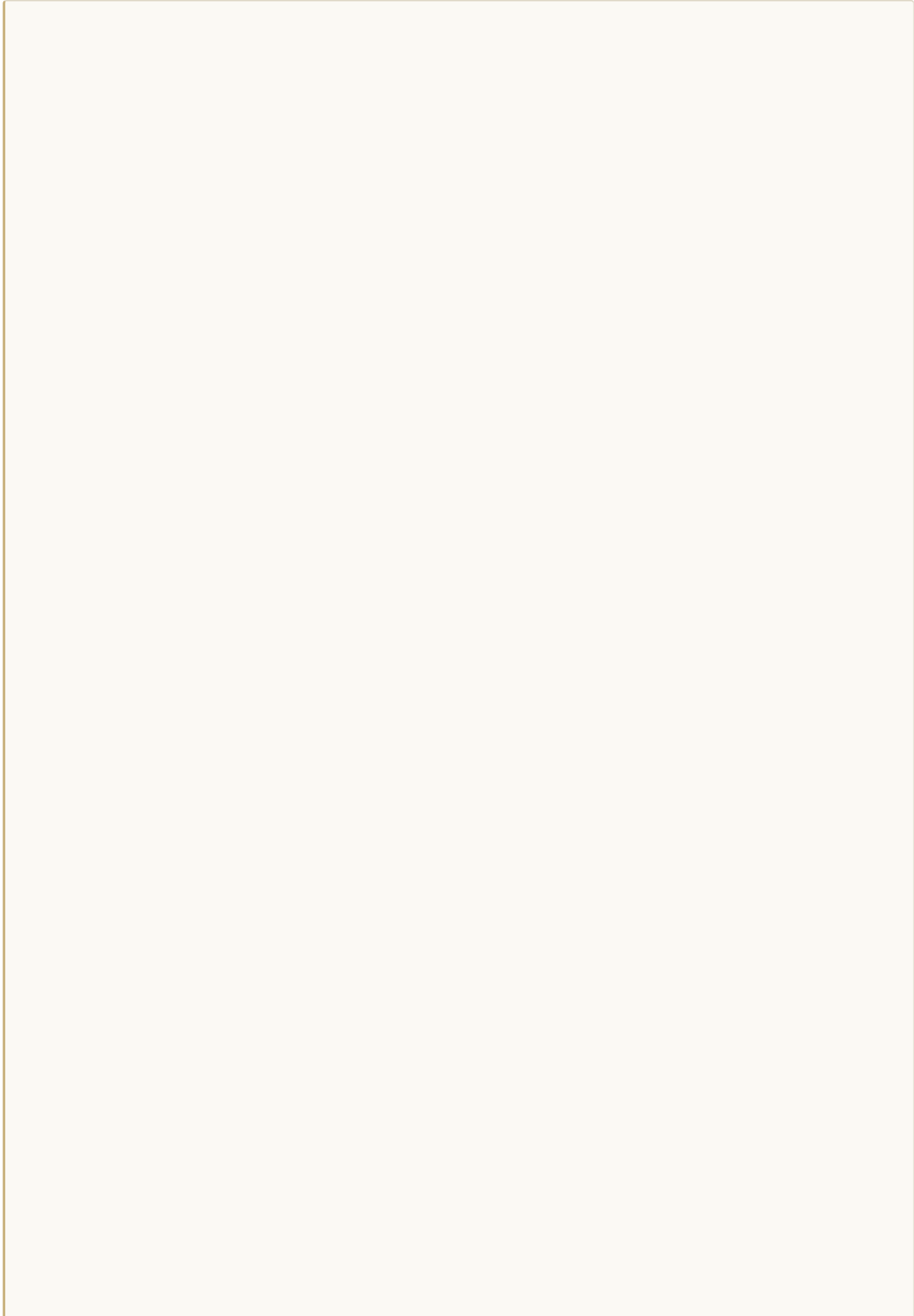
6.2 INTEGRATION OPTIONS

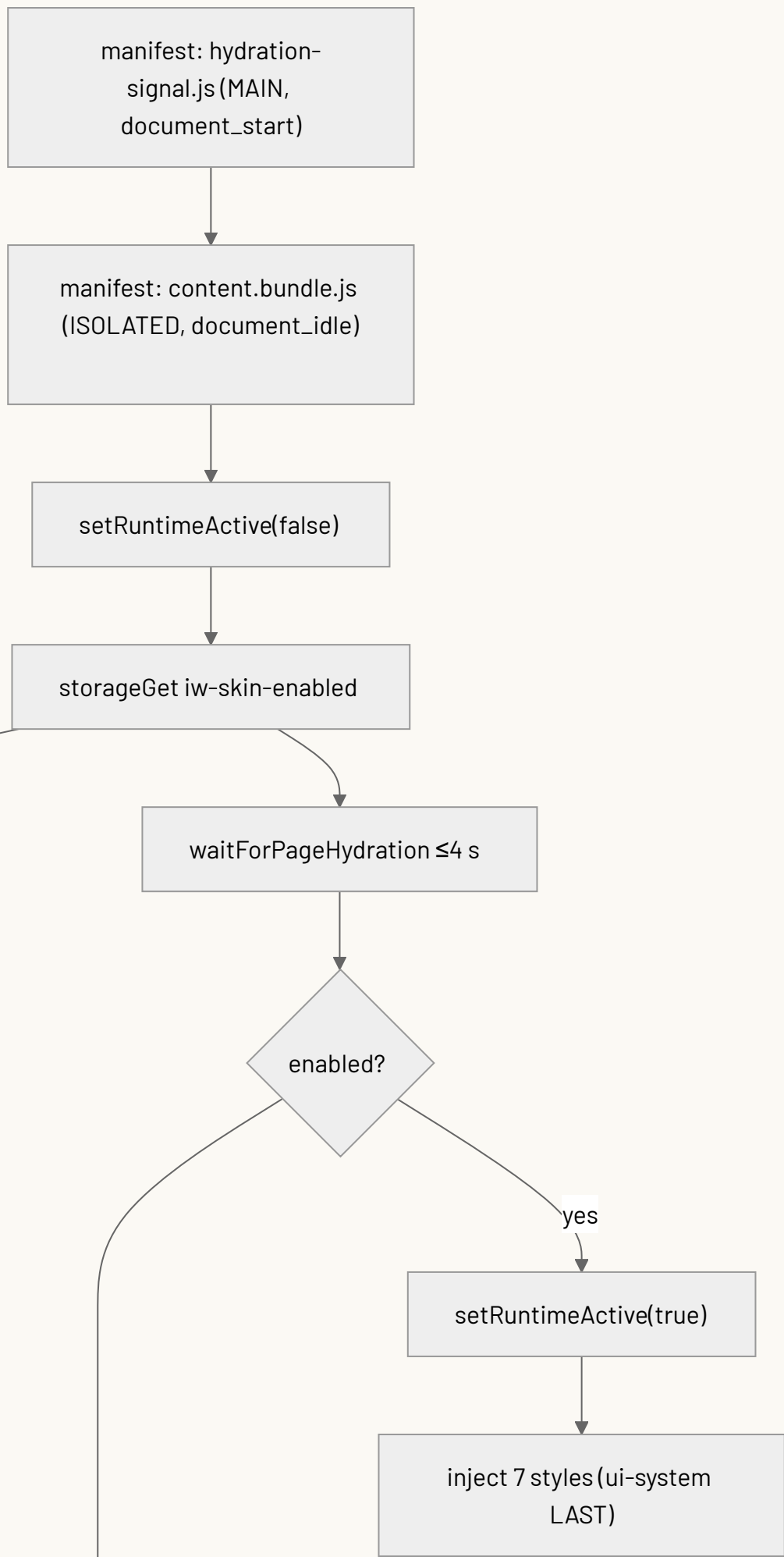
Option	What it is	Pros	Cons	When to choose
A. Keep the extension, with light cooperation from IdleWorlds	The skin stays a Chrome extension. IdleWorlds adds stable hooks (<code>data-*</code> roles on panels, cards, controls), a public hydration signal , and optionally an approved village endpoint	Minimal game work; removes the biggest fragilities (C3, C5, C6, SEC-04)	Still a runtime overlay (C2, C7, C9); Chrome-only; players must install it	Community theme, opt-in
B. Official opt-in theme loaded by the game	The game ships the stylesheets and a trimmed runtime as a selectable theme	Available to all players and browsers the game supports; no extension permissions	Keeps the classifier approach inside the app, which is not idiomatic for React; the runtime adapters must change: • <code>chrome.storage</code> → app settings; • <code>assetUrl</code> → CDN paths; • no hydration gate; • no kill switch → theme toggle.	A stepping stone while components are migrated
C. Port into the IdleWorlds codebase (recommended long-term)	Theme tokens and component variants implement the look. The classifiers, observer and overlays are deleted	Idiomatic, fast, accessible, testable; removes most findings	The largest effort (§6.7)	The skin becomes the official look

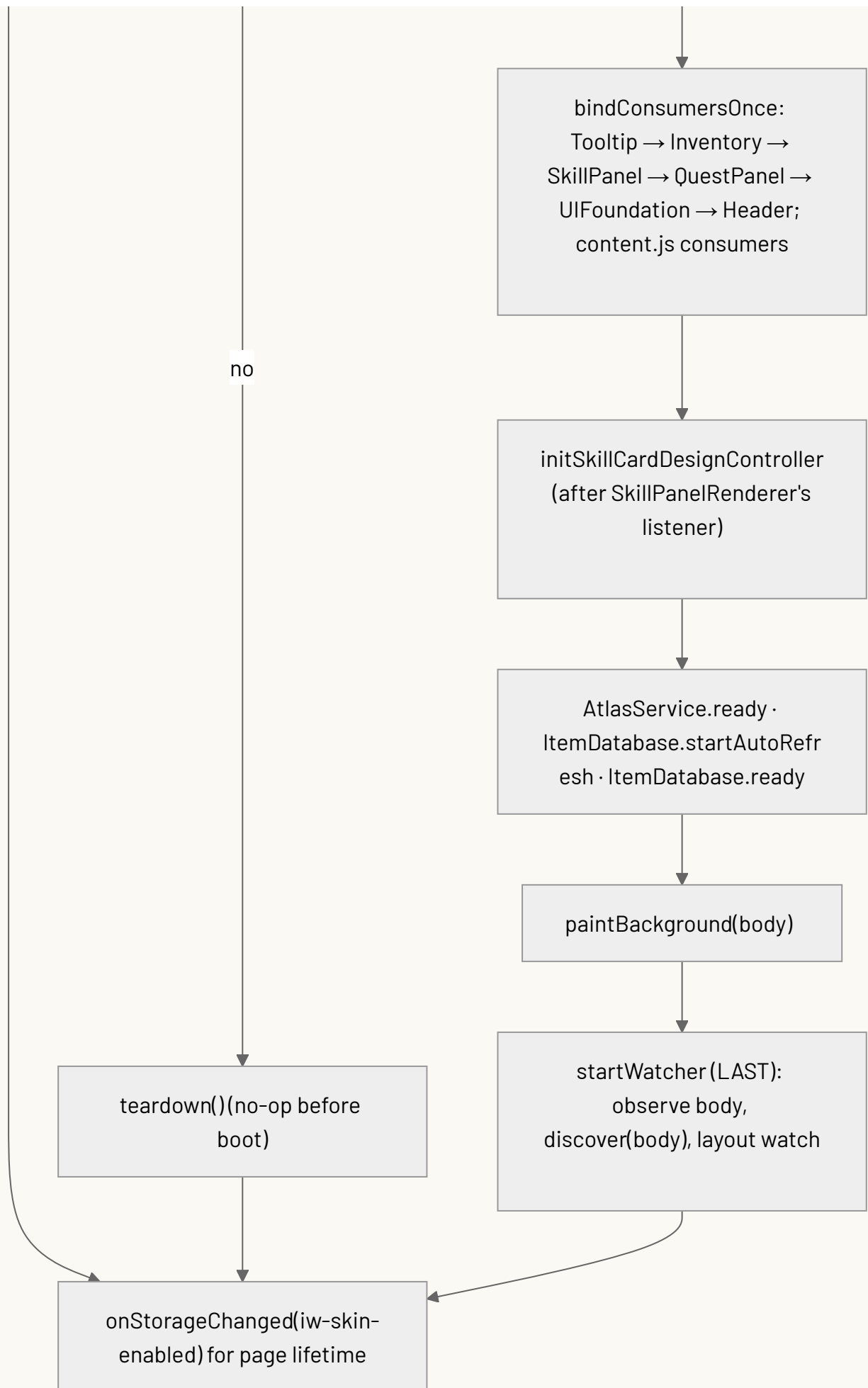
Option B adapter sketch (only the runtime seams change):

Seam	Extension implementation	Option B replacement
Storage	<code>Runtime.storageGet/Set/Remove</code> , <code>onStorageChanged</code> (<code>chrome.storage.local</code>)	App settings store (or <code>localStorage</code> with a prefixed namespace)
Assets	<code>Runtime.assetUrl</code> → <code>chrome.runtime.getURL</code>	<code>new URL(path, CDN_BASE) ;</code> <code>StyleInjector.rewriteAssetUrls</code> points to the CDN
Styles	<code>StyleInjector</code> runtime <code><style></code> elements	Bundle the CSS in the app build; drop the injector
Boot	<code>content.js</code> <code>start()</code> with hydration gate	Call <code>boot()</code> from a top-level <code>useEffect</code> after hydration; <code>teardown()</code> on theme switch
Village data	<code>VillageScene</code> fetch	Read from the app's player store
Kill switch	Storage listener	Theme setting

6.3 INITIALISATION ORDER AND DEPENDENCIES



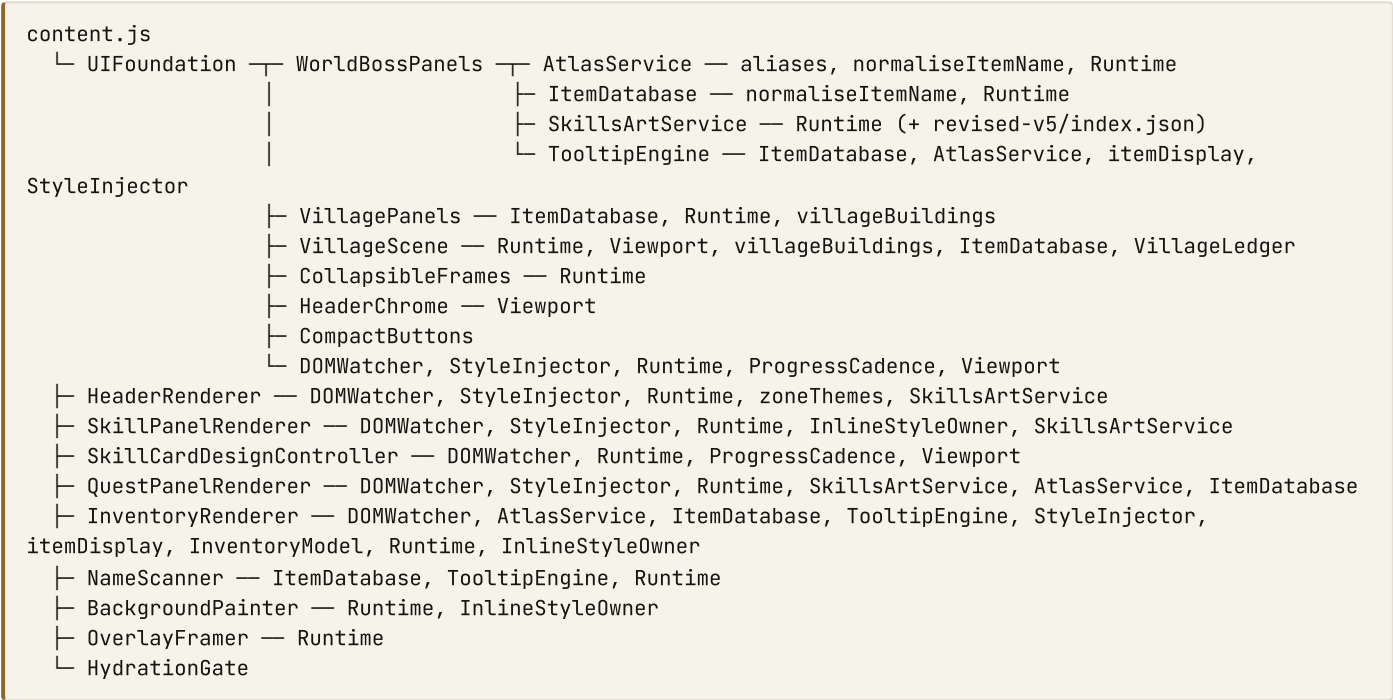




Hard ordering constraints (breaking one causes a real defect):

Constraint	Why	Enforced by
Hydration signal before the first boot	React #418 when appending mid-hydration(C3)	start() awaits the gate
setRuntimeActive(true) first; setRuntimeActive(false) last in teardown	guard skips work while inactive, and teardown's own guarded cleanup must still run	content.js
ui-system injected last	Its generic rules must win ties	static-invariants.test.mjs
Consumers bound before startWatcher	The initial discovery flush would otherwise be lost	content.js comment; smoke.test.mjs
SkillPanelRenderer listener before QuestPanelRenderer and SkillCardDesignController	The quest renderer rejects fs-skill-panel cards; V2 requires the class the renderer adds	Registration order in bindConsumersOnce and boot
Classify pass order(12 steps, §1.5.4)	Later steps read earlier marks (collapse needs frames; compact buttons need nav, zone and inventory marks; header chrome needs nav and zone bar)	UIFoundation.queueClassify
iw:item-db-updated / iw:atlas-updated consumers are idempotent	Data can arrive at any time, including while disabled	guard + isRuntimeActive

Module dependency layers (imports only flow downward):



There are no import cycles [Generated: tools/js-api-census.md "Import graph": 35 runtime files, 0 cycles].

6.4 CONFIGURATION AND CUSTOMISATION

6.4.1 Constants

None of these are user settings. Change them in source, rebuild, and run the listed tests.

Constant	File:line	Value	Effect	Safe range and notes	Tests
ENABLED_KEY	content.js:51	'iw-skin-enabled'	Kill-switch storage key	Changing it resets everyone to enabled	smoke

Constant	File:line	Value	Effect	Safe range and notes	Tests
VERSION	content.js:52	'1.6.0'	Logged version	Must equal manifest.json and package.json (enforced by npm run package)	package step
HYDRATION_GATE_ENABLED	HydrationGate.js:21	true	Wait for hydration	false = immediate boot (risk of #418)	hydration-gate
TIMEOUT_MS / POLL_MS	HydrationGate.js:23-24	4000 / 50	Maximum wait / poll interval	2000-8000 / 25-100	hydration-gate
GIVE_UP_MS	hydration-signal.js:26	15000	Signal polling limit	≥ TIMEOUT_MS	hydration-gate
FLUSH_BUDGET	DOMWatcher.js:42	60	Elements per animation frame	30-150; higher means longer frames on bursts	flush-quiescence, smoke
SEL_INV_ROW / SEL_SKILL_PANEL	DOMWatcher.js:29-30	.compact-row, [class*="item-row"] / .compact-panel	Which game nodes are observed as rows and cards	Must match the game's markup	many
BREAKPOINTS	Viewport.js:66	640, 768, 1024, 1280, 1536	Layout epoch triggers	Must match the game's Tailwind breakpoints	viewport-swap
API_URL	ItemDatabase.js:13	https://idleworlds.com/items.json	Catalogue source	See SEC-03	data-services
CACHE_MAX_AGE_MS / REFRESH_RETRY_MS	ItemDatabase.js:20-21	6 h / 15 min	Cache freshness / retry	≥1 h / ≥5 min to limit load	data-services
SHOW_DELAY / EDGE_GAP / EDGE_MARGIN	TooltipEngine.js:35-37	120 ms / 0 / 10 px	Hover delay / trigger-card gap / viewport margin	EDGE_GAP must stay 0: the hover handoff relies on the surfaces touching	tooltip-virtual-anchor
MIN_NAME_LENGTH	NameScanner.js:43	3	Minimum matched name length	Lower values cause false matches	smoke
PROGRESS_EPSILON / PROGRESS_LEAD_BIAS / PROGRESS_DURATION_EPSILON_MS	ProgressCadence.js:22, 35, 40	0.05 / 1.12 / 15	Ignore tiny changes / lead factor / write threshold	Bias 1.0-1.2 (lower lags, higher overshoots)	action-progress-theme, skill-card-v2-render
LABEL_CEILING_PX / LABEL_FLOOR_PX	SkillCardDesignController.js:184, 187	9.5 / 4	V2 action label font range	Raise the floor for legibility (B-11)	skill-card-v2-render
LONG_ACTION_SECONDS	SkillCardDesignController.js:244	11	Countdown threshold	—	skill-action-countdown
ACTION_CAP_PX	SkillsArtService.js:58	60	Three-slice cap width in atlas pixels	Measured from the art; change only with new art	quest-command-width, sprite audit
ZONE_SURFACE_MAX	HeaderRenderer.js:58	34	Highest zone with header art	Bump when new zone art is imported	zone-headers
NAV_LABELS	UIFoundation.js:25	game, market, leaderboards, village, dungeon	Route tabs styled	Add new routes here	header-compact, menu-rows
TOOLKIT_URL	UIFoundation.js:289	https://idleworldstoolkit.com	Outbound link	B-07	header-compact
SECTION_FRAME_NAMES	UIFoundation.js:439	Regex of panel headings	Orphan heading frames	Add panels here	panel-frame-nesting
DAILY_BOOST	UIFoundation.js:1325	Regex	Boost chip detection	Update if the copy changes	collapsible-frames

Constant	File:line	Value	Effect	Safe range and notes	Tests
REFRESH_MS / RETRY_MS / ROUTE_REFRESH_MS	VillageScene.js:14, 15, 17	5 min / 30 s / 10 s	Village API cadence	Keep ≥ current values (server load)	village-scene
STORE_KEY	CollapsibleFrames.js:29; VillageLedger.js:45	iw-collapsed-frames; iw-village-ledger	Preference keys	Changing them loses saved choices	collapsible-frames, village-ledger
GAME_NAVIES	BackgroundPainter.js:19	20 hex colours	Colours repainted	Add the game's new surface colours	smoke
BUNDLE_SOFT_BUDGET	build-tools/build.mjs:117	300,000 bytes	Build warning threshold	Advisory only	—
ASSET_REVISION	build-tools/vendor-assets.mjs:25	sprite repo commit	Pinned atlas source	Bump, then <code>npm run vendor</code> , then commit the assets	data-services

6.4.2 Customisation procedures

Change the palette.

- Edit the `--iw-th-*` defaults in `base.css` :root (L74–193), or a theme block (L216–318).
- Derived edges update automatically through `color-mix` (L320–368).
- Check contrast (§7.5) and run `npm run fixtures`.

Add or re-map a zone theme (a new zone number, or a zone moving to another theme):

1. Update `assets/skills-ui/zone-theme-map.json` in the sprite repository.
2. Run `npm run import:skill-themes`. This regenerates `src/modules/zoneThemes.js` and copies `theme_<t>.webp`, `panel_corners_<t>.webp` and `separator_flourish_<t>.webp`.
3. A **new theme name** also needs:
 - a palette block in `base.css`;
 - `card-button-atlas.css` windows(rerun `build-tools/build-card-buttons.mjs` with the new art);
 - revised-v5 and compact-ghost-v3 atlases for the theme (and its name in `revised-v5/index.json` → `themes`, or the unreachable `exact-v3` branch is taken, and it has no art);
 - `--fs-quest-frame-position` in `skillpanel.css` L2706–2735.
4. Test: `zone-themes`, `revised-button-atlas`, `compact-button-atlas`, `card-button-art`.

Add header art for a new zone.

1. Run `npm run import:zone-headers` and `npm run import:zone-headers:mobile` (or `npm run art:zones` for placeholders).
2. Raise `ZONE_SURFACE_MAX`.
3. Test: `zone-headers`.

Support a new skill discipline. Five places, per the project record (`docs/traps/classification.md`):

1. `DOMWatcher.js`: `SKILL_IDENTITY_ALIASES`, plus the verb and anchored-label branches in `detectSkillType`.
2. `SkillPanelRenderer.js`: a `SKILL_META` row(`actions` exact button texts; `titleActions` title prefixes).
3. `skillpanel.css`: an `accent(.fs-skill--<type>)`.
4. `QuestPanelRenderer.js`: a `DISCIPLINE_STYLE` row.
5. The test fixture maps(`SKILL_GLYPHS`, `SKILL_ART`, `SKILL_LEVELS`).

Optionally:

- a V2 action icon: `npm run import:action-icons` with updated artwork, plus a per-type rule in `skillcard-v2.css` L376–385;
- `SkillsArtService.ICON_ALIASES` for the medallion.

Test: `skill-card-system`, `skill-card-v2*`, `quest-card-detection`, `smoke`.

Frame a new game panel.

- Panels rendered as `.panel` are framed automatically.
- A heading-only panel needs its name in `SECTION_FRAME_NAMES`.
- Add a collapse key only if the default title key is unsuitable.
- Test: `panel-frame-nesting`, `collapsible-frames`.

Opt a new self-painting control out of the generic button rule. Add its hook to the `:not()` chain of `ui-system.css:1019`, `:1033` and `:1065`, and to the matching `base.css` chains (L428–552). Never raise specificity instead (§3.2.2).

Add a stylesheet. Prefer appending to an existing injection group: the smoke test pins the count at 7. If a new sheet is unavoidable:

1. import it in `content.js` (or `UIFoundation.injectUIFoundationStyles`) at the right cascade position;
2. update `tests/static-invariants.test.mjs` and `tests/smoke.test.mjs`;
3. update the `INJECTION` map in `tools/css-inventory.mjs` and `FILE_ORDER` in `tools/css-superseded.mjs` (both fail otherwise).

Add a test suite. Append it to `test:run` in `package.json` by hand, or it never runs in CI [[Source](#): `CLAUDE.md`].

6.5 ASSUMPTIONS ABOUT THE APPLICATION

Every item below is something the skin **relies on** in the game. When one changes, the corresponding surface usually falls back to native (fail-safe). The **Detection** column says how the break shows up.

6.5.1 Platform and framework

#	Assumption	Used by	If it changes	Detection
A1	React root exposes <code>__reactContainer\$....</code> → <code>stateNode.current.memoizedState.isDehydrated</code>	<code>hydration-signal.js</code>	Boot waits 4 s ("timeout") or reports "unknown"	Console hydration line
A2	Tailwind breakpoints 640/768/1024/1280/1536	<code>Viewport.js</code> , CSS media queries	Hidden-column mismatches; layouts switch at the wrong widths	<code>viewport-swap</code> ; mobile audit
A3	The panel stack is duplicated(<code>hidden xl:grid / xl:hidden</code>)	<code>Viewport</code> , all classifiers	If removed: no harm. If a third copy appears: the wrong copy may be decorated	Visual check at 1279/1280 px
A4	The game's design tokens are named <code>--background</code> , <code>--card</code> , <code>--primary</code> , ...	<code>base.css</code> L74+	Unclassified components keep stock colours	Visual
A5	Progress fills use inline <code>width: N%</code> (or <code>aria-valuenow</code>)	<code>UIFoundation</code> progress, V2 fill, CSS <code>[style*="width"]</code>	No smoothing; fill band lost	<code>action-progress-theme</code>
A6	The game fetch client adds <code>x-idleworlds-league</code> from a <code>/ssf</code> path prefix	<code>VillageScene</code>	Wrong league's village, or an error	Village scene on an SSF character

6.5.2 Markup hooks

#	Hook	Used by	If it changes
A7	<code>.panel</code>	Section frames, overlay card choice, village and boss roots	Frames missing

#	Hook	Used by	If it changes
A8	<code>.compact-panel</code>	Skill, quest, boss and village cards; <code>base.css</code> ground; 561 CSS rules	Cards unskinned
A9	<code>.compact-row</code> , <code>[class*="item-row"]</code>	Inventory rows, market rows, row ground	Inventory overlay missing
A10	<code>#current-action-panel</code>	Activity panel fallback; V2 countdown host	Countdown and progress styling missing
A11	<code><header></code> with player text; <code>h1.header-player-name</code> ; <code>.min-w-0.overflow-hidden</code> ; <code>button.header-icon-btn</code> ; <code>.grid.grid-cols-2</code> / <code>.stat-chip</code>	<code>HeaderRenderer</code>	Header unskinned or partly skinned
A12	<code>.feed-panel</code>	Chat and log background CSS	Stock background
A13	Tailwind class substrings: <code>hover:underline</code> , <code>chat-name-</code> , <code>decoration-dotted</code> , <code>rounded-*</code> , <code>opacity-30</code> , <code>px-</code> , <code>border</code> , <code>tab</code> , <code>text-left</code> , <code>flex-1</code> , <code>bg-ember/orange/primary/accent</code> , danger or warning text hues	<code>base.css</code> / <code>ui-system.css</code> exclusions; button state; requirement state; market	Player names gain plates; wrong primary button state; unmet requirements not red
A14	Active state via <code>aria-selected</code> , <code>aria-current="page"</code> , <code>aria-pressed</code> , <code>data-state="active"</code> , or active-looking classes	Nav, filters, loadout toggle	Active tab or filter not highlighted
A15	Lucide <code><svg></code> inside pager buttons	V2 pager CSS	Double chevrons
A16	Portalled dialogs are <code>position: fixed</code> , <code>z-index ≥ 20</code> , viewport-covering, with blur or a dark translucent background	<code>OverlayFramer</code>	Modal frames missing
A17	Card inner structure (identity, title, readout, buttons, material line, requirement line, reward line in one card)	<code>SkillPanelRenderer.annotateStructure</code>	<code>three-zone</code> not applied; V2 incomplete
A18	Header chrome sibling layout (nav, optional notice, zone bar with exactly 2 children)	<code>HeaderChrome</code>	Unmerged header (fail-safe)
A19	Village slot card shape (head row first, picker second; names as direct <code><p></code>)	<code>VillagePanels</code>	Slot art and roles missing

6.5.3 English copy

#	Text pattern	Used by
A20	Panel headings: "Inventory", "Market", "Leaderboards", "Quests", "World Bosses", "Village", "Village Add-ons", "Salvaging", "Skill Actions"/"Actions", "Zone Selector", "Current Action", "Action Log", "World Chat"	Frames, activity panels, Village, bosses, market, <code>VillageScene</code> anchor
A21	Nav labels: Game, Market, Leaderboards, Village, Dungeon	<code>NAV_LABELS</code>
A22	Zone bar: "Zone N:", "Zones", "Next zone", "Change Zone"	Zone bar, compact buttons, header zone
A23	Header: "Combat Lv N", "Players online: N", "ATK N ... DEF N ... HP N", "IdleWorlds"	<code>HeaderRenderer</code>
A24	Skill verbs: Fight, Mine, Prospect, Cut, Smelt, Forge, Brew, Enchant, Tailor, Sew, Weave, Fish, Chop, Craft Parts, Build, Craft, Gather, Harvest; identities: "coming soon"/"upcoming skill"	<code>DOMWatcher.detectSkillType</code>
A25	Readout "Lv N - X% • ... to go"; "Base reward: "; "N/M" material counts; "Requires ..."	<code>SkillPanelRenderer</code>
A26	Quests: "Reward: ", "Turn In", "Skip (n)", "N% complete", "Work Order"	<code>QuestPanelRenderer</code> , <code>DOMWatcher</code>
A27	Inventory: "Equip", "Equipped", "Unequip", "List", filter names (All, Gear, Materials, Consumables, Drops), "Prev"/"Previous"/"Next", "n / m"	<code>InventoryRenderer</code>
A28	Activity: "View All", <code>[\\d,.] + xp / hr</code> , <code>hh:mm:ss</code> timestamps, "system", "queued", placeholder "Message World Chat", "Send"	<code>UIFoundation</code>
A29	Boost: "Daily XP Boost ... +N% XP"	<code>classifyDailyBoost</code>
A30	Bosses: boss names, "Prejoin"/"Prejoined", "controls Zone N", "Race to capture", "contested", team colour words	<code>WorldBossPanels</code>
A31	Village: "Current tier: N", building and house names matching <code>villageBuildings.js</code> , action verbs (Install, Upgrade, Uninstall, Destroy, Cancel)	<code>VillagePanels</code> , <code>VillageScene.readHousingPerks</code>

#	Text pattern	Used by
A32	Timer formats <code>h:mm:ss</code> , <code>m:ss</code> , <code>1h 2m 3s</code>	V2 countdown

6.5.4 Data contracts

#	Contract	Used by
A33	<code>items.json</code> : { generatedAt, items: [{ item_id, name, category, subcategory, tier, effects_raw, acquisition_type, acquisition_summary, acquisition_detail, craft_skill, craft_level, source_zone, drop_rate, drop_boosted_by, req_text, req_level, req_skill, wiki_slug, base_value, work_order_turn_in_gold, work_order_turn_in_note, stat fields... }] };ETag/Last-Modified	ItemDatabase, itemDisplay, TooltipEngine, WorldBossPanels, VillageLedger (validated manually by <code>npm run audit:items</code>)
A34	Building items are <code>construction_building_tier_<N></code>	VillageScene, VillageLedger
A35	<code>/api/player?section=...&scope=core</code> → <code>player.housing.tier</code> (0-5), <code>player.villageAddons.totalSlots</code> (0-5), <code>installed[]</code> {slot, itemKey, name}	VillageScene.normaliseVillage
A36	Item names in the DOM match catalogue names after normalisation (with cloak aliases)	Inventory, quests, name scanner, atlas
A37	The wiki path <code>https://idleworlds.com/wiki/items/<slug></code>	Tooltip link

6.6 WHEN A GAME UPDATE BREAKS THE SKIN: DIAGNOSIS

6.6.1 Symptom → cause → fix

Symptom	Likely cause	Diagnose	Fix location
Skin appears 4 s late on every load	Hydration probe cannot find React's root (A1)	Console: boot after page hydration: timeout (4000ms)	<code>hydration-signal.js</code> container list or field path; or ask for a public signal
Skin never appears	Extension not loaded; kill switch <code>false</code> ; host changed; Chrome <111; exception at boot	<code>chrome://extensions</code> errors; the content-script context shows [IW Fantasy Skin] logs; <code>chrome.storage.local.get('iw-skin-enabled')</code>	Manifest <code>matches</code> ; enable the key
A whole panel is unframed	<code>.panel</code> removed or renamed (A7); heading renamed (A20)	Page context: <code>document.querySelectorAll('.panel').length</code> ; check <code>[data-iw-ui="section-frame"]</code>	<code>UIFoundation.classifySectionFrames</code> , <code>SECTION_FRAME_NAMES</code>
Skill cards look stock	<code>.compact-panel</code> change (A8); verbs or identities changed (A24)	In page context, inspect the card for <code>data-fs-skill</code> ; if absent → detection; if present but no <code>data-iw-skill-layout</code> → structure (A17)	<code>DOMWatcher.detectSkillType</code> ; <code>SkillPanelRenderer.annotateStructure</code>
Skill card partly broken (misplaced title, wrong button)	Role mis-resolution after a markup change	List roles: <code>[...card.querySelectorAll('[data-iw-skill-role]')].map(e => [e.dataset.iwSkillRole, e.textContent.trim()])</code>	<code>annotateStructure</code> resolvers; the candidate memo (H-09)
Inventory rows show raw text and overlay together	New native child not suppressed; row class change (A9)	Check <code>data-fs-suppressed</code> on native children; the <code>data-fs-inv</code> signature	<code>InventoryRenderer.hideOriginalChildren</code>
Inventory shows ? icons everywhere	Atlas metadata failed, or names changed (A36)	Content-script console: [AtlasService] warnings	Atlas vendoring; <code>aliases.js</code> ; <code>AtlasService.resolve</code>
No tooltips, and plain item names	Catalogue failed (network, CORS on <code>www</code> , schema)	Content-script console: [ItemDatabase] warnings; Network panel for <code>items.json</code>	ItemDatabase (SEC-03, A33)
Quests unskinned	"Reward:"/"Turn In" copy changed (A26)	Card lacks <code>fs-quest-panel</code>	<code>QuestPanelRenderer.isQuestCard</code>

Symptom	Likely cause	Diagnose	Fix location
Header unskinned	Header text probes fail (A23) or classes changed (A11)	No <code>[data-iw-header="root"]</code>	<code>HeaderRenderer.findLiveHeader / classifyProfile</code>
Header shows three stacked boxes instead of one	<code>HeaderChrome</code> refused the structure (A18)	No <code>[data-iw-chrome="shell"]</code>	<code>HeaderChrome.resolve</code>
Theme stuck on "default"	Zone label copy changed (A22)	<code>document.documentElement.dataset.iwZoneTheme ; [data-iw-ui="zone-title"]</code> present?	<code>classifyZoneBar , currentZoneNumber</code>
Active tab or filter not highlighted	Active-state mechanism changed (A14)	Inspect native ARIA and classes on the active tab	<code>deriveTabActive ; inventory filter state ; compact selection</code>
A player name suddenly has a button plate	Text-link class substring changed (A13)	Inspect the name button's classes	Add a new exclusion to the generic control <code>:not()</code> chains
Progress bar jumps instead of gliding	Fill no longer uses inline <code>width %</code> (A5)	Inspect the fill element's <code>style</code>	<code>findCurrentActionProgress ; progressPercent ; V2 FILL</code> selector
Village scene says "could not be loaded"	API shape, status or league header changed (A35, A6)	Network: <code>/api/player</code> status and JSON	<code>normaliseVillage ; leagueOf ; B-08</code>
Modal pop-ups unframed	Overlay layer shape changed (A16)	<code>document.querySelectorAll('[data-iw-overlay]')</code>	<code>OverlayFramer.isScrim</code> thresholds
Page becomes slow after a game update	A classifier cache stopped validating (the <code>[]</code> . <code>every()</code> trap), or a new mutation loop	Perf harness; Performance panel shows <code>flushPending</code> every frame on an idle page; <code>flush-quiescence</code> fixture with the new shape	Compare-before-write; cache coverage checks (<code>docs/traps/performance.md</code>)
Everything looks right, but a game control stopped working	Skin CSS hides or blocks a new control (<code>pointer-events , display</code>), or a mirror covers it	Kill switch: does the control work with the skin off? DevTools → inspect the element at the click point	The responsible CSS rule (Appendix B lookup by selector); rule 1 fix

6.6.2 Diagnostic toolkit

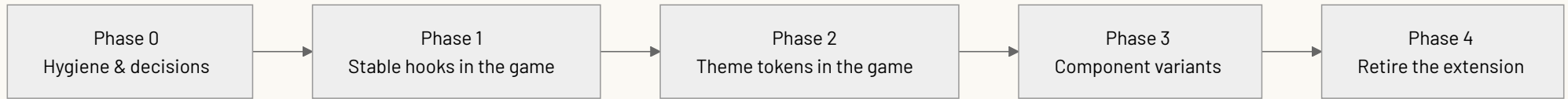
Tool	Use
Kill switch (§6.1.4)	A/B test the same live state with and without the skin
Content-script console context	Skin logs and warnings (<code>[IW Fantasy Skin] <label>: <message></code> , once per label)
Attribute inspection	All skin marks are readable <code>data-iw-*</code> / <code>data-fs-*</code> attributes (Appendix D lists every one with its owner)
<code>claude/capture-*.js</code> DevTools snippets (for example <code>capture-inventory.js</code> , <code>capture-skill-cards.js</code> , <code>capture-header.js</code>)	Read-only dumps of live DOM and computed styles to JSON, for reproducing live shapes in fixtures. Review a snippet before pasting it into a logged-in console , and never share dumps that include player data
<code>node claude/audit-mobile.mjs --widths 360,390,430,768</code>	Phone and tablet layout audit to <code>tmp/mobile-audit/</code> [Project record: <code>docs/traps/mobile.md</code>]
<code>node claude/perf-harness.mjs --runs 3 [--noskin --bundle <file>]</code>	Main-thread cost on a ticking 15k-node page to <code>tmp/perf/</code>
<code>npm run audit:items</code>	Validates the live <code>items.json</code> contract (network)
<code>build-tools/audit-header-live.mjs</code>	Live header audit using a local browser profile (SEC-07: use a test account)
Appendix B/C	Find which rules target a selector or hook; check specificity and <code>!important</code>

6.6.3 Troubleshooting FAQ

- "The skin works on apex but not on `www.idleworlds.com`." Check the `items.json` request in the Network panel (CORS). Check whether `www` redirects to apex. See SEC-03.
- "My CSS change does nothing."
 1. Is the rule in a sheet that is imported, and did you rebuild and reload the extension?

2. Is it beaten by a later rule (Appendix B, sort by selector), a renderer's inline `!important` (steer via `var()`), or the generic control rule (add a `:not()` exclusion)?
 3. Is the element's role mark present?
- **"A test passes locally, but the live page differs."** Fixtures omit live shapes or sheets [**Project record:** `docs/traps/test-harness.md`]. Capture the live DOM with a `claude/capture-*.js` snippet and model that shape.
 - **"Suites fail intermittently on my machine."** `smoke`, `skill-card-system` and `quest-command-width` are timing-sensitive under load (project memory). Re-run them alone before debugging.
 - **"The bundle check fails in CI."** Run `npm run build` and commit `dist/content.bundle.js`.
 - **"Icons blank after bumping ASSET_REVISION."** Run `npm run vendor`, commit the assets, and confirm `item_icons_index.csv` columns (the `AtlasService` validation warns).
-

6.7 LONG-TERM IMPROVEMENT AND MIGRATION PLAN



Phase	Goal	Work	Exit criteria
0. Hygiene and decisions	Make the current skin safe to share and cheaper to maintain	- H-20 (remove secrets from the tree); - H-01, H-18, H-02, H-06-H-10, H-12; - delete dead CSS (§3.13 step 1); - owner decisions on SEC-04, CONT-01, SEC-09, FUN-01; - fix doc drift D-01...D-10.	npm test green; css-superseded.md shows 0 rows; zip < 90 MB; decisions recorded
1. Stable hooks (game change, small)	Remove copy-based detection	The game renders <code>data-*</code> hooks on: - panels(<code>data-panel="inventory"</code> , ...)and cards(<code>data-card="skill"</code> , <code>data-skill="mining"</code> , <code>data-card="quest"</code>); - roles inside cards (identity, title, readout, action, pager, materials, requirement, reward); - nav tabs(<code>data-route</code>), zone bar parts, activity panel parts, village slots, boss cards; plus a public hydration event. Update the classifiers to prefer hooks (keeping text fallbacks for one release)	Detection works with localisation; <code>hydration-signal.js</code> deleted; A20-A32 no longer required
2. Theme tokens (game change, medium)	Remove <code>!important</code> and specificity escalation	- Move the Ashen Iron and per-zone palettes into the game's theme variables. - Set <code>data-zone-theme</code> from game state. - Use cascade layers or the Tailwind theme so skin CSS wins without <code>!important</code> . - Serve art from the game CDN. - Split the atlases (PERF-07).	<code>!important</code> count below 5% of declarations; BackgroundPainter deleted; no text scan for zones
3. Component variants (game change, large)	Move the classifier work into React components	Implement the variants(<code>Panel variant="forged"</code> , <code>SkillCard</code> , <code>QuestCard</code> , <code>InventoryRow</code> , <code>BossCard</code> , <code>VillageSummary</code> , <code>ItemTooltip</code> , <code>CollapsiblePanel</code> , <code>Button variant="compact"</code>), using the existing CSS as the visual spec. Resolve A11Y-01..03 with real markup	Each surface renders the fantasy look with the extension disabled; per-component tests and visual regression
4. Retire the extension	One implementation	Ship a theme setting; migrate stored preferences if desired; archive the repository	The extension is no longer needed; the store listing (if any) is unpublished

Suggested mapping from skin marks to component props (the CSS in `src/styles` can keep working during migration if components emit the same attributes):

Skin mark (today)	Component prop or attribute (future)
<code>data-iw-ui="section-frame" + section-title</code>	<code><Panel variant="forged" title="..."></code>
<code>fs-skill-panel</code> , <code>data-fs-skill</code> , <code>data-iw-skill-role / -zone</code>	<code><SkillCard skill="mining" slots={{ identity, title, commands, materials }}></code>
<code>data-iw-skill-v2-*</code>	<code>SkillCard</code> internal state (level, percent, fill duration from the tick interval, timer)
<code>fs-quest-panel</code> , <code>data-iw-quest-*</code>	<code><QuestCard state="ready" discipline="..." objective={item}></code>
<code>.fs-inv-row overlay + data-fs-*</code>	<code><InventoryRow item={item} owned={ownedDetails} actions={...}></code>
<code>data-iw-panel="current-action" + parts</code>	<code><CurrentAction progress={p} tickMs={t} queue={...}></code>
<code>data-iw-collapse*</code>	<code><CollapsiblePanel persistKey="..."></code>
<code>data-iw-compact-button + layers</code>	<code><Button variant="compact" selected></code>
<code>data-iw-overlay</code>	<code><Modal variant="forged"></code>
<code>html[data-iw-zone-theme]</code> + inline art variables	<code><ThemeRoot zoneTheme={theme}></code> with CSS variables

6.8 PACKAGE-SIZE CLEAN-UP (SIZE-01)

- In `build-tools/package-release.mjs` , extend `EXCLUDE` with:
 - `/^assets\/skills-ui\/buttons\/exact-v3\/` ;

- `/(^|\\)(registration|verification)\\.json$/;`
 - the non-runtime index and provenance files: `assets/skills-ui/buttons/card-v6/index.json`, `assets/skills-ui/buttons/compact-ghost-v3/index.json`, `assets/quest-frames/index.json`, `assets/skills-ui/zone-theme-map.json`, `assets/village/buildings.json`, `assets/skills-ui/action-icons/index.json` [Verify each is not fetched: the census lists the only runtime fetches, N3–N5];
 - `assets/skills_xp_plaque_wide.webp`.
2. Remove the matching `web_accessible_resources` glob for `exact-v3` in `manifest.json`. The package step fails if a pattern matches no file.
 3. Optionally delete the `else` branch in `SkillsArtService.applyThemeVariables` and the folder itself (keep the generator script if the art might return).
 4. Run `npm run build && npm test && npm run package`. Load the zip unpacked, open each zone theme (or force `data-iw-zone-theme` per theme in DevTools), and confirm button art renders.

Expected result: the zip shrinks by about 60 MB (about 42% of 142.8 MB).

6.9 DECISION CHECKLIST FOR IDLEWORLDS

#	Decision	Context
1	May the skin (or its successor) read <code>/api/player</code> , and at what rate? Or will IdleWorlds provide village data another way?	SEC-04, B-08
2	Are the World Boss notice, orb explanation, drop-rate texts and terms ("Dominion Ward", "Queued") acceptable, or should they come from the game?	CONT-01, CONT-02, B-06
3	Keep, move or remove the Toolkit link?	SEC-09, B-07
4	Should the XP-cycle button be restored in the new skill card design?	FUN-01, B-01
5	Is re-hosting the game's item, gear, village and housing art inside the extension permitted?	§1.8, C23
6	Is AI-generated art acceptable under the game's content policy?	§1.8
7	Which reduced-motion behaviour is intended for the game's own animations?	FUN-03, B-03
8	Which integration option (A/B/C) and timeline?	§6.2, §6.7
9	Distribution channel (unpacked, private store listing, public listing) and who owns updates?	PKG-01
10	Font licence attribution (SIL OFL) in releases	§1.8

6.10 PROVING THAT A CHANGE PRESERVES VISUALS AND FUNCTIONALITY

Definition of done for any change to `src/`:

1. `npm run build`, then `npm test`, **green**. Commit `dist/content.bundle.js`.
2. `npm run fixtures`: compare the rendered fixtures at 320/360/390/430/600/768 px (and desktop) with the previous run (keep the last good screenshots). Any difference must be intended.
3. For CSS changes, regenerate `tools/css-inventory.mjs` and `tools/css-superseded.mjs`, and review the diff of Appendix B and `css-superseded.md`. No new host-hook dependency without a note.
4. For classifier changes:
 - add or extend a suite with **a negative control** (break the fixture on purpose and confirm the test fails) [**Project record**: `CLAUDE.md` "Verification discipline"];
 - include the flush-quiescence check if the change writes to the DOM.
5. **Live verification by a person** (no automated session can do this). Use the checklist in §7.8, with the kill switch for A/B comparison. Record exactly which kind of verification was done: "built", "fixture-rendered", or "live-verified".
6. Functionality spot-checks with the skin on:
 - every control on the changed surface works with mouse, keyboard (Tab, Enter, Space) and touch;
 - disabled states still look disabled;

- active, equipped, met/unmet and team colours still differ.

7. TESTING AND VERIFICATION

This chapter covers:

- what the existing automated tests prove (and do not prove);
- the results of running them for this handover;
- a manual test plan for every major feature;
- responsive, cross-browser, accessibility, performance, regression and security testing;
- recommended new tests;
- a final verification checklist.

Kinds of verification, stated honestly. For this handover:

- the test suite was **run** (§7.2);
- the source was **read**;
- nothing was **live-verified**: no session could load the real game with a real account.

Every "expected result" below is what the source and fixtures predict. A person must confirm it on the live game [**Project record**: CLAUDE.md "Say which kind of verification you did"].

7.1 TEST ARCHITECTURE

- **No framework.** Each `tests/*.test.mjs` is a standalone Node script that fails with a non-zero exit code or a thrown assertion. Run one with `node tests/<name>.test.mjs`.
- `npm test` rebuilds the bundle (`--allow-missing-assets`), runs `node --check dist/content.bundle.js`, then runs `npm run test:run`: the 47 suites in a fixed order, then `build-tools/audit-sprite-windows.mjs`.
- **A new suite must be appended to `test:run` by hand**, or it never runs in CI.
- **Engines:**
 - **jsdom**: DOM logic and classification, fast; no layout, so rects are zero.
 - **Playwright/Chromium**: real layout, cascade, computed styles, animations, the extension loading. Requires `npm playwright install chromium`.
 - **node**: pure logic and static invariants.
- **Fixtures** model live DOM shapes captured with the `claude/capture-*.js` snippets. A harness that boots the bundle must put `data-iw-page-hydrated="1"` on `<html>`, or boot waits out the 4 s gate [**Project record**: docs/traps/test-harness.md].
- **Loading the built bundle.** Most suites load `dist/content.bundle.js`, not `src/`, so build before running a single suite.
- **Negative controls.** Several suites prove they can fail. Examples from the log: "previous implementation wrong on 2/4 cases (control works: this test can fail)", and the sprite audit fails if it parses zero windows [**Test run**].
- **CI** runs `npm test` on Node 24 with Playwright Chromium, then `git diff --exit-code dist/content.bundle.js`.

7.1.1 Suite catalogue

#	Suite	Engine	What it pins	Features
1	data-services	node + stubs	ItemDatabase cache freshness, stale fallback, validators, legacy localStorage eviction; AtlasService partial loads and backoff	F-42, F-43
2	inventory-model	node	Owned-item detail parsing (upgraded cloaks, rolled stats)	F-40
3	item-display	node	Category and tier classes, stat chips and rows	F-40, F-44

#	Suite	Engine	What it pins	Features
4	static-invariants	node	Architecture rules asserted against source (injection order and count, one observer, breakpoints aligned to Tailwind, bundle budget warning, ...)	F-03, F-04, F-55
5	hydration-gate	jsdom	Gate protocol: already hydrated, late latch, timeout, disabled, latch removal	F-01
6	extension-e2e	Playwright	The real unpacked extension: manifest, both content-script worlds, web_accessible_resources, assets load, no page errors or skin console errors	F-01, F-03
7	smoke	jsdom	Boots the built bundle on a synthetic IdleWorlds-shaped DOM: mounts, Mine → Fish cache invalidation, native inline styles restored, all styles removed on disable, re-enable without duplicates, repeated disable, game text nodes never detached	F-02, F-04, F-07–F-45
8	route-swap-reclassify	jsdom	Surfaces re-classify when routes swap panels	F-12, F-16, F-19, F-21–F-23, F-49
9	overlay-framer	jsdom	Scrim/card detection, negative controls, teardown	F-53
10	zone-themes	node	Generated zone map in step with JSON; every theme asset exists	F-09
11	button-art-states	Playwright	Themed button art: all themes, asset exposure, hover and clicked switching	F-09, F-27, F-32
12	revised-button-atlas	Playwright	Nine revised atlases, exact alpha registration, sprite windows, stable geometry, disabled controls, with all sheets plus inline !important shorthands	F-09, F-32, F-36, F-38
13	compact-button-atlas	Playwright	Compact ghost atlas windows and states	F-11
14	compact-button-reconcile	Playwright	Native text replacement, ARIA-only selection, quiet idle, teardown	F-11
15	compact-button-boot	Playwright	Real bundled startup and reactivation path for compact buttons	F-02, F-11
16	corner-filigree-density	Playwright	1× fallback and 2× themed corner sheets select the same logical corners	F-12
17	separator-symmetry	Playwright	The inventory title ornament is truly bilateral	F-41
18	action-progress-theme	Playwright	Current Action and Zone Control rails, zone palette, animation, mobile, reduced motion	F-21, F-48, F-54
19	skill-action-countdown	Playwright	Long-action countdown lifecycle	F-34
20	card-button-art	Playwright	V2 card button art: nine themes, three states, both arrows, stable geometry, native content and progress, disabled	F-32, F-33, F-36
21	zone-headers	node	All zone header files present and referenced	F-15
22	inventory-root	jsdom	Inventory root resolution on the live shell shape	F-41
23	header-hidden-duplicate	jsdom	The header resolves the visible copy, not the hidden duplicate	F-05, F-14
24	skill-structure-signature	jsdom	Real structureSignature (extracted from source): insensitive to ticks, sensitive to structure	F-26

#	Suite	Engine	What it pins	Features
25	skill-card-v2	jsdom	V2 derivations from roles (22 checks)	F-31-F-35
26	skill-card-v2-wiring	node	V2 wiring invariants	F-31-F-35
27	skill-card-v2-render	Playwright	V2 layout and rendering defects (32 checks)	F-31-F-36
28	skill-card-system	Playwright	Cross-checked card rules with all sheets (124 checks)	F-26-F-37
29	ingredient-entries	jsdom	Material entry boundaries (textContent joining)	F-30
30	quest-structure-signature	jsdom	Quest signature behaviour. Tests a stale copy of the logic (TEST-01)	F-38
31	quest-card-detection	jsdom	Quest recognition on whitespace-free markup	F-38
32	world-boss-panels	jsdom	Boss decoration, rewards, control state, idempotence, teardown	F-46-F-48
33	zone-control-motion	Playwright	Ward motion, crossfades, mobile, reduced motion	F-48, F-54
34	tooltip-virtual-anchor	jsdom	Leaving a card opened on a virtual anchor	F-44, F-45
35	boss-action-label	Playwright	QUEUED/PREJOIN label centring	F-46
36	boss-participation-hit	Playwright	Participation link hit area and status row	F-46
37	village-panels	jsdom	Building art, slot roles, install picker, housing hero, teardown; generated table pinned to <code>buildings.json</code>	F-50
38	village-scene	jsdom	Placement, real slots, no-house state, rendered-route read, snapshot across tabs, league reset, idempotence	F-51
39	village-ledger	jsdom	Item join, per-building effects, building-only totals, disclosure, rebuild, late storage read	F-52
40	village-scene-layout	Playwright	The plot composition actually lays out	F-51
41	collapsible-frames	Playwright	Every parent frame folds alone; persistence; teardown (38 checks)	F-13, F-24
42	panel-frame-nesting	Playwright	Leaf-versus-container frame nesting	F-12
43	viewport-swap	Playwright	Decorations follow the visible copy across 1280 px	F-05
44	quest-command-width	Playwright	Quest command rail frame follows the label	F-38
45	header-compact	Playwright	Nav rail, announcement, zone bar, merged chrome (34 checks)	F-14, F-16-F-20

#	Suite	Engine	What it pins	Features
46	<code>menu-rows</code>	Playwright	Single-row menus on phones and tablets	F-16, F-17, F-41, F-55
47	<code>flush-quiescence</code>	Playwright	A quiet page costs nothing: zero role writes and flushes when idle; busy toggle does not strip roles	F-04, all renderers
—	<code>audit-sprite-windows.mjs</code>	node	Every pixel sprite window lands on a named atlas cell with isotropic scale and a matching box; fails on zero windows	F-41

7.1.2 What the suites do not prove

- **The live game's DOM matches the fixtures.** Fixture drift is the main historical source of false passes [**Project record**].
- **Real server behaviour.** `items.json` on apex and `www`, CORS, and the `/api/player` statuses and payload.
- **CSP handling** on the live origin.
- **Assistive technology output.** Screen readers are not exercised.
- **Browsers other than Chromium**, and real phones (touch, GPU memory).
- **Operability of every game control** with the skin on. FUN-01 exists because no test asserts that a hidden game control stays reachable.
- **Long-session memory and performance on real hardware.** PERF-01, PERF-06 and PERF-07 are unmeasured.
- **Content accuracy** of skin-authored copy (CONT-01).

7.2 CURRENT RESULTS

Item	Result
Date	2026-09-17
Commit	059081671bf3af4490a025121f25f13ada142b7e (the source tree was clean)
Command	<code>npm test</code>
Exit status	0
Wall time	3 min 26 s
Build	<code>dist/content.bundle.js</code> : 668,309 bytes, 48 source modules (including the CSS and JSON modules)
Syntax check	<code>node --check</code> passed
Suites	All 47 printed their pass markers, and the sprite-window audit passed ("2 pixel sprite window(s) checked against 12 atlas cells")
Check lines	826 lines beginning <code>ok</code> , plus the per-suite summaries (e.g. <code>skill-card-system</code> (124 checks), <code>skill-card-v2-render</code> (32 checks), <code>skill-card-v2</code> (22 checks))
Warnings	Bundle over the 300,000-byte soft budget (advisory, printed twice: by the build and by <code>static-invariants</code>)
Expected console noise	Suites that simulate offline or failing networks log <code>[ItemDatabase] Using stale cache after refresh failure: offline</code> , <code>[AtlasService] Partial load: ... 503</code> , "network disabled in smoke test" and similar. These are the scenarios under test, not failures
<code>dist</code> after the run	Unchanged (the rebuilt bundle is byte-identical to the committed one)
Evidence	<code>evidence/npm-test-2026-09-17.log</code> (local path replaced by <code><repo></code>)

Known flakiness [**Project record**: memory note on this machine]: `smoke` (burst timing), `skill-card-system` (timing) and `quest-command-width` can fail under heavy machine load, independent of code changes. All three passed in this run. Re-run a failing one alone before investigating.

7.3 MANUAL TEST PLAN

Setup for every case:

- Chrome ≥111 with the extension loaded unpacked (§6.1.2) and a logged-in test account.
- DevTools open. The kill switch is ready in the content-script console context (§6.1.4).
- Test at a desktop width (≥1280 px) unless the case says otherwise.

Failure indicators apply to every case. Any of these is a failure:

- a red error in the page console mentioning the extension;
- a `[IW Fantasy Skin]` warning in the content-script console;
- a game control that no longer works;
- a state (active, equipped, disabled, met/unmet, team colour) that is no longer visually distinct.

ID	Feature(s)	Steps	Expected result	Failure indicators
MT-01	F-01	Hard-reload the game 3 times. Watch the content-script console	boot after page hydration: 1 (<ms>) each time, typically well under 4000 ms; skin appears; no React #418 error in the page console attributable to the skin	timeout (4000ms) or unknown ; #418 errors that disappear with the skin disabled
MT-02	F-02	Set <code>iw-skin-enabled</code> to <code>false</code> . Inspect the DOM. Set it to <code>true</code>	Off: no <code>style[data-iw-style]</code> , no <code>[data-iw-ui]</code> , no <code>.fs-inv-row</code> ; page looks stock; game works. On: skin returns and nothing is duplicated (one <code>#iw-tip</code> , one Toolkit link, 7 styles)	Leftover marks or styles; duplicated nodes; collapsed panels missing titles after re-enable (FUN-04)
MT-03	F-03, F-07	Page context: list <code>style[data-iw-style]</code> . Check fonts in DevTools → Rendering → font list	7 styles in order; Cinzel, Barlow and Crimson Text served from <code>chrome-extension://</code>	Missing or out-of-order styles; fonts 404
MT-04	F-04	Leave the game idle with an action running for 60 s. Performance panel recording for 10 s	Smooth progress; no long tasks from the skin; <code>flushPending</code> only when the game mutates	Flush every frame while idle; growing memory
MT-05	F-05	Resize across 1279 ↔ 1281 px, then 1023 ↔ 1025 px	Skins follow the visible layout immediately; no unskinned copy appears	Stock panels after crossing; decorations on the hidden copy
MT-06	F-08	Compare surface colours with the skin on and off; open a menu or dialog	Dark warm surfaces; no navy panels left; translucent overlays still translucent [Verify: FUN-11]	Navy remnants; opaque overlays hiding content
MT-07	F-09, F-15	Travel to zones in at least 3 different themes (e.g. zones 1, 4, 22). Then open Market	Palette, corners, separators, button art and header painting change per zone; Market keeps the last zone's theme	<code>data-iw-zone-theme="default"</code> on the Game route; stale header art
MT-08	F-10, F-11	Hover, press and tab through: nav tabs, inventory filters and pager, loadout I/II, Change Zone, chat Send, a modal's buttons	Themed plates; hover and pressed art; focus ring visible; selected loadout and active filter remain distinct; disabled look distinct	A control with no focus ring; selection indistinguishable; misaligned layers
MT-09	F-12, F-13	For each panel: check the frame. Collapse, reload, expand. Collapse two panels	Frames on every panel; collapse leaves a single bar with the title; state persists across reload; panels fold independently	Title missing on the collapsed bar; wrong panel folds; toggle overlaps content
MT-10	F-14, F-18	Inspect the header at 1440, 1024, 860, 768 and 390 px. Click "Players online" and each utility button	Crest, glass identity, utility plates and status grid; announcement bar; all buttons work	Overlapping tiles; player name recoloured; dead buttons
MT-11	F-16, F-17	Visit each route. Check the active tab. Click Toolkit. Narrow to <768 px	Active tab highlighted on each route; Toolkit opens a new tab without navigating the game; hidden below 768 px	Wrong tab active; the game tab navigates away
MT-12	F-19, F-20	On the Game route, check the zone bar and merged chrome. Use Zones and Next zone	One merged box (zone title and notice; nav and actions); buttons work; tones distinct (teal Zones, ember Next)	Stacked boxes (the structure changed); tone swapped
MT-13	F-21	Start a skill action; watch Current Action for 3 cycles. Queue an action	Bar glides continuously and snaps back at completion; queue row styled; queue button works	Bar steps or slides backwards; jitter

ID	Feature(s)	Steps	Expected result	Failure indicators
MT-14	F-22, F-23	Open Action Log, click View All; send a chat message with Enter and with Send	Log rows styled with timestamps; chat scrolls; composer works with keyboard and button	Messages not sent; feed height broken
MT-15	F-24	With a Daily XP Boost active, look at the Skill Actions heading	Green boost chip	Missing chip (copy changed)
MT-16	F-25–F-30	For each discipline card (Combat, Mining, Smithing, Herbalism, Alchemy, Jewelcrafting, Spellcrafting, Tailoring, Woodcutting, Construction, Crafting, Fishing, locked): check identity, title, materials and buttons; page recipes	Correct accent and medallion per discipline; title without emoji; met materials green, unmet amber; pager works	Wrong discipline art; unmet shown as met; pager dead
MT-17	F-29 (FUN-01)	On a card, try to click the level line "Lv N ..." or its V2 equivalent	Current code: the game's cycle-XP control cannot be reached (FUN-01). Record whether the game offers another way to cycle	Confirms FUN-01; decide B-01
MT-18	F-31–F-35	Run an action; watch the V2 button fill; run a >11 s action; view a card with an unmet requirement; narrow the card column	XP ring matches the percentage; fill glides; countdown mirrors Current Action; ⚠ requirement note; label fits (not below readable size, A11Y-03)	Countdown differs from Current Action; label overflow or 4 px text
MT-19	F-38, F-39	On quests: check the sigil, reward chip and command rail with Turn In All (n). Hover and focus the objective; press Enter, then Escape	Sigil shows the objective item art and %; the rail frame fits the label; hovercard opens and closes, focus returns	Stretched frame ends; hovercard won't close; Turn In not clickable
MT-20	F-40, F-41	Inventory: each filter; pages; equip and unequip; set actions; upgraded gear; items with sockets; phone width	Rich rows with correct icon, "+N" badge, chips and details; native actions work; equipped marked; active filter lit; single-row filter bar	Raw text visible next to the overlay; ? icons for known items; action clicks miss
MT-21	F-42–F-44	Hover icons and names; keyboard-open a card; click Wiki ↗; check the footer source label; test near viewport edges and on a short window	Card within the viewport; Wiki opens in a new tab; "live data" or "cached data" label; Escape closes	Card off-screen; stale label persists; clicks swallowed
MT-22	F-45	In Action Log or chat, rest the pointer on an item name mentioned in text; on a phone, tap it	Hovercard opens for the exact word. No visual underline (FUN-02)	Card for the wrong item; game click cancelled on tap
MT-23	F-46, F-47	World Bosses: art, labels, notice; toggle Possible Rewards; reload; focus a reward tile, press Enter	Portrait per boss; QUEUED or PREJOIN label matches the game state; disclosure persists; tile opens its card	Label contradicts the game state; notice text wrong (CONT-01)
MT-24	F-48	Zone Control during red, blue and contested states (or fixtures)	Crest, meter and factions match the game's numbers; impact plays on change (not under reduced motion)	Team colours swapped (FUN-12); meter disagrees with numbers
MT-25	F-49	Market: listings, empty slots, name buttons, steppers	Glass rows; empty slots dim; stepper colours unchanged	Red/green steppers recoloured
MT-26	F-50	Village route: housing hero; each slot; open the install picker; hover and focus building names; Install, Uninstall, Upgrade, Destroy (on a test account)	Art per building and house; tier pips; picker option art stays visible across updates (FUN-06); actions work	Option art flickers or disappears; wrong art
MT-27	F-51, F-52	Dashboard: Village scene after visiting the Village route; after 5 min; on an SSF character; with network blocked (DevTools offline)	Correct plots and house; ledger totals equal the sum of installed buildings; SSF shows the SSF village; offline shows the "last loaded" or "could not be loaded" copy	Wrong league's village; totals include vacant or house
MT-28	F-53	Open Players Online and any confirmation modal; scroll inside	Dim blurred scrim; forged frame on the card; frame does not scroll away; modal controls work	Frame on an inner sub-panel (FUN-07); scrim opaque
MT-29	F-54	Enable OS "reduce motion"; reload	No skin animations or transitions; ward loops stop; note whether game spinners also stop (FUN-03)	Animations still running
MT-30	F-55	Run MT-09, MT-10, MT-11, MT-16, MT-19, MT-20 and MT-27 at 360, 390, 430, 768 and 1024 px, and on a real phone	No horizontal overflow; tap targets usable (collapse and pager); menus in one row; Toolkit hidden on phones	Overflow; overlapping controls; unreachable buttons

7.4 RESPONSIVE TESTING

Width / device	Why	Focus
360, 390, 430 px (phones)	Skin-specific $\leq 430/420/400/384/380$ rules	Skill and quest cards, inventory rows, tooltip full width, composer stacking, Village scene
640/641 px	Tailwind <code>sm</code> ; skill and quest phone layouts	Card rails, activity panel gaps
760/761, 767/768 px	Boss and village phone rules; mobile header art; Toolkit hidden	Boss grid, header art swap, nav
860/861 px	Header stacking; merged chrome stacking	Header grid, chrome rows
1023/1024/1025 px	Frame padding and corners; inventory density	Frames, inventory chips
1279/1280/1281 px	Hidden duplicate column swap; nav and filter font fitting	Viewport correctness (MT-05)
≥ 1536 px	Largest breakpoint	Wide layouts
Touch device (Android Chrome)	<code>(pointer: coarse)</code> and <code>(hover: none)</code>	Tap-to-toggle hovercards, collapse hit area, pager height
Browser zoom 200%	Reflow	Text readable; no clipped controls

Tools:

- `npm run fixtures` (320–768 px renders, short-viewport tooltip scroll);
- `node claude/audit-mobile.mjs --widths 360,390,430,768` (live-site audit, see `docs/traps/mobile.md`; use a test account);
- DevTools device toolbar.

Expected: no horizontal scroll; no overlapping interactive elements; single-row menus below 1280 px.

7.5 ACCESSIBILITY TESTING

Test	How	Expected	Known issues
Keyboard only	Tab through the header, nav, a skill card, a quest, inventory, the World Bosses rewards, the Village ledger and collapse toggles; operate with Enter/Space; Escape closes hovercards	Every native control reachable in a sensible order; focus visible; hovercards return focus	FUN-01 (XP-cycle control unreachable); prose item names not reachable (by design)
Focus visibility	Check the ring contrast on dark plates	Visible ring on all controls	The 1px <code>base.css</code> ring may be faint on some plates [Verify]
Screen reader (NVDA + Chrome on Windows; TalkBack on Android)	Read skill cards, quest cards, boss action buttons, the Village ledger	Each control's name matches its visible purpose; content read once	A11Y-01 (duplicates; "QUEUED" shown but "Prejoined" read); A11Y-02 (focusable <code><p></code> without a role)
Automated scan	axe DevTools or Lighthouse on Game, Inventory, Village and World Bosses with the skin on and off	No new critical violations with the skin on	Expect findings for small text (A11Y-03) and possibly contrast [Verify]
Contrast	Measure dim and faint text tokens on grounds (<code>--iw-dim</code> , <code>--iw-faint</code> over <code>--iw-ink-*</code>)	WCAG AA 4.5:1 for body text	[Verify]: not measured for this handover
Target size	Measure collapse toggles, pager arrows and reward tiles on touch	$\geq 24 \times 24$ px (WCAG 2.2 AA 2.5.8)	Covered by <code>(pointer: coarse)</code> rules; verify on devices
Reduced motion	MT-29	No skin motion	FUN-03 affects the game's motion
Zoom and reflow	200% and 400% zoom, 320 px CSS width	Content usable	—

7.6 RECOMMENDED NEW AUTOMATED TESTS

ID	Test	Type	Pins	Negative control
T-01	Game-control operability: with all sheets loaded on the skill card fixture, every native <code>button</code> in a card is visible (<code>checkVisibility</code>), has <code>pointer-events: auto</code> , and receives a synthetic click via <code>elementFromPoint</code> at its centre	Playwright	FUN-01; future rule-1 regressions	Fails today on the <code>level-progress</code> button (confirm it fails, then decide B-01)
T-02	Tooltip escaping: a record with markup in every text field (including <code>craft_level</code> , <code>tier</code> , <code>wiki_slug</code>) opens a card; assert no element children beyond the template and the literal text is present	jsdom	SEC-01, SEC-08	Revert H-01 and confirm failure
T-03	Catalogue validation: malformed records (null, numbers as names, objects as tiers) are skipped with one warning; valid records are indexed	node	SEC-02	—
T-04	Kill-switch round trip with a collapsed panel: collapse → disable → enable → assert the title is visible and <code>data-iw-collapse-spine</code> exists	Playwright	FUN-04	Fails before H-08
T-05	Village picker art stability: decorate a slot with the picker open, bump the DB revision twice, and assert option art is present after each pass	jsdom	FUN-06	Fails before H-10
T-06	Same-panel re-annotation: annotate a card, replace its title element (new node), re-render the same card, and assert the role is on the new node	jsdom	FUN-08	Fails before H-09
T-07	Kill switch during the hydration wait: set <code>false</code> 1s into a 4s wait, and assert the skin does not boot	jsdom	H-07	Fails today
T-08	Forged events ignored: dispatch <code>iw:skill-panel</code> and <code>iw:dom-flush</code> from page-world script with junk <code>detail</code> , and assert no warnings and no work (flush counter unchanged)	Playwright	SEC-05, H-23	—
T-09	Quest signature on the real module: extract <code>textLeaves / structureSignature</code> from <code>QuestPanelRenderer.js</code> (like <code>skill-structure-signature</code>) and update the expectations to the current contract (<code>disabled</code> excluded, (n) stripped)	jsdom	TEST-01	Mutate the source function and confirm failure
T-10	web_accessible_resources completeness: every <code>url(.. /assets/...)</code> in the CSS and every <code>assetUrl()</code> constant path is covered by a manifest pattern and exists in the package	node	H-18, H-19	Remove a pattern and confirm failure
T-11	Pointer-move cost: synthetic mouse moves over a large text container on the perf fixture; assert the per-frame cost of <code>handlePointer</code> stays below a budget	Playwright + perf harness	PERF-06	Disable H-15
T-12	Zone scan cost on non-Game routes: a Market-shaped page with no zone bar; count <code>querySelectorAll('div,span,p,strong')</code> calls per flush	Playwright	PERF-02	—
T-13	Dead-CSS gate: <code>tools/css-superseded.mjs</code> reports 0 rows (after clean-up)	node	MAINT-01	Add a duplicate declaration
T-14	Trigger attribute ownership: a game <code><p></code> with pre-existing <code>tabindex="-1"</code> and <code>aria-describedby="x"</code> becomes a trigger, opens and closes, is torn down; the native values are restored	jsdom	FUN-09, H-11	Fails today
T-15	Village API statuses: mock 401, 429 with <code>Retry-After</code> , 500 and malformed JSON; assert backoff, messages and no tight retry	jsdom	SEC-04, H-14	—
T-16	Reduced-motion scope: with reduced motion, a game element with <code>animation: spin 1s infinite</code> keeps animating while skin animations stop	Playwright	FUN-03 after B-03	Fails today by design
T-17	Font tokens resolved: computed <code>font-family</code> of the quest brief and boss reward name equals the intended token	Playwright	FUN-05 after B-04	Fails today
T-18	Atlas memory smoke: load the inventory fixture and record <code>performance.measureUserAgentSpecificMemory()</code> (cross-origin-isolated harness) or Chrome's task-manager value manually	Manual or Playwright	PERF-07	—

7.7 CROSS-BROWSER, PERFORMANCE, REGRESSION AND SECURITY TESTING

7.7.1 Cross-browser

Browser	Status	What to check
Chrome stable (Windows, macOS, Linux)	Target	Full manual plan
Chrome 111 (minimum)	Effective floor	Smoke: MT-01, MT-03, MT-07, MT-16, MT-20 (<code>color-mix</code> , <code>world: "MAIN"</code>)
Chrome for Android	Target (touch)	MT-21, MT-22, MT-30
Edge, Brave, Opera (Chromium)	Untested [Verify]	MT-01-MT-03 smoke; extension install path differs
Firefox, Safari	Unsupported (MV3 <code>world: "MAIN"</code> , <code>chrome.*</code>)	Not applicable

7.7.2 Performance

Test	Command / method	Expected	Failure indicator
Main-thread cost on a ticking page	<code>node claude/perf-harness.mjs --runs 3 vs --noskin</code>	Skin overhead within the budget recorded in <code>docs/traps/performance.md</code> [Project record]	Regression versus the previous run's numbers (keep them in <code>tmp/perf/</code>)
Quiescence	<code>node tests/flush-quiescence.test.mjs</code>	0 flushes and 0 role writes on an idle page	Any non-zero
Boot cost	DevTools Performance: reload with a 200-row inventory	No long task > 200 ms attributable to the skin [Assumption: target]	Style recalc storms (per-row <code>getComputedStyle</code>)
Scroll smoothness	Scroll inventory and chat on a phone	No jank; no scroll-anchoring jumps on busy toggles	Jumps when the game disables buttons
Memory	Chrome Task Manager after 30 min on inventory and Village	Stable; atlas decode $\approx$ one-time (PERF-07)	Continuous growth
Animations	Performance monitor with Zone Control visible vs collapsed	Paint count stops when collapsed or hidden (after H-17)	Constant paints off-screen (PERF-01, today)

7.7.3 Regression

- Before a change:** run `npm test` and `npm run fixtures`; keep the fixture screenshots and `tmp/perf/` numbers.
- After a change:** repeat, compare the screenshots, and review the diffs of `tools/css-inventory.json` , Appendix B and `tools/css-superseded.md` .
- Live A/B:** use the kill switch on the same game state; run the MT cases for the affected features.
- After each IdleWorlds update:** run the §6.6.1 symptom scan (a 10-minute smoke over MT-01, MT-07, MT-09, MT-16, MT-19, MT-20, MT-23, MT-27).

7.7.4 Security

Test	How	Expected
API census gate	<code>node handoff/reskin-technical-handover/tools/js-api-census.mjs</code> , then diff <code>tools/js-api-census.md</code> in review	No new network, storage, HTML-sink, dynamic-code or navigation call sites without review
Escaping	T-02	Literal text only
Network destinations	DevTools Network panel over a 10-minute session (Game, Inventory, Village, dashboard)	Only <code>chrome-extension://...</code> , <code>items.json</code> and <code>/api/player</code> from the skin
Storage contents	Content-script console: <code>chrome.storage.local.get(null).then(o => Object.keys(o))</code>	Only the keys in §5.2.3
Page storage	Page context: <code>Object.keys(localStorage)</code> before and after enabling the skin	No new keys; the legacy <code>iw-item-db-cache</code> removed if present

Test	How	Expected
Forged events	T-08	No effect beyond work
CSP	If IdleWorlds tightens its CSP, run MT-01, MT-03 and MT-20 with the new header (§5.2.9 [Verify])	Styles, images and fonts still load
Packaging	<code>npm run package</code> ; unzip and list files	No secrets, no <code>.md</code> / <code>.html</code> , no profile data, no source maps
Dependency audit (dev tooling)	<code>npm audit --omit=optional</code>	No critical vulnerabilities in build or test dependencies (none ship)

7.8 FINAL VERIFICATION CHECKLIST

Use before a release or a handover acceptance. Each item records **how** it was verified (built, fixture, live).

Build and automated

- ☐ `npm ci` completes; `npm run build` produces `dist/content.bundle.js`, and `git diff dist` is empty after committing.
- ☐ `npm test` exits 0; there are no unexpected warnings beyond the bundle-size advisory.
- ☐ `npm run fixtures` renders; screenshots reviewed against the previous release.
- ☐ `tools/css-inventory.mjs`, `tools/css-superseded.mjs` and `tools/js-api-census.mjs` regenerated; diffs reviewed.
- ☐ `npm run package` succeeds. The zip contains no unreachable art (after H-18), no secrets and no source maps. Icons are present if distributing.

Live: activation and safety

- ☐ MT-01: boot after hydration without React errors attributable to the skin.
- ☐ MT-02: kill switch round trip clean (no residue except the documented `#iw-tip`, and no missing collapsed titles).
- ☐ Network: only the expected destinations (§7.7.4).
- ☐ Storage: only the expected keys.
- ☐ No game control lost (T-01 or a manual sweep); FUN-01 resolved or accepted.

Live: features

- ☐ Themes across at least 3 zones (MT-07); header art per zone.
- ☐ Frames and collapse on every panel (MT-09).
- ☐ Header, nav, Toolkit and zone bar (MT-10–MT-12).
- ☐ Activity panels (MT-13–MT-15).
- ☐ All skill disciplines and V2 behaviours (MT-16–MT-18).
- ☐ Quests (MT-19); inventory (MT-20); hovercards (MT-21–MT-22).
- ☐ World Bosses and Zone Control (MT-23–MT-24); Market (MT-25).
- ☐ Village route, scene and ledger, including an SSF character (MT-26–MT-27).
- ☐ Modals (MT-28); reduced motion (MT-29).

Live: responsive and accessibility

- ☐ MT-30 at 360/390/430/768/1024 px and one real phone.
- ☐ Keyboard-only pass (§7.5); screen-reader spot check on skill, quest and boss cards.
- ☐ Contrast and target-size checks recorded.

Decisions and documentation

- ☐ Owner decisions recorded for §6.9 items 1–5.

- ☐ Documentation drift D-01...D-10 fixed or accepted.
- ☐ SEC-07 : no browser profile or business documents inside any shared copy of the repository.

APPENDIX D — DOM CONTRACT

Everything the skin writes into, or reads back from, the shared page, grouped by owning module:

- attributes and classes;
- elements and IDs;
- inline custom properties;
- events;
- storage keys;
- CSS highlight registrations.

Use this appendix to answer "who owns this mark, what does it mean, and what removes it?".

Conventions

- **On:**
 - **game** — a node React renders;
 - **skin** — a node the skin created;
 - `<html>` — the document element, which is outside the observed `<body>`.
- **Cleared by:** the teardown function that removes the item.
- **Completeness.** The 141 distinct `data-iw-*/data-fs-*` names found by `tools/hook-census.mjs` all appear in this appendix. A script cross-checked this at the time of writing (§D.12).

D.1 LIFECYCLE, RUNTIME AND PAGE-LEVEL MARKS

Name	On	Written by	Values	Meaning	Cleared by
<code>data-iw-page-hydrated</code>	<code><html></code>	<code>hydration-signal.js</code> (MAIN world)	<code>1</code> , <code>unknown</code>	React root hydrated / shape unreadable	<code>HydrationGate</code> on release; <code>clearHydrationLatch</code> on teardown
<code>data-iw-style</code>	<code>skin</code> <code><style></code> in <code><head></code>	<code>StyleInjector.inject</code>	<code>base</code> , <code>tooltip-engine</code> , <code>inventory</code> , <code>skillpanel</code> , <code>header</code> , <code>overlay</code> , <code>ui-system</code>	Injected stylesheet identity	<code>StyleInjector.remove</code> <code>All</code>
<code>data-iw-zone-theme</code>	<code><html></code>	<code>HeaderRenderer.applyZoneTheme</code>	9 theme names or <code>default</code>	Current zone palette	<code>clearHeaderRenderer</code>
<code>data-iw-button-atlas</code>	<code><html></code>	<code>SkillsArtService.applyThemeVariables</code>	<code>revised-v5</code>	Button art set in use	<code>clearThemeVariables</code> (via <code>clearHeaderRenderer</code>)
<code>data-iw-compact-atlas</code>	<code><html></code>	<code>SkillsArtService.applyThemeVariables</code>	<code>compact-ghost-v3</code>	Compact button art set in use	same
<code>data-iw-skill-card-design</code>	<code><html></code>	<code>SkillCardDesignController.initSkillCardDesignController</code>	<code>new</code>	V2 card design active	<code>clearSkillCardDesignController</code>
<code>data-iw-skill-design-toggle</code>	(retired)	—	—	Legacy design toggle node; removed on sight	<code>initSkillCardDesignController</code> / <code>clear...</code> remove any
<code>data-iw-painted</code>	<code>game</code> surfaces	<code>BackgroundPainter.paintElement</code>	<code>1</code>	Surface colours repainted	<code>clearBackgroundPainter</code>

Name	On	Written by	Values	Meaning	Cleared by
data-iw-overlay	game scrim / card	OverlayFramer.tagScrim	scrim, panel	Modal layer / its content card	clearOverlayFramer
data-iw-order-handle	(none)	—	—	Retired Rearrange feature. It remains only inside CSS :not() chains, for specificity	n/a

Inline custom properties on <html> (all written by HeaderRenderer.applyZoneTheme and SkillsArtService.applyThemeVariables, and cleared by clearHeaderRenderer):

- iw-zone-atlas, --iw-corner-filigree, --iw-zone-separator;
- iw-compact-atlas;
- iw-<kind> and --iw-<kind>-paint for 12 kinds: action-idle, action-hover, action-clicked, action-secondary-idle, action-secondary-hover, action-secondary-clicked, chevron-prev-idle, chevron-prev-hover, chevron-prev-clicked, chevron-next-idle, chevron-next-hover, chevron-next-clicked.

D.2 UIFoundation (CLASSIFIER HUB)

Name	On	Values	Meaning	Cleared by
data-iw-ui	game	main-nav, main-nav-shell, nav-tab, zone-bar, zone-title, zone-action, section-frame, section-title (also skill-panel, written by SkillPanelRenderer; also on the skin Toolkit link and Village scene)	Surface role	clearUIFoundation (document-wide)
data-iw-tab	game nav tab	tab label(game, market, ...)	Which route tab	clearUIFoundation
data-iw-state	game nav tab	active	Active route (from game semantics, then route, then colour)	clearUIFoundation
data-iw-zone-action	game zone buttons	zones, prev, next	Zone control tone	clearUIFoundation
data-iw-zone-link	game zone buttons	1	Secondary zone buttons rendered as text links	clearUIFoundation
data-iw-panel	game activity panel host; skin Village scene	current-action, action-log, world-chat, village-scene	Activity panel identity	clearUIFoundation; scene removal
data-iw-panel-part	game	header, header-title, header-tools, panel-control, progress, queue, feed, feed-row, chat-input, composer, send	Parts of activity panels	clearUIFoundation
data-iw-panel-header	game	split	Header layout variant	clearUIFoundation
data-iw-progress-reset	game progress track	1 (one frame)	Suppress the transition on completion	rAF; clearUIFoundation
data-iw-skill-boost	game	1	Daily XP Boost line	clearDailyBoost
data-iw-boss	game .compact-panel	card	World Boss card	untagBossCards
data-iw-market	game	root, row, namebtn	Market frame, listing row, name button	untagMarket

Name	On	Values	Meaning	Cleared by
data-iw-nav-link	skin <a>	toolkit	Toolkit link	cClearUIFoundation removes the element

Inline: `--iw-progress-duration` on the game Current Action track (removed by `cClearUIFoundation`).

Element: `a[data-iw-nav-link="toolkit"]` appended into the game nav track.

D.3 HeaderRenderer AND HeaderChrome

Name	On	Values	Meaning	Cleared by
data-iw-header	game	root, layout, identity-region, profile, profile-name, profile-meta, profile-online, brand, profile-title, utilities, utility-button, status-grid, status-card, announcement, zone-shell	Header parts	cClearHeaderRenderer
data-iw-header-card	game status card	1-based index	Tile position	cClearHeaderRenderer
data-iw-header-stat	game status card	combat, boost, timer, gold, other	Tile kind	cClearHeaderRenderer
data-iw-zone	game header root	zone number or fallback	Header painting key	cClearHeaderRenderer
data-iw-chrome	game	shell, nav, notice, zone-bar, zone-text, zone-actions	Merged header chrome placement	cClearHeaderChrome

- Inline** (game nodes), removed by `cClearHeaderRenderer`, which sweeps every element:
 - header root: `--iw-header-surface`, `--iw-header-surface-mobile`, `--iw-header-frame`, `--iw-header-frame-bar` (dead: no asset), `--iw-header-crest`, `--iw-header-divider`, `--iw-utility-frame`, `--iw-status-frame`;
 - nav: `--iw-nav-rail`, `--iw-nav-active`, `--iw-nav-idle`;
 - announcement: `--iw-announcement-frame`;
 - zone bar: `--iw-zone-frame`, `--iw-zone-scene`, `--iw-zone-button-active`, `--iw-zone-button-idle`, `--iw-zone-button-teal`.
- Element:** `span.fs-header-crest[aria-hidden]` **prepended** into the game identity region.

D.4 SkillPanelRenderer

Name	On	Values	Meaning	Cleared by
class fs-skill-panel, fs-skill--<type>	game .compact-panel	discipline	Skill card and accent	cClearPanelChrome
class fs-skills-section-frame	game Skill Actions frame	—	Framed Skill Actions panel	cClearSkillPanels
data-fs-skill	game card	discipline	Detected skill	cClearPanelChrome
data-fs-skill-label	game card	display label	Discipline label	cClearPanelChrome
data-fs-skill-flavour, data-fs-skill-rune	(retired)	—	Legacy marks, only deleted	cClearPanelChrome

Name	On	Values	Meaning	Cleared by
data-iw-skill	game card	discipline	Skill identity for CSS	clearPanelChrome
data-iw-skill-glyph	game card	glyph character	Discipline glyph	clearPanelChrome
data-iw-skill-role	game	identity, identity-level, identity-icon, action-title, level-progress, action-button, nav-button, nav-group, xp-gain, requirement, reward, action-detail, progress-track, progress-fill, ingredient	Semantic parts	clearStructu reRoles
data-iw-skill-zone	game	identity, content, commands	Three-zone grouping	clearStructu reRoles
data-iw-skill-layout-shell	game	1	Common shell of the zones	clearStructu reRoles
data-iw-skill-layout	game card	three-zone	Layout applied	clearStructu reRoles
data-iw-nav-direction	game pager button	prev, next	Pager direction	clearStructu reRoles
data-iw-req-state	game requirement	met, unmet	From the game's text colour classes (rule 5)	clearStructu reRoles
data-iw-btn-state	game button	primary, secondary, disabled, icon	Plate variant	clearPanelIn lineTreatmen t
data-iw-readout	game readout branch	1	Neutralised readout node	neutraliseRe adouts / clear
data-iw-progress-display	game level-progress	display text	Clean readout for CSS content	clear
data-iw-clean-text	game identity or title	clean label	Text for CSS ::before (A11Y-01)	clear
data-iw-ingr	game material text	1	Neutralised material node	neutraliseIn gredients / clear
data-iw-ingredient-list-source	game material line	1	Native line hidden by CSS; mirrored by the grid	clearIngredi entLists
data-iw-skill-ingredient-list	skin grid	1	Material grid	clearIngredi entLists
data-iw-ingredient-signature	skin grid	signature	Rebuild guard	removed with the grid
data-iw-ingredient-state	skin grid item	met, unmet	Material state	removed with the grid
data-iw-base-exp	skin chip	value	Base XP chip	removed with the chip
data-iw-skill-art	skin medallion span	discipline	Medallion art key	removed
data-iw-skill-art-ready, data-iw-skill-art-pending	skin medallion span	1	Art painted / waiting for the index	removed

- **Inline** (game nodes; !important, owned by 3 InlineStyleOwner's and restored exactly): button plate properties, readout neutralisation, material neutralisation.

- **Skin elements:** `span.fs-skill-medallion-art`, `span.fs-skill-identity-percent`, `span.fs-skill-identity-progress` > `span.fs-skill-identity-progress-fill`, `span.fs-skill-base-exp`, `div.fs-skill-ingredient-grid[role=list]` > `span.fs-skill-ingredient-item[role=listitem]` (inserted **after** the native material line).
- **Highlight:** `CSS.highlights['iw-skill-ingredient-met']`.

D.5 SkillsArtService (ON CARDS, FRAMES AND BOSS BUTTONS)

Name	On	Values	Meaning	Cleared by
<code>data-iw-skills-ui-ready</code>	game card, frame, boss action button	<code>1</code>	UI atlas variables present	<code>cClearPanel</code>
<code>data-iw-skills-atlas</code> , <code>data-iw-skills-atlas-index</code>	skin medallion span	atlas file, cell index	Painted sprite	removed with the span

Inline (game nodes), removed by `cClearUiVariables`:

- `--fs-skills-panel-texture`, `--fs-skills-ui-atlas`, `--fs-skills-nav-prev`, `--fs-skills-nav-next`;
- `--fs-ui<token>-size / -position` for `medallion-frame`, `nav-idle`, `nav-active`, `action-idle`, `action-disabled`, `corner`, `xp-plaque`, `separator`, `flourish`;
- `--fs-ui<action-idle|action-disabled>-cap-size|cap-l-position|cap-r-position|mid-size|mid-position`;
- `--fs-ui-action-cap-ratio`.

D.6 SkillCardDesignController (V2 CARD)

Name	On	Values	Meaning	Cleared by
<code>data-iw-skill-v2</code>	game card	<code>1</code>	V2 applied	<code>cClearSkillCardV2</code>
<code>data-iw-skill-v2-type</code>	game card	discipline	Glyph selection	<code>cClearSkillCardV2</code>
<code>data-iw-skill-v2-state</code>	game card	expanded	Card state (collapsed variants retired)	<code>cClearSkillCardV2</code>
<code>data-iw-skill-v2-section</code>	game content nodes	<code>requirements</code> , <code>materials</code> , <code>details</code> , <code>rewards</code> , <code>queue</code> , <code>sources</code>	Section classification (clip-hidden)	<code>cClearSkillCardV2</code>
<code>data-iw-skill-v2-label-fit</code>	game card	fit key	Label measurement cache key	<code>cClearSkillCardV2</code>
<code>data-iw-skill-v2-fill-reset</code>	game card	<code>1</code> (one frame)	Fill snap-back	<code>rAF</code> ; clear
<code>data-iw-skill-v2-long-action</code>	game card	<code>1</code>	Countdown mode	<code>cClearLongActionTimer</code>
<code>data-iw-skill-v2-level-readout</code>	skin span	<code>1</code>	Level badge	removed
<code>data-iw-skill-v2-action-glyph</code>	skin span	<code>1</code>	Discipline glyph (sibling of the button)	removed
<code>data-iw-skill-v2-action-timer</code>	skin glyph	<code>1</code>	Glyph shows the countdown	<code>cClearLongActionTimer</code>
<code>data-iw-skill-v2-action-label</code>	skin span	<code>1</code>	<code>aria-hidden</code> label mirror	removed
<code>data-iw-skill-v2-body</code>	skin div	<code>1</code>	Detail body	removed
<code>data-iw-skill-v2-row</code>	skin div	<code>body</code> , <code>tabs</code>	Grid row of body or foot	removed
<code>data-iw-skill-v2-body-signature</code>	skin body	signature	Rebuild guard	removed
<code>data-iw-skill-v2-body-state</code>	skin body row	<code>met</code> , <code>unmet</code> , <code>neutral</code>	Row state	removed

Name	On	Values	Meaning	Cleared by
data-iw-skill-v2-body-kind	skin body row	material	Row kind	removed
data-iw-skill-v2-controls	skin div	1	Foot row	removed
data-iw-skill-v2-req-note	skin span	1	Unmet requirement note (role=note)	removed
data-iw-skill-v2-summary, data-iw-skill-v2-break, data-iw-skill-v2-expand, data-iw-skill-v2-tabs, data-iw-skill-v2-tab	(retired)	–	Earlier designs' nodes, removed on sight	removeRetiredNodes
data-iw-skill-v2-summary-state	(none)	–	Dead CSS selector only (skillcard-v2.css:216)	n/a

Inline (game card), removed by clearSkillCardV2: --iw-skill-v2-progress, --iw-skill-v2-fill-duration, --iw-skill-v2-action-label-font.

D.7 QuestPanelRenderer

Name	On	Values	Meaning	Cleared by
class fs-quest-panel	game .compact-panel	–	Quest card	clearCard
data-fs-quest	game card	1	Decorated quest	clearCard
data-iw-quest-role	game	kicker, title, brief, objective, reward, progress-label, progress-track, progress-fill, turn-in, skip	Parts	clearRoles
data-iw-quest-zone	game	commands, content, row, body	Zones	clearRoles
data-iw-quest-state	game card	ready, work-order, active	Quest state (from Turn In disabled / Skip presence)	clearCard
data-iw-quest-reward	game reward line	reward text	Chip text for CSS content	clearCard
data-iw-quest-percent	game card	number	Completion % for the sigil	clearCard
data-iw-quest-glyph	skin sigil	glyph	Discipline glyph (::before content)	removed
data-iw-quest-icon	skin sigil	1	Objective sprite painted	removed
data-iw-quest-icon-pending, data-iw-quest-art-pending	game card	1	Waiting for the atlas or art index	clearCard

- **Objective trigger** (on the game objective <p>): see D.9. Removed by clearObjectiveTrigger (FUN-09).
- **Inline** (game card): --fs-quest-accent, --fs-quest-cmd-w, plus the D.5 variables.
- **Skin elements**: span.fs-quest-sigil[aria-hidden] > span.fs-quest-sigil-icon + span.fs-quest-sigil-pct (inserted as the **first child** of the game body zone).

D.8 InventoryRenderer AND AtlasService

Name	On	Values	Meaning	Cleared by
data-fs-inv	game row	row signature	Rendered-overlay guard	clearInventoryRenderer

Name	On	Values	Meaning	Cleared by
<code>data-fs-hidden</code>	(legacy)	—	Pre-1.6 mark; only removed	<code>clearInventoryRenderer</code>
<code>data-fs-preserved-action</code>	game	<code>control</code> , <code>host</code>	Native action kept live / its host	<code>restoreOriginalChildren</code>
<code>data-fs-action-host</code>	game	<code>1</code>	Host branch of a preserved action (<code>display: contents</code>)	same
<code>data-fs-action-kind</code>	game control	<code>equipped</code> , <code>equip</code> , <code>set</code> , <code>secondary</code> , <code>icon</code>	Action plate variant (rule 5)	same
<code>data-fs-suppressed</code>	game	<code>1</code>	Native text branch hidden (plus inline <code>display: none</code>)	same; the inline style owner restores <code>display</code>
<code>data-iw-inventory-root</code>	game panel	<code>1</code>	Inventory frame	<code>clearInventoryRenderer</code>
<code>data-iw-inventory-title</code>	game heading	<code>1</code>	Title	same
<code>data-iw-inventory-list</code>	game list	<code>1</code>	Recessed list frame	same
<code>data-iw-inventory-control</code>	game	<code>filter</code> , <code>page</code> , <code>icon</code> , <code>glyph</code> , <code>page-count</code>	Tool row parts	same
<code>data-iw-inventory-filter-state</code>	game filter	<code>active</code>	Active filter (from game semantics)	same
<code>data-iw-inventory-filters</code>	game filter row	<code>1</code>	Single-row container	same
<code>data-iw-atlas</code>	skin icon host	<code>gear</code> , <code>item</code>	Which atlas painted	removed with the overlay
<code>data-iw-badge</code>	skin badge span	level	"+N" badge	removed

- **Skin elements:** `div.fs-inv-row` (overlay, **appended** into the game row); `div.fs-inv-rule[aria-hidden]` (after the header row); `span.iw-icon-badge`; `span.iw-item-ref[role=button][tabindex=0]` (inside the overlay).
- **Classes on skin nodes:** `fs-cat-*`, `fs-inv-orb`, `has-details`, `has-requirements`, `is-equipped`, `has-set-action`.

D.9 ITEM HOVERCARD TRIGGERS AND `TooltipEngine`

Name	On	Written by	Values	Meaning	Cleared by
<code>data-iw-item</code>	skin <code>.iw-item-ref</code> , skin inventory icon, skin boss reward tile, game quest objective <p>	InventoryRenderer, QuestPanelRenderer, WorldBossPanels, <code>TooltipEngine</code> <code>itemRef</code>	item id	Catalogue lookup key	owner's clear
<code>data-iw-item-name</code>	same, plus the game village building name <p>	same + <code>VillagePanels</code>	item name	Lookup key when there is no id	owner's clear
<code>data-iw-tooltip-trigger</code>	skin icon or tile; game objective and building <p>	same	<code>1</code>	Delegated hovercard trigger	owner's clear
<code>tabindex</code> , <code>aria-</code> <code>haspopup</code> , <code>aria-</code> <code>controls</code> , <code>aria-</code> <code>expanded</code>	game objective and building <p> (and skin triggers)	<code>QuestPanelRenderer</code> , <code>VillagePanels</code> (write); <code>TooltipEngine</code> (<code>aria-</code> <code>controls</code> / <code>aria-haspopup</code> / <code>aria-</code> <code>expanded</code> on show and hide)	<code>0</code> , <code>dialog</code> , <code>iw-</code> <code>tip</code> , <code>true</code> / <code>false</code>	Keyboard access and popup semantics	<code>clearObjectiveTrigger</code> , <code>clearTrigger</code> (unconditional , FUN-09)

Name	On	Written by	Values	Meaning	Cleared by
aria-describedby	any anchor	TooltipEngine <code>cleanupAnchor</code> (removal only)	—	Residue from pre-hovercard builds	— (FUN-09)
#iw-tip.iw-tip	skin div in <code><body></code>	initTooltipEngine	<code>role=dialog</code> , <code>aria-modal=false</code> , <code>tabindex=-1</code> , <code>aria-label</code> , <code>class is-open</code>	The shared hovercard	Not removed on teardown (FUN-10)

D.10 FRAMES, CONTROLS, OVERLAYS: CollapsibleFrames, CompactButtons

Name	On	Values	Meaning	Cleared by
data-iw-collapse	skin <code><button type=button></code> appended into the game head	1	Collapse toggle (<code>aria-expanded</code> , <code>aria-label</code> , <code>title</code>)	<code>clearCollapsibleFrames</code>
data-iw-collapse-head	game head row	1	Toggle container and gutter	same
data-iw-collapse-title	game title	1	Title kept in a collapsed bar	same
data-iw-collapse-spine	game ancestors of the title	1	Path kept visible when collapsed	same (the cache is not reset: FUN-04)
data-iw-collapsed	game panel or frame	1	Folded	same
data-iw-compact-button	game control	<code>text</code> , <code>icon</code>	Compact art shape	<code>clearCompactButtons</code>
data-iw-compact-selected	game control	<code>true</code>	Selected loadout (from game semantics)	same
data-iw-compact-layer	skin spans appended into game controls	<code>idle</code> , <code>hover</code> , <code>clicked</code>	Crossfade art layers (<code>aria-hidden</code>)	same

D.11 WORLD BOSSES, VILLAGE, VILLAGE SCENE AND LEDGER

Name	On	Values	Meaning	Cleared by
data-iw-encounter	game boss card	boss key or <code>zone</code>	Encounter identity	<code>clearWorldBossPanel</code>
data-iw-boss-role	game	<code>header</code> , <code>title</code> , <code>action</code> , <code>participation</code> , <code>difficulty</code> , <code>buff</code> , <code>timer</code> , <code>status</code> , <code>progress</code> , <code>details</code> , <code>control-strength</code> , <code>control-hp</code> , <code>control-progress</code>	Card parts	same
data-iw-boss-action-state	game action button	<code>active</code> , <code>idle</code>	Join state	same
data-iw-boss-action-label	game action button	<code>Queued</code> , <code>Prejoin</code>	Visual relabel via CSS content (CONT-02)	same
data-iw-boss-art-ready	game action button	1	Atlas variables applied	same (<code>clearPanel</code>)

Name	On	Values	Meaning	Cleared by
data-iw-boss-owned	skin nodes	1	Notice, art, rewards, Dominion	removed
data-iw-control	game zone card	red, blue, contested	Controlling team	same
data-iw-control-crest	skin crest	state	Crest art state	removed
data-iw-control-source	skin meter	factions	Meter derived from faction numbers	removed
data-iw-village	game	housing, house, addons, slot	Village panel kind	clearVillagePanel
data-iw-village-state	game slot	vacant, installed	Slot state	same
data-iw-village-role	game	head, copy, index, name, effects, vacant, actions, action, picker, option, option-owned, option-name, house-name, house-tier, house-salvage, house-next, house-max, house-note, intro, note	Parts	same
data-iw-village-action	game button	install, upgrade, uninstall, destroy, cancel, other	Action tone	same
data-iw-village-art	skin art	building, house, generic	Art kind	same
data-iw-village-tier	game housing card	0-5	House tier	same
data-iw-village-pip	skin pip	held, open	Tier pips	same
data-iw-village-owned	skin nodes	1	VillagePanels' own nodes	removed
data-iw-village-scene	skin section	1	Dashboard Village frame	clearVillageScene
data-iw-village-scene-owned	skin nodes	1	Scene and ledger nodes (distinct namespace, to avoid the VillagePanels sweep)	same
data-slot, data-plot-state	skin plot elements	1-5; empty, installed, locked	Plot placement and state	same
data-iw-village-ledger	skin	1	Ledger root	clearVillageLedger (with the scene)
data-iw-vs-toggle	skin buttons	1, section	Ledger disclosure buttons	same
data-iw-vs-open	skin entries	1, 0	Open state	same

Name	On	Values	Meaning	Cleared by
data-iw-vs-entry	skin entries	entry key	Building or home entry	same
data-iw-vs-slot-state	skin vacant lines	state	Vacant slot line	same
data-iw-vs-stat-kind	skin stat lines	skill-level, ...	Stat styling	same
data-iw-village-totals	skin totals block	1	Overall stats	same

Inline:

- `--iw-village-sprite` on skin art elements;
- `width` and `--iw-control-split` on the skin Dominion meter;
- the D.5 variables on game boss action buttons.

D.12 EVENTS, STORAGE, HIGHLIGHTS AND IDS

Kind	Name	Direction	Details
Event	iw:inventory-row, iw:skill-panel, iw:dom-flush, iw:name-scan-flush	DOMWatcher → consumers	§1.5.5
Event	iw:atlas-updated	AtlasService → consumers	{revision, gearReady, itemReady}
Event	iw:item-db-updated	ItemDatabase → consumers	{revision, generatedAt, source, count}
Storage (chrome.storage.local)	iw-skin-enabled, iw-item-db-cache, iw-item-db-cache-checked, iw-collapsed-frames, iw-village-ledger, iw-boss-rewards-collapsed:<bossKey>	—	§5.2.3
Storage (page localStorage)	iw-item-db-cache	removed once if present	Legacy
CSS highlight	iw-item-name	NameScanner	No stylesheet renders it (FUN-02)
CSS highlight	iw-skill-ingredient-met	SkillPanelRenderer	skillpanel.css:458
Element ID	iw-tip	TooltipEngine	Shared hovercard
Element IDs	iw-vs-entry<n>	VillageLedger	aria-controls targets

Completeness check (run at the time of writing): every name in `tools/hook-census.json` appears in this appendix. To repeat it:

```
node -e "const n=require('./handoff/reskin-technical-handover/tools/hook-census.json').map(e⇒e.name);const t=require('fs').readFileSync('./handoff/reskin-technical-handover/appendix-d-dom-contract.md','utf8');const m=n.filter(x⇒!new RegExp('(^[^a-z0-9-])'+x+'([a-z0-9-]|$)').test(t));console.log(m.length?'MISSING:'+m.join(', '):'all '+n.length+' names documented')"
```

D.13 GENERATED CENSUS

For each attribute name, the JavaScript files that mention it and the stylesheets that select on it: `tools/hook-census.md` (141 names). Regenerate it with `node handoff/reskin-technical-handover/tools/hook-census.mjs`.